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Chapter 1

Rings and Fields

1.1 DVR and Dedekind Domains

1.1.1 Absolute Values and Discrete Valuation Ring

Definition 1.1.1. *An absolute value on a field k is a map $|\cdot| : k \to \mathbb{R}_+$ s.t. for all $x, y \in k$,*

1. $|x| = 0$ *iff* $x = 0$
2. $|xy| = |x||y|$
3. $|x + y| \leq |x| + |y|$

If $|x + y| \leq \max(|x|, |y|)$ also holds, then the absolute value is nonarchimedean; otherwise it is archimedean.

e.g. The trivial absolute value on k is the map $|\cdot| : k \to \mathbb{R}_+$ s.t. $|x| = \begin{matrix} 1 & x \neq 0 \\ 0 & x = 0 \end{matrix}$.

Lemma 1.1.1. *An absolute value $|\cdot|$ on a field k is nonarchimedean iff $|1 + \cdots + 1| \leq 1, \forall n \geq 1$.*

pf. If $|\cdot|$ is nonarchimedean, $|1+\cdots+1| \leq \max(|1+\cdots+1|, |1|) \leq 1$ by the inductive hypothesis. Hence for all $n \geq 1$, $|1 + \cdots + 1| \leq 1$.
If $|\cdot|$ is not nonarchimedean, $\exists a, b$ s.t. $|a + b| > \max(|a|, |b|)$. Then $|1 + a/b| > 1$ so the inequality does not hold for all $n \geq 1$.

Corollary 1.1.1. *In a field of positive characteristic every absolute value is nonarchimedean, and the only absolute value on a finite field is the trivial one.*

Definition 1.1.2. *Two absolute values $|\cdot|, |\cdot|'$ on the same field k are equivalent if there exists $\alpha \in \mathbb{R}_{++}$ for which $|x|' = |x|^\alpha$, $\forall x \in k$.*

Definition 1.1.3. *For a prime p, the p-adic valuation $v_p : \mathbb{C} \to \mathbb{Z}$ is defined by $v_p(\pm\Pi_q q^{e_q}) \equiv e_q$ and $v_p(0) \equiv \infty$. The p-adic absolute value on $\mathbb{Q}$ is defined by $|x|_p \equiv p^{-v_p(x)}$ where $|0|_p = p^{-\infty} = 0$.*

Theorem 1.1.1 (Ostrowski's Theorem)**.** *Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_p$ for some $p \leq \infty$.*

pf. Let $|\cdot|$ be any nontrivial absolute value on $\mathbb{Q}$. Since it is nontrivial, $\exists p \in \mathbb{Z}$ prime s.t. $|p| \neq 1$. Then by the definition, given $|p| = c, |p^n| = |p|^n = c^n$. Then $|\cdot| = (|\cdot|_p)^\alpha$ s.t. $\alpha = -\log_p c \in \mathbb{R}_{++}$.

Theorem 1.1.2 (Product Formula)**.** *For every $x \in \mathbb{Q}^\times$, we have $\Pi_{p\leq\infty}|x|_p = 1$.*

pf. $\Pi|x|$ over all kinds of absolute value on $\mathbb{Q}$ is equivalent to $\Pi_{p\leq\infty}|x|_p^{\sum_{r\in\mathbb{R}_{++}} r}$ by the Ostrowski's theorem. Then $\Pi_{p\leq\infty}|x|_p =$ (products of all absolute values on $\mathbb{Q}$ for $x)^{-\sum_{r\in\mathbb{R}_{++}} r} \to 1$

Definition 1.1.4. *A valuation on a field k is a group homomorphism $k^\times \to \mathbb{R}$ s.t. for all $x, y \in k$, we have $v(x+y) \geq \min(v(x), v(y))$. We may extend v to a map $k \to \mathbb{R} \cup \infty\}$ by defining $v(0) \equiv \infty$. For any $0 < c < 1$, defining $|x|_v \equiv c^{v(x)}$ yields a nonarchimedean absolute valur. The image of v in $\mathbb{R}$ is the value group of v. We say that v is a discrete valuation if its value group is equal to $\mathbb{Z}$. Given a field k with valuation v, the set $A \equiv \{x \in k : v(x) \geq 0\}$ is the valuation ring of k. A discrete valuation ring(DVR) is an integral domain that is the valuation ring of its fraction field w.r.t. a discrete valuation. Such a ring A cannot be a field since $v(FracA) = \mathbb{Z} \neq \mathbb{Z}_+ = v(A)$.*

$x \in k^\times, v(1/x) = v(1) - v(x) = -v(x)$ so at least one of x and $1/x$ has nonnegative valuation and lies in A. Thus $x \in A$ is invertible iff $v(x) = 0 \leftrightarrow A^\times = \{x \in k : v(x) = 0\}$.

Definition 1.1.5. *A valuation ring is an integral domain A with fraction field k with the property that $\forall x \in k$, either $x \in A$ or $x^{-1} \in A$.*

For a valuation ring A, $\exists \pi \in A$ s.t. $v(\pi) = 1$ called a uniformiser(since $v(A) = \mathbb{Z}_+$). If we fix a uniformiser π, $\forall x \in k^\times, x = u\pi^n$ for unique u, n, i.e. A is a UFD. Every nonzero ideal of A is equal to $(\pi^n) = \{a \in A : v(a) \geq n\}$ for some integer $n \geq 0$, i.e. A is a PID. Thus ideals of A are totally ordered and the ideal $\mathbf{m} = (\pi) = \{a \in A : v(a) > 0\}$ is the unique maximal ideal of A.

Definition 1.1.6. *A local ring is a commutative ring with a unique maximal ideal. The residue field of a local ring A with maximal ideal $\mathbf{m}$ is the field $A/\mathbf{m}$.*

e.g.

- For the p-adic valuation $v_p\mathbb{Q} \to \mathbb{Z} \cup \{\infty\}, \mathbb{Z}_{(p)} \equiv \{\frac{a}{b} : a, b \in \mathbb{Z}, p \not| b\}$ is the valuation ring with maximal ideal $\mathbf{m} = (p)$. The residue field $\mathbf{Z}_{(p)}/p\mathbb{Z}_{(p)} \cong \mathbb{Z}/p\mathbb{Z} \cong \mathbb{F}_p$.
- For any field k, the valuation $v : k((t)) \to \mathbb{Z} \cup \{\infty\}$ on the field of Laurent series over k defined by $v(\sum_{n \geq n_0} a_n t^n) = n_0$ where $a_{n_0} \neq 0$ has valuation ring $k[[t]]$.

Theorem 1.1.3. *For an integral domain A, TFAE*

- *A is a DVR*
- *A is a noetherian valuation ring that is not a field.*
- *A is a local PID that is not a field*
- *A is an integrally closed noetherian ring of dimension 1.*
- *A is a regular noetherian local ring of dimension 1.*
- *A is a noetherian local ring whose maximal ideal is nonzero and principal.*
- *A is a maximal noetherian ring of dimension 1.*

Definition 1.1.7. *Given a ring extension $A \subseteq B$, an element $b \in B$ is integral over A if it is a root of a monic polynomial in $A[x]$. The ring B is integral over A if all of its elements are integral over A.*

Proposition 1.1.1. *Let $\alpha, \beta \in B$ be integral over $A \subseteq B$. Then so are $\alpha + \beta, \alpha\beta$.*

pf. Let $\begin{matrix} f \in A[x] & f(x) = a_0 = a_1 x + \cdots + x^m. \\ g \in A[y] & g(y) = b_0 + b_1 y + \cdots + b^n. \end{matrix}$ s.t. $f(\alpha) = g(\beta) = 0$. Consider the case $A = \mathbb{Z}[a_0, \ldots, a_{m-1}, b_0, \ldots, b_{n-1}]$, $B = \frac{A[x,y]}{(f,g)}$ with α, β equal to the images of x and y in B respectively. Let k be the algebraic closure of the fraction field A, and let $\alpha_1, \ldots, \alpha_m$ be the roots of f in k and $\beta_1, \ldots, \beta_n$ be the roots of g in k. The polynomial $h(z) = \Pi_{i,j}(z - (\alpha_i + \beta_j))$ has coefficients that may be expressed as polunomials in the symmetric functions of the α_i, β_j equivalently the coefficients a_i, b_j of f and g respectively. Thus $h \in A[z]$ and $h(x + y) = 0$ so $x + y$ are integral over A. Similarly for $h(z) = \Pi_{i,j}(z - \alpha_i\beta_j), xy$ can be shown to be integral over A, Then any $A' \subseteq B'$ with homomorphisms $A \to A', B \to B'$ defined by $a_i \mapsto a_i, b_i \mapsto b_i, x \mapsto \alpha, y \mapsto \beta$, if $x + y, xy \in B$ are integral over A, then so are $\alpha + \beta, \alpha\beta \in B'$. Thus the proposition follows.

Definition 1.1.8. *Given a ring extension B/A, the ring $\tilde{A} = \{b \in B : b$ is integral over $A\}$ is the integral closure of A in B. When $\tilde{A} = A$, A is said to be integrally closed in B. For a domain A, its integral closure is its integral closure in its fraction field and A is integrally closed if it is integrally closed in a fraction field.*

Proposition 1.1.2. *If $C/B/A$ is a tower if ring extensions in which B is integral over A and C is integral over B, then C is integral over A.*

Corollary 1.1.2. *If B/A is a ring extension, then the integral closure of A in B is integrally closed in B.*

Proposition 1.1.3. *The ring $\mathbb{Z}$ is integrally closed.*

pf. Let $r/s \in \mathbb{Q}$ be a root of a polynomial $f \in \mathbb{Z}[x]$ s.t. $(r,s) = 1$. Then $(r/s)^n + a_{n-1}(r/s)^{n-1} + \cdots + a_0 = 0 \to r^n + a_{n-1}sr^{n-1} + \cdots + a_0 s^n = 0 \leftrightarrow r^n = -s(a_{n-1}r^{n-1} + \cdots + a_0 s^{n-1})$, which implies $s|r$. But they are coprime, so $s = \pm 1$ and therefore $r/s \in \mathbb{Z}$.

Corollary 1.1.3. *Every UFD is integrally closed. In particular every PID is integrally closed.*

e.g. Consider the ring $\mathbb{Z}[\sqrt{5}]$. Then $x^2 - x - 1 \in \mathbb{Z}[\sqrt{5}][x]$ has a root $\phi = \frac{1+\sqrt{5}}{2} \in Frac\mathbb{Z}[\sqrt{5}]$ but not in $\mathbb{Z}[\sqrt{5}]$ so it is not integrally closed and therefore it is not a UFD.

Proposition 1.1.4. *Every valuation ring is integrally closed.*

pf. Let A be a valuation ring with fraction field k and let $\alpha \in k$ be integral over A. Then $\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0, a_1 \ldots a_{n-1} \in A$. If $\alpha \notin A, \alpha^{-1} \in A$ since A is a valuation ring. Multiplying the equation by $\alpha^{-(n-1)} \in A, \alpha = -a_{n-1} - a_{n-1}\alpha^{-1} - \cdots - a_0\alpha^{1-n} \in A$, a contradiction. Thus A is integrally closed.

Definition 1.1.9. *A number field K is a finite extension of $\mathbb{Q}$. The ring of integers $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}$ in K.*

Proposition 1.1.5. *Let A be an integrally closed domain with fraction field K. Let α be an element of a finite extension L/K, and let $f \in K[x]$ be its minimal polynomial over K. Then α is integral over A iff $f \in A[x]$.*

pf. If $f \in A[x]$, α is integral over A.
Suppose α is integral over A and let $g \in A[x]$ be a monic polynomial s.t. $g(\alpha) = 0$. In $\bar{K}[x]$, we may factor $f(x) = \Pi(x - \alpha_i)$. For each α_i, we have a field embedding $K(\alpha) \to \bar{K}$ that sends α to α_i and fixes K. As elements of $\bar{K}$, we have $g(\alpha_i) = 0$, so $\forall \alpha_i \in \bar{K}$ is integral over A and lies in the integral closure $\tilde{A}$ of A in $\bar{K}$. Each coefficient of $f \in K[x]$ can be expressed as a sum of products of the α_i and is therefore an element of the ring $\tilde{A}$ that also lies in K. But $A = \tilde{A} \cap K$, since A is integrally closed in its fraction field K.

- Localisation of Rings

Let A be a commutative ring and let S be a multiplicative subset of A. The **localisation** of A w.r.t. S is a ring $S^{-1}A$ equipped with a ring homomorphism $\iota : A \to S^{-1}A$ that maps S into $(S^{-1}A)^\times$ and satisfied the universal property :

$$\begin{tikzcd} A \arrow[rr, "\phi"] \arrow[dr, "\iota"'] & & B \\ & S^{-1}A \arrow[ur, "\exists!"'] & \end{tikzcd}$$

if $\phi : A \to B$ is a ring homomorphism with $\phi(S) \subseteq B^\times$, then there is a unique ring homomorphism $S^{-1}A \to B$ that makes the diagram commutes.
$\to$Construct $S^{-1}A$ as the quotient of $A \times S$ modulo the equivalence relation $\sim$ s.t.
$(a,s) \sim (b,t) \leftrightarrow \exists u \in S$ s.t. $(at - sb)u = 0$. Then use a/s to denote the equivalence class of (a,s) and define $\iota(a) \equiv a/1$. If $\phi : A \to B$ is a ring homomorphism with $\phi(S) \subseteq B^\times . \phi = \pi \circ \iota$ where π is defined by $\pi(a/s) \equiv \phi(a)\phi(s)^{-1}$. Then such homomorphism $\phi(a)\phi(s)^{-1} = \pi(\iota(a))\pi(\iota(s))^{-1} = \pi(\iota(a)\iota(s)^{-1}) = \pi((a/1)(1/s)) = \pi(a/s)$ so π is unique.

If A is an integral domain, the map $\iota : A \to S^{-1}A$ is injective, allowing to identify A with its image $\iota(A) \subseteq S^{-1}A$. So we can view $S^{-1}A$ as an intermediate ring that lies between A and its fraction field $FracA$.

If $\phi : A \to B$ is a ring homomorphism and $\mathbf{b}$ is a B-ideal, then $\phi^{-1}(\mathbf{b})$ is an A-ideal called the contraction of $\mathbf{b}$ to A, denoted $\mathbf{b^c}$. If A is a subring of B and ϕ is an inclusion map, then $\mathbf{b^c} = \mathbf{b} \cap A$.
If $\mathbf{a}$ is an A-ideal, the B-ideal generated by $\phi(\mathbf{a})$ is the extension of $\mathbf{a}$ to B, denoted $\mathbf{a^e}$.
$\to$ if A is an integral domain, and $B = S^{-1}A$ and if $a_1, \ldots, a_n \in A$ generate $\mathbf{a}$ as an A-ideal, then they also generate $\mathbf{a^e} = \mathbf{a}B$ as a B-ideal since $\mathbf{a^e} = \mathbf{a}B \equiv \{ab : a \in \mathbf{a}, b \in B\}$.
$\to$If A is a noetherian ring/PID, then so is its localisation.

Theorem 1.1.4. *Let S be a multiplicative subset of an integral domain A. There is a one-to-one correspondence between the prime ideals of $S^{-1}A$ and the prime ideas of A that do not intersect S given by the inverse map $\mathbf{q} \mapsto \mathbf{q} \cap A, \mathbf{p} \mapsto \mathbf{p}S^{-1}A$.*

e.g.

- For a field k, let $A = k[x]$ and $\mathbf{p} = (x-2)$. Then $A_\mathbf{p} = \{f \in k(x) : f$ is defined at $2\}$. Since A is a PID, so is $A_\mathbf{p}$ with unique maximal ideal $\mathbf{p}A_\mathbf{p} = \{f \in k(x) : f(2) = 0\}$, hence it is a DVR. The residue field is $A_\mathbf{p}/\mathbf{p}A_\mathbf{p} \cong k$.

- Let $p \in \mathbb{Z}, \mathbb{Z}_{(p)} = \{a/b : a, b \in \mathbb{Z}, p \not| b\}$. Since $\mathbb{Z}$ is a PID so is $\mathbb{Z}_{(p)}$ and therefore it is a DVR. The residue field is $\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \cong \mathbb{F}_p$.

- Localisation of Modules

  Let A be a commutative ring, let S be a multiplicative subset of A, and let M be an A-module. Then the **localisation** $S^{-1}M$ of M w.r.t. S is an $S^{-1}A$-module equipped with an A-module homomorphism $\iota : M \to S^{-1}M$ with the universal property: If N is an $S^{-1}A$-module and $\phi : M \to N$ is an A-module homomorphism, then ϕ factors uniquely through $S^{-1}M$.
  $\to$ Construct $S^{-1}M$ as the quotient of the $S^{-1}A$-module $M \times S$ modulo the same equivalence relation above. Or $S^{-1}M$ can be constructed as the base change of M from A to $S^{-1}A$, i.e. $S^{-1}M \equiv M \otimes S^{-1}A$.

  If A lies in a field K and M is an A-module that is contained in a K-vector space, $\forall s \in A, s \neq 0$ is injective and we can view M as an A-submodule of any of its localisations $S^{-1}M$.

  Proposition 1.1.6. *Let A be a subring of a field K, and let M be an A-module contained in a K-vector space V. Then $M = \cap_{\mathbf{m}} M_{\mathbf{m}} = \cap_{\mathbf{p}} M_{\mathbf{p}}$ where $\mathbf{m}$ ranges over the maximal ideals of A, $\mathbf{p}$ ranges over the prime ideals of A and the intersection takes place in V.*

  pf. $M \subseteq \cap_{\mathbf{m}} M_{\mathbf{m}}$ is immediate.
  Suppose $x \in \cap_{\mathbf{m}} M_{\mathbf{m}}$ and consider the A-ideal $\mathbf{a} = \{a \in A : ax \in M\{$. For each maximal ideal $\mathbf{m}, x = m/s$ for some $m \in M, a \in A \setminus \mathbf{m}$. Then $sx \in M, s \in \alpha$ and $s \notin \mathbf{m}$ so $\mathbf{a} \not\subseteq \mathbf{m}$. Thus α must be a unit ideal, i.e. $1 \in \mathbf{a}$ and $x \cdot 1 = x \in M \to \cap_{\mathbf{m}} M_{\mathbf{m}} \subseteq M$.
  Since maximals are primes and for all prime ideal is contained in maximal ideals, $\cap_{\mathbf{m}} M_{\mathbf{m}} = \cap_{\mathbf{p}} M_{\mathbf{p}}$.
  $\to$ When $K = FracA, V = K$, M is an A-submodule of K. Every ideal I of A is an A-submodule of K, and can be localised as above. Thus the localisation of I at a prime ideal $\mathbf{p}$ of A is the same thing as the extension of I to the localisation of A at $\mathbf{p} \to I_{\mathbf{p}} = \{i/s : i \in I, s \in A \setminus \mathbf{p}\} = \{ia/s : i \in I, s \in A \setminus \mathbf{p}\} = IA_{\mathbf{p}}$,

  Corollary 1.1.4. *Let A be an integral domain. Every ideal I of A is equal to the intersection of its localisations at the maximal ideals of A, and also to the intersection of its localisations at the prime ideals of A.*

1.1.2 Dedekind Domains

Proposition 1.1.7. *Let A be a noetherian domain. TFAE*

1. *For every nonzero prime ideal* $\mathbf{p} \subset A$*, the local ring* $A_{\mathbf{p}}$ *is a DVR.*

2. *The ring* A *is integrally closed and* $\dim A \leq 1$.

pf. If A is a field, both 1 and 2 hold, so assume A is not a field, and $K = FracA$. $\dim A$ is the supremum of the length of all chains of prime ideals. Thus by the first theorem, every chain of prime ideals $(0) \subsetneq \mathbf{p}_1 \subsetneq \cdots \subsetneq \mathbf{p}_1$ extends to a corresponding chain in $A_{\mathbf{p}}$ of the same length. Conversely, every chain in $A_{\mathbf{p}}$ contracts to a chain in A of the same length. Thus $\dim A = \sup\{\dim A_{\mathbf{p}} : \mathbf{p} \in SpecA\} = 1$, since every $A_{\mathbf{p}}$ is a DVR or a field, in which case $\dim A_{\mathbf{p}} = 1$ and $\dim A_{\mathbf{p}} = 0$ respectively. Any $x \in K$ that is integral over A is integral over every $A_{\mathbf{p}}$, since they all contain A, and $A_{\mathbf{p}}$ is integrally closed since they are DVRs or fields. So $x \in \cap_{\mathbf{p}} A_{\mathbf{p}} = A$ and therefore A is integrally closed.
Suppose $x \in K$ is integral over $A_{\mathbf{p}}$. Then $x^n + \frac{a_{n-1}}{s_{n-1}} s^{n-1} + \cdots + \frac{a_0}{s_0} = 0$ for some $a_0, \ldots, a_{n-1} \in A, s_0, \ldots, s_{n-1} \in A \setminus \mathbf{p}$. Define $s = s_0 \ldots s_{n-1} \in S$. Multiplying both sides by s^n shows that sx is integral over A, hence $sx \in A$, since A is integrally closed. But $sx/s = x \in A_{\mathbf{p}}$ so $A_{\mathbf{p}}$ is integrally closed, given A is integrally closed. Also noetherian property, dimension, and reduced property are inherited from A. Thus $A_{\mathbf{p}}$ is an integrally closed noetherian local domain of dimension at most 1. Thus it is a DVR.

Definition 1.1.10. *A noetherian domain satisfying either of the equivalent properties of the previous proposition is called a Dedekind domain.*

Corollary 1.1.5. *Every PID is a Dedekind domain. In particular* $\mathbb{Z}$ *is a Dedekind domain, as is* $k[x]$ *for any field* k.

Definition 1.1.11. *A fractional ideal of a noetherian domain* A *is finitely generated* A*-submodule of its fraction field.*

Lemma 1.1.2. *Let* A *be a noetherian domain with fraction field* K*, and let* $I \subseteq K$ *be an* A*-module. Then* I *is finitely generated iff* $aI \subseteq A$ *for some nonzero* $a \in A$.

pf. If $r_1/s_1, \ldots, r_n/s_n$ generate I as an A-module, then $\exists a = s_1 \ldots s_n \in A$ s.t. $aI \subseteq A$.
If $aI \subseteq A$, then aI is an ideal of A, hence finitely generated, given A is noetherian. If $a_1, \ldots, a_n$ generate aI as an ideal, $a_1/a, \ldots, a_n/a$ generate I.
$\rightarrow$fractional ideals are A-submodule of K for which there exists a nonzero $r \in A$ s.t. $rI \subseteq A$ even when A is not noetherian.

Corollary 1.1.6. *Every fractional ideal of* A *can be written in the form* $\frac{1}{a} I$ *for some nonzero* $a \in A$ *and the ideal* I.

Definition 1.1.12. *A fractional ideal of* A *is principal if it is generated by one element, i.e. it has the form* xA *for some* $x \in A$.

Ideal Operations

- $I + J \equiv (i + j : i \in I, j \in J)$
- $IJ \equiv (ij : i \in I, j \in J)$
- $(I : J) \equiv \{x \in K : xJ \subseteq I\}$

Lemma 1.1.3. *Let I and J be fractional ideals of a noetherian domain A with $J \neq 0$. Then $(I : J)$ is a fractional ideal of A.*

pf. Suppose $I, J \subseteq A$ are ideals. Then for any nonzero $j \in J, j(I : J) \subseteq I \subseteq A$ so $(I : J)$ is finitely generated by the previous lemma. If I, J are not ideals, pick $a, b \in A$ s.t. $aI \subseteq A, bJ \subseteq A$ by the previous lemma. Then $(I : J) = (abI : abJ), abI, abJ \subseteq A$. Then by the previous argument it is finitely generated. Thus $(I : J)$ is an A-submodule of the fraction field of A s.t. it is finitely generated. So the lemma follows.

Definition 1.1.13. *A fractional ideal I is invertible if $IJ = A$ for some fractional ideal J.*

Lemma 1.1.4. *A fractional ideal I of A is invertible iff $I(A : I) = A$.*

pf. Suppose I is invertible with $IJ = A$. Then $jI \subseteq A$ for all $j \in J$ so $J \subseteq (A : I)$. Also $A = IJ \subseteq I(A : I) \subseteq A$ thus $I(A : I) = A$
If $I(A : I) = A$, by the previous lemma $(A : I)$ is a fractional ideal, it is invertible.

Definition 1.1.14. *The ideal group $\mathscr{I}_A$ of a noetherian domain A is the group of invertible fractional ideals.*

$\rightarrow$ every nonzero fractional ideal (x) is invertible, since $(x)^{-1} = (x^{-1})$. The set of nonzero principal ideals $\mathscr{P}_A$ is a subgroup of $\mathscr{I}_A$.

Definition 1.1.15. *Let A be a noetherian domain. The quotient $cl(A) \equiv \mathscr{I}_A/\mathscr{P}_A$ is the ideal class group of A; it is also called the Picard group of A and denoted $Pic(A)$.*

e.g. If A is a DVR with uniformizer π, then its nonzero fractional ideals are the principal fractional ideals $(\pi^n), n \in \mathbb{Z}$. Also $(\pi^m)(\pi^n) = (\pi^{m+n})$. Thus the ideal group of A is isomorphic to $\mathbb{Z}$ and $\mathscr{P}_A = \mathscr{I}_A \rightarrow cl(A)$ is trivial.

Lemma 1.1.5. *Let I, J be fractional ideals of a noetherian domain A, and let $\mathbf{p}$ be a prime ideal of A. Then $I_{\mathbf{p}}, J_{\mathbf{p}}$ are fractional ideals of $A_{\mathbf{p}}$ as are $(I + J)_{\mathbf{p}} = I_{\mathbf{p}} + J_{\mathbf{p}}$, $(IJ)_{\mathbf{p}} = I_{\mathbf{p}} J_{\mathbf{p}}$, $(I : J)_{\mathbf{p}} = (I_{\mathbf{p}} : J_{\mathbf{p}})$. The same applies if we localise w.r.t. any multiplicative subset S of A.*

pf. $I_{\mathbf{p}} = IA_{\mathbf{p}}$ is a finitely generated $A_{\mathbf{p}}$-module, since I is a finitely generated A-module. Thus a fractional ideal of $A_{\mathbf{p}}$ and similarly for $J_{\mathbf{p}}$
$(I+J)_{\mathbf{p}} = (I+J)A_{\mathbf{p}} = IA_{\mathbf{p}} + JA_{\mathbf{p}} = I_{\mathbf{p}} + J_{\mathbf{p}}$
$(IJ)_{\mathbf{p}} = (IJ)A_{\mathbf{p}} = I_{\mathbf{p}}J_{\mathbf{p}}$
$(I:J)_{\mathbf{p}} = \{x \in K : xJ \subseteq I\}_{\mathbf{p}} = \{x \in K : xJ_{\mathbf{p}} \subseteq I_{\mathbf{p}}\} = (I_{\mathbf{p}} : J_{\mathbf{p}})$.

Theorem 1.1.5. *Let I be a fractional ideal of a noetherian domain A. Then I is invertible iff its localisation at every maximal ideal of A is invertible, equivalently, iff its localisation at every prime ideal of A is invertible.*

pf. Suppose I is invertible. Then $I(A:I) = A$ and for any maximal ideal $\mathbf{m}$, $I_{\mathbf{m}}(A_{\mathbf{m}} : I_{\mathbf{m}}) = A_{\mathbf{m}}$ by the previous lemma. Thus $I_{\mathbf{m}}$ is invertible.
Suppose $I_{\mathbf{m}}$ is invertible for every maximal ideal $\mathbf{m}$. Then $I_{\mathbf{m}}(A_{\mathbf{m}} : I_{\mathbf{m}}) = A_{\mathbf{m}}$ for every maximal $\mathbf{m}$. Then we get $I(A:I) = \cap_{\mathbf{m}}(I(A:I))_{\mathbf{m}} = \cap_{\mathbf{m}} I_{\mathbf{m}}(A_{\mathbf{m}} : I_{\mathbf{m}}) = \cap_{\mathbf{m}} A_{\mathbf{m}} = A$ by the previous lemma. So I is invertible.

Corollary 1.1.7. *In a Dedekind domain, every nonzero fractional ideal is invertible.*

pf.If A is a Dedekind domain, then all of its localisation at maximal ideals are DVRs, hence PIDs and in a PID every nonzero fractional ideal is invertible. Thus by the previous theorem every nonzero fractional ideal of A is invertible.

Lemma 1.1.6. *A nonzero fractional ideal I in a noetherian local domain A is invertible iff it is principal.*

pf. If I is principal, it is invertible.
Let I be an invertible fractional ideal, and let $\mathbf{m}$ be the maximal ideal of A. Then we have $II^{-1} = A$ so $\sum_i a_i b_i = 1, a_i \in I, b_i \in I^{-1}, a_i b_i \in II^{-1} = A$. Then one of the products $a_i b_i$ must be a unit, otherwise the sum would not be a unit. Wlog, $a_1 b_1$ is a unit. $\forall x \in I, a_1 b_1 x \in (a_1)$ since $b_1 x \in A$. Thus $x = (a_1 b_1)^{-1} a_1 b_1 x \in (a_1)$ since $(a_1 b_1)^{-1} \in A$. Thus $I \subseteq (a_1) \subseteq I$, which implies $I = (a_1)$ is principal.

Corollary 1.1.8. *A nonzero fractional ideal in a noetherian domain A is invertible iff it is locally principal, that is its localisation at every maximal ideal of A is principal.*

An integral domain s.t. every nonzero ideal is invertible is a Dedekind domain.
pf. Let $(a_1) \subsetneq (a_1, a_2) \subsetneq \dots$ be an ascending chain of ideals. Then there exists $(a_0)^{-1} \supsetneq (a_1, a_2)^{-1} \supsetneq \dots$ which implies it terminates, i.e. it is Noetherian. For any prime $\mathbf{p}$, $A_{\mathbf{p}}$ is a local noetherian domain, and therefore all the invertible ideal is principal, which implies $A_{\mathbf{p}}$ is a DVR. Hence A is a Dedekind domain.

Lemma 1.1.7. *Let A be a Dedekind domain and let $a \in A$ be nonzero. The set of prime ideals that contain a is finite.*

pf. Define $\begin{array}{ll} S \equiv & \{I \in \mathscr{I}_A : (a) \subseteq I \subseteq A\} \\ T \equiv & \{I \in \mathscr{I}_A : A \subseteq I \subseteq (a)^{-1}\} \end{array}$, subsets of the ideal group $\mathscr{I}_A$. Then since $A \in S \cap T$, they are both nonempty and partially ordered by inclusion. The elements of S are all ideals and we have bijections $\begin{array}{rcl} \phi_1 : S & \to & T \\ I & \mapsto & I^{-1} \end{array}$, $\begin{array}{rcl} \phi_2 : T & \to & S \\ I & \mapsto & aI \end{array}$ with ϕ_1 order-reversing and ϕ_2 order-preserving. The composition $\phi \equiv \phi_2 \circ \phi_1$ is thus an order-reversing permutation of S. Since A is noetherian, the set S satisfies the a.c.c. By applying the order-reversing permutation ϕ, S also satisfies d.c.c.. If a lies in infinitely many distinct prime ideals $\mathbf{p_1}, \mathbf{p_2}, \ldots$, then $\mathbf{p_1} \supseteq \mathbf{p_1} \cap \mathbf{p_2} \supseteq \ldots$ is a descending chain of ideals of S that must stabilise. Thus for n sufficiently large, $\mathbf{p_1} \ldots \mathbf{p_{n-1}} \subseteq \mathbf{p_1} \cap \cdots \cap \mathbf{p_{n-1}} = \mathbf{p_1} \cap \ldots \mathbf{p_n} \subseteq \mathbf{p}_n$. The prime ideal $\mathbf{p}_n$ contains the product $\mathbf{p}_1 \ldots \mathbf{p}_{n-1}$ so it must contain one of the factors of $\mathbf{p}_1 \ldots \mathbf{p}_{n-1}$, which contradicts $\dim A \leq 1$.

Corollary 1.1.9. *Let I be a nonzero ideal of a Dedekind domain A. The number of prime ideals of A that contain I is finite.*

pf. Apply the previous lemma to all $a \in I$.

e.g. The Dedekind domain $A = \mathbb{C}[t]$ contains uncountably many nonzero prime $\mathbf{p}_r = (t - r), r \in \mathbb{C}$. But any nonzero $f \in \mathbb{C}[t]$ lies in only finitely many of them, i.e. f has finitely many roots.

Let $\mathbf{p}$ be a nonzero prime ideal in a Dedekind domain A with fraction field K, let π be a uniformizer for the discrete valuation ring $A_{\mathbf{p}}$, and let I be a nonzero fractional ideal of A. The localisation $I_{\mathbf{p}}$ is a nonzero fractional ideal of $A_{\mathbf{p}}$, hence of the form $(\pi^n), n \in \mathbb{Z}$ that does not depend on the choice of uniformizer. Extend the valuation $v_{\mathbf{p}} : K \to \mathbb{Z} \cup \{\infty\}$ to fractional ideals by defining $v_{\mathbf{p}}(I) \equiv n, v_{\mathbf{p}}((0)) \equiv \infty$. The map $v_{\mathbf{p}} : \mathscr{I}_A \to \mathbb{Z}$ is a group homomorphism (If $I_{\mathbf{p}} = (\pi^m), J_{\mathbf{p}} = (\pi^n)$, then $(IJ)_{\mathbf{p}} = (\pi^m)(\pi^n) = (\pi^{m+n})$).

Lemma 1.1.8. *Let $\mathbf{p}$ be a nonzero prime ideal in a Dedekind domain A. If I is an ideal of A, then $v_{\mathbf{p}}(I) = 0$ iff $\mathbf{p}$ does not contain I. In particular, if $\mathbf{q}$ is any nonzero prime ideal different from $\mathbf{p}$, then $v_{\mathbf{q}}(\mathbf{p}) = v_{\mathbf{p}}(\mathbf{q}) = 0$.*

pf. If $I \subseteq \mathbf{p}, v_{\mathbf{p}}(I) \geq v_{\mathbf{p}}(\mathbf{p}) = 1$ is nonzero.
If $I \not\subseteq \mathbf{p}, \exists a \in I \setminus \mathbf{p}$. Then $0 = v_{\mathbf{p}}(a) \geq v_{\mathbf{p}}(I) \geq v_{\mathbf{p}}(A) = 0$ since $(a) \subseteq I \subseteq A$.
The prime ideals $\mathbf{p}, \mathbf{q}$ are nonzero, hence maximal, given $\dim A \leq 1$. Thus neither contains the other, so by the previous argument, the lemma follows.

Corollary 1.1.10. *Let A be a Dedekind domain with fraction field K. For each nonzero fractional ideal I, we have $v_{\mathbf{p}}(I) = 0$ for all but finitely many prime ideals $\mathbf{p}$. In particular, if $x \in K^{\times}, v_{\mathbf{p}}(x) = 0$ for all but finitely many $\mathbf{p}$.*

pf. For $I \subseteq A$, by the previous lemma and the previous corollary, the corollary follows. For $I \not\subseteq A, I = \frac{1}{a}J, a \in A, J$ an A-ideal. Then $v_{\mathbf{p}}(I) = v_{\mathbf{p}}(J) - v_{\mathbf{p}}(a) = 0 - 0 = 0$ for all but finitely many prime ideals $\mathbf{p}$. This holds in particular for $I = (x), \forall x \in K^{\times}$.

Theorem 1.1.6. *Let A be a Dedekind domain. The ideal group $\mathscr{I}_A$ of A is the free abelian group generated by its nonzero prime ideal $\mathbf{p}$. The isomorphism $\mathscr{I}_A \cong \oplus_{\mathbf{p}}\mathbb{Z}$ is given by the inverse maps* $\begin{array}{ccc} I & \mapsto & (\dots, v_{\mathbf{p}}(I), \dots) \\ \Pi_{\mathbf{p}}\mathbf{p}^{e_{\mathbf{p}}} & & (\dots, e_{\mathbf{p}}, \dots) \end{array}$.

pf. By the previous corollary, the first map is well-defined. For each nonzero prime ideal $\mathbf{p}$, the maps $I \mapsto v_{\mathbf{p}}(I)$ and $e_{\mathbf{p}} \mapsto \mathbf{p}^{e_{\mathbf{p}}}$ are group homomorphisms, and it follows that the maps in the theorem are both group homomorphisms.
If $v_{\mathbf{p}}(I) = v_{\mathbf{p}}(J), I_{\mathbf{p}} = J_{\mathbf{p}}$. So if this holds for every prime $\mathbf{p}$, $I = \cap_{\mathbf{p}} I_{\mathbf{p}} = \cap_{\mathbf{p}} J_{\mathbf{p}} = J$. So the first map is injective.
By the previous lemma, for any vector $(\dots, e_{\mathbf{p}}, \dots)$ in the image, we have $v_{\mathbf{q}}(\Pi_{\mathbf{p}}\mathbf{p}^{e_{\mathbf{p}}}) = \sum_{\mathbf{p}} e_{\mathbf{p}} v_{\mathbf{q}}(\mathbf{p}) = e_{\mathbf{q}}$, which implies $\Pi_{\mathbf{p}}\mathbf{p}^{e_{\mathbf{p}}}$ is the pre-image of $(\dots, e_{\mathbf{p}}, \dots)$. Thus the first map is surjective and its inverse is the second map.
$\rightarrow$ When A is a DVR, the isomorphism in the previous theorem becomes a discrete valuation map $v_{\mathbf{p}} : \mathscr{I}_A \rightarrow \mathbb{Z}$ where $\mathbf{p}$ is the unique maximal ideal of A.

Corollary 1.1.11. *In a Dedekind domain, every nonzero fractional ideal I has a unique factorisation $I = \Pi_{\mathbf{p}}\mathbf{p}^{v_{\mathbf{p}}(I)}$ into nonzero prime ideals $\mathbf{p}$.*

Every integral domain with UFD is a Dedekind domain
pf. Let A be any integral domain with UFD. Since every ideal is a finite product of prime ideals, a.c.c. holds and hence noetherian. If $\dim A > 1, \exists \mathbf{p}, \mathbf{q}$ s.t. $(0) \subsetneq \mathbf{p} \subsetneq \mathbf{q}$. Since $\mathbf{q}$ contains $\mathbf{p}$ and both are primes, $\mathbf{q} = \mathbf{p}$ a contradiction. Hence it is a Dedekind domain.

A Noetherian domain A with "to contain is to divide" is a Dedekind domain.
pf. If $\dim A > 1, \exists \mathbf{p}, \mathbf{q}$ s.t. $(0) \subsetneq \mathbf{p} \subsetneq \mathbf{q}$. Then $\mathbf{q}|\mathbf{p}$. Then $\exists a \in \mathbf{q} \setminus \mathbf{p}$ nonzero s.t. $\exists b \in A$ with $ab \in \mathbf{p}$ a contradiction since $\mathbf{p}$ is a prime. Hence $\dim A \leq 1$. Let x be integral over A. If $x \notin A, x = \frac{a}{s}, a, s \in A$, $(a,s) = 1$. Then there is a minimal polynomial for $x \rightarrow (a/s)^n + \alpha_{n-1}(a/s)^{n-1} + \dots + \alpha_0 = 0. \forall \alpha_i \in A$. By multiplying both sides by s^n and rearranging it, the above equation becomes $a^n = -s(\alpha_{n-1}a^{n-1} + \alpha_{n-2}a^{n-2}s + \dots + \alpha_0 s^{n-1})$, which implies $s|a$. But $(s,a) = 1$ so $s = 1$ and $x \in A$ and it is integrally closed. Thus it is a Dedekind domain.

$x \in I \leftrightarrow (x) \subseteq I \leftrightarrow v_{\mathbf{p}}(x) \geq e_{\mathbf{p}}, \forall \mathbf{p}$ and therefore $I = \{x \in K : v_{\mathbf{p}}(x) \geq e_{\mathbf{p}}, \forall \mathbf{p}\}$.

Corollary 1.1.12. *A Dedekind domain is a UFD iff it is a PID, equivalently iff its class group is trivial.*

pf. Every PID is a UFD.
Let $\mathbf{p}$ be a nonzero prime ideal in a Dedekind domain A that is a UFD. Let $a \in \mathbf{p}$, nonzero, and let $a = p_1 \dots p_n$ be the unique factorisation of a into irreducible elements. Then $\mathbf{p}$ contains and therefore divides $(a) = (p_1)\dots(p_n)$ so $\mathbf{p}$ divides some (p_i), which is necessarily a prime ideal. But A has dimension 1, so $\mathbf{p} = (p_i)$. Given the fact that every prime ideals are principal and that A is a UFD, A is a PID.

Definition 1.1.16. *Two A-ideals I, J are said to be relatively prime or coprime if $I + J = A$.*

Lemma 1.1.9. *Let A be a Dedekind domain and let I, I' be nonzero ideals. There exists an ideal J copriem to I' s.t. Ij is principal.*

pf. Let $\mathbf{p_1}, \dots, \mathbf{p_n}$ be the nonzero prime ideals dividing I', which is finite by the previous corollary. For $1 \leq i \leq n$, define $\mathbf{a}_i \equiv \mathbf{p_1} \dots \mathbf{p_{i-1}} \mathbf{p_{i+1}} \dots \mathbf{p_n}$ and choose $a_i \in I$ so that $a_i \in \mathbf{a}_i I, a_i \notin \mathbf{p}_i I$. Then $\mathbf{a}_i I \cap \mathbf{p}_i I \subsetneq \mathbf{a}_i I$ because $v_{p_i}(\mathbf{a}_i I \cap \mathbf{p}_i I) = v_{p_i}(\mathbf{p}_i I) > v_{p_i} = v_{p_i}(\mathbf{a}_i I)$ so such a_i exists. Each a_i is necessarily nonzero, and satisfies $v_{p_i}(a_i) = v_{p_i}(I)$ since $v_{p_i}(a_i) \geq v_{p_i}(\mathbf{a}_i I) = v_{p_i}(I), v_{p_i}(a_i) < v_{p_i}(\mathbf{p}_i I) = v_{p_i}(I) + 1$ and for $i \neq j, v_{p_j}(a_i) \geq v_{p_j}(\mathbf{p}_j I) > v_{p_j}$. Define $a \equiv \sum_i a_i$, so that $v_{p_i}(a) = v_{p_i}(a_i) = v_{p_i}(I), \forall i \in [1, n]$ by the nonarchimedean triangular equality. Thus $v_p(a) = v_p(I)$ for all prime $\mathbf{p} | I'$. $(a) \subseteq I$ and therefore it is divisible by I given A a Dedekind domain. So $(a) = IJ$ for some ideal J. For each prime $\mathbf{p} | I'$, $v_p(J) = v_p(a) - v_p(I) = 0$ so J is coprime to I' and $IJ = (a)$.

NONARCHIMEDEAN TRIANGLE EQUALTIY : if $|\cdot|$ is a nonarchimedean absolute value on a field k and $|x| \neq |y|$, then $|x + y| = \max(|x|, |y|)$.
pf. Assume the contrary. Then $|x+y| < \max(|x|, |y|) = |x|$ by assume wlog $\max(|x|, |y|) = |x|$. Then $1 < \frac{|x|}{|x+y|}$ since it is a field. Then $1 > \frac{|x+y|}{|x|} = 1 + \frac{|y|}{|x|} \leftrightarrow \frac{|y|}{|x|} < 0$ a contradiction. Hence the equality follows.

An integral domain, with the property that $\forall I$ ideals of A, $\exists J$ another ideal s.t. IJ is principal, is a Dedekind domain.
pf. $IJ = (a)$. Then since every principal ideal is invertible, $\exists (a)^{-1}$. Multiplying both sides gives $IJ(a)^{-1} = A$. Thus every nonzero ideal I in A is invertible. Hence it is a Dedekind domain.

Corollary 1.1.13 (Finite approximation)**.** *Let I be a nonzero fractional ideal in a Dedekind domain A and let $\mathbf{p_1},\dots,\mathbf{p_n}$ be a finite set of nonzero prime ideals of A. Then I contains an element x for which $v_{p_i}(x) = v_{p_i}(I)$ for $1 \le i \le n$.*

pf. Let $I = \frac{1}{s}J, s \in A, J$ an A ideal. Then we can pick $a \in J$ so that $v_{p_i}(a) = v_{p_i}(J)$ for all $i \in [1,n]$. If we let $x = a/s$, $v_{p_i}(x) = v_{p_i}(a) - v_{p_i}(s) = v_{p_i}(J) - v_{p_i}(s) = v_{p_i}(I)$.

Corollary 1.1.14. *Let I be a nonzero ideal in a Dedekind domain A. The quotient ring A/I is a PID.*

pf. Let $\phi : A \to A/I$ be the quotient map, let $\bar{J}$ be an (A/I)-ideal and let $J \equiv \phi^{-1}(\bar{J})$ be its inverse image in A. Then $I \subseteq J, \bar{J} \cong J/I$ as (A/I)-modules. Then by the finite approximation, $\exists a \in J$ s.t. $v_p(a) = v_p(J)$ for all nonzero prime ideals $\mathbf{p}|I$. For every nonzero prime ideal $\mathbf{p}$,
$v_p(J) \le v_p(I)$, $v_p((a)+I) = \begin{matrix} \min(v_p(a), v_p(I)) = v_p(a) = v_p(J) & \mathbf{p}|I \\ \min(v_p(a), v_p(I)) = v_p(I) = 0 = v_p(J) & \mathbf{p} \nmid I \end{matrix}$. So $(a)+I = J$. Thus $\bar{J} \cong J/I = ((a)+I)/I = \phi((a)) = (\phi(a))$.

If every quotient of A by a nonzero ideal is a PID for an integral domain A, then it is a Dedekind domain.
pf. For any ideals $I \subsetneq J, A/J \subseteq A/I$. Let $(x),(y)$ be maximal principal ideals for $A/I, A/J$ respectively. Then $(x) \subseteq (y)$. This implies $y|x$ so to contain is to divide holds.

Definition 1.1.17. *A ring that has only finitely many maximal ideals is called semilocal.*

Corollary 1.1.15. *Every semilocal Dedekind domain is a PID.*

pf. If we let I' be the product of all prime ideals in A, then by the previous lemma, any ideal I will have $J = A$ s.t. $IJ = I$ principal.

Theorem 1.1.7. *Let I be a nonzero ideal in a Dedekind domain A and let $a \in I$ be nonzero. Then $I = (a,b)$ for some $b \in I$.*

pf. $(a) \subseteq I \to I$ divides (a) so $II' = (a)$ for some nonzero ideal I'. By the previous lemma, there is an ideal J coprime to I' s.t. IJ is principal, so $IJ = (b)$ for some $b \in I$. then $((a),(b)) = (II'IJ) = I$ so $(a,b) = I$.

If for every nonzero ideal I in $A, 0 \ne a \in I, I = (a,b)$ for some $b \in I$, then A is a Dedekind domain.
pf. It is noetherian, since any nonzero ideal I in A are generated by two elements. If $\dim A > 1$, $\exists p,q$ prime s.t. $(0) \subsetneq p \subsetneq q$. Since p,q both are nonzero ideals in

$A, p = (a_p, b_p), q = (a_q, b_q)$. Since A is an integral domain and both are primes, a_q or b_q divides both a_p, b_p. But then p is not a prime ideal, a contradiction. Hence $\dim A \leq 1$ and therefore it is an integral domain.

Chapter 2

Extensions

2.1 Algebraic Extensions

2.1.1 Etale Algebra

Definition 2.1.1. *A polynomial f in $K[x]$ is separable if $(f, f') = (1)$. Otherwise f is inseparable.*

→ If f is separable, then it splits into distinct linear factors over $\bar{K}$, where it has $\deg f$ distinct roots.

Definition 2.1.2. *Let L/K be an algebraic field extension. An element $\alpha \in K$ is separable over K if it is the root of a separable polynomial in $K[x]$. The extension L/K is separable if $\forall \alpha \in K$ is separable over K; otherwise it is inseparable.*

Lemma 2.1.1. *An irreducible polynomial $f \in K[x]$ is inseparable iff $f' = 0$.*

pf. Let $f \in L[x]$ be irreducible. Then f is neither zero nor a unit, so $\deg f > 0$. If $f' = 0, (f, f') = f \notin K^\times$ and f is inseparable.
If f is inseparable, then $g \equiv (f, f')$ is a nontrivial divisor of f and f'. Given f irreducible, $\deg g = \deg f$, but $\deg f' < \deg g$ so g cannot divide f' unless $f' = 0$.

Corollary 2.1.1. *Let $f \in K[x]$ be irreducible and let $p \geq 0$ be the characteristic of K. Then $f(x) = g(x^{p^n})$ for some irreducible separable polynomial $g \in K[x]$ and integer $n \geq 0$ that are uniquely determined by f.*

pf. If f is separable, $g = f$ and the theorem holds.
Otherwise, $f(x) \equiv \sum f_r x^r$ is inseparable and $f'(x) = \sum r f_r x^{r-1} = 0$ by the previous lemma. This can occur only if $p > 0, f_r = 0, \forall r \geq 0$ not divisible by p. So $f(x) = g(x^p)$ for some $g \in K[x]$. If g is separable, nothing to prove. Otherwise, we can repeat this procedure until g is separable.

Corollary 2.1.2. *If $charK = 0$, then every algebraic extension of K is separable.*

Lemma 2.1.2. *Let $L = K(\alpha)$ be an algebraic field extension contained in an algebraic closure $\bar{K}$ of K and let $f \in K[x]$ be the minimal polynomial of α over K. Then $\#Hom_K(L,\bar{K}) = \#\{\beta \in \bar{K} : f(\beta) = 0\} \leq [L:K]$, with equality iff α is separable over K.*

pf. For each element of $Hom_K(L,\bar{K})$ is uniquely determined by the image of α, which must be a root β of $f(x)$ in $\bar{K}$. The number of these roots is equal to $[L:K] = \deg f$ precisely when f and therefore α is separable over K.

Definition 2.1.3. *Let L/K be a finite extension of fields. The separable degree of L/K is*
$[L:K]_s \equiv \#Hom_K(L,\bar{K})$. The inseparable degree of f is $[L:K]_i \equiv [L:K]/[L:K]_s$.

Theorem 2.1.1. *Let L/K be an algebraic field extension and let $\phi_K : K \to \Omega$ be a homomorphism to an algebraically closed field Ω. Then ϕ_K extends to a homomorphism $\phi_L : L \to \Omega$.*

pf. Define a partial ordering on the set $\mathscr{F}$ of pairs (F,ϕ_F) for which F/K is a subextension of L/K and $\phi_F : F \to \Omega$ extends ϕ_K by defining $(F_1,\phi_{F_1}) \leq (F_2,\phi_{F_2})$ whenever $F_1 \subseteq F_1$ and ϕ_{F_2} extends ϕ_{F_1}. Given any totally ordered subset $\mathscr{C}$ of $\mathscr{F}$, let E be the field $\cup\{F : (F,\phi_F) \in \mathscr{C}\}$ and define $\begin{array}{rcl} \phi_E : E & \to & \Omega \\ x & \mapsto & \phi_F(x) \end{array}$ for $x \in F \subseteq E$. Then (E,ϕ_E) is a maximal element of $\mathscr{C}$ and therefore by Zorn's lemma, $\mathscr{F}$ contains a maximal element (M,ϕ_M).
CLAIM:$M = L$
pf. Assume contrary. Then $\exists \alpha \in L \setminus M$, so consider the field $F = M(\alpha) \subseteq L$ properly containing M, and extend ϕ_M to $\phi_F : F \to \Omega$ be letting $\phi_F(\alpha)$ be any root of $\alpha_M(f)$ in Ω where $f \in M[x]$ is a minimal polynomial of α over M and $\alpha_M(f)$ is the image of f in $\Omega[x]$ obtained by applying ϕ_M to each coefficient. Then (M,ϕ_M) is strictly dominated by (F,ϕ_F), a contradiction to the maximality of M. Hence $M = L$.

Lemma 2.1.3. *Let $L/F/K$ be a tower of finite extensions of fields and $\bar{K}$ be an algebraic closure of K that contains L. Then $\#Hom_K(L,\bar{K}) = \#Hom_K(F,\bar{K})\#Hom_F(L,\bar{K})$.*

pf. When $F = K$ or $L = F$, lemma trivially holds. So assume $K \subsetneq F \subsetneq L$ and decompose the extensions $L/F, F/K$ into finite towers of non-trivial simple extensions

$$K = K_0 \subsetneq K_1 \subsetneq \cdots \subsetneq K_m = F = K_m \subsetneq K_{m+1} \subsetneq \cdots \subsetneq K_m = L$$

where $K_i = K_{i-1}(\alpha), \forall i \in [1,n]$. Then $\#Hom_K(L,\bar{K}) = \#Hom_{K_0}(K_n,\bar{K})$
$= \Pi_i \#Hom_{K_{i-1}}(K_i,\bar{K}) = \#Hom_{K_0}(K_m,\bar{K})\#Hom_{K_m}(L,\bar{K}) = \#Hom_K(F,\bar{K})\#Hom_F(L,\bar{K})$
if the second equality holds.
CLAIM: $\#Hom_{K_0}(K_n,\bar{K}) = \Pi_i \#Hom_{K_{i-1}}(K_i,\bar{K})$**.**

pf. Consider the map $\begin{array}{rcl} \Phi : Hom_K(K_n, \bar{K}) & \to & \bar{K}^n \\ \phi & \mapsto & (\phi(\alpha_1), \ldots, \phi(\alpha_n)) \end{array}$ This map is injective since $\phi : K_n = K_0(\alpha_1, \ldots, \alpha_n) \to \bar{K}$ is uniquely determined by the images $\phi(\alpha_1), \ldots, \phi(\alpha_n)$. So given $h_i \equiv \#Hom_{K_{i-1}}(K_i, \bar{K})$, if the image of Φ has the cardinality of $\Pi_i h_i$, then the CLAIM holds. For any polynomial $f \in \bar{K}[x]$, let $r(f) \equiv \{\beta \in \bar{K} : f(\beta) = 0\}$ be the set of its $\bar{K}$-roots. $\forall \sigma \in Aut_K(\bar{K})$ induces a bijection $r(f) \to r(\sigma(f))$ with $\#r(f) = \#r(\sigma(f))$. Let f_i be the minimal polynomial of α_i over K_{i-1}. By the previous lemma $h_i = \#r(f_i)$ and $\forall \phi \in Hom_K(K_n, \bar{K})$, $h_i = \#r(\phi(f_i))$ since by the previous theorem, we can extend ϕ to an element of $Hom_K(\bar{K}, \bar{K}) = Aut_K(\bar{K})$. Let $\phi_0 : K \to \bar{K}$ be the inclusion map. $\forall \phi \in Hom_K(K_n, \bar{K})$, we may define a compatible sequence of pairs (ϕ_i, β_i) with $\phi_i \in Hom_K(K_i, \bar{K})$, satisfying $\phi_i|_{K_{i-1}} = \phi_{i-1}$, $\beta \in r(\phi_{i-1}(f_i))$ by putting $\phi_i = \phi|_{K_i}$ and $\beta_i = \phi(\alpha_i)$. Then $(\beta_1, \ldots, \beta_n)$ uniquely determine ϕ and the sequence of pairs (α_i, β_i), so the set of the compatible sequences of pairs is in bijection with the image of Φ. So initially choose $\beta_1 \in r(f_1)$ and let $\phi_1(\alpha_1) = \beta_1$, which uniquely determine $\phi_1 \in Hom_K(K_1, \bar{K})$. There are exactly h_1 choices. Extend this sequence by choosing $\beta_2 \in r(\phi_1(f_2))$ and let $\phi_2(\alpha_2 = \beta_2, \phi_2|_{K_1} = \phi_1$. They uniquely determine $\phi_2 \in Hom_K(K_2, \bar{K})$. There are exactly h_2 such choices. By repeating this procedure, we find there are exactly $\Pi_i h_i$ sequences of pairs (α_i, β_i) corresponding to some $\phi \in Hom_K(K_n, \bar{K})$. So the claim follows.

Corollary 2.1.3. *Let $L/F/K$ be a tower of finite extensions of fields. Then $[L:K]_s = [L:F]_s[F:K]_s$ and $[L:K]_i = [L:F]_i[F:K]_i$.*

pf. The first equality follows by the previous lemma. The second follows by the first equality and the identities $[L:K] = [L:F][F:K], [L:K] = [L:K]_i[L:K]_s$.

Theorem 2.1.2. *Let L/K be a finite extensions of fields. TFAE*

1. *L/K is separable*
2. *$[L:K]_s = [L:K]$*
3. *$K = K(\alpha)$ for some $\alpha \in K$ separable over K*
4. *$L \cong K[x]/(f)$ for some monic irreducible separable polynomial $f \in K[x]$.*

pf.

- $iii \leftrightarrow iv$
  Let f be the minimal polynomial of α. Then $iii \to iv$.
  Let α be the image of x in $K[x]/(f)$. Then $iv \to iii$.
- $ii \leftrightarrow iii$ by the previous lemma

- $i \to iii$ by the primitive element theorem
- $iii \to i$
  Let $L = K(\alpha)$ with α separable over K. $\forall \beta \in K$, $L = K(\beta)(\alpha)$. Then α is separable over $K(\beta)$ since its minimal polynomial over $K(\beta)$ divides the minimal polynomial over K, which is separable. Then by the previous lemma, $[L:K]_s = L:K], [L:K(\beta]_s = [L:K(\beta)]$. Then $[K(\beta):K]_s = [K(\beta):K]$ so β is separable over K. Since β is arbitrary, i follows.

Corollary 2.1.4. *Let L/K be a finite extension of fields. The $[L:K]_s \leq [L:K]$ with equality iff L/K is separable*

pf. Decompose L/K into a finite tower of simple extensions and proceed by induction on the number of extensions. Then it follows.

Corollary 2.1.5. *Let $L/F/K$ is a tower of finite extensions of fields. Then L/K is separable iff both $L/F, F/K$ separable.*

pf. If $L/K, K/F$ are separable, by the previous corollary $[L:F]_s = [L:F], [F:K]_s = [F:K]$ and therefore $[L:K]_s = [L:K]$ and the corollary follows.
The converse is trivial.

Corollary 2.1.6. *Let $L/F/K$ be a tower of algebraic field extensions. Then L/K is separable iff both $L/F, F/K$ are separable.*

pf. Assume $L/F, F/K$ are separable. $\forall \beta \in L$, if $\beta \in F$, nothing to prove. Otherwise, Let M/K be the subextention of F/K generated by the coefficients of the minimal polynomial $f \in F[x]$ of β over F. Then it is a finite separable extension of K and $M(\beta)$ is also a finite separable extension of M since the minimal polynomial of β over $M(\beta)$ is f, which is separable. Then by the previous corollary, $M(\beta)$ is separable over K.
The converse is trivial.

Corollary 2.1.7. *Let L/K be an algebraic field extension and let $F = \{\alpha \in L : \alpha$ is separable over $K\}$. Then F is a separable field extension of K.*

pf. If α, β are separable over K, then $K(\alpha), K(\alpha, \beta)$ are separable extensions of K by the previous corollary. Thus for all elements of $K(\alpha, \beta)$ including $\alpha\beta, \alpha+\beta$ are separable over K and lies in F. So it is a field and by construction it is separable.

Definition 2.1.4. *Let L/K be an algebraic field extension. The field F in the above corollary is the separable closure of K in L. When L is an algebraic closure of K, it is simply called a separable closure of K.*

Definition 2.1.5. *A field K is perfect if every algebraic extension of K is separable.*

Theorem 2.1.3. *A field K of characteristic $p > 0$ is perfect iff $K = K^p$, i.e. every element of K is a pth power, equivalently the map $x \mapsto x^p$ is an automorphism.*

pf. If $K \neq K^p$, $\forall \alpha \in K \setminus K^p$, $x^p - \alpha$ is irreducible and the extension $K[x]/(x^p - \alpha)$ is inseparable and therefore K is not perfect.
If $K = K^p$, let $f \in L[x]$ be irreducible. Then $f(x) = g(x^{p^n})$ for some separable $g \in K[x]$. If $n > 0$, $f(x) = g(x^{p^n}) = \tilde{g}(x^{p^{n-1}})^p$, where $\tilde{g}$ is the polynomial obtained from g by replacing each coefficient with its pth root, a contradiction to the irreducibility of f. So $n = 0, f = g$ is separable. Since very irreducible polynomial in $K[x]$ is separable, every algebraic extension of K is separable, so K is perfect.

Corollary 2.1.8. *Every finite field is a perfect field.*

pf. If a field K has cardinality p^n, $\#K^\times = p^n - 1$, thus $\alpha = \alpha^{p^n} = (\alpha^{p^{n-1}})^p$ for all $\alpha \in K$. So every element of K is a pth power.

Definition 2.1.6. *A field K is separably closed if K has no nontrivial finite separable extensions. Equivalently, K is equal to its separable closure in any algebraic closure of K.*

Definition 2.1.7. *An algebraic extension L/K is purely inseparable, if $[L : K]_s = 1$.*

e.g. If $K = \mathbb{F}_p(t)$, $L = K[x]/(x^p - t) = \mathbb{F}_p(t^{1/p})$, then L/K is a purely inseparable extension of degree p.

Proposition 2.1.1. *Let K be a field of characteristic $p > 0$. If L/K is purely inseparable of degree p, then $L = K(a^{1/p}) \cong K[x]/(x^p - a)$ for some $a \in K \setminus K^p$.*

pf. $\forall \alpha \in L \setminus K$, inseparable over K, its minimal polynomial over K is of the form $f(x) = g(x^{p^n})$ with f monic. Then $1 < \deg f \leq [L : K] = p$, so g must be a monic polynomial of degree 1, i.e. $g(x) = x - a$. Then $f(x) = x^p - a, a \notin K^p$, given f irreducible. Then $[L : K(\alpha)] = 1 \to L = K(\alpha) \cong K[x]/(x^p - a)$.

Theorem 2.1.4. *Let L/K be an algebraic extension and let F be the separable closure of K in L. Then L/F is purely inseparable.*

pf. If L/K is separable, $F = L$ and the theorem follows. Otherwise, $charK = p \neq 0$. Fix an algebraic closure $\bar{K}$ of K that contains L. Let $\alpha \in L \setminus F$ have minimal polynomial f over F. Then $f(x) = g(x^{p^n}), g \in F[x]$ irreducible and separable. Then we must have $\deg g = 1$ since otherwise the roots of g would be separable over F and therefore over K, but not lie in the separable closure F of K in L. Thus $f(x) = x^{p^n} - a$ for some $a \in F$. Since $p > 0$, $f(x) = x^{p^n} - \alpha^{p^n} = (x - \alpha)^{p^n}$. There is thus only one F-homomorphism from $F(\alpha)$ to $\bar{K}$. Since α is arbitrary, $\#Hom_F(L, \bar{K}) = 1$, so L/F is purely inseparable.

Corollary 2.1.9. *Every algebraic extension L/K can be uniquely determined into a tower of algebraic extensions $L/F/K$ with F/K separable and L/F purely inseparable.*

Corollary 2.1.10. *The inseparable degree of any finite extension of fields is a power of the characteristic.*

Definition 2.1.8. *Let K be a field. An etale K-algebra is a K-algebra L that is isomorphic to a finite product of separable field extensions of K. The dimension of an etale K-algebra is its dimension as a K-vector space. When the dimension is finite, L is said to be a finite etale K-algebra. A homomorphism of etale K-algebras is a homomorphism of K-algebras.*

→ for any separable polynomial $f \in K[x], L = K[x]/(f)$. Then L becomes a commutative K-algebra whose elements are all separable over K. By the CRT, if f is separable, then the K-algebra L is isomorphic to a direct product of finite separable extensions of $K \to f = f_1 \dots f_n$ is a factorisation of f into irreducibles in $K[x], \frac{K[x]}{(f)} = \frac{K[x]}{(f_1 \dots f_n)} \cong \frac{K[x]}{(f_1)} \times \dots \times \frac{K[x]}{(f_n)}$ and by the separability of f, f_is are pairwise coprime. Thus L is a K-algebra that is isomorphic to a finite product of separable field extensions $K[x]/(f_i)$ of K.

Definition 2.1.9. *A ring R is simple if it is nonzero and has no nonzero proper ideals. A ring R is semisimple if it is isomorphic to a nonempty finite product of simple rings ΠR_i.*

→ If R is commutative, it is simple iff it is a field and it is semisimple iff it is a finite product of fields. Thus etale algebras are semisimple algebras.

Proposition 2.1.2. *Let $A = \Pi K_i$ be a K-algebra written as a product of field extensions K_i/K. Every surjective homomorphism $\phi : A \to B$ of K-algebras corresponds to the projection of A onto a subproduct of its factors.*

pf. $\ker\phi$ is a subproduct of ΠK_i and therefore $A \cong \ker\phi \times im\phi$ and $B = im\phi$ is isomorphic to the complementary subproduct.

Corollary 2.1.11. *The decomposition of an etale algebra into field extensions is unique upto permutation and isomorphism of factors.*

pf. Let A be an etale K-algebra and suppose A is isomorphic to two products of field extensions of $K \to \Pi_{i=1}^m K_i \cong A \cong \Pi_{j=1}^n L_j$. Composing with isomorphisms yields surjective K-algebra homomorphisms $\begin{matrix} \pi_i : \Pi L_j \to K_i \\ \pi_j \Pi K_i \to L_j \end{matrix}$. By the previous proposition, each K_i must be isomorphic to one of the L_j and vice versa. Thus the uniqueness follows.

Definition 2.1.10. *Let $\phi : A \to B$ be a homomorphism of rings, so B is an A-module. Let M be any A-module. The tensor product of A-modules $M \otimes_A B$ is a B-module called the base change of M from A to B. If M is an A-algebra, then its base change to B is a B-algebra.*

$\to \forall \phi : A \to B$ determines a functor from the category of A-modules to the category of B-modules. It has an adjoint functor called the restriction of scalars that converts B-module M into an A-module by the rule $am = \phi(a)m$.

If a ring homomorphism $\phi : A \to B$ is an inclusion, we have a ring extension B/A.

Proposition 2.1.3. *Suppose L is a finite etale K-algebra and K'/K is any field extension. Then $L \otimes_K K'$ is a finite etale K'-algebra of the same dimension as L.*

pf. If L is not a field, it is a product of fields. Then by the surjection, we can focus on one factor of L, which is a field. So wlog assume L is a field. Then $L \cong K[x]/(f)$ for some separable $f \in K[x]$ and if $f = f_1 \dots f_m$ is the factorisation of f in $K'[x]$, $L \otimes_K K' \cong K'[x]/(f) \cong \Pi_i K'[x]/(f_i)$, in which each factor $K'[x]/(f_i)$ is a finite separable extension of K'. Thus $L \otimes_K K'$ is a finite etale K'-algebra and $\dim_K L = \deg f = \dim_{K'} K'[x]/(f)$ so the dimension is preserved.

e.g. Any finite real vector space V is a finite etale $\mathbb{R}$-algebra, and its base change from $\mathbb{R}$ to $\mathbb{C}$ complex vector space $V \otimes_{\mathbb{R}} \mathbb{C}$ is a finite etale $\mathbb{C}$-algebra of the same dimension.

Corollary 2.1.12. *Let $L \cong K[x]/(f)$ be a finite separable extension of a field K defined by an irreducible separable polynomial $f \in K[x]$. Let K'/K be any field extension and let $f = f_1 \dots f_m$ be the factorisation of f into distinct irreducible polynomials $f_i \in K'[x]$. Then we have an isomorphism of finite etale K'-algebras $L \otimes_K K' \cong \Pi_i K'[x]/(f_i)$, where each $K'[x]/(f_i)$ is a finite separable field extension of K'.*

Definition 2.1.11. *An element $\alpha \in R$ is nilpotent if $\alpha^n = 0$. A ring R is reduced if it contains no nonzero nilpotents.*

Theorem 2.1.5. *Let L be a commutative K-algebra of finite dimension and assume that the dimension of L is less than the cardinality of K. Then TFAE:*

1. *L is a finite etale K-algebra*
2. *Every element of L is separable over K*
3. *$L \otimes_K K'$ is reduced for every extension K'/K*

4. *$L \otimes_K K'$ is semisimple for every extension K'/K*

5. *$L = K[x]/(f)$ for some separable $f \in K[x]$*

pf.

- $i \to ii$
  Let $L = \Pi K_i$ with K_i/K separable and consider $\alpha = (\alpha_1, \ldots, \alpha_n) \in L = \Pi K_i$ is separable over K with separable minimal polynomial $f_i \in K[x]$ and α is a root of $f \equiv LCM(f_1, \ldots, f_n)$, which is separable. Thus α is separable.

- $ii \to iii$
  If $\alpha \in L$ is nonzero and separable over K, it cannot be nilpotent, since otherwise the minimal polynomial of the nilpotent x^n, $n > 1$ is not separable. Separability is preserved under the base change so the iii follows.

- $iii \leftrightarrow iv$

  Lemma 2.1.4. *Let K be a field. A commutative K-algebra of finite dimension is semisimple iff it is reduced.*

  pf. If A is semisimple, it is reduced, since otherwise we could project a nonzero nilpotent of A to a nonzero nilpotent in a field, a contradiction.
  Every ideal of a commutative K-algebra A is also a K-vector space, and therefore when $\dim_K A$ is finite, A satisfies both a.c.c. and d.c.c. and is therefore noetherian and artinian. Thus it has finitely many maximal ideals $M_1, \ldots, M_n$ and the intersection of maximal ideals is equal to the nilradical of A. Taking the product of the projection map $A \to A/M_i$ yields a surjective ring homomorphism $\phi : A \to \oplus A/M_i$ from A to a product of fields. If A is reduced, $\ker \phi = \cap M_i = \{0\}$, i.e. it is an isomorphis. Thus A is semisimple.

- $iv \to i$
  Assume L is semisimple by taking $K' = K$. Wlog assume L is a field. By base-changing to the separable closure of K in L, we can reduce to the case that L/K is a purely inseparable field extension. If $L = K$, nothing to prove. Otherwise $\exists \alpha \in L$ inseparable and the minimal polynomial for α has the form $f(x) = x^{p^n} - a$ for some $a \in K, n \geq 1$. Consider $\gamma \equiv \alpha \otimes 1 - 1 \otimes \alpha \in L \otimes_K L$. Then $\gamma \neq 0$ since $\gamma \notin K$, but $\gamma^{p^n} = \alpha^{p^n} \otimes 1 - 1 \otimes \alpha^{p^n} = a \otimes 1 - 1 \otimes a = 0$ so γ is a nilpotent and $L \otimes_K L$ is not reduced, contradiction to iii.

- $v \to i$ by the previous corollary

- $i \to v$
  Suppose $L = \Pi L_i$ with each L_i/K finite separable extensions of K. Pick a monic irreducible separable polynomial $f_1(x)$ so that $L_1 \cong K[x]/(f_1)$ and do the same for all $i \in [2,n]$, ensuring each polynomial f_j, we pick is not euqal to f_i $\forall i < j$. This can be done by replacing $f_j(x)$ with $f_j(x+a)$, $a \in K^\times$, if necessary. The existence is guaranteed by the caridnality condition. The polynomials $f_1,\dots,f_n$ are the coprime and separable and therefore their product f is separable and $L = K[x]/(f)$.

Proposition 2.1.4. *Suppose L is a finite etale K-algebra and Ω is a separably closed field extension of K. There is an isomorphism of finite etale Ω-algebras* $\begin{array}{ccc} L \otimes_K \Omega & \to & \Pi_{\sigma \in Hom_K(L,\Omega)}\Omega \\ \beta \otimes 1 & \mapsto & (\sigma(\beta))_\sigma \end{array}$ *for each $\beta \in L$.*

pf. Reduce to the case where $L = K[x]/(f)$ is a separable field extension and then we may factor $f(x) = \Pi(x-\alpha_i)$ over Ω with all α_i are distinct. Then there is a bijection between $Hom_K(K[x]/(f),\Omega)$ and the set $\{\alpha_i\}$ since $\forall \sigma \in Hom_K(K[x]/(f),\Omega)$ is determined by $\sigma(x) \in \{\alpha_i\}$ and for each α_i, $x \mapsto \alpha_i$ determines a K-algebra homomorphism $\sigma_i \in Hom_K(K[x]/(f),\Omega)$. Then we have Ω-algebra isomorphisms

$$\frac{K[x]}{(f)} \otimes_K \Omega \to \frac{\Omega[x]}{(f)} \to \Pi_i \frac{\Omega[x]}{(x-\alpha_i)} \to \Pi_i \Omega$$

which maps $x \otimes 1 \mapsto x \mapsto (\alpha_1,\dots,\alpha_n) \mapsto (\sigma_1(x),\dots,\sigma_n(x))$. Since the element $x \otimes 1$ generates $L \otimes_K \Omega, \beta \otimes 1 \mapsto (\sigma(\beta))_\sigma$ for every $\beta \in L$.

Definition 2.1.12. *Let B/A be a commutative ring extension in which B is a free A-module of finite rank. The norm $N_{B/A}(b)$ and trace $T_{B/A}(b)$ of b aare the determinant and trace of the A-linear multiplicaton-by-b map $B \to B$ defined by $x \mapsto bx$.*

If A is a field and B is a finite A-algebra, B is a finite A-vector space, hence a free A-module of finite rank. Then we can compute the norm and trace by picking a basis for B as an A-module and computing the matrix of the multiplication-by-b map w.r.t. this basis.

If $B_1/A, B_2/A$ are two ring extensions that are free A-modules of finite rank, then
$N_{B_1 \times B_2/A}(x) = N_{B_1/A}(x_1)N_{B_2/A}(x_2)$, $T_{B_1 \times B_2/A} = T_{B_1/A}(x_1) + T_{B_2/A}(x_2)$, $\forall x = (x_1,x_2) \in B_1 \times B_2$.

Lemma 2.1.5. *Let B/A be ring extension with B free of rank n over A, and let $\phi : A \to A'$ be a ring homomorphism. The base change $B' = B \otimes_A A'$ of B to A' is a free A'-module of rank n and for every $b \in B, \phi(N_{B/A}(b)) = N_{B'/A'}(b \otimes 1)$, $\phi(T_{B/A}(b)) = T_{B'/A'}(b \otimes 1)$.*

pf. Let $b \in B, (b_1, \ldots, b_n)$ be a basis for B as an A-module, and $M = (m_{ij}) \in A^{n \times n}$ be the matrix of $B \to_{\times b} B$ w.r.t. this basis. Then $(b_1 \otimes 1, \ldots, b_n \otimes 1)$ is a basis for B' as an A'-module and $M' = (\phi(m_{ij})) \in A'^{n \times n}$ is the batrix of $B' \to_{\times b \otimes 1} B'$ and therefore

$$\begin{aligned} \phi(N_{B/A}(b)) &= \phi(\det M) = \det M' = N_{B'/A'}(b \otimes 1) \\ \phi(T_{B/A}(b)) &= \phi(trM) = trM' = N_{B'/A'}(b \otimes 1) \end{aligned}.$$

Theorem 2.1.6. *Let K be a field with separable closure Ω and let L be a finite etale K-algebra. $\forall \alpha \in L$, $N_{L/K}(\alpha) = \Pi_{\sigma \in Hom_K(L,\Omega)} \sigma(\alpha)$, $T_{L/K}(\alpha) = \sum_{\sigma \in Hom_K(L,\Omega)} \sigma(\alpha)$.*

pf. Let n be the rank of L as a K-module. Then $N_{L/K}(\alpha) = N_{(L \otimes_K \Omega)/\Omega}(\alpha \otimes 1) = N_{\Omega^n/\Omega}(\sigma_1(\alpha), \ldots, \sigma_n(\alpha))$
$= \Pi \sigma_i(\alpha)$ by the previous lemma with isomorphism $L \otimes_K \Omega \to \Pi_\sigma \Omega = \Omega^n$ sends $\alpha \otimes 1$ to $(\sigma_1(\alpha), \ldots, \sigma_n(\alpha))$. Using the standard basis for Ω^n, the matrix of multiplication-by-$(\sigma_1(\alpha), \ldots, \sigma_n(\alpha))$ is the diagonal matrix with $\sigma_i(\alpha)$ in the ith diagonal entry. Similarly, $T_{L/K}(\alpha) = T_{(L \otimes_K \Omega)/\Omega}(\alpha \otimes 1) = T_{\Omega^n/\Omega}(\sigma_1(\alpha), \ldots, \sigma_n(\alpha)) = \sum \sigma_i(\alpha)$.

Proposition 2.1.5. *Let L/K be a finite extension, let $\bar{K}$ be an algebraic closure of K containing L. Let $\alpha \in L^\times$ have minimal polynomial $f \in K[x]$ with factorisation $f(x) = \Pi_{i=1}^d (x - \alpha_i)$ in $\bar{K}[x]$, and let $e = [L : K(\alpha)]$. Then*

$$N_{L/K}(\alpha) = \Pi \alpha_i^e, T_{L/K} = e \sum \alpha_i$$

In particular, if $f(x) = \sum a_i x^i$, $N_{L/K}(\alpha) = (-1)^{de} \alpha_0^e$, $T_{L/K}(\alpha) = -e a_{d-1}$.

pf. By the previous theorem $\begin{matrix} N_{K(\alpha)/K}(\alpha) = \Pi \sigma(\alpha) = \Pi \alpha_i \\ T_{K(\alpha)/K} = \sum \sigma(\alpha) = \sum \alpha_i \end{matrix}$. Since $e = [L : K(\alpha)]$, we can view the matrix as a block diagonal matrix s.t. one block for $K(\alpha)$ is repeated by e times(given not necessarily separable). Then $N_{L/K}(\alpha) = \Pi \alpha_i^e, T_{L/K} = e \sum \alpha_i$ hold.
For a polynomial $f(x) = x^d + a_{d-1} x^{d-1} + \cdots + (-1)^d a_0, \Pi \alpha_i = -a_{d-1}, \sum \alpha_i = (-1)^d a_0$. Then by the previous statement, $N_{L/K}(\alpha) = (-1)^{de} a_0^e$, $T_{L/K} - e a_{d-1}$ follows.

Corollary 2.1.13. *Let A be an integrally closed domain with fraction field K and let L/K be a finite extension. If $\alpha \in L$ is integral over A, then $N_{L/K}(\alpha) \in A, T_{L/K}(\alpha) \in A$.*

Theorem 2.1.7 (Transitivity of Norm and Trace)**.** *Let $A \subseteq B \subseteq C$ be rings with C free of finite rank over B, and B free of finite rank over A. Then C is free of finite rank over A and $N_{C/A} = N_{B/A} \circ N_{C/B}$, $T_{C/A} = T_{B/A} \circ T_{C/B}$.*

2.1.2 Dedekind Extensions

Definition 2.1.13. *Let A be a commutative ring and M an A-module. The dual module $M^\vee$ is the A-module $Hom_A(M, A)$ with scalar multiplication $(af)(m) =$*

$af(m), a \in A$, $f \in Hom_A(M,A)$, $m \in M$. If $\phi : M \to N$ is an A-module homomorphism, the dual homomorphism. $\begin{array}{rcl} \phi^\vee : N^\vee & \to & M^\vee \\ g & \mapsto & g \circ \phi \end{array}$
$\forall g \in N^\vee, m \in M$.

Lemma 2.1.6. *Let A be a commutative ring. For all A-module M,N, the A-modules $(M \oplus N)^\vee$ and $M^\vee \oplus N^\vee$ are canonically isomorphism.*

pf. $\begin{array}{l} \phi \mapsto (m \mapsto \phi(m,0), n \mapsto \phi(0,n)) \\ (\phi,\psi) \mapsto ((m,n) \mapsto \phi(m) + \psi(n)) \end{array}$ are inverses of each other, hence the lemma follows.

Definition 2.1.14. *A A-module M is reflexive if $M^{\vee\vee} \cong M$.*

If A is a field and M is a finitely generated A-module, then M is a vector space of finite dimension, it is reflexive. (e.g. invertible fractional ideals)

Proposition 2.1.6. *Let A be an integral domain with fraction field K and let M. be a nonzero A-submodule of K. Then $M^\vee \cong (A : M) \equiv \{x \in K : xM \subseteq A\}$; in particular if M is an invertible fractional ideal, then $M^\vee \cong M^{-1}, M^{\vee\vee} \cong M$.*

pf. For any $x \in (A : M)$, $m \mapsto xm$ is an A-linear map from M to A, hence $x \in M^\vee$ and this defines an A-module homomorphism $\phi : (A : M) \to M^\vee$, since the map $x \mapsto (m \mapsto xm)$ is itself A-linear. Since $M \subseteq K$ is a nonzero A-module, it contains $0 \neq a \in A$. If $f \in M^\vee, m = b/c \in M$, $f(m) = f(\frac{b}{c}) = \frac{ac}{ac} f(\frac{b}{c}) = \frac{b}{ac} f(\frac{ac}{c}) = \frac{b}{ac} f(a) = \frac{f(a)}{a} m$ by the A-linearity of f. Thus f corresponds to multiplication by $x = \frac{f(a)}{a}$, which lies in $(A : M)$ since $xm = f(m) \in A$, $\forall m \in M$. This map $f \mapsto f(a)/a$ defines an A-module homomorphism $M^\vee \to (A : M)$ inverses to ϕ, and therefore it is an isomorphism. When M is an invertible fractional ideal, $M^\vee \cong (A : M) = M^{-1}$, so the last statement holds.

Theorem 2.1.8. *Let A be a commutative ring and let M be a free A-module of rank n. Then $M^\vee$ is also a free A-module of rank n, and each basis $(e_1, \ldots, e_n)$ of M uniquely determines a dual basis $(e_1^\vee, \ldots, e_n^\vee)$ of $M^\vee$ with the property $e_i^\vee = \delta_{ij} \equiv$* $\begin{array}{ll} 1 & i = j \\ 0 & otherwise \end{array}$.

pf. If $n = 0$, $M = M^\vee = 0$ and therefore the theorem holds.
$n \geq 1$, fix an A-basis $\mathbf{e} \equiv (e_1, \ldots, e_n)$ for M. Then for all $\mathbf{a} \equiv (a_1, \ldots, a_n) \in A^n$, define $f_a \in M^\vee$ by setting $f_a(e_i) = a_i$ and extending A-linearly. The map $\mathbf{a} \mapsto f_a$ gives an A-module homomorphism $A^n \to M^\vee$ with inverse $f \mapsto (f(e_1), \ldots, f(e_n))$ and is therefore an isomorphism. Thus $M^\vee \cong A^n$ is a free A-module or rank n.
Let $e_i^\vee \equiv f_{\hat{\imath}}$, where $\hat{\imath} \equiv (0, \ldots, 0, 1, 0, \ldots, 0) \in A^n$ has a 1 in the ith position. Then $\mathbf{e}^\vee \equiv (e_1^\vee, \ldots, e_n^\vee)$ is a basis for $M^\vee$, since $(\hat{1}, \ldots, \hat{n})$ is a basis for A^n and $e_i^\vee(e_j) = \delta_{ij}$. The basis $\mathbf{e}^\vee$ is uniquely determined by $\mathbf{e}$ since it must be the image of $(\hat{1}, \ldots, \hat{n})$ under the isomorphism $\mathbf{a} \mapsto f_a$ determined by $\mathbf{e}$.

Definition 2.1.15. *Let A be a commutative ring and M an A-module. A bilinear pairing on M is an A-linear $<\cdot,\cdot>: M\times M\to A$. Explicitly* $\forall u,v,w\in M,\lambda\in A$, $\begin{array}{l} <u+v,w>=<u,w>+<v,w> \\ <u,v+w>=<u,v>+<u,w> \\ <\lambda u,v>=<u,\lambda v>=\lambda<u,v> \end{array}$.

$\to$ If $<v,w>=<w,v>$, then it is symmetric, if $<v,w>=-<w,v>$, it is skew-symmetric, and if $<v,v>=0$, it is alternating.

The pairing $<\cdot,\cdot,>$ induces an A-module homomorphism $\begin{array}{rcl} \phi: M & \to & M^\vee \\ m & \mapsto & (n\mapsto <m,n>) \end{array}$. If $\ker\phi=\{0\}$, it is nondegenerate, and if ϕ is an isomorphism, it is perfect.

Proposition 2.1.7. *Let A be a commutative ring and let M be a free A-module of rank n with a perfect pairing $<\cdot,\cdot>$. For each A-basis $(e_1,\dots,e_n)$ of M, there is a unique basis $(e_1.,\dots,e'_n)$ for M s.t. $<e'_i,e_j>=\delta_{ij}$.*

pf. If M is a free A-module with basis $(e_1,\dots,e_n)$ and $<\cdot,\cdot>$ is a perfect pairing, by the inverse of the isomorphism $\phi: M\to M^\vee$ applied to the dual basis $(e_1^\vee,\dots,e_n^\vee)$, we can obtain a basis $(e'_1,\dots,e'_n)$ for M s.t. $<e'_i,e_j>=\delta_{ij}$; $e'_i=\phi^{-1}(e_i^\vee)$. Then $e_i^\vee=\phi(e'_i)=m\mapsto <e'_i,m>$ and $<e'_i,e_j>=\phi(e'_i)(e_j)=e_i^\vee(e_j)=\delta_{ij}$.
If there is another basis $(f'_1,\dots,f'_n)$ for M with the same property, then for each i, $<e'_i-f'_i,e_j>=\delta_{ij}-\delta_{ij}=0$, $\forall e_j$ and therefore $<e'_i-f'_i,m>=0$, $\forall m\in M$. Then $e'_if'_i\in\ker\phi=\{0\}$ and therefore uniqueness follows.

Definition 2.1.16. *Let A be an integral domain with fractional field K and let V be a K-vector space of finite dimension. A A-lattice in V is a finitely generated A-submodule M of V that spans V as a K-vector space.*

Definition 2.1.17. *Let A be a noetherian domain with fraction field K, and let V be a K-vector space of finite dimension with a perfect pairing $<\cdot,\cdot>$. If M is an A-lattice in V, its dual lattice is the A-module $M^*\equiv\{x\in V:<x,m>\in A,\ \forall m\in M\}$.*

Theorem 2.1.9. *Let A be a noetherian domain with fraction field K, let V be a K-vector space with a perfect pairing $<\cdot,\cdot>$ and let M be an A-lattice in V. The dual lattice M^* is an A-lattice in V isomorphic to $M^\vee$.*

pf. $\mathbf{e}\equiv(e_1,\dots,e_n)$ be a K-basis for V that lies in M and let $\mathbf{e}'\equiv(e'_1,\dots,e'_n)$ be the unique K-basis for V given by the previous proposition that satisfies $<e'_i,e_j>=\delta_{ij}$. Write a finite set S of generators for M in terms of the basis $\mathbf{e}$ with coefficients in K and let d be the product of all denominators that appear.
CLAIM: de' **lies in** M^*
pf. For each $e'_i,m\in S$ if we put $m=\sum_j m_je_j$, then $<de'_i,m>=d<e'_i,\sum_j m_je_j>=d\sum_j m_j<e'_i,e_j>=d\sum_j m_j\delta_{ij}=dm_i\in A$ and therefore $de'_i\in M^*$.

Thus M^* contains a basis $d\mathbf{e}'$ for V.
CLAIM: M^* is finitely generated
pf. Define $N \equiv \{\sum a_i e_i, \forall a_i \in A\} \cong A^n$ to be the free A-submodule of M spanned by $\mathbf{e}$. The A-module N contains a basis for V and is finitely generated and therefore it is an A-lattice in V, The K-basis $\mathbf{e}'$ for V lies in N^*, since $< e'_i, e_j >= \delta_{ij} \in A$. Given $x \in N^*$ write $x = \sum x_i e'_i$, then $< x, e_i >= x_i < e'_i, e_j >= x_i \in A$ since $x \in N^*$. Thus x lies in the A-span of $\mathbf{e}'$. Thus N^* is a free A-module of rank n and in particular a finitely generated module over a noetherian ring, i,e. it is a noehterian module. Since $N \subseteq M$, $M^* \subset N^*$. Since every submodule of noetherian module is finitely generated, so is M^*.
Hence M^* is an A-lattice.
CLAIM: $M^* \cong M^\vee$.
pf. There are A-module homomorphism $\begin{array}{rcl} \phi : M^* & \to & M^\vee \\ x & \mapsto & (M \mapsto < x, m >) \end{array}$ and the $\begin{array}{rcl} \psi : M^\vee & \to & M^* \\ f & \mapsto & \sum_i f(e_i) e'_i \end{array}$, which are inverses to each other.
Indeed, $\forall x = \sum x_i e'_i \in M^*, \psi(\phi(x)) = \sum \phi(x)(e_i) e'_i = \sum < x, e_i > e'_i = \sum_{i,j} x_j < e'_j, e_i > e'_i = \sum_i x_i e'_i = x$
$\forall f \in M^\vee, \phi(\psi(f))(m) = \phi(\sum f(e_i) e'_i)(m) = \sum \phi(f(e_i) e'_i)(m) = \sum < f(e_i) e'_i, \sum_j m_j e_j >= f(m)$.

Corollary 2.1.14. *Let A be a noetherian domain with fraction field K. If M_1, M_2 are A-lattices in K-vector spaces V_1, V_2 with perfect pairing $< \cdot, \cdot >_1, < \cdot, \cdot >_2$ respectively, then $< \cdot, \cdot >_1 + < \cdot, \cdot >_2$ defines a perfect pairing on $V_1 \oplus V_2$ and $(M \oplus N)^* \cong M^* \oplus N^*$.*

Corollary 2.1.15. *Let A be a noetherian domain with fraction field K, let V be a K-vector space with a perfect pairing $< \cdot, \cdot >$, and let M be a free A-lattice in V with A-basis $(e_1, \ldots, e_n)$. The dual lattice M^* is a free A-lattice in V that has a unique A-basis $(e_1^*, \ldots, e_n^*)$ that satisfies $< e_i^*, e_j >= \delta_{ij}$.*

pf., By the previous theorem with $N = M, e_i^* = e'_i$.

Lemma 2.1.7. *Let A be a noetherian domain with fraction field K, let V be a K-vector space of finite dimension with a perfect pairing $< \cdot, \cdot >$, let M be an A-lattice in V, and let S be a multiplicative subset of A. Then $S^{-1}M$ and $S^{-1}M^*$ are $(S^{-1}A)$-lattices in V satisfying $(S^{-1}M)^* = S^{-1}M^*$.*

pf. $S^{-1}M, S^{-1}M^*$ contain a basis for V since M, M^* do and both are finitely generated as $S^{-1}A$-modules since M, M^* are finitely generated A-modules. Thus both are $S^{-1}A$-lattices.
Let $m_1, \ldots, m_n$ be A-module generated for M. Then they are also generators of $S^{-1}M$ as an $S^{-1}A$-module. If $x \in (S^{-1}M)^*$, then fir each $m_i, < x, m_i >= a_i/s_i$ for

some $a_i \in A, s_i \in S$ and if we put $s = s_1 \dots s_n$, $< sx, m_i > \in A, \forall m_i$ and therefore $\forall m \in M$. Thus $sx \in M^*, x \in S^{-1}M^*$.
If $x = y/s \in S^{-1}M^*, y \in M^*, s \in S$, then $< y, m_i > \in A, \forall m_i$ and $< x, m_i > = < y, m_i > /s \in S^{-1}A$ for all m_i, hence for all $m \in S^{-1}M$ and therefore $x \in (S^{-1}M)^*$.

Proposition 2.1.8. *Let A be a Dedekind domain with fraction field K, let V be a K-vector space of finite dimension with a symmetric perfect pairing $< \cdot, \cdot >$ and let M be an A-lattice in V. Then $M^{**} = M$.*

pf. Let $\mathbf{p}$ be a maximal ideal of A. Then $(M^{**})_{\mathbf{p}} = M^{**}_{\mathbf{p}}$ by the previous lemma. So wlog replace A with $A_{\mathbf{p}}$, which is a DVR. Then A is a PID and M, M^* are both torsion-free modules over PID, hence free A-modules. Choose an A-basis $(e_1, \dots, e_n)$ for M, and let $(e_1^*, \dots, e_n^*)$ be the unique dual A-basis for M^* that satisfies $< e_i^*, e_j > = \delta_{ij}$ by the previous corollary. If we let $(e_1^{**}, \dots, e_n^{**})$ be the unique A-basis for M^{**} that satisfies $< e_i^{**}, e_j^* > = \delta_{ij}$. By the uniqueness $e_i^{**} = e_i, \forall i$ and therefore $M^{**} = M$.

Proposition 2.1.9. *Let A be a Dedekind domain with fraction field K, let L/K be a finite extension, and let B be the integral closure of A in L. Then every element of L can be written as $b/a, a \in A, b \in B$. In particular, B spans L as a K-vector space and L is the fraction field of B.*

pf. Let $\alpha \in L$. By multiplying the minimal polynomial of α in $K[x]$ by the product of the denominators of its coefficients, we obtain a polynomial in $A[x] \to g(x) = a_n x^n + \dots + a_1 x + a_0$ with $a_n \neq 0$, that has α as a root. By Replacing x with x/a_n, and multiplying both sides by a_n^{n-1}, we get $a_n^{n-1} g(x/a_n) = x^n + a_{n-1} x^{n-1} + \dots + a_n^{n-2} a_1 x + a_n^{n-1} a_0$, a monic polynomial with coefficients in A that has $a_n \alpha \in L$ as a root. Thus $a_n \alpha B$, given B integral closure of A in L and $\alpha = b/a_n, b \in B, a_n \in A$ as claimed. Thus B generates L as K-vector space and $B \subseteq L \subseteq FracB$ implies $L = FracB$, since no smaller field can contain B.

Proposition 2.1.10. *Let A be a Dedekind domain with fraction field K, let L/K be a finite extension of fields, and let B be the integral closure of A in L. Then $N_{L/K}(b) \in A, T_{L/K}(b) \in A$, $\forall b \in B$.*

pf. The minimal polynomial $f = \sum_{i=0}^{d} a_i x^i \in K[x]$ of b has coefficients in A and therefore $N_{L/K}(b) = (-1)^{de} a_0^e \in A, T_{L/K}(b) = -e a_{d-1} \in A$.

Definition 2.1.18. *Let B/A be a ring extension with B a free A-module of finite rank. The trace pairing on B is the map $B \times B \to A$ defined by $< x, y >_{B/A} \equiv T_{B/A}(xy)$.*

Theorem 2.1.10. *Let L be a commutative K-algebra of finite dimension. The trace pairing*
$< \cdot, \cdot >_{L/K}$ is a symmetric bilinear pairing. It is a perfect pairing iff L is a finite etale K-algebra.

pf. Bilinearity follows from the K-linearaity of the trace map $T_{L/K}$ and symmetry is immediate.
If L is not reduced, the proposition holds, since it is not etale and the trace pairing is degenerate (for x nilpotent, the map $y \mapsto T_{L/K}(xy)$ must be a zero map, since every xy is also nilpotent and the trace of any nilpotent element z is zero).
If L is reduced, then it is semisimple and thus a product of fields. Wlog we can reduce to the case where L is a field since the trace pairing on a product of field extensions is nondegerate iff the trace pairing on each factor is nondegerate and a product of field extensions is etale iff each factor is etale. $T_{L/K}$ is a zero map iff the field extension L/K is inseparable. If $T_{L/K}$ is a zero map, then the trace pairing is clearly degenerate. Otherwise, pick $z \in L$ for which $T_{L/K}(z) \neq 0$. Then $\forall x \in L^{\times}, < x, z/x >_{L/K} = T_{L/K}(z) \neq 0$ so $x \mapsto < x, y >_{L/K}$ is not the zero map and the trace pairing is nondegenerate.
($T_{L/K}$ **is a zero map iff the field extension** L/K **is inseparable.**
pf. If L/K is inseparable, for $\alpha \in L \setminus K$, its minimal polynomial in K is $x^{p^n} - a$ and therefore $T_{L/K} = \sum \alpha_i = -a_{d-1} = 0$.
Conversely, if L/K is separable, $\exists \alpha \in L \setminus K$ with its minimal polynomial f s.t. $f(\alpha) = 0$. Then $\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0 \Leftrightarrow a_{n-1} = -(\alpha + a_{n-2}\alpha^{-1} + \ldots a_0\alpha^{n-1}) \in K$ is not 0).

AKLB means A is a Dedekind domain, $FracA = K$, L/K is a finite separable extension of K, and B is the integral closure of A in L.

Proposition 2.1.11. *Assume AKLB. Then B is an A-lattice in L, and in particular B is finitely generated A-module.*

pf. By the proposition 2.2.9, B spans L as a K-vector space, so it contains a basis $(e_1, \ldots, e_n)$ for L as a K-vector space. Let $M \subseteq B$ be the A-span of $(e_1, \ldots, e_n)$. /then M is an A-lattice in L contained in B and it has a dual lattice M^* that contains the A-module $B^* \equiv \{x \in L :< x, b >_{L/K} \in A \ \forall b \in B\}$. Then by the proposition 2.2.10, $B \subseteq B^*$ and therefore $M \subseteq B \subseteq B^* \subseteq M^*$. Then M^* is an A-lattice in L and hence finitely generated, hence noetherian. Thus its A-submodule B is finitely generated and thus an A-lattice in L by the theorem 2.2.9.
$\rightarrow$ If L/K is inseparable, B needs not be finitely generated as an A-module.

Lemma 2.1.8. *Let B/A be an extension of domains with B integral over A and let $\mathbf{q_0} \subsetneq \mathbf{q}_1$ be primes of B. Then $\mathbf{q}_0 \cap A \subsetneq \mathbf{q}_1 \cap A$ and* $\dim A \geq \dim B$.

pf. Wlog replace B with $B/\mathbf{q}_0$ and replace $A, \mathbf{q_0}, \mathbf{q_1}$ with their image in $B/\mathbf{q}_0$. Then $\mathbf{q}_0 = (0)$ and $\mathbf{q}_1$ is a nonzero prime ideal. Let $\alpha \in \mathbf{q}_1$ be nonzero. Its minimal polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_0$ over K has coefficients in A, since $\alpha \in \mathbf{q}_1 \subseteq B$ is integral over A. $a_0 \neq 0$ since if $a_0 = 0$, that polynomial is not irreducible. Thus

we have $a_0 = -\alpha a_1 - \cdots - \alpha^n \in \mathbf{q}_1$ thus $0 \neq \mathbf{q}_1 \cap A$. So $\mathbf{q}_1 \cap A$ is not the zero ideal and therefore properly contains $\mathbf{q}_0 \cap A = \{0\}$. We can apply this result repeatedly to any chain of distinct prime ideals in B to get a corresponding chain of distinct ideals in A. Then $\dim A \geq \dim B$.

Theorem 2.1.11. *Let A be a Dedekind domain with fraction field K, let L/K be a finite separable extension and let B be the integral closure of A in L. Then B is a Dedekind domain.*

pf. B is an integrally closed domain by definition. Also B is finitely generated over the noetherian ring A by the proposition 2.2.11. and therefore it is noetherian. Also B has dimension at most 1 since $\dim B \leq \dim A \leq 1$ by the previous lemma. Hence B is a Dedekind domain.

Corollary 2.1.16. *The ring of integers of a number field is a Dedekind domain.*

Assume AKLB. Then B is a Dedekind domain by the previous theorem. Every nonzero ideal in a Dedekind domain can be uniquely factored into prime ideals. If A has dimension 0, so does B so assume $\dim A = 1$, in which case $\dim B = 1$ as well. Thus In AKLB setup, we implicitly assume $A \neq K$.

Given a prime $\mathbf{p}$ of A, consider the ideal $\mathbf{p}B$, which may not be a prime, But since B is a Dedekind domain, it can be uniquely factored into nonzero prime ideals. Thus $\mathbf{p}B = \Pi_{\mathbf{q}}\mathbf{q}^{e_{\mathbf{q}}}$ where $\mathbf{q}$ ranges over primes of B and $e_{\mathbf{q}} \geq 0$ are zero for all but finitely many primes $\mathbf{q}$. The prime ideal $\mathbf{q}$ for which $e_{\mathbf{q}} > 0$ are said to lie over or above the prime ideal $\mathbf{p}$.

Lemma 2.1.9. *Let A be a ring of dimension one contained in a Dedekind domain B. Let $\mathbf{p}$ be a prime of A and let $\mathbf{q}$ be a prime of B. Then $\mathbf{q}|\mathbf{p}$ iff $\mathbf{q} \cap A = \mathbf{p}$.*

pf. If $\mathbf{q}$ divides $\mathbf{p}B$, then it contains $\mathbf{p}B$, since B is a Dedekind domain. Thus $\mathbf{q} \cap A$ contains $\mathbf{p}B \cap A$ which contains $\mathbf{p}$. The ideal $\mathbf{p}$ is maximal and $\mathbf{q} \cap A \neq A$, $\mathbf{q} \cap A = \mathbf{p}$
If $\mathbf{q} \cap A = \mathbf{p}$, $\mathbf{q} = \mathbf{q}B$ contains $(\mathbf{q} \cap A)B = \mathbf{p}B$. Thus given B a Dedekind domain, $\mathbf{q}$ divides $\mathbf{p}B$.
$\rightarrow \exists \begin{array}{ccc} SpecB & \rightarrow & SpecA \\ \mathbf{q} & \mapsto & \mathbf{q} \cap A \end{array}$, a surjective map.

The primes of A are all maximal, given $\dim A = 1$, and therefore each has an associated residue field $A/\mathbf{p}$ and similarly for primes $\mathbf{q}$ of B. If $\mathbf{q}$ lies above $\mathbf{p}$, we may regard the residue field $B/\mathbf{q}$ as a field extension of $A/\mathbf{p}$. A ring homomorphism of fields $A/\mathbf{p} = A/(\mathbf{q} \cap A) \rightarrow B/\mathbf{q}$ is injective since the map $A \rightarrow B \rightarrow B/\mathbf{q}$ has the kernel $\mathbf{q} = A \cap \mathbf{q}$.

Definition 2.1.19. *Assume AKLB and let $\mathbf{p}$ be a prime of A. The exponent $e_{\mathbf{q}}$ in the factorisation $\mathbf{p}B = \Pi_{\mathbf{q}|\mathbf{p}}\mathbf{q}^{e_{\mathbf{q}}}$ is the ramification index of $\mathbf{q}$ and the degree*

$f_{\mathbf{q}} = [B/\mathbf{q} : A/\mathbf{p}]$ *of the corresponding residue field extension is the residue degree of* $\mathbf{q}$.

Lemma 2.1.10. *Let A be a Dedekind domain with fraction field K, let $M/L/K$ be a tower of finite separable extension and let B and C be the integral closure of A in L and M respectively. Then C is the integral closure of B in M and if τ is a prime of M lying above a prime $\mathbf{q}$ of L lying above a prime $\mathbf{p}$ of K, then $e_{\tau/p} = e_{\tau/q} e_{q/p}, f_{\tau/p} = f_{\tau/q} f_{q/p}$.*

pf. If $\mathbf{q}$ divides $\mathbf{p}B$, then it contains $\mathbf{p}B$ by the "to divide is to contain", and therefore $\mathbf{p} \subseteq \mathbf{p}B \cap A \subseteq \mathbf{q} \cap A$. The ideal $\mathbf{p}$ is maximal and $\mathbf{q} \cap A \neq A$, $\mathbf{a} \cap A = \mathbf{p}$ if $\mathbf{p} \cap A = \mathbf{p}$, $\mathbf{q} = \mathbf{q}B$ certainly contains $(\mathbf{q} \cap A)B = \mathbf{p}B$ and B is a Dedekind domain, so $\mathbf{q}$ divides $\mathbf{p}B$.
$\rightarrow \exists \begin{array}{ccc} SpecB & \rightarrow & SpecA \\ \mathbf{q} & \mapsto & \mathbf{q} \cap A \end{array}$ which is surjective (for any nonzero $\mathbf{p} \in SpecA, \mathbf{p}B \neq 0$ and it is not a unit ideal, and therefore divisible by at least one $\mathbf{q} \in SpecB$).

Definition 2.1.20. *Assuem AKLB, and let $\mathbf{p}$ be a prime of A. The exponent $e_{\mathbf{q}}$ in the factorisation $\mathbf{p}B = \Pi_{\mathbf{q}|\mathbf{p}} \mathbf{q}^{e_{\mathbf{q}}}$ is the ramification index of $\mathbf{q}$ and the degree $f_{\mathbf{q}} = [B/\mathbf{q} : A/\mathbf{p}]$ of the corresponding residue field extension is the residue degree of* $\mathbf{q}$.

Lemma 2.1.11. *Let A be a Dedekind domain with fraction field K, let $M/L/K$ be a tower of finite separable extension, and let B, C be the integral closures of A in L, M respectively. Then C is the integral closure of B in M and if τ is a prime of M lying above a priem $\mathbf{q}$ of L lying above a prime $\mathbf{p}$ of K, then $e_{\tau/\mathbf{p}} = e_{\tau/\mathbf{q}} e_{\mathbf{q}/\mathbf{p}}, f_{\tau/\mathbf{p}} = f_{\tau/\mathbf{q}} f_{\mathbf{q}/\mathbf{p}}$.*

pf. The integral closure of B in M lies in C and it contains C since $A \subseteq B$. Thus we have a tower of Dedekind extensions $C/B/A$. If $\tau|\mathbf{q}|\mathbf{p}$, then the factorisation of $\mathbf{p}C$ in C refines the factorisation of $\mathbf{p}B$ in B, so $e_{\tau/\mathbf{p}} = e_{\tau/\mathbf{q}} e_{\mathbf{q}/\mathbf{p}}$. The residue field embedding $A/\mathbf{p} \rightarrow C/\tau$ factors as $A/\mathbf{p} \rightarrow B/\mathbf{q} \rightarrow C/\tau, f_{\tau/\mathbf{p}} = f_{\tau/\mathbf{q}} f_{\mathbf{q}/\mathbf{p}}$.

Lemma 2.1.12. *Assuem AKLB, and let $\mathbf{p}$ be a prime of A. The dimension of $B/\mathbf{q}B$ as an $A/\mathbf{p}$-vector space is equal to the dimension of L as a K-vector space.*

pf. Let $A_{\mathbf{p}} \equiv S^{-1}A$, $B_{\mathbf{q}} \equiv S^{-1}B$ be localisations of A and B respectively, where $S = A \setminus \mathbf{p}$. Then $A_{\mathbf{p}}/\mathbf{p}A = S^{-1}A/\mathbf{p}S^{-1}A \cong A/\mathbf{p}$ and $B_{\mathbf{p}}/\mathbf{p}B_{\mathbf{p}} \cong S^{-1}B/(\mathbf{p}S^{-1}B) \cong B/\mathbf{p}B$. Assume wlog A is a DVR, and therefore it is a PID. B is a finitely generated as an A-module and as an integral domain containing A it must be torsion-free. By the structure theorem for finitely generated modules over a PID, B is free of finite rank over A. Also B spans L as a K-vector space, so any A-basis for B is a K-basis for L. Thus $B \cong A^n$. Then $\mathbf{p}B \cong \mathbf{p}A^n = (\mathbf{p}A)^n \rightarrow B/\mathbf{p}B \cong A^n/(\mathbf{p}A)^n \cong (A/\mathbf{p})^n$ is a free $A/\mathbf{p}$-module of dimension n.

Theorem 2.1.12. *Assume AKLB. For each prime* $\mathbf{p}$ *of* A, $\sum_{\mathbf{q}|\mathbf{p}} e_{\mathbf{q}} f_{\mathbf{q}} = [L:K]$.

pf. $B/\mathbf{p}B \cong \Pi_{\mathbf{q}|\mathbf{p}} B/\mathbf{q}^{e_{\mathbf{q}}}$. Then by the previous proposition
$[L:K] = [B/\mathbf{p}B : A/\mathbf{p}] = \sum_{\mathbf{q}|\mathbf{p}} [B/\mathbf{q}^{e_{\mathbf{q}}} : A/\mathbf{p}]$ (by the CRT)
$= \sum_{\mathbf{q}|\mathbf{p}} e_{\mathbf{q}} [B/\mathbf{q} : A/\mathbf{p}]$ (given $B/\mathbf{q}^{e_{\mathbf{p}}}$ a $B/\mathbf{q}$vector space of dimension $e_{\mathbf{q}}$)
$= \sum_{\mathbf{q}|\mathbf{p}} e_{\mathbf{a}} f_{\mathbf{q}}$.
For $\pi \in \mathbf{q}$ a uniformiser of $B_{\mathbf{q}}$, the images of $(\pi^0, \ldots, \pi^{e_{\mathbf{q}}-1})$ in $B/\mathbf{q}^{e_{\mathbf{q}}}$ are a $B/\mathbf{q}$-basis for $B/\mathbf{q}^{e_{\mathbf{q}}}$.

Corollary 2.1.17. *Assume AKLB, and let* $\mathbf{p}$ *be a prime of* A. *Let* $g_{\mathbf{p}} \equiv \#\{\mathbf{q}|\mathbf{p}\}$ *denote the cardinality of the fibre above* $\mathbf{p}$. *Then* $g_{\mathbf{p}}$ *is an integer in the interval* $[1,n]$, *where* $n = [L:K]$ *as are* $e_{\mathbf{q}}, f_{\mathbf{q}}$ *for each* $\mathbf{q}|\mathbf{p}$.

Definition 2.1.21. *Assume AKLB, let* $\mathbf{p}$ *be a prime of* A.

- L/K *is totally ramified at* $\mathbf{q}$ *if* $e_{\mathbf{q}} = [L:K]$
- L/K *is unramified at* $\mathbf{q}$ *if* $e_{\mathbf{q}} = 1$ *and* $B/\mathbf{q}$ *is a separable extension of* $A/\mathbf{p}$.
- L/K *is unramified above* $\mathbf{p}$ *if it is unramified at all* $\mathbf{q}|\mathbf{p}$, *equivalently if* $B/\mathbf{p}B$ *is a finite etale algebra over* $A/\mathbf{p}$

. *When* L/K *is unramified above* $\mathbf{p}$, *we say that*

- $\mathbf{p}$ *remains inert in* L *if* $\mathbf{q} = \mathbf{p}B$ *is prime*
- $\mathbf{p}$ *splits completely in* L *if* $g_{\mathbf{p}} = [L:K]$

e.g. Let $A \equiv \mathbb{R}[x], K = FracA = \mathbb{R}(x), L = \mathbb{R}\sqrt{x^3+3x}$. The integral closure of A in L is the Dedekind domain $B = \mathbb{R}[x,y]/(y^2 - x^3 - 3x)$. Then $[L:K] = 2$.
-The prime $(x-1)$ factors in B into $(x-1) = (x-1, y-2)(x-1, y+2)$ since $y^2 - 4 = x^3 + 3x - 5 \in (x-1)$. Thus $e_{(x-1,y-2)} = 1, f_{(x-1,y-2)} = [B/(x-1,y-2) : A/(x-1)] = [\mathbb{R}:\mathbb{R}] = 1$. so L/K is unramified above $(x-1)$ and $(x-1)$ splits completely.
-The prime $(x+1)$ remains prime in B since $y^2 = -1$ has no solution in $\mathbb{R}$. Thus $e_{(x+1)} = 1, f_{(x+1)} = [B/(x+1) : A/(x+1)] \cong [\mathbb{C}:\mathbb{R}] = 2$. so L/K at $(x+1)$ is unramified above $\mathbf{p}$ and $\mathbf{p}$ remains inert.
-The prime (x) factors in B as $(x) = (x,y)^2, e_{(x,y)} = 2, f_{(x,y)} = 1$. so L/K is totally ramified at (x).

For a ring extension B/A in which B is a free A-module of finite rank. the norm map is defined as $\begin{array}{rcl} N_{B/A} : B & \to & A \\ b & \mapsto & det(B \to B) \end{array}$. If A is a Dedekind domain, even when M is not free, $A_{\mathbf{p}}$ is a DVR and therefore a PID, $M_{\mathbf{p}}$ is a torsion-free

$A_{\mathbf{p}}$-module, given it lies in a K-vector space and $A_{\mathbf{p}} \subseteq K$, and any finitely generated torion-free module over a PID is free. Thus its localisation $M_{\mathbf{p}}$ at any prime $\mathbf{p}$ of A is a free $A_{\mathbf{p}}$-module, i.e. M is a locally free module.

Definition 2.1.22. *Let A be a Dedekind domain with fraction field K, let V be an n-dimenstional K-vector space, let M,N be A-lattices in V and let $\mathbf{p}$ be a prime of A. Then $A_{\mathbf{p}}$ is a PID and we must have $M_{\mathbf{p}} \cong A_{\mathbf{p}}^n \cong N_{\mathbf{p}}$ by the above statement. Choose an $A_{\mathbf{p}}$-module isomorphism $\phi_{\mathbf{p}} : M_{\mathbf{p}} \to N_{\mathbf{p}}$ and let $\hat{\phi}_{\mathbf{p}}$ denote the unique K-linear map $V \to V$ extending $\phi_{\mathbf{p}}$. The linear map $\hat{\phi}_{\mathbf{p}}$ is an isomorphism and therefore has nonzero determinant. The module index $[M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}} \equiv (\det \hat{\phi}_{\mathbf{p}})$.*

$\to$ this does not depend on the choice of $\phi_{\mathbf{p}}$, since any other choice can be written as $\phi_1 \phi_{\mathbf{p}} \phi_2$, where $\phi_1 : M_{\mathbf{p}} \to M_{\mathbf{p}}, \phi_2 : N_{\mathbf{p}} \to N_{\mathbf{p}}$ that necessarily have the unit determinant.

The module index $[M : N]_A$ is the A-module $[M : N]_A \equiv \cap_{\mathbf{p}} [M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}}$, where $\mathbf{p}$ ranges over primes of A and the intersection takes places in K. $\to [M : N]_A$ is an A-submodule of K.

Proposition 2.1.12. *Let A be a Dedekind domain with fraction field K and let M,N be A-lattices in a K-vector space of finite dimension. The module index $[M : N]_A$ is a nonzero fractional ideal of A whose localisation at each prime $\mathbf{p}$ of A is equal to the local module index $[M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}}$.*

pf. The finitely generated A-module M is locally free, then there exists nonzero $a_1, \ldots, a_r \in A$ generating unit ideal s.t. each $M[1/a_i]$ is a free $A[1/a_i]$-module. Similarly we have $b_1, \ldots, b_s \in A$ generating the unit ideal s.t. each $N[1/b_j]$ is a free $A[1/b_j]$-module. For any pair a_i, b_j, if we localise at the multiplicative set $S_{ij} \equiv \{a_i^m b_j^n : m,n \in \mathbb{Z}_+\}$, then $S_{ij}^{-1}M, S_{ij}^{-1}N$ are free $S_{ij}^{-1}A$-module and $([S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}A})_{\mathbf{p}} = [M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}}$, for all primes $\mathbf{p}$ of A that do not contain either a_i or b_j, since we can fix a global $S_{ij}^{-1}A$-module isomorphism $\phi : S_{ij}^{-1}M \to S_{ij}^{-1}M$ that induces $A_{\mathbf{p}}$-module isomorphisms $\phi_{\mathbf{p}} : M_{\mathbf{p}} \to N_{\mathbf{p}}$ with $(\det \hat{\phi})_{\mathbf{p}} = (\det \hat{\phi}_{\mathbf{p}})$. If a_i or b_j is contained in $\mathbf{p}$, $[S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}}$ is a unit ideal. Additionally, since $\{a_i\}, \{b_j\}$ both generate the unit ideal, for every prime $\mathbf{p}$, $\exists a_i, b_j$ s.t. they do not lie in $\mathbf{p}$. Thus
$[M : N]_A = \cap_{\mathbf{p}} [M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}} = \cap_{ij} [S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}A}$.
Let $I_{ij} \equiv [S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}A}$. Each I_{ij} is a nonzero principal fractional $S_{ij}^{-1}A$-ideal, and therefore we can choose a single $\alpha \in K^\times$ s.t. αI_{ij} is an $S_{ij}^{-1}A$-ideal. The intersection of the αI_{ij} lies in $\cap_{ij} S_{ij}^{-1}A = A$ and therefore it is an A-submodule of A, hence an ideal and finitely generated because A is noetherian. Thus $[M : N]_A$

is a fractional ideal of A. It is nonzero since it contains the product of the generators of the I_{ij}.
The localisation of the intersection of a finite set of A-modules is equal to the intersection of their localisation, thus $([M:N]_A)_{\mathbf{p}} = (\cap_{ij}[S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}A})_{\mathbf{p}} =$
$\cap_{ij}([S_{ij}^{-1}M : S_{ij}^{-1}M]_{S_{ij}^{-1}A})_{\mathbf{p}}$
$= [M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}}$.
$\rightarrow [M:N]_A \in \mathscr{I}_A$
$\rightarrow [M:N]_A[N:M]_A = [M:M]_A = A$ thus $[M:N]_A$ and $[N:M]_A$ are inverses in the ideal group $\mathscr{I}_A$.
$\rightarrow$ When $N \subseteq M$, $[M:N]_A \subseteq A$ is actually an ideal since in the case we can express a basis for $N_{\mathbf{p}}$ as $A_{\mathbf{p}}$-linear combinations of a basis for $M_{\mathbf{p}}$ and the matrix for $\hat{\phi}_{\mathbf{p}}$ will have entries in $A_{\mathbf{p}}$.

If M,N are nonzero fractional ideals of A, then $M[M:N]_A = M(N:M) = N$ and if $N \subseteq M$, $I \equiv [M:N]_A \subseteq A$ is an ideal and $MI = N = NA \leftrightarrow M/N \cong A/I$ as the quotients of A-modules. Then $I = \{a \in A : aM \subseteq N\}$ is an annihilator of M/N, which is a cyclic A-module. Conversely, if $M/N \cong A/I$ for nonzero fractional ideals $N \subseteq M$, $I = [M:N]_A$.

Theorem 2.1.13. *Let A be a Dedekind domain with fraction field K, and let $N \subseteq M$ be A-lattices in a K-vector space V of dimension r for which the quotient module M/N is a direct sum of cyclic A-modules : $M/N \cong A/I_1 \oplus \cdots \oplus A/I_n$ where $I_1,\ldots,I_n$ are nonzero ideals of A. Then $[M:N]_A = I_1 \ldots I_n$.*

pf. Let $\mathbf{p}$ be a prime of A, let π be a uniformiser for $A_{\mathbf{p}}$ and let $e_j = v_{\mathbf{p}}(I_j)$ for $j \in [1,n]$. Pick a basis for $M_{\mathbf{p}}$ and an isomorphism $\phi_{\mathbf{p}} : M_{\mathbf{p}} \rightarrow N_{\mathbf{p}}$ so that $M_{\mathbf{p}}/N_{\mathbf{p}}$ so that $M_{\mathbf{p}}/N_{\mathbf{p}} = coker\phi_{\mathbf{p}}$. The matrix of $\phi_{\mathbf{p}}$ is an $r \times r$ matrix over the PID $A_{\mathbf{p}}$ with nonzero determinant. Thus it has UDV form with $U,V \in GL_r(A_{\mathbf{p}}), D = diag(\pi^{d_1},\ldots,\pi^{d_r})$ for some uniquely determined nonnegative integers $d_1 \leq \cdots \leq d_r$. Then $A_{\mathbf{p}}/(\pi^{e_1}) \oplus \cdots \oplus A_{\mathbf{p}}/(\pi^{e_n}) \cong M_{\mathbf{p}}/N_{\mathbf{p}} = coker\phi \cong A_{\mathbf{p}}/(\pi^{d_1}) \oplus \cdots \oplus A_{\mathbf{p}}/(\pi^{d_r})$. By the structure theorem for modules over a PID, the nontrivial summands on each side are precisely the invariant factors of $M_{\mathbf{p}}/N_{\mathbf{p}}$, possibly in different orders. Thus $\sum_j e_j = \sum_i d_i$ and by the definition of the module index $[M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}} = (\det\phi_{\mathbf{p}}) = (\det D) = (\pi^{\sum d_i}) = (\pi^{\sum e_i}) = (\pi_{\mathbf{p}}^{e_1})\ldots(\pi_{\mathbf{p}}^{e_n}) = (I_1 \ldots I_n)_{\mathbf{p}}$. Thus $[M:N]_A = I_1 \ldots I_n$, since the localisations are $([M:N]_A)_{\mathbf{p}} = [M_{\mathbf{p}} : N_{\mathbf{p}}]$, and $(I_1 \ldots I_n)_{\mathbf{p}}$ coincide for every prime $\mathbf{p}$.

Definition 2.1.23. *Assume AKLB. The ideal norm is the map* $\begin{array}{rcl} N_{B/A} : \mathscr{I}_B & \rightarrow & \mathscr{I}_A \\ I & \mapsto & [B:I]_A \end{array}$.
We extend $N_{B/A}$ to the zero ideal by defining $N_{B/A}((0)) = (0)$.

Proposition 2.1.13. *Assume AKLB and let $\alpha \in L$. Then $N_{B/A}((\alpha)) = (N_{L/K}(\alpha))$.*

pf. The case $\alpha = 0$ is immediate. So assume $\alpha \in L^\times$. Then $N_{B/A}((\alpha)) = [B : \alpha B]_A = \cap_{\mathbf{p}}[B_p : \alpha B_p]_{A_p} = (\det(L \to L)) = (N_{L/K}(\alpha))$, since each $B_p \to \alpha B_p$ is an isomorphism of free A_p-modules that are A_p-lattices in L.

Proposition 2.1.14. *Assume AKLB. The map $N_{B/A}\mathscr{I}_B \to \mathscr{I}_A$ is a group homomorphism.*

pf. Let $\mathbf{p}$ be a maximal ideal of A. Then A_p is a DVR and B_p is a semilocal Dedekind domain, hence a PID. Thus every element of $\mathscr{I}_{B_p}$ is a principal ideal (α) for some $\alpha \in L^\times$, and by the previous proposition, $N_{B_p/A_p} : \mathscr{I}_B \to \mathscr{I}_{A_p}$ is a group homomorphism, since $N_{L/K}$ is. For any $I, J \in \mathscr{I}_B, N_{B/A}(IJ) = \cap_{\mathbf{p}} N_{B_p/A_p}(I_p J_p) = \cap_{\mathbf{p}} N_{B_p/A_p})N_{B_p/A_p}(I_{\mathbf{p}})J_{\mathbf{p}}) = N_{B/A}(I)N_{B/A}(J)$.

Corollary 2.1.18. *Assume AKLB. For all $I, J \in \mathscr{I}_B, [I : J]_A = N_{B/A}(I^{-1}J) = N_{B/A}((J : I))$.*

pf. Because B is a Dedekind domain, $(J : I) = I^{-1}J$ and therefore the second equality follows. Also $[I : J]_A = [I : B]_A[B : J]_A = [B : I]_A^{-1}[B : J] = N_{B/A}(I^{-1})N_{B/A}(J) = N_{B/A}(I^{-1}J)$ by the previous proposition.

Corollary 2.1.19. *Assume AKLB and let I be a fractional ideal of B. The ideal norm of I is the fractional ideal of A generated by the image of I under the field norm $N_{L/K}$, that is, $N_{B/A}(I) = (N_{L/K}(\alpha) : \alpha \in I)$.*

pf. Let $J \equiv (N_{L/K}(\alpha) : \alpha \in I)$. Then for any nonzero prime $\mathbf{p}$ of A, the localisation of the ideal $N_{B/A}(I) = [B : I]_A$ at $\mathbf{p}$ is $[B_p : I_p]_{A_p} = N_{B_p/A_p}(I_{\mathbf{p}})$. The fractional ideal $N_{B_p/A_p}(I_p)$ of $A_{\mathbf{p}}$ is principal, so $N_{B_p/A_p}(I_{\mathbf{p}}) = J_{\mathbf{p}}$ by the proposition, and $N_{B/A}(I) = \cap_{\mathbf{p}} N_{B_p/A_p}(I_{\mathbf{p}})N_{B_p/A_p}(J_{\mathbf{p}}) = \cap_{\mathbf{p}} J_{\mathbf{p}} = J$.

Theorem 2.1.14. *Assume AKLB and let $\mathbf{q}$ be a prime lying above $\mathbf{p}$. Then $N_{B/A}(\mathbf{q}) = \mathbf{p}^{f_q}$ where $f_{\mathbf{q}} = [B/\mathbf{q} : A/\mathbf{p}]$ is the residue field degree of $\mathbf{q}$.*

pf. The $(A/\mathbf{p})$-vector space $B/\mathbf{q}$ has dimension $f_{\mathbf{q}}$; as a quotient of A-modules, we have $B/\mathbf{q} \cong A/\mathbf{p} \oplus \cdots \oplus A/\mathbf{p}$, an $f_{\mathbf{q}}$-fold direct sum of cyclic A-modules $A/\mathbf{p}$, and therefore we can apply the previous theorem. Thus $N_{B/A}(\mathbf{q}) = [B : \mathbf{q}]_A = \mathbf{p} \ldots \mathbf{p} = \mathbf{p}^{f_{\mathbf{q}}}$.

Corollary 2.1.20. *Assume AKLB. For $I \in \mathscr{I}_A$, $N_{B/A}(IB) = I^n$ where $n = [L : K]$.*

pf. Since $N_{B/A}$ and $I \mapsto IB$ are group homomorphisms, we can reduce to the case where $I = \mathbf{p}$ is a nonzero prime. Then $N_{B/A}(\mathbf{p}B) = N_{B/A}(\Pi_{\mathbf{q}|\mathbf{p}}\mathbf{q}^{e_q}) = \Pi_{\mathbf{q}|\mathbf{p}} N_{B/A}(\mathbf{q})^{e_q} = \Pi_{\mathbf{q}|\mathbf{p}}\mathbf{p}^{e_q f_q} = \mathbf{p}^{\sum_{\mathbf{q}|\mathbf{p}} e_q f_q} = \mathbf{p}^n$. e.g.

- The Ideal Norm in Number Fields

  Let $A = \mathbb{Z}, K = \mathbb{Q}, B = \mathcal{O}_L$ the ring of integers of the number field L. Then for $\mathbf{q}$ a nonzero prime ideal of $\mathcal{O}_L$, $N(\mathbf{q}) = (\mathbf{p}^f)$, where $p \in \mathbb{Z}$ is the unique prime in $\mathbf{q} \cap \mathbb{Z}$ and f is the degree of the finite field $B/\mathbf{q}$ as an extension of $\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$.

  Proposition 2.1.15. *Let L be a number field with ring of integers $\mathcal{O}_L$. For any nonzero $\mathcal{O}_L$-ideal α, $N(\alpha) = ([\mathcal{O}_L : \alpha])$. If $\beta \subseteq \alpha$ are nonzero fractional ideals of $\mathcal{O}_L$, then $[\alpha : \beta]_{\mathbb{Z}} = ([\alpha : \beta])$.*

  pf. The ring $\mathcal{O}_L$ is a free $\mathbb{Z}$ module of rank $n \equiv [L : \mathbb{Q}]$. It is free because it is a torsion-free module over $\mathbb{Z}$, which is a PID with rank n because it contains a $\mathbb{Q}$-basis for L. The same is true of any nonzero fractional ideal of $\mathbb{O}_L$; thus $\alpha\mathcal{O}_L$ spans L, because the multiplication by α-map $L \to_{\times\alpha} L$ is an invertible $\mathbb{Q}$-linear transformation. Fix $\mathbb{Z}$-basis for $\mathcal{O}_L$ and the nonzero $\mathcal{O}_L$-ideal α. Let $\Phi \in \mathbb{Z}^{n\times n}$ be the matrix whose columns express each basis element of α in terms of the basis for $\mathcal{O}_L$. Multiplication by Φ defines a $\mathbb{Z}$-module isomorphism from $\mathcal{O}_L$ to α, since it maps the basis for $\mathcal{O}_L$ to the basis for α. Thus $[\mathcal{O}_L : \alpha]_{\mathbb{Z}} = (\det\Phi)$; for every prime $p \in \mathbb{Z}, \Phi$ is defined as a $\mathbb{Z}_{(p)}$-module isomorphism $\phi_{(p)} : (\mathcal{O}_L)_{(p)} \to \alpha_{(p)}$ with $\det\hat{\phi}_{(p)} = \det\Phi$, given the fact that any $\mathbb{Z}$-basis for a free $\mathbb{Z}$-module M is also a $\mathbb{Z}_{(p)}$-basis for the free $\mathbb{Z}_{(p)}$-module $M_{(o)}$. Then if we identify $\mathcal{O}_L$ with $\mathbb{Z}^n$, $|\det\Phi|$ is the volume of a fundamental parallelepiped for α, viewed as a sublattice of $\mathbb{Z}^n$, the absolute value of the determinant of Φ is equal to the index of α in $\mathcal{O}_L$. Thus $([\mathcal{O}_L : \alpha]) = (\det\Phi) = [\mathcal{O}_L : \alpha]_{\mathbb{Z}} = N(\alpha)$, so the proposition follows.
  For any $\alpha \in L^\times, [\alpha : \beta] = [\alpha\alpha : \alpha\beta]$, $[\alpha : \beta]_{\mathbb{Z}}$. Then we have a tower of free $\mathbb{Z}$-modules $\beta \subseteq \alpha \subseteq \mathcal{O}_L \to [\mathcal{O}_L : \alpha][\alpha : \beta] = [\mathcal{O}_L : \beta]$ so by replacing both sides with the $\mathbb{Z}$-ideals they generate, $N(\alpha)([\alpha : \beta]) = N(\alpha)N(\alpha^{-1}\beta) = N(\beta)$. so the rest of the proposition follows.

  $\to \mathbb{Z}$ has only ± 1 as units, so we can identify each fractional ideal with a positive rational number and view the absolute norm $N : \mathcal{I}_{\mathcal{O}_{\mathcal{L}}} \to \mathcal{I}_{\mathbb{Z}}$ as a homomorphism $N : \mathcal{I}_{\mathcal{O}_L} \to \mathbb{Q}^\times_{++}$ from the ideal group of $\mathcal{O}_L$ to the multiplicative group of positive rational numbers.

In the usual AKLB setup, by the primitive element theorem, when L/K is a finite separable extension, $L = K(\alpha)$, $\exists\alpha \in L$ and also $\alpha \in B$ but this does not mean $B = A[\alpha]$.

Definition 2.1.24. *Extensions L/K for which $B = A[\alpha]$ for some $\alpha \in B$ are said to be monogenic.*

→ then B is a free A-module and A-basis for B is an integral power basis $\{1, \alpha, \ldots, \alpha^{n-1}\}$.

Theorem 2.1.15 (Dedekind-Kummer)**.** *Assuem AKLB with $L = K(\alpha)$, $\alpha \in B$. Let $f \in A[x]$ be the minimal polynomial for α, let $\mathbf{p}$ be a prime of A and let $\bar{f} = \bar{g}_q^{e_1} \ldots \bar{g}_r^{e_r}$ be its factorisation into monic irreducibles in $(A/\mathbf{p})[x]$. Let $\mathbf{q}_i \equiv (\mathbf{p}, g_i(\alpha))$, where $g_i \in A[x]$ is any lift of $\bar{g}_i$ in $(A/\mathbf{p})[x]$ under the reduction map $A[x] \to (A/\mathbf{p})[x]$. If $B = A[\alpha]$, then $\mathbf{p}B = \mathbf{q}_1^{e_1} \ldots \mathbf{q}_r^{e_r}$, is the prime factorisation of $\mathbf{p}B$ in B and the residue field degree of $\mathbf{q}_i$ is* $\deg \bar{g}_i$.

pf. We have $B = A[\alpha] \cong A[x]/(f)$ and therefore $\frac{B}{\mathbf{q}_i} = \frac{A[\alpha]}{(\mathbf{p}, g_i(\alpha))} \cong \frac{A[x]}{(f, \mathbf{p}, g_i)} \cong \frac{(A/\mathbf{p})[x]}{(\bar{f}, \bar{g}_i)} \cong \frac{(A/\mathbf{p})[x]}{\bar{g}_i)}$, which is a field, given $\bar{g}_i$ irreducible. Thus $\mathbf{q}_i$ is a prime above $\mathbf{p}$ with residue field degree $f_{\mathbf{q}_i} = \deg \bar{g}_i$ as claimed.
The ideal $\Pi_i \mathbf{q}_i^{e_i} = \Pi_i (\mathbf{p}, g_i(\alpha))^{e_i} = \Pi(\mathbf{p}B + (g_i(\alpha)))^{e_i}$ is divisible by $\mathbf{p}B$ since $\Pi_i (g_i(\alpha)^{e_i}) \equiv (f(\alpha)) \equiv (0) \; mod \; \mathbf{p}B$. The $\bar{g}_i(x)$ are distinct elements of $(A/\mathbf{p})[x]/(f) \cong A[x]/(\mathbf{p}, f) \cong A[\alpha]/\mathbf{p}A[\alpha]$ and therefore $g_i(\alpha)$ are distinct module $\mathbf{p}B$. Thus the prime ideals $\mathbf{q}_i$ are distinct fand $e_i \geq e_{\mathbf{q}_i}$ and $\{\mathbf{q}|\mathbf{p}\} \subseteq \{\mathbf{q}_i\}$ in order for $\Pi \mathbf{q}_i^{e_i}$ to be divisible by $\mathbf{p}B$. Thus $\{\mathbf{q}_i\} = \{\mathbf{q}|\mathbf{p}\}$ since $\mathbf{q}_i$ is a prime above $\mathbf{p}$. Now $N_{B/A}(\Pi \mathbf{q}_i^{e_i}) = \Pi N_{B/A}(\mathbf{q}_i)^{e_i} = \Pi(\mathbf{p}^{f_{q_i}})^{e_i} = \mathbf{p}^{e_i \deg \bar{g}_i} = \mathbf{p}^{\deg f} = \mathbf{p}^{[L:K]}$ so $e_i = e_{q_i}$.

Definition 2.1.25. *Let S/R be an extension of commutative rings. The conductor of R in S is the largest S-ideal that is also an R-ideal, equivalently it is the largest ideal of S contained in R. It can be written as $\mathbf{c} \equiv \{\alpha \in S : \alpha S \subseteq R\} = \{\alpha \in R : \alpha S \subseteq R\}$. If R is an integral domain, the conductor of R is the conductor of R in its integral closure.*

Lemma 2.1.13. *Let R be a noetherian domain. The conductor of R in its integral closure S is nonzero iff S is finitely generated as an R-module.*

If R has nonzero conductor, its integral closure S is a fractional ideal of R that is also a ring, i.e. $S = \frac{1}{r} I$, $r \in R$, I is an R-ideal, and the conductor $\mathbf{c}$ is the set of denominator $r \in R$ for which $S = \frac{1}{r} I$.

Definition 2.1.26. *An order $\mathcal{O}$ is a noetherian domain of dimension one whose conductor is nonzero, equivalently, whose integral closure is finitely generated as an $\mathcal{O}$-module.*

Every Dedekind domain that is not a field is an order. The integral closure of an order is a Dedekind domain, but not every ring whose integral closure is a Dedekind domain is an order. But in AKLB, B is finitely generated over A and therefore over every intermediate ring between A and B, including all those whose integral closure is B.

Definition 2.1.27. *Let A be a noetherian domain with fraction field K, and let L be a K-algebra of finite dimension. An A-order in L is an A-lattice that is also a ring.*

Proposition 2.1.16. *Assume AKLB and let $\mathcal{O}$ be a subring of L. Then $\mathcal{O}$ is an A-order in L iff it is an order with integral closure of B.*

pf. Under AKLB assumption, $\dim A = 1 = \dim B$ since $A = B \cap K, \mathcal{O} \subseteq L$ is an A-module containing 1, so it contains A. Suppose $\mathcal{O}$ is an A-order in L. Then $\mathcal{O}$ is an A-lattice, hence finitely generated as an A-module, and therefore integral over A. Thus $\mathcal{O}$ lies in the integral closure B of A in L. The fraction field of $\mathcal{O}$ is a K-vector space spanning 4L,$henceequalto4L$, so $\mathcal{O}$ and B have the same fraction field and B is the integral closure of $\mathcal{O}$. Thus $\mathcal{O}$ is a domain of dimension 1, and it is noetherian because it is a finitely generated module over the noetherian ring A. The integral closure B of $\mathcal{O}$ is finitely generated over A, hence over $\mathcal{O}$; hence it is on order.
Suppose $\mathcal{O}$ is an order with the integral closure B. It is an A-submodule of the noetherian A-module B, hence finitely generated over A. It contains a K-basis for L because L is its fraction field. Thus $\mathcal{O}$ is an A-lattice in L that is also a ring and therefore is an A-order in L.

Definition 2.1.28. *An A-order in L is a maximal order if it is not properly contained in any other A-order in L.*

$\rightarrow$ if A is a Dedekind domain, every A-order in L lies in a maximal order, and under the AKLB assumption, B is the unique maximal order.

Proposition 2.1.17. *In any order $\mathcal{O}$, only finitely many primes contain the conductor.*

pf. The map $\begin{array}{ccc} SpecB & \rightarrow & Spec\mathcal{O} \\ \mathbf{q} & \mapsto & \mathbf{q} \cap \mathcal{O} \end{array}$ is surjective, since the fact that B is integrally closed ensures every prime $\mathbf{p}$ of $\mathcal{O}$ has at least one prime $\mathbf{q}$ lying above it. If a prime $\mathbf{q}$ of B contains the conductor $\rfloor$, so does $\mathbf{p} = \mathbf{q} \cap \mathcal{O}$ and conversely. In B, we can factor $\mathbf{c}$ into a product of powers of finitely many primes $\mathbf{q}$, and therefore only finitely many prime $\mathbf{p}$ of $\mathcal{O}$ contain $\mathbf{c}$.

Lemma 2.1.14. *Let $\mathcal{O}$ be an order with integral closure B and conductor $\mathbf{c}$ and let $\mathbf{p}$ be a prime of $\mathcal{O}$ not containing $\mathbf{c}$. then $\mathbf{p}B$ is prime of B.*

pf. Let $\mathbf{q}$ be a prime of B, lying above $\mathbf{p}$, so that $\mathbf{p} = \mathbf{q} \cap \mathcal{O}$, and pick an element $s \in \mathbf{c}$ not in $\mathbf{p}$. Then if $a/b \in \mathcal{O}_{\mathbf{p}}$ with $a \in \mathcal{O}, b \in) \setminus \mathbf{p}$, then $b \in B \setminus \mathbf{q}$ and therefore $a/b \in B_{\mathbf{q}} \rightarrow \mathcal{O}_{\mathbf{p}} \subseteq B_{\mathbf{q}}$
If $a/b \in B_{\mathbf{q}}$ with $a \in B, b \in B \setminus \mathbf{q}$, then $sa \in \mathcal{O}, sb \in \mathcal{O} \setminus \mathbf{p}$ so $(sa)/(sb) = a/b \in \mathcal{O}_{\mathbf{p}} \rightarrow$

$B_{\mathbf{q}} \subseteq \mathcal{O}_{\mathbf{p}}$ and therefore $\mathcal{O}_{\mathbf{p}} = B_{\mathbf{q}}$. Note that $\mathbf{q}'|\mathbf{p}$ implies $B_{\mathbf{q}'} = \mathcal{O}_{\mathbf{p}} = B_{\mathbf{q}}$ which also implies $\mathbf{q}' = \mathbf{q}$. So there is only one prime $\mathbf{q}$ lying above $\mathbf{p}$. Thus $\mathbf{p}B = \mathbf{q}^e$, $e \geq 1$. Since we must have $\mathbf{p}\mathcal{O}_{\mathbf{p}} = \mathbf{q}B_{\mathbf{q}}$, $\mathbf{q}^e B_{\mathbf{q}} = \mathbf{q}B_{\mathbf{q}} \to e = 1$.

Corollary 2.1.21. *Let $\mathcal{O}$ be an order with integral closure B and conductor $\mathbf{c}$. The restriction of the map $SpecB \to Spec\mathcal{O}$ defined by $\mathbf{q} \mapsto \mathbf{q} \cap \mathcal{O}$ to the prime ideals not containing $\mathbf{c}$ is a bijection with inverse $\mathbf{p} \to \mathbf{p}B$.*

Theorem 2.1.16. *Let $\mathcal{O}$ be an order with integral closure B and conductor $\mathbf{c}$, and let $\mathbf{p}$ be a prime of $\mathcal{O}$. TFAE*

1. *$\mathbf{p}$ does not contain $\mathbf{c}$*
2. *$\mathcal{O} = \{x \in B : x\mathbf{p} \subseteq \mathbf{p}\}$*
3. *$\mathbf{p}$ is invertible*
4. *$\mathcal{O}_{\mathbf{p}}$ is a DVR*
5. *$\mathbf{p}\mathcal{O}_{\mathbf{p}}$ is principal.*

If any of these equivalent properties hold, then $\mathbf{p}B$ is a prime of B.

pf.

- $i \to ii$
  $\forall x \in \mathcal{O}, x\mathbf{p} \subseteq \mathbf{p}$ and therefore $x \in \{x \in B : x\mathbf{p} \subseteq \mathbf{p}\}$.. If $x \in \{x \in B : x\mathbf{p} \subseteq \mathbf{p}\}, x = a/b$ with $a, b \in \mathcal{O},, (a,b) =$ s.t. there exists a minimal monic polynomial satisfying $(a/b)^n + \alpha_{n-1}(a/b)^{n-1} + \ldots \alpha_0 = 0 \leftrightarrow a^n = -b(\alpha_{n-1}a^{n-1} + \cdots + b^{n-1}\alpha_0) \leftrightarrow b|a$ this is only possible when $b = 1$ given $\mathbf{p}$ does not contain $\mathbf{c}$ and therefore $x \in \mathbf{O}$ and therefore $\mathcal{O} = \{x \in B : x\mathbf{p} \subseteq \mathbf{p}\}$

- $ii \to i$
  If $\mathbf{c} \subseteq \mathbf{p}, \forall x \in B, xB = \mathbf{c}$ which implies $B = \mathcal{O}$ a contradiction. hence $\mathbf{c} \not\subseteq \mathbf{p}$.

- $i \to iii$
  $\exists x \in \mathbf{c} \setminus \mathbf{p}$ by i. Also there exists $\mathcal{O}$-ideal I s.t. $\frac{1}{x}I = B$ by definition. Then $\mathcal{O} = \{x' \in B : x'\mathbf{p} \subseteq \mathbf{p}\}$ and therefore $\mathcal{O} = \{x' \in \frac{1}{x}I : x'\mathbf{p} \subseteq \mathbf{p}\}$. Then $I\mathbf{p} \subseteq \mathcal{O}$ and $\forall y \in \mathcal{O} = \{x' \in \frac{1}{x}O : x'\mathbf{p} \subseteq \mathbf{p}\}$, $y = \frac{r}{x}$ with $r \in \mathcal{O}$ w.t. $\frac{r}{x}\mathbf{p} \subseteq \mathbf{p}$. Thus $x = \frac{rp'}{x} = a\bar{p}, a \in \mathcal{O}$ and $r \in I, \frac{p'}{x} \in \mathbf{p}$ and therefore $\mathbf{p}$ is invertible.

- $iii \to iv$
  Since $\mathcal{O}$ is an order, $\mathcal{O}_{\mathbf{p}}$ is a local ring and noetherian with dimension 1. Its unique maximal ideal is $\mathbf{p}$. Assume it is not principal. Then $\mathbf{p} = (a,b)$, $a, b \in \mathcal{O}$. Since $\mathbf{p}$ is invertible, $\exists J$ s.t. $J\mathbf{p} = \mathcal{O}$. Then $\exists c \in J$ s.t. $c(ra + r'b) = u$ where

u is a unit in $\mathcal{O}$. Thus $u^{-1}cra + u^{-1}cr'b = 1$ which means a, b are coprime andtherefore $0 \subsetneq (a) \subsetneq (a,b) = \mathbf{p}$ a contradiction. Hence it is principal and therefore it is a DVR.

- $iv \leftrightarrow v$
  If the unique maximal of $\mathcal{O}_{\mathbf{p}} = \mathbf{p}\mathcal{O}_{\mathbf{p}}$ is principal, $\mathcal{O}_{\mathbf{p}}$ is a PID. Thus it is a local PID and therefore is a DVR.
  $\mathcal{O}_{\mathbf{p}}$ is a DVR and therefore it is a PID so its maximal is principal.

- $iv \rightarrow i$
  Assume the contrary, i.e. $\mathbf{c} \subseteq \mathbf{p}$. If $\mathbf{c}$ is prime, then $\mathbf{p} = \mathbf{c}$ and $\mathcal{O}_{\mathbf{p}} = B$. If not, $\exists a \in \mathbf{p} \setminus \mathbf{c}$. For such an $a, (a) \neq \mathbf{p}$. Also $\forall b \in \mathbf{c}$, $(b) = \mathbf{p}$, i.e. $\mathbf{p}\mathcal{O}_{\mathbf{p}}$ is not principal hence $\mathcal{O}_{\mathbf{p}}$ is not a DVR, a contradiction.

Orders in Dedekind domains also have a geometric interpretation. If $\mathcal{O}$ is an order, the curve $X = Spec\mathcal{O}$ have a singularity at each closed point P corresponding to a maximal of $\mathcal{O}$ that contains the conductor. Taking the integral closure B of $\mathcal{O}$ yields a smooth curve $Y = SpecB$ with the same function field as X and a morphism $Y \rightarrow X$ that looks like a bijection above non-singular point. The curve Y is called the normalisation of X.

Definition 2.1.29. *Let A be a noetherian domain and let J be an ideal of A. A fractional ideal I of A is prime to J if $IA_{\mathbf{p}} = A_{\mathbf{p}}$ for all prime ideals $\mathbf{p}$ that contain J. The set of invertible fractional ideals prime to J is denoted $\mathcal{I}_A^J$, which is a subgroup of the ideal group $\mathcal{I}_A$.*

Theorem 2.1.17. *Let $\mathcal{O}$ be an order with integral closure B. Let $\rfloor$ be any ideal of B contained in the conductor of $\mathcal{O}$. The map $\mathbf{q} \mapsto \mathbf{q} \cap \mathcal{O}$ induces a group isomorphism from $\mathcal{I}_B^{\mathbf{c}}$ to $\mathcal{I}_{\mathcal{O}}{}^{\mathbf{c}}$ and both groups are isomorphic to the free abelian group generated by their prime ideals. In particular, every fractional ideal of $\mathcal{O}$ prime to the conductor has a unique factorisation into prime ideals $\Pi \mathbf{p}_i^{e_i}$ which matches the factorisation $IB = \Pi \mathbf{q}_i^{e_i}$ with $\mathbf{p}_i = \mathbf{q}_i \cap \mathcal{O}$.*

pf. The B-ideal $\mathbf{c}$ lies in the conductor of $\mathcal{O}$ and is therefore also an $\mathbf{O}$-ideal, so the subgroups $\mathcal{I}_B^{\mathbf{c}}$ and $\mathcal{I}_{\mathcal{O}}{}^{\mathbf{c}}$ are well defined and the map $\mathbf{q} \mapsto \mathbf{q} \cap \mathcal{O}$ gives a bijection between the sets of prime ideals contained in these subgroups by the previous corollary. So the theorem follows.

2.1.3 Galois Extensions

In AKLB setup, add another assumption that L/K is normal, hence Galois, and let $G \equiv Gal(L/K)$ to denote the Galois group. Then we denote it by AKLBG.

Definition 2.1.30. *Let G be a group. A left G-module is an abelian group M equipped with a left G-action that commutes with its group operation; in additive notation, we have $\sigma(a+b) = \sigma(a) + \sigma(b), \forall \sigma \in G$, $a, b \in M$. One similarly defines a right G-module as an abelian group with a right G-action that commutes with the group operation.*

Theorem 2.1.18. *Assume AKLBG. For each fractional ideal I of B and $\sigma \in G$, define $\sigma(I) \equiv \{\sigma(x) : x \in I\}$. The set $\sigma(I)$ is a fractional ideal of B, and this defines a group action on $\mathscr{I}_B$ that makes it a left G-module. Moreover, the restriction of this action to $SpecB$ makes it a G-set.*

pf. **CLAIM:** $\sigma(B) = B, \forall \sigma \in G$
pf. $\forall b \in B$ is integral over A hece $f(b) = 0$ for some monic polynomial $f \in A[x]$ and $0 = \sigma(0) = \sigma(f(b)) = f(\sigma(b)) \to \sigma(b) \in B$, given B is an integral closure of A. Thus $\sigma(B) \subseteq B$. Similarly, $\sigma^{-1}(B) \subseteq B \leftrightarrow B \subseteq \sigma(B)$ and therefore the claim follows.
Each $\sigma \in G = Gal(L/K)$ is a field automorphism of L and thus commutes with addition and multiplication. Thus if $I \subseteq I$ is a finitely generated B-module, $\sigma(I)$ is a finitely generated $\sigma(B)$-module and therefore B-module, i.e. I is a fractional ideal. Clearly $\sigma((0)) = (0), \forall \sigma \in G$ so G permutes $\mathscr{I}_B$, the group of nonzero fractional ideals. Also $(\sigma\tau)(I) = \{(\sigma\tau)(x) : x \in I\} = \{\sigma(\tau(x)) : x \in I\} = \{\sigma(y) : y \in \tau(I)\} = \sigma(\tau(I))$ so G is a left action of $\mathscr{I}_B$.
Let $I, J \in \mathscr{I}_B, \sigma \in G$. Each $x \in IJ$ has the form $x = \sum_i a_i b_i$, $a_i \in I, b_i \in J$ and $\sigma(x) = \sum_i \sigma(a_i)\sigma(b_i) \in \sigma(I)\sigma(J)$ thus $\sigma(IJ) \subseteq \sigma(I)\sigma(J)$. Simiralarly, $\sigma^{-1}(\sigma(I)\sigma(J)) \subseteq IJ \leftrightarrow \sigma(I)\sigma(J) \subseteq \sigma(IJ)$ and therefore $\sigma(IJ) = \sigma(I)\sigma(J)$ implying that $\mathscr{I}_B$ is a left G-module.
Let $\mathbf{p}$ be a prime of B and let $\sigma(\mathbf{p}) = \mathbf{q_1^{e_1}} \dots \mathbf{q_n^{e_n}}$ be the unique factorisation of $\sigma(\mathbf{p})$ in B. Applying σ^{-1} to both sides yields $\mathbf{p} = \sigma^{-1}(\mathbf{q}_1)^{e_1} \dots \sigma^{-1}(\mathbf{q}_n)^{e_n}$ and therefore $n = 1$ and $e_1 = 1$, given $\mathbf{p}$ prime and the G-action of $\mathscr{I}_B$ restricts to a G-action on $SpecB$ and $MaxSpecB$.

Corollary 2.1.22. *Assume AKLBG. For each prime $\mathbf{p}$ of A, the group G acts transitively on the set $\{\mathbf{q}|\mathbf{p}\}$; in other words, the orbits of the G-action on $SpecB$ are the fibres of the contraction map $SpecB \to SpecA$.*

pf. Consider any $\sigma \in G$. For $\mathbf{q}|\mathbf{p}$, we have $\mathbf{p}B \subseteq \mathbf{q}$ and $\sigma(\mathbf{p}B) \subseteq \sigma(\mathbf{q})$ so $\sigma(\mathbf{q})|\mathbf{p}$. Thus $\{\mathbf{q}|\mathbf{p}\}$ is closed under the action of G. Let $\{\mathbf{q}|\mathbf{p}\} = \{\mathbf{q_1}, \dots, \mathbf{q_n}\}$ and suppose $\mathbf{q_1}, \mathbf{q_2}$ lie in distinct G-orbits. The primes $\mathbf{q}_i$ are maximal ideals, hence pairwise coprime, so by the CRT there is a ring isomorphism $\frac{B}{\mathbf{q_1} \dots \mathbf{q_n}} \cong \frac{B}{\mathbf{q_1}} \times \dots \times \frac{B}{\mathbf{q}_n}$ and we may choose $b \in B$ s.t. $b \equiv 0 \mod \mathbf{q}_2$, $b \equiv 1 \mod \sigma^{-1}(\mathbf{q_1})$, $\forall \sigma \in G$. Then $b \in \mathbf{q}_2$ and $N_{L/K}(b) = \Pi_{\sigma \in G} \sigma(b) \equiv 1 \mod \mathbf{q}_1$ hence $N_{L/K}(b) \notin A \cap \mathbf{q}_1 = \mathbf{p}$, But $N_{L/K}(b) \in N_{L/K}(\mathbf{q}_2) = \mathbf{p}^{f_{q_2}} \subseteq \mathbf{p}$, a contradiction.
$\to \forall \sigma \in G = Gal(L/K)$, it restricts to a ring automorphism of B that fixes every

element of the subring $A = B \cap K$ and thus every element of any prime $\mathbf{p}$ of A. So it induces an isomorphism of residue field extensions $\bar{\sigma} \in Hom_{A/\mathbf{p}}(B/\mathbf{q}, B/\sigma(\mathbf{q}))$ defined by $\bar{\sigma}(x + \mathbf{q}) \equiv \sigma(x) + \sigma(\mathbf{q})$ for $x \in B$.

Corollary 2.1.23. *Assume AKLBG and let* $\mathbf{p}$ *be a prime of* A*. The residue field degrees* $f_{\mathbf{q}} \equiv [B/\mathbf{q} : A/\mathbf{p}]$ *are the same for every* $\mathbf{q}|\mathbf{p}$ *as are the ramification indices* $e_{\mathbf{q}} \equiv v_{\mathbf{q}}(\mathbf{p}B)$.

pf. $\forall \sigma \in G$, we have an isomorphism of the residue field $B/\mathbf{q}$ and $B/\sigma(\mathbf{q})$ that fixes $A/\mathbf{p}$, so they clearly have the same degree $f_{\mathbf{q}} = f_{\sigma(\mathbf{q})}$ and G acts transitively on $\{\mathbf{q}|\mathbf{p}\}$ by the previous corollary, so the function $\mathbf{q} \mapsto f_{\mathbf{q}}$ must be constant on $\{\mathbf{q}|\mathbf{p}\}$. $\forall \sigma \in G$, $\sigma(\mathbf{p}) = \mathbf{p}, \sigma(B) = B$, so $\sigma(\mathbf{p}B) = \mathbf{p}B$ and for each $\mathbf{q}|\mathbf{p}$, $e_{\mathbf{q}} = v_{\mathbf{q}}(\mathbf{p}B) = v_{\mathbf{q}}(\sigma(\mathbf{p}B)) = v_{\mathbf{q}}(\sigma(\Pi_{\tau|\mathbf{p}}\tau^{e_\tau})) = v_{\mathbf{q}}(\Pi\sigma(\tau)^{e_\tau}) = v_{\mathbf{q}})\Pi\tau^{e_{\sigma^{-1}(\tau)}}) = e_{\sigma^{-1}(\mathbf{q})}$. The transitivity of the G-action on $\{\mathbf{q}|[\}$ again implies $\mathbf{q} \mapsto e_{\mathbf{q}}$ is constant on $\{\mathbf{q}|\mathbf{p}\}$.

Corollary 2.1.24. *Assume AKLBG. For each prime* $\mathbf{p}$ *of* A, $e_{\mathbf{p}} f_{\mathbf{p}} g_{\mathbf{p}} = [L : K]$*, where* $g_{\mathbf{p}} \equiv \#\{\mathbf{q}|\mathbf{p}\}$.

Assume AKLBG. When $n \equiv [L : K]$ is a prime, there are three ways a prime $\mathbf{p}$ of

	$e_{\mathbf{p}} = n, f_{\mathbf{p}} = g_{\mathbf{p}} = 1$	$\mathbf{p}$ is totally ramified in L	
A can split in $B \rightarrow$	$f_{\mathbf{p}} = n, e_{\mathbf{p}} = g_{\mathbf{p}} = 1$	$\mathbf{p}$ remains inert in L	if $B/\mathbf{p}B$ is finite etale
	$g_{\mathbf{p}} = n, e_{\mathbf{p}} = f_{\mathbf{p}} = 1$	$\mathbf{p}$ splits completely in L	if $B/\mathbf{p}B$ is finite etale

Definition 2.1.31. *Assume AKLBG. For each prime* $\mathbf{q}$ *of* B*, the decomposition group* $D_{\mathbf{q}}$ *denoted* $D_{\mathbf{q}}(L/K)$ *is the stabiliser of* $\mathbf{q}$ *in* G.

Lemma 2.1.15. *Assume AKLBG and let* $\mathbf{p}$ *be a prime of* A*. The decomposition groups* $D_{\mathbf{q}}$ *for* $\mathbf{q}|\mathbf{p}$ *are all conjugates in* G*, with* $\#D_{\mathbf{q}} = e_{\mathbf{p}} f_{\mathbf{p}}$, $[G : D_{\mathbf{q}}] = g_{\mathbf{p}}$.

pf. Points in an orbit of group action have conjugate stabilisers, so the $D_{\mathbf{q}}$ for $\mathbf{q}|\mathbf{p}$ are all conjugate by the corollary 2.2.22. The orbit stabiliser theorem implies $[G : D_{\mathbf{q}}] = \#\{\mathbf{q}|\mathbf{p}\} = g_{\mathbf{p}}$. So $\#G = [L : K] = e_{\mathbf{p}} f_{\mathbf{p}} g_{\mathbf{p}}$ by the previous corollary. so $\#D_{\mathbf{q}} = \#G/[G : D_{\mathbf{q}}] = e_{\mathbf{p}} f_{\mathbf{p}}$.

$\forall \sigma \in G$ induces a residue field isomorphism $\bar{\sigma} \in Hom_{A/\mathbf{p}}(B/\mathbf{q}, B/\sigma(\mathbf{q}))$. If $\sigma \in D_{\mathbf{q}}, \sigma(\mathbf{q}) = \mathbf{q}$, $\bar{\sigma} \in Aut_{a/\mathbf{p}}(B/\mathbf{q})$ and the map $\sigma \mapsto \bar{\sigma}$ defines a group homomorphism $\pi_{\mathbf{q}} : D_{\mathbf{q}} \rightarrow Aut_{A/\mathbf{p}}(B/\mathbf{q})$.

Proposition 2.1.18. *Assume AKLBG and let* $\mathbf{q}|\mathbf{p}$ *be a prime of* B*. The group homomorphism* $\begin{array}{rcl} \pi_{\mathbf{q}} : D_{\mathbf{q}} & \rightarrow & Aut_{A/\mathbf{p}}(B/\mathbf{q}) \\ \sigma & \mapsto & \bar{\sigma} \end{array}$ *is surjective and* $B/\mathbf{q}$ *is normal over* $A/\mathbf{p}$.

pf. Let F be the separable closure of $A/\mathbf{p}$ in $B/\mathbf{q}$ and for $\bar{b} \in F$, $\exists b \in B$ s.t. $b \equiv \bar{b} \mod \mathbf{q}$, $b \equiv 0 \mod \sigma^{-1}(\mathbf{q})$, $\forall \sigma \in G \setminus D_{\mathbf{q}}$; by the CRT, such an b exists since $\forall \sigma \in G \setminus D_{\mathbf{q}}$, the ideal $\mathbf{q}, \sigma(\mathbf{q})$ are distinct and therefore coprime. Now define $g(x) \equiv \Pi_{\sigma \in G}(x - \sigma(b)) \in A[x]$ and let $\bar{g}$ denote the image of g in $(A/\mathbf{p})[x]$. Then $\bar{b}$ is the root of a polynomial $\bar{g} \in (A/\mathbf{p})[x]$ that splits completely in $(B/\mathbf{q})[x]$ and since $\bar{b}$ was chosen arbitrarily, this applies to all $\bar{b} \in F^{\times}$. Thus F is normal extension of $A/\mathbf{p}$ and $Gal(F/(A/\mathbf{p})) \cong Aut_{A/\mathbf{p}}(B/\mathbf{q})$, since F is the separable closure of $A/\mathbf{p}$ in $B/\mathbf{q}$. For each $\sigma \in G \setminus D_{\mathbf{q}}$, $\bar{\sigma}(\bar{b}) = 0$ so 0 is a root of $\bar{g}(x)$ with multiplicity at least $m = \#(G \setminus D_{\mathbf{q}})$ and the remaining roots are $\bar{\sigma}(\bar{b})$ for $\sigma \in D_{\mathbf{q}}$, all of which are $Gal(F/(A/\mathbf{p}))$-conjugates of $\bar{b}$. Thus $\bar{g}(x)/x^m$ divides a power of the minimal polynomial $f(x)$ of $\bar{b}$, but $f(x)$ is irreducible in $(A/\mathbf{p})[x]$, so $\bar{g}(x)/x^m$ is a power of $f(x)$ and every $Gal(F/(A/\mathbf{p}))$-conjugate of $\bar{b}$ has the form $\bar{\sigma}(\bar{b})$ for some $\sigma \in D_{\mathbf{q}}$. Applying this to $\bar{b}$ chosen so that $F = (A/\mathbf{p})(\bar{b})$ shows that the map $\pi_{\mathbf{q}} : D_{\mathbf{q}} \to Aut_{A/\mathbf{p}}(B/\mathbf{q}) \cong Gal(F/(A/\mathbf{p}))$ is surjective.
For each $b \in B$, define $g \in A[x]$ and $\bar{g} \in (A/\mathbf{p})[x]$ as above. Then $\forall b \in B/\mathbf{q}$ is the root of a polynomial in $(A/\mathbf{p})[x]$ that splits completely in $(B/\mathbf{q})[x]$.

Definition 2.1.32. *Assume AKLBG and let $\mathbf{q}|\mathbf{p}$ be a prime of B. The Kernel of the surjective homomorphism $\pi_{\mathbf{q}} : D_{\mathbf{q}} \to Aut_{A/\mathbf{p}}(B/\mathbf{q})$ is the inertial group $I_{\mathbf{q}}$ of $\mathbf{q}$.*

Corollary 2.1.25. *Assume AKLBG and let $\mathbf{q}|\mathbf{p}$ be a prime of B. We have an exact sequence*

$$0 \to I_{\mathbf{q}} \to D_{\mathbf{q}} \to Aut_{A/\mathbf{p}}(B/\mathbf{q}) \to 0$$

and $\#I_{\mathbf{q}} = e_{\mathbf{p}}[B/\mathbf{q} : A/\mathbf{p}]_i$.

Proposition 2.1.19. *Assume AKLBG, let $\mathbf{q}|\mathbf{p}$ be a prime of B and suppose $B/\mathbf{q}$ is a separable extension of $A/\mathbf{p}$. We have a tower of field extensions $K \subseteq L^{D_{\mathbf{q}}} \subseteq L^{I_{\mathbf{q}}} \subseteq L$ with $e_{\mathbf{p}} = [L : L^{I_{\mathbf{q}}}] = \#I_{\mathbf{q}}$, $f_{\mathbf{p}} = [L^{I_{\mathbf{q}}} : L^{D_{\mathbf{q}}}] = \#D_{\mathbf{q}}/\#I_{\mathbf{q}}$, $g_{\mathbf{p}} = [L^{D_{\mathbf{q}}} : K] = \#\{\mathbf{q}|\mathbf{p}\}$. The fields $L^{D_{\mathbf{q}}}, L^{I_{\mathbf{q}}}$ are the decomposition field and inertial field associated to $\mathbf{q}$.*

pf. The third equality follows by the previous lemma. The second equality follows by the previous proposition ahd the separability of $(B/\mathbf{a})/(A/\mathbf{p})$, since $D_{\mathbf{q}}/I_{\mathbf{q}} \cong Gal((B/\mathbf{q})/(A/\mathbf{p}))$ has cardinality $f_{\mathbf{p}} = [B/\mathbf{q} : A/\mathbf{p}]$. Then the first equality follows by the previous two equalities.

Consider an intermediate field E lying between K and L. Fix a prime $\mathbf{q}|\mathbf{p}$ of B and let $\mathbf{q}_E \equiv \mathbf{q} \cap E$ so that $\mathbf{q}_E|\mathbf{p}$. Use $\bar{G}_{\mathbf{q}}(L/K) \equiv Aut_{A/\mathbf{p}}(B/\mathbf{q}), \bar{G}_{\mathbf{q}}(L/E) \equiv Aut_{(B\cap E)/\mathbf{q}_E}(B/\mathbf{q}), \bar{G}_{\mathbf{q}_E}(E/K) \equiv Aut_{A/\mathbf{p}}((B\cap E)/\mathbf{q}_E)$ to denote the automorphism groups of the residue field extensions associated to the tower $K \subseteq E \subseteq L$.

Proposition 2.1.20. *Assume AKLBD, and let E be an intermediate field between K and L. Let $\mathbf{q}$ be a prime of B and let $\mathbf{q}_E = \mathbf{q} \cap E, \mathbf{p} = \mathbf{q} \cap K$. Then*

$$I_{\mathbf{q}}(L/E) = I_{\mathbf{p}}(L/K) \cap Gal(L/E),\ D_{\mathbf{q}}(L/E) = D_{\mathbf{q}}(L/K) \cap Gal(L/E)$$

If E/K is Galois, we have the commutative diagram of exact sequences

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & I_{\mathbf{q}}(L/E) \arrow[r] \arrow[d] & I_{\mathbf{q}}(L/K) \arrow[r] \arrow[d] & I_{\mathbf{q}_E}(E/K) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & D_{\mathbf{q}}(L/E) \arrow[r] \arrow[d] & D_{\mathbf{q}}(L/K) \arrow[r] \arrow[d] & D_{\mathbf{q}_E}(E/K) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \bar{G}_{\mathbf{q}}(L/E) \arrow[r] \arrow[d] & \bar{G}_{\mathbf{q}}(L/K) \arrow[r] \arrow[d] & \bar{G}_{\mathbf{q}_E}(E/K) \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

pf. $D_{\mathbf{q}}(L/E) \subseteq Gal(L/E) \subseteq Gal(L/K)$. An element $\sigma \in Gal(L/K)$ lies in $D_{\mathbf{q}}(L/E)$ iff it fixes E hence lies in $Gal(L/E)$ and satisfies $\sigma(\mathbf{q}) = \mathbf{q}$ hence lies in $D_{\mathbf{q}}(L/K)$. Thus the second equality follows.
The restriction of $\pi_{\mathbf{q}}(L/K) : D_{\mathbf{q}}(L/K) \to \bar{G}_{\mathbf{q}}(L/K)$ to $D_{\mathbf{q}}(L/E)$ is the map $\pi_{\mathbf{q}} : D_{\mathbf{q}}(L/E) \to \bar{G}_{\mathbf{q}}(L/E)$ hence the kernels agree after intersecting with $Gal(L/E)$ and thus the first equality follows.
Exactness of columns follows by the previous corollary. Each row corresponds to an inclusion followed by a restriction in which the inclusion is precisely the kernel of restriction. Thus the exactness of the first two groups in each row follows.
Surjectivity of the restriction maps follows from the bijection used before.
($Hom_K(L,\Omega) \to Hom_E(L,\Omega) \times Hom_K(E,\Omega)$ whose second factor is restriction and we may view this as a bijection $\phi : Gal(L/K) \to Gal(L/E) \times Gal(E/K)$. If $\sigma \in Gal(E/K)$ stabilises $\mathbf{q}_E$, $\phi^{-1}(1,\sigma) \in Gal(L/K)$ stabilises $\mathbf{q}$ and restricts to σ. Hence the surjectivity of restriction map of the first two rows follows).
The upper left square commute because all the maps are inclusions. The upper right square commutes because inclusion and restriction commute. the lower left square commutes because the horizontal maps are inclusions and the vertical maps coincide on $D_{\mathbf{q}}(L/E)$. The lower right square commutes because the horizontal maps are restrictions and the vertical maps agree after restricting to E.

Corollary 2.1.26. *Assume AKLBG, let E be an intermediate field between K and L. Let $\mathbf{q}$ be a prime of B and let $\mathbf{q}_E = \mathbf{q} \cap E, \mathbf{p} = \mathbf{q} \cap K$. Then* $\begin{array}{ll} e_{\mathbf{q}_E/\mathbf{p}} = 1 & E \subseteq L^{I_{\mathbf{q}}} \\ e_{\mathbf{q}_E/\mathbf{p}} = f_{\mathbf{q}_E/\mathbf{p}} = 1 & E \subseteq L^{D_{\mathbf{q}}} \end{array}$ *where $I_{\mathbf{q}}, D_{\mathbf{q}}$ are the interita and decomposition groups of $\mathbf{q}$.*

pf. $I_{\mathbf{q}}(L/E) = I_{\mathbf{q}}(L/K) \cap Gal(L/E)$ and for $F = L^{I_{\mathbf{q}}}, I_{\mathbf{q}}(L/F) = I_{\mathbf{q}}(L/K) = Gal(L/F)$. Also $Gal(L/EF) = Gal(L/E) \cap Gal(L/F)$ Then we get $I_{\mathbf{q}}(L/E) = I_{\mathbf{q}}(L/K) \cap Ga;(L/E) = Gal(L/F) \cap Gal(L/EF) = Gal(L/EF) = I_{\mathbf{q}}(L/EF)$. Now $e_{\mathbf{q}_E/\mathbf{q}} = \#I_{\mathbf{q}}(L/E) = \#I_{\mathbf{q}}(L/EF) =$

$e_{\mathbf{q}/\mathbf{q}_{EF}}$ is equal to $e_{\mathbf{q}/\mathbf{q}_F} = \#I_{\mathbf{q}}(L/F)$ iff $E \subseteq F$. Thus the first claim follows since $I_{\mathbf{q}}(L/F) = I_{\mathbf{q}}(L/K)$ implies $e_{\mathbf{q}/\mathbf{q}_F} = 1$ since $e_{\mathbf{q}/\mathbf{q}_F} e_{\mathbf{q}_F/\mathbf{p}} = e_{\mathbf{q}/\mathbf{p}}$.
The second follows mutatis mutandis.

Corollary 2.1.27. *Assume AKLBG and let* $\mathbf{p}$ *be a prime of* K*. The fields* $L^{I_{\mathbf{p}}}, L^{D_{\mathbf{p}}}$ *are Galois extensions fo* K *and for any intermediate field* E*, we have* $e_{\mathbf{q}_E/\mathbf{p}} = 1$ *for all* $\mathbf{q}_E|\mathbf{p}$ *iff* $E \subseteq L^{I_{\mathbf{p}}}$ *and* $e_{\mathbf{q}_E/\mathbf{p}} = f_{\mathbf{q}_E/\mathbf{p}} = 1$ *for all* $\mathbf{q}_E|\mathbf{p}$ *iff* $E \subseteq L^{D_{\mathbf{p}}}$*. When* $A/\mathbf{p}$ *is a perfect field, the inertial field is the largest subfield of* L *in which* $\mathbf{p}$ *is unramified, and the decomposition field is the largest subfield in which* $\mathbf{p}$ *splits completely.*

pf. Since G acts transitively on $\{\mathbf{q}|\mathbf{p}\}$, $I_{\mathbf{p}}$ is generated by a complete set of conjugate subgroups $I_{\mathbf{q}}$ and is therefore stable under conjugation, hence normal, and similarly for $D_{\mathbf{p}}$. Thus the corresponding fixed fields are Galois extensions of K. The rest follows by the previous corollary.

Assume further that the residue fields $A/\mathbf{p}$ are finite for all primes $\mathbf{p}$ of K. (e.g. when K is a global field, such assumption holds). Here $B/\mathbf{q}$ is necessarily a Galois extension of $A/\mathbf{p}$ since finite extensions of finite fields are always Galois. Denote $\mathbb{F}_{\mathbf{q}} \equiv B/\mathbf{q}, \mathbb{F}_{\mathbf{p}} \equiv A/\mathbf{p}$l there are finite fields of p-powers order, where p is the characteristic of $\mathbb{F}_p$.

Let $\mathbf{q}|\mathbf{p}$ be a prime of B. Then we get an exact sequence

$$0 \to I_{\mathbf{q}} \to D_{\mathbf{q}} \to_{\pi_{\mathbf{q}}} Gal(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}}) \to 0$$

If $\mathbf{p}$ is unramified, $e_{\mathbf{p}} = e_{\mathbf{q}} = 1$ and $I_{\mathbf{q}}$ is trivial. Then we have an isomorphism $\pi_{\mathbf{q}} : D_{\mathbf{q}} \to Gal(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$. Here Galois group $Gal(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ is the cyclic group of order $f_{\mathbf{p}} = [\mathbb{F}_{\mathbf{q}} : \mathbb{F}_{\mathbf{p}}]$ generated by the Frobenius automorphism $x \mapsto x^{\#\mathbb{F}_{\mathbf{p}}}$.

Definition 2.1.33. *Assume AKLBG with finite residue fields and* $\mathbf{q}|\mathbf{p}$ *unramified. The inverse image of the Frobenius automorphism of* $Gal(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ *under* $\pi_{\mathbf{q}} : D_{\mathbf{q}} \to Gal(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ *is the Frobenius element* $\sigma_{\mathbf{q}} \in D_{\mathbf{q}} \subseteq G$*.*

Proposition 2.1.21. *Assume AKLBG with finite residue fields and* $\mathbf{q}|\mathbf{p}$ *unramified. The Frobenius element* $\sigma_{\mathbf{q}}$ *is the unique* $\sigma \in G$ *s.t.* $\forall x \in B$, $\sigma(x) \equiv x^{\#\mathbb{F}_{\mathbf{p}}} \mod \mathbf{p}$.

pf. Clearly $\sigma_{\mathbf{q}}$ has this property. So existence follows. Suppose $\sigma \in G$ also has this property. For any $x \in \mathbf{q}$, $x \equiv \mod \mathbf{q}$, $\sigma(x) \equiv x^{\#\mathbb{F}_{\mathbf{p}}} \mod \mathbf{q}$ implies $\sigma(x) \equiv 0 \mod \mathbf{q}$, so $\sigma(x) \in \mathbf{q}$. Thus $\sigma(\mathbf{q}) = \mathbf{q}$ and therefore $\sigma \in D_{\mathbf{q}}$. The isomorphism $\pi_{\mathbf{q}} : D_{\mathbf{q}} \to Gal(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ maps both σ and $\sigma_{\mathbf{q}}$ to the Frobenius automorphism $x \mapsto x^{\#\mathbb{F}_{\mathbf{p}}}$ so $\sigma = \sigma_{\mathbf{q}}$.

Proposition 2.1.22. *Assume AKLBG with finite residue fields and* $\mathbf{q}|\mathbf{p}$ *unramified. For all* $\mathbf{q}'|\mathbf{p}$ *the Frobenius elements* $\sigma_{\mathbf{q}}, \sigma_{\mathbf{q}'}$ *are conjugate in* G*.*

pf, G acts transitively on $\{\mathbf{q}|\mathbf{p}\}$ so let $\tau \in G$ be s,t, $\mathbf{q}' = \tau(\mathbf{q})$. Then for any $x \in B$, we have

$$\begin{aligned} \sigma_{\mathbf{q}}(x) &\equiv x^{\#\mathbb{F}_{\mathbf{p}}} \mod \mathbf{q} \\ \tau(\sigma_{\mathbf{q}}(x)) &\equiv \tau(x^{\#\mathbb{F}_{\mathbf{p}}}) \mod \tau(\mathbf{q}) \\ (\tau\sigma_{\mathbf{q}})(x) &\equiv \tau(x)^{\#\mathbb{F}_{\mathbf{p}}} \mod \mathbf{q}' \\ (\tau\sigma_{\mathbf{q}})(\tau^{-1}(x)) &\equiv \tau(\tau^{-1}(x))^{\#\mathbb{F}_{\mathbf{p}}} \mod \mathbf{q}' \\ (\tau\sigma_{\mathbf{q}}\tau^{-1})(x) &\equiv x^{\#\mathbb{F}_{\mathbf{p}}} \mod \mathbf{q}' \end{aligned}$$

By the previous proposition, the uniqueness of $\sigma_{\mathbf{q}'}$ follows and therefore $\sigma_{\mathbf{q}'} = \tau\sigma_{\mathbf{q}}\tau^{-1}$.

Definition 2.1.34. *Assume AKLBG with finite residue fields and* $\mathbf{q}|\mathbf{p}$ *unramified. The conjugacy class of the Frobenius element* $\sigma_{\mathbf{q}} \in G$ *is the Frobenius class of* $\mathbf{p}$*, denoted* $Frob_{\mathbf{p}}$.

Definition 2.1.35. *Assume AKLBG with finite residue fields. For each unramified prime* $\mathbf{q}$ *of* L *we define the Artin symbol* $(\frac{L/K}{\mathbf{q}}) \equiv \sigma_{\mathbf{q}}$.

Proposition 2.1.23. *Assume AKLBG with finite residue fields and* $\mathbf{q}|\mathbf{p}$ *unramified. Then* $\mathbf{p}$ *splits completely iff* $\frac{L/K}{\mathbf{q}} = 1$.

pf. If $\mathbf{p}$ splits, then $e_{\mathbf{p}}f_{\mathbf{p}} = 1$ and $D_{\mathbf{q}} =< \sigma_{\mathbf{q}} >= \{1\}$.
Conversely, if $D_{\mathbf{q}} =< \sigma_{\mathbf{q}} >= \{1\}$, then $e_{\mathbf{p}}f_{\mathbf{p}} = 1$ and $\mathbf{p}$ splits completely.

Proposition 2.1.24. *Assume AKLBG with finite residue fields and let* $\mathbf{q}|\mathbf{p}$ *be unramified. Let* E *be an intermediate field between* K *and* L *and define* $\mathbf{q}_E \equiv \mathbf{q} \cap E$. *Then*

$$(\frac{L/E}{\mathbf{q}}) = (\frac{L/K}{\mathbf{q}})^{[\mathbb{F}_{\mathbf{q}_E}:\mathbb{F}_{\mathbf{p}}]}$$

and if E/K *is Galois, then* $(\frac{E/K}{\mathbf{q}_E})$ *is the restriction of* $(\frac{L/K}{\mathbf{q}})$ *to* E.

pf. $\#\mathbb{F}_{\mathbf{q}_E} = (\#\mathbb{F}_{\mathbf{p}})^{[\mathbb{F}_{\mathbf{q}_E}:\mathbb{F}_{\mathbf{p}}]}$. So the first claim follows.
By the commutativity of the diagram in the previous proposition 2.2.19, the Frobenius automorphism $x \mapsto x^{\#\mathbb{F}_{\mathbf{p}}}$ of $Gal(\mathbb{F}_{\mathbf{q}_E}/\mathbb{F}_{\mathbf{p}})$ is the restriction of the Frobenius automorphism $x \mapsto x^{\#\mathbb{F}_{\mathbf{p}}}$ of $Gal(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ to $\mathbb{F}_{\mathbf{q}_E}$.

2.2 Extensions of Complete DVR

2.2.1 Completions

A metric on a set X is a function $d : X \times X \to \mathbb{R}_+$ that satisfies

- $d(x,y) = 0$ iff $x = y$
- $d(x,y) = d(y,x)$

- $d(x,z) \leq d(x,y) + d(y,z)$

A metric also satisfies

- $d(x,z) \leq \max(d(x,y), d(y,z))$

is an ultrametric and is said to be nonarchimedean.
$\rightarrow$every absolute value on a ring induces a metric $d(x,y) \equiv |x-y|$.
$\rightarrow$ the metric d defines a topology on X generated by open balls $B_{<r}(x) \equiv \{y \in X : d(x,y) < r\}$ where $r \in \mathbb{R}_{++}, x \in X$ and we call X a metric space.

Definition 2.2.1. *Let X be a metric space. A sequence (x_n) of elements of X converges to x if there is an $x \in X$ s.t. $\forall \epsilon > 0$, $\exists N \in \mathbb{Z}_{++}$ s.t. $d(x_n, x) < \epsilon$, $\forall n \geq N$. The sequence (x_n) is Cauchy if $\forall \epsilon > 0$, $\exists N \in \mathbb{Z}_{++}$ s.t. $d(x_m, x_n) < \epsilon$ for all $m, n \geq N$. Every convergent sequence is Cauchy but not vice versa. If every Cauchy sequence is convergent, a metric space is said to be complete.*

Definition 2.2.2. *An abelian group G is a topological group if it is a topological space in which the map $\begin{array}{ccc} G \times G & \rightarrow & G \\ (g,h) & \mapsto & g+h \end{array}$ and the map $\begin{array}{ccc} G & \rightarrow & G \\ g & \mapsto & -g \end{array}$ are both continuous. A commutative ring R is a topological ring if it is a topological space in which the maps $\begin{array}{ccc} R \times R & \rightarrow & R \\ (r,s) & \mapsto & r+s \text{ or } rs \end{array}$ are both continuous; the additive group of R is a topological group, since $(-1,s) \mapsto -s$ is continuous, but the unit group $R^\times$ need not be a topological group. A field K is a topological field if it is a topological ring whose unit group is a topological group.*

If R is a ring with an absolute value, it is a topological ring under the induced topology and tis unit group is also a topological group by the triangular inequality and the multiplicative property of absolute value.

Definition 2.2.3. *Let X be a metric space. Cauchy sequences $(x_n), (y_n)$ are equivalent if $d(x_n, y_n) \rightarrow 0$ as $n \rightarrow \infty$; this defines an equivalence relation on the set of Cauchy sequences in X and we use $[(x_n)]$ to denote the equivalence class of (x_n). The completion of X is the metric space $\hat{X}$ whose elements are equivalence classes of Cauchy sequences with the metric $d([x_n],[y_n]) \equiv \lim d(x_n, y_n)$. We may canonically embed X in its completion $\hat{X}$ via the map $x \mapsto \hat{x} = [(x,x,\ldots)]$.*

When X is a topological ring, we can extend the ring operation to $\hat{X}$ by defining $[(x_n)] + [(y_n)] \equiv [(x_n + y_n)]$, $[(x_n)][(y_n)] \equiv [(x_n y_n)]$. When the metric on X is induced by an absolute value $|\cdot|$, we extend it to an absolute value on $\hat{X}$ via $|[(x_n)]| \equiv \lim |x_n|$.

When X is a field with a metric induced by an absolute value, the completion is also a field.
(Given $[(x_n)] \neq 0$, we pick each x_n s.t. $x_n \neq 0$ for all n and by the multiplicative property $(1/x_n)$ is Cauchy. Then $1/[(x_n)] = [(1/x_n)]$ so its completion is a field).

Proposition 2.2.1. *Let K be a field with an absolute value $|\cdot|$ viewed as a topological field under the induced topology, and let $\hat{K}$ be the completion of K. The field $\hat{K}$ is compete, and has the following universal property : every embedding of K as a topological field into a complete field L can be uniquely extended to an embedding of $\hat{K}$ into L which is an isomorphism whenever k is dense in L. Up to a canonical isomorphism, $\hat{K}$ is the unique topological field with this property.*

pf. Assume $\hat{K}$ is not complete. Then there exists a Cauchy sequence (x_n) which is not convergent in $\hat{X}$. But since it is Cauchy, $\forall m,n \geq N, d(x_m,x_n) < \epsilon$

$$\begin{tikzcd} K \arrow[rr, "\phi"] \arrow[dr, "\iota"'] & & L \\ & \hat{K} \arrow[ur, "\exists!"'] & \end{tikzcd}$$

Theorem 2.2.1 (Weak Approximation)**.** *Let K be a field and let $|\cdot|_1,\ldots,|\cdot|_n$ be pairwise inequivalent nontrivial absolute values on K. Let $a_1,\ldots,a_n \in K$ and let $\epsilon_1,\ldots,\epsilon_n$ be positive real numbers. Then there exists an $x \in K$ s.t. $|x-a_i|_i < \epsilon_i, 1 \leq i \leq n$.*

pf. **CLAIM : Given inequivalent nontrivial absolute values $|\cdot|_1,\ldots,|\cdot|_n$ on k, $\exists x \in k^\times$ s.t. $|x|_1 > 1$ and $|x|_i < 1$, $1 < i \leq n$.**
pf. $n = 1$, since k is a field either $a \in k$ or $a^{-1} \in k$ has $|x|_1 > 1$ Pick such an element x.
$n = 2$, if for the previously chosen x, $|x|_2 < 1$, then nothing to be done. Otherwise, pick an element $y \in k$ s.t. $|y|_2 = |x|_2, |y|_1 = 1$(e.g. $y = x/|x|_1$), which exists due to the pairwise coprimeness. Then xy^{-2} is the desired element.
By repeating this procedure, we get an element x with the desired property.
CLAIM : Given inequivalent nontrivial absolute values $|\cdot|_1,\ldots,|\cdot|_n$ on k, there is a sequence $(x_1,x_2,\ldots)$ of elements $x_j \in k$ that converges to 1 w.r.t. $|\cdot|_1$ and to 0 w.r.t. $|\cdot|_i, \forall 1 < i \leq n$.
pf. by the previous claim we have x s.t. $|x|_1 > 1, |x|_i < 1, \forall 1 < i \leq n$. Then By setting $x' = \frac{x}{|x|_1}$, we get a sequence (x'^n) with the desired property.
Then $\forall i \in (1,n)$, $\exists x_i$ with $|x_i|_i$ converges to 1 and $|x_i|_j$ converges to 0 for $j \neq i$. Fix $x_n = \sum_i a_i x_i^n$. Then by the triangular inequalty the sequence (x_n) is the sequence with the desired property.

Corollary 2.2.1. *Let K be a field with absolute values $|\cdot|_1, |\cdot|_2$. The induced topologies on K coincide iff $|\cdot|_1$ and $|\cdot|_2$ are equivalent.*

pf. If they are not equivalent, by the previous theorem $\exists x_1, x_2 \in K$ s.t. x_1 goes to 1 for $|\cdot|_1$, and 0 for $|\cdot|_2$. Then for an open ball $x_1 \in B^1_{<\epsilon}(x-1)$ but $x_1 \notin B^2_{<\epsilon'}(x-1)$ for any ϵ'. Since open balls induce topology, but they are different for those absolute values and therefore induced topologies are different.
If they are equivalent, $\exists c \in \mathbb{R}_{++}$ s.t, $|\cdot|_1^c = |\cdot|_2$ and we can get a corresponding open balls for each absolute values. Similarly open balls induce the topologies and we get the same open balls. Thus they have the same induced topologies.

PROPERTIES of TOPOLOGY induced by a NONARCHIMEDEAN ABSOLUTE VALUE

- Every point in an open ball is a centre : $B_{<r}(y) = B_{<r}(x)\ \forall y \in B_{<r}(x)$
  pf. $\forall z \in B_{<r}(y), d(y,z) < r, d(y,x) < r \to d(z,x) \leq d(z,y) + d(y,x) \leq \min(d(z,y), d(y,x)) < r$.

- Any pair of open balls are either disjoint or concentric.
  pf. If they are concentric, nothing to prove. Otherwise, assume the contrary, so $\exists x' \in B_{<r}(x) \cap B_{<r}(y)$ not concentric. Since they are not concentric, $y \notin B_{<r}(x), x \notin B_{<r}(y)$. But $x' \in B_{<r}(x), B_{<r}(x) = B_{<r}(x')$, and similarly $B_{<r}(x') = B_{<r}(y)$, a contradiction. Thus they are disjoint.

- Every open ball is closed and every closed ball is open.
  pf. $B_{<r}(x) = \{y : d(y,x) < r\}$. Pick a maximum value of $k = d(x,b)$, $b \in B_{<r}(x)$. Then $B_{<r}(x) = B_{\leq k}(x)$ a closed ball.
  Let $B_{\leq k}(x)$ be a closed ball. Then for any $b \in B_{\leq k}(x)^c$, set $r = \min(d(b,x))$. Then $B_{\leq k}(x) = B_{<r}(x)$.

- X is totally disconnected.
  pf. Let x, y be two disjoint points. Let $r = d(x,y)/2$. Then $B_{<r}(x)$ is an open nhood of x not containing y. Then its complement is the closed ball and by the previous statement, it is an open ball. Since $y \in B_{<r}(x)^c$ by the previous statement it is a centere. Thus we get the desired two open balls.

For an additive topological group G, every translation by h map $g \mapsto g + h$ is a homeomorphism, since it is continuous and has a continuous inverse (translation by $-h$). Thus we can focus on nhoods of the identity to understand the topology of the topological space.
By construction K is dense in $\hat{K}$.

Definition 2.2.4. *A directed set is a set I with a relation $\leq$ that is reflexive, antisymmetric($i \leq j \leq i$ implies $i = j$) and transitive, in which every finite subset has an upper bound.*

Definition 2.2.5. *An inverse system in a category is a family of objects $\{X_i : i \in I\}$ indexed by a directed set I and a family of morphisms $\{f_{ij} : X_i \leftarrow X_j : i \leq j\}$ s.t. each f_{ii} is the identity and $f_{ik} = f_{ij} \circ f_{jk}$ for all $i \leq j \leq k$.*

Definition 2.2.6. *Let (X_i, f_{ij}) be an inverse system in a concrete category with products. The inverse limit of (X_i, f_{ij}) is the object $X \equiv \lim X_i \equiv \{x \in \Pi_{i \in I} X_i : x_i = f_{ij}(x_j), \forall i \leq j\} \subseteq \Pi X_i$. The restriction $\pi_i : X \to X_i$ of the projection $\Pi X_i \to X_i$ satisfy $\pi_i = f_{ij} \circ \pi_j, i \leq j$.*

$\to$it has the universal property that if Y is another object with morphisms $\psi_i : Y \to X_i$ that satisfy $\psi_i = f_{ij} \circ \psi_j, \forall i \leq j$, then there is a unique morphism $Y \to X$ for which all of the diagram

$$\begin{tikzcd}
 & Y \arrow[d, "\exists!"] \arrow[ddl, "\psi_i"'] \arrow[ddr, "\psi_j"] & \\
 & X \arrow[dl, "\pi_i"] \arrow[dr, "\pi_j"'] & \\
X_i & & X_j \arrow[ll, "f_{ij}"]
\end{tikzcd}$$

commutes.

Proposition 2.2.2. *Let (X_i, f_{ij}) be an inverse system of Hausdorff topological spaces. Then $X \equiv \lim X_i$ is a closed subset of ΠX_i and if the X_i are compact, then X is compact.*

pf. The set X is the intersection of the sets $Y_{ij} \equiv \{x \in \Pi X_i : x_i = f_{ij}(x_j)\}$, $i \leq j$, each of which can be written as $Y_{ij} = \Pi_{k \neq i,j} X_k \times Z_{ij}$, where Z_{ij} is the preimage of the diagonal $\Delta_i \equiv \{(x_i, x_i) : x_i \in X_i\} \subseteq X_i \times X_i$ under the continuous map $X_i \times X_j \to X_i \times X_i$ defined by $(x_i, x_j) \mapsto (x_i, f_{ij}(x_j))$. Each Δ_i is closed in $X_i \times X_i$ because X_i is Hausdorff, so each Z_{ij} is closed in $X_i \times X_j$ and each Y_{ij} is closed in ΠX_i. Thus X is closed in ΠX_i.
By Tychonoff's theorem, if X_i are compact, so is $X = \Pi X_i$.

Definition 2.2.7. *A local field is a field with a nontrivial absolute value $|\cdot|$ that is locally compact (if every point has a compact nhood) under the topology induced by $|\cdot|$.*

Lemma 2.2.1. *Let K be a field with a nontrivial absolute value $|\cdot|$. Then K is a local field iff every closed ball in K is compact.*

pf. Suppose K is a local field. Then $0 \in K$ lies in a compact nhood that contains a closed ball $B_{\leq s}(0)$, which is compact. Fix $\alpha \in K^\times$ with $|\alpha| > 1$. The map $x \mapsto \alpha x$ is continuous and $|\cdot|$ is multiplicative, so $B_{\leq |\alpha|^n s}(0)$ is compact for every $n \in \mathbb{Z}_{++}$ since the continuous image of a compact set is compact. Thus we have compact balls about 0 of arbitrarily large radii, implying that every closed ball $B_{\leq r}(0)$ is a closed subset of a compact set, hence compact. For every $z \in K$, the translation

map $x \mapsto x+z$ is continuous, so every closed ball $B_{\leq r}(z)$ is compact.
By the definition of the local compactness, the converse holds.

Corollary 2.2.2. *Let K be a local field with nontrivial absolute value $|\cdot|$. Then K is complete.*

pf. Suppose the contrary. Then there exists a Cauchy sequence (x_n) in K that converges to a limit $x \in \hat{K} \setminus K$. Pick $N \in Z_{++}$ s.t. $|x_n - x| \leq 1/2, \forall n \geq N$ and consider the closed ball $S \equiv B_{\leq 1}(x_N)$ in K, which is compact by the previous lemma. The Cauchy sequence $(x_n)_{n \geq N}$ in S has a convergent subsequence whose limit lies in $S \subseteq K$, since S is compact and therefore sequentially compact. But this limit must be equal to $x \notin K$, a contradiction.

Proposition 2.2.3. *Let K be a field with absolute value $|\cdot|_v$ induced by a discrete valuation v with valuation ring A and uniformiser π. Then K is a local field iff K is complete and the residue field $A/\pi A$ is finite.*

pf. If K is a local field, K is complete by the previous corollary and the valuation ring $A = \{x \in K : v(x) \geq 0\} = \{x \in K : |x|_v \leq\} = B_{\leq 1}(0)$ is a closed ball, hence by the previous lemma compact. The cosets $x + \pi A$ of the subgroup $\pi A \subseteq A$ are open balls $B_{<1}(x)$ since $y \in x + \pi A$ iff $|x - y|_v \leq |\pi|_v < 1$. The collection $\{x + \pi A : x \in A\}$ of cosets of πA is an open cover of A and therefore it must have a finite subcover given A compact. Thus $A/\pi A$ is finite.
Suppose K is complete and $A/\pi A$ is finite. The valuation ring $A \subseteq K$ is also complete and we have an isomorphism of topological rings $A = \hat{A} \cong \lim \frac{A}{\pi^n A}$. Each quotient $A/\pi^n A$ is finite since $A/\pi A$ is finite and therefore compact. Thus its inverse limit A is also compact. Then by the previous lemma K is a local field, since it contains a compact closed ball $B_{\leq 1}(0) = A$ and $|\cdot|_v$ is nontrivial.

Corollary 2.2.3. *Let L be a global field with a nontrivial absolute value $|\cdot|_v$. Then the completion L_v of L w.r.t. $|\cdot|_v$ is a local field.*

pf. Let L/K be a finite extension with $K = \mathbb{Q}$ or $K = \mathbb{F}_q(t)$ and $A = \mathbb{Z}$ or $\mathbb{F}_q[t]$, so that $K = FracA$. Then A is a Dedekind domain, as is its integral closure B in L.

- $|\cdot|_v$ is archimedean
  $K = \mathbb{Q}$, and the completion of L w.r.t. $|\cdot|_v$ must contain the completion of $\mathbb{Q}$ w.r.t. the restriction of $|\cdot|_v$ to $\mathbb{Q}$, which must be isomorphic to $\mathbb{R}$. Thus L_v is a finite extension of $\mathbb{R}$ and must be isomorphic to either $\mathbb{R}$ or $\mathbb{C}$, both of which are local fields.

- $|\cdot|_v$ is nonarchimedean
  Let $C \equiv \{x \in L : |x|_v \leq 1\}$ be the valuation ring of L w.r.t. $|\cdot|_v$ and let $\mathbf{m} \equiv \{x \in$

$L : |x|_v < 1\}$ be its maximal ideal, which is nonzero because $|\cdot|_v$ is nontrivial. The restriction of $|\cdot|_v$ to K is a nonarchimdedean absolute value and from the classification of absolute values on $\mathbb{Q}$(**) and $\mathbb{F}_q[t]$ we can assume it is induced by a discrete valuation on A; in particular $|x|_v \leq 1$, $\forall x \in A$ and therefore $A \subseteq C$. Like all valuation rings, C is integrally closed in its fraction field L, and $A \subseteq C$, and therefore $B \subseteq C$, given B the integral closure of A in L. The ideal $\mathbf{q} = \mathbf{m} \cap B$ is maximal and the DVR $B_{\mathbf{q}}$lies in C and must equal C, since there are no intermediate rings between a DVR and its fraction field (Since C is not a field $C \neq L$). Thus the absolute value induced by $v_{\mathbf{q}}$ is equivalent to $|\cdot|_v$ since they have the same valuation rings. By choosing $0 < c < 1$, we can assume $|\cdot|_v = c^{v_{\mathbf{q}}(\cdot)}$ is induced by $v_{\mathbf{q}}$. Thus $|\cdot|_v$ is induced by a discrete valuation. The residue field $B_{\mathbf{q}}/\mathbf{q}B_{\mathbf{q}} \cong B/\mathbf{q}$ is finite since $B/\mathbf{q}$ is a finite extension of finite field $A/\mathbf{q}$ where $\mathbf{q} = \mathbf{q} \cap A$. If we consider the completion L_v with valuation ring B_v, we can take any uniformiser π for $\mathbf{q} \subseteq B \subseteq B_v$ as a uniformiser for B_v, $\frac{B}{\mathbf{q}} \cong \frac{B_{\mathbf{q}}}{\mathbf{q}B_{\mathbf{q}}} = \frac{B_{\mathbf{q}}}{\pi B_{\mathbf{q}}} \cong \frac{B_v}{\pi B_v}$, so $B_v/\pi B_v$ is finite. Thus L_v is a complete field with an absolute value induced by a discrete valuation and finite residue field, and therefore a local field by the previous proposition.

**For each prime $\pi \in \mathbb{F}_q[t]$ and any nonzero $f \in \mathbb{F}_q[t]$, let $v_\pi(f)$ be the largest integer for which $\pi^n | f$, equivalently the largest n for which $f \in (\pi^n)$. For each $f/g \in \mathbb{F}_q(t)^\times$, define $v_\pi(f/g) \equiv v_\pi(f) - v_\pi(g)$ and let $v_\pi(0) \equiv \infty$, and also define $\deg 0 \equiv -\infty, \deg(f/g) \equiv \deg f - \deg g$.

For each prime $\pi \in \mathbb{F}_q[t]$, define $|r|_\pi = (q^{\deg \pi})^{-v_\pi(r)}$, $\forall r \in \mathbb{F}_q(t)$. Then $|\cdot|_\pi$ is a nonarchimedean absolute value on $\mathbb{F}_q[t]$.

pf. $|r+s|_\pi = (q^{\deg \pi})^{-v_\pi(r+s)} = (q^{\deg \pi})^{-v_\pi(r)}(q^{\deg \pi})^{-v_\pi(s)} = |r|_\pi |s|_\pi \leq \max(|r|_v, |s|_v)$ hence nonarchimedean.

Define $|r|_\infty \equiv q^{\deg r}$, $\forall r \in \mathbb{F}_q(t)$. Then $|\cdot|_\infty$ is a nonarchimedean absolute value on $\mathbb{F}_q(t)$.

pf. $|r+s|_\infty = q^{\deg(r+s)} = q^{\max(\deg r, \deg s)} \leq \max(|r|_\infty, |s|_\infty)$, hence a nonarchimedean absolute value.

The residue field of $\mathbb{F}_q(t)$ w.r.t. $|\cdot|_\pi$.

pf. $\mathbb{F}_q(t)/\mathbf{m}\mathbb{F}_q(t) = \{r + \mathbf{m} : r \in \mathbb{F}_q(t) : ||r||_\pi = 1\}$ a coset of polynomial coprime to π.

Every nontrivial absolute value on $\mathbb{F}_q[t]$ is equivalent to $|\cdot|_\infty$ or $|\cdot|_\pi$ for some prime $\pi \in \mathbb{F}_q[t]$.

pf. Any element f in $\mathbb{F}_q(t)$ can be expressed as a finite product of irreducible polynomials $f_1, f_2, \ldots$. Prime ideals of $\mathbb{F}_q[t]$ consists of principal ideals, generated by an irreducible polynomial. Since $||\cdot||$ is nontrivial, pick any irreducible polynomial s.t. $||f_1|| = a > 1$. Since $||\cdot||$ is multiplicative, $\forall f \in \mathbb{F}_q(t)$ s.t. $f = \Pi f_i^{e_i}$ and $e_1 \neq 0$, $|f| = a^{e_1}||f/f_i^{e_i}||$. For another $j \neq i$, if $||f_j|| \neq 1$, wlog assume $||f_j|| = b > 1($ or take an inverse). Then $||f_j||^{\log_b a} = ||f_i|| \leftrightarrow f_i = f_j^{\log_b a}$ a contradiction. Hence

$\forall j \neq i, ||f_j|| = 1$. This implies it is equivalent to $||\cdot||_\pi$ where π is a PID generated by the irreducible polynomial s.t. $||f_i|| > 1$.
If $||t|| = a > 1$, t is an irreducible polynomial with the above definition, and therefore it is equivalent to $||\cdot||_{(t)}$, which is equivalent to $||\cdot||_\infty$. If $||t|| \leq 1, \pi \neq t$ and is equivalent to some prime generated by an irreducible polynomial with no root 0.

Proposition 2.2.4. *A locally compact topological vector space over a nondiscrete locally compact field has finite dimension.*

Theorem 2.2.2. *Let L be a local field. If L is archimedean, then it is isomorphic to $\mathbb{R}$ or $\mathbb{C}$; otherwise L is isomorphic to a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$.*

pf. Let L be a local field with nontrivial absolute value $|\cdot|$; then L is complete by the previous corollary.

- L has characteristic zero
  the prime field of L is $\mathbb{Q}$ and L contains the completion of $\mathbb{Q}$ w.r.t. the restriction of $|\cdot|$ to $\mathbb{Q}$. By the Ostrowski's theorem, the restriction of $|\cdot|$ to $\mathbb{Q}$ is equivalent to eitehr the standard archimedean absolute value (the completion is $\mathbb{R}$) or to a p-adic absolute value (the completion is $\mathbb{Q}_p$)Thus L contains a subfield K isomorphic to $\mathbb{R}$ or to $\mathbb{Q}_p$ for some prime p.

- L has positive characteristic p
  the prime field of L is $\mathbb{F}_p$ and L must contain a transcendental element s since no algebraic extension of $\mathbb{F}_p$ has a nontrivial absolute value. Thus L contains $\mathbb{F}_p(s)$ and therefore contains the completion of $\mathbb{F}_p(s)$ w.r.t. $|\cdot|$. Every completioin of $\mathbb{F}_p(s)$ is isomorphic to $\mathbb{F}_q((t))$ for some q a power of p and t transcendental over $\mathbb{F}_q$(**). Thus L contains a subfield K isomorphic to $\mathbb{F}_q((t))$.

If K is archimedean, $K = \mathbb{R}$ is a local feild and if K is nonarchimedean the $K = \mathbb{Q}_p$ or $K = \mathbb{F}_q((t))$ is a complete field with a discrete valuation and finite residue field, hence a local field by the proposition 2.3.3. Thus K is locally compact and it is nondiscrete because its absolute value is nontrivial. Then by the previous proposition, L has finite degree over K. If K is archimedean, $K = \mathbb{R}$ and L must be $\mathbb{R}$ or $\mathbb{C}$; otherwise L is a finite extension of $\mathbb{Q}_p$ ofr $\mathbb{F}_q((t))$ as claimed.

**

$\mathbb{F}_q(t)_\pi$ is isomorphic to $\mathbb{F}_\pi((T))$ where $\mathbb{F}_\pi$ denotes the residue field of $\mathbb{F}_q(t)$ w.r.t. $|\cdot|_\pi$ and $T \in \mathbb{F}_q(t)$ is a uniformiser. Let π_1, π_2 be two discrete valuation for which $\mathbb{F}_{\pi_1} \cong \mathbb{F}_{\pi_2} \cong \mathbb{F}_q$ and consider the completions $\mathbb{F}_q(t)_{\pi_1}, \mathbb{F}_q(t)_{\pi_2}$. Then they are isomorphic as fields/topological fields/$\mathbb{F}_q(t)$-algebras.
pf For any polynomial $f \in \mathbb{F}_q(t)_\pi$ can be expressed as $\pi \to f(t) = \cdots + f_{-1}\pi^{-1} + f_0 +$

$f_1\pi + f_2\pi^2 + \dots$. Let $T = \pi$ be a uniformiser, which is an irreducible polynomial. Then we can get a homomorphism $\phi : \mathbb{F}_q(t)_\pi \to \mathbb{F}_\pi((T))$ by sending any polynomial in the domain to an expansion in terms of $\pi = T$. It is injective, since every different polynomial in the domain has different unique $T = \pi$ expansion. Also it is surjective since for any $f \in \mathbb{F}_\pi((T))$, send T to π and the coefficients unchanged. Then we can similarly say that for any irreducible polynomial π inducing the absolute value with same residue field, have isomorphisms form $\mathbb{F}_\pi((T)) \to \mathbb{F}_{\pi_2}((T'))$ as a field and $\mathbb{F}_q(t)$-algebras.

The structure of the abelian group $\mathbb{F}_q(t)_\pi^\times / \mathbb{F}_q(t)_\pi^{\times n}$ where q, n are primes and π is a discrete valuation on $\mathbb{F}_p(t)$.

pf. a free abelian group of dimension n

Definition 2.2.8. *Let R be a commutative ring, and let $f(x) = \sum f_i x^i \in R[x]$ be a polynomial. The formal derivative f' of f is the polynomial $f'(x) \equiv \sum i f_i x^{i-1} \in R[x]$.*

The formal derivative satisfies the usual identities

$$\begin{cases} (af + bg)' = af' + bg' & linearity \\ (fg)' = f'g + fg' & Leibniz\ rule \\ (f \circ g)' = (f' \circ g)g' & chain\ rule \end{cases}$$

Lemma 2.2.2. *Let R be a ring, let $f = \sum f_i x^i \in R[x]$ be a polynomial, and let $a \in R$. Then $f(x) = f(a) + f'(a)(x-a) + g(x)(x-a)^2$ for a unique $g \in R[x]$.*

pf. $f(x) = f(a + (x-a)) = \sum_{i \geq 0} f_i (a + (x-a))^i = \sum_{i \geq 0} f_i \sum_{0 \leq j \leq i} \binom{i}{j} a^j (x-a)^{i-j}$ $= f(a) + \sum_{i \geq 1} f_i \sum_{0 \leq j < i} \binom{i}{j} a^j (x-a)^{i-j} = f(a) + f'(a)(x-a) + \sum_{i \geq 2} f_i \sum_{0 \leq j \leq i-2} \binom{i}{j} a^j (x-a)^{i-j} = f(a) + f'(a)(x-a) + g(x)(x-a)^2$ where $g(x) = \sum_{i \geq 2} f_i \sum_{0 \leq j \leq i-2} \binom{i}{j} a^j (x-a)^{i-j-2} \in R[x]$.

$\to$ the binomial coefficients are integers, hence well-defined elements of R under the canonical homomorphism $\mathbb{Z} \to R$, even when $j!$ is divisible by the $charR$.

Corollary 2.2.4. *Let R be a ring, $f \in R[x]$ and $a \in R$. Then $f(a) = f'(a) = 0$ iff a is at least a double root of f, that is $f(x) = (x-a)^2 g(x)$ for some $g \in R[x]$.*

Definition 2.2.9. *Let $f \in R[x]$ be a polynomial over a ring R and let $a \in R$. If $f(a) = 0, f'(a) \neq 0$, then a is a simple root of f.*

If R is a ring and I is an R-ideal, a lift of an element $\bar{r}$ of the quotient R/I means a preimage of $\bar{r}$ under the quotient map $R \to R/I$.

Lemma 2.2.3 (Hensel's Lemma I)**.** *Let A be a complete DVR with maximal ideal $\mathbf{p}$ and residue field $k \equiv A/\mathbf{p}$. Suppose $f \in A[x]$ is a monic polynomial whose reduction to $k[x]$ has a simple root $\bar{a} \in k$. Then $\bar{a}$ can be lifted to a root of f in A.*

pf. Let a_0 be any lift of $\bar{a}$ to A, then it is not necessarily a root of f but it is a root modulo $\mathbf{p}$. Fix $0 < c < 1$ and define the absolute value $|\cdot| \equiv c^{v_p(\cdot)}$. Since $\bar{a}$ is a simple root, $f(a_0) \in \mathbf{p}, f'(a_0) \notin \mathbf{p} \to |f(a_0)| \leq c < 1, |f'(a_0)| = 1$. Define $\epsilon \equiv \frac{|f(a_0)|}{|f'(a_0)|^2} < 1$.

$$\textbf{CLAIM :} \quad \begin{aligned} |a_n| &\leq 1 \\ |a_n - a_0| &\leq \epsilon < 1 \\ |f'(a_n)| &= |f'(a_0)| \\ |f(a_n)| &\leq \epsilon^{2^n} |f'(a_0)|^2 \end{aligned} \quad \textbf{for all } n \in \mathbb{Z}_+.$$

pf. when $n = 0$, the claim trivially holds. Assume the claim holds for all n and consider the case $n+1$.

- $a_{n+1} - a_n| = |f(a_n)/f'(a_n)| \leq \epsilon^{2^n} |f'(a_0)|^2/|f'(a_0)| = \epsilon^{2^n} |f'(a_0)| \leq \epsilon^{2^n} \to |a_{n+1}| = |a_{n+1} - a_n + a_n| \leq \max(|a_{n+1} - a_n|, |a_n|) \leq 1$
- $|a_{n+1} - a_0| \leq \max(|a_{n+1} - a_n|, |a_n - a_0|) \leq \max(\epsilon^{2^n}, \epsilon) = \epsilon$
- $f'(a_{n+1}) = f'(a_n) - f''(a_m)\frac{f(a_n)}{f'(a_n)} + g(a_{n+1})(\frac{f(a_n)}{f'(a_n)})^2$ where $f''(a_n), g(a_{n+1}) \in A$ so $|f''(a_n)|, |g(a_{n+1})| \leq 1, |f(a_n)/f'(a_n)| = |f(a_n)|/|f'(a_0)| \leq \epsilon^{2^n} |f'(a_0)|$ so the absolute values of the last two terms on the RHS are strictly smaller than the first term $|f'(a_n)| = |f'(a_0)| \to |f'(a_{n+1})| = |f'(a_n)| = |f'(a_0)|$
- $f(a_{n+1}) = f(a_n) - f'(a_n)\frac{f(a_n)}{f'(a_n)} + h(a_{n+1})(\frac{f(a_n)}{f'(a_n)})^2 = h(a_{n+1})(\frac{f(a_n)}{f'(a_n)})^2$ for some $h \in A[x]$. Since $|h(a_{n+1})| \leq 1, |f(a_{n+1})| \leq |f(a_n)|^2/|f'(a_n)|^2 = |f(a_n)|^2/|f'(a_0)|^2 \leq \epsilon^{2^{n+1}} |f'(a_0)|^2$

hence the claim follows by the induction.
$|a_{n+1} - a_n| \leq \epsilon^{2^n} \to 0$ as $n \to \infty$ and for a nonarchimedean absolute value this implies (a_n) is a Cauchy. Thus $\exists a \equiv \lim a_n \in A$, given A complete. We have $f(a) = \lim f(a_n) = 0$ so a is a root of f and $|a - a_0| = \lim |a_n - a_0| < 1 \to a \equiv a_0 \mod \mathbf{p}$ is a lift of $\bar{a}$.

Actually, the above proof proves more general version of the Hensel's Lemma.

Lemma 2.2.4 (Hensel's Lemma II)**.** *Let A be a complete DVR. Let $f \in A[x]$ and suppose $a_0 \in A$ satisfies $|f(a_0)| < |f'(a_0)|^2$ and for $n \geq 0$, define $a_{n+1} \equiv a_n - f(a_n)/f'(a_n)$. The sequence (a_n) is well-defined and converges to the unique root $a \in A$ of f for which $|a - a_0| \leq \epsilon \equiv |f(a_0)|/|f'(a_0)|^2$. Moreover, $|f(a_n)| \leq \epsilon^{2^n} |f'(a_0)|^2, \forall n \geq 0$.*

Both of the Hensel's Lemmas does not require for A to be complete. A local ring A for which the Hensel's lemma I holds is called a henselian ring. One can define the henselisation of a noetherian local ring R as the minimal extension of R that is henselian. Loosely speaking, in henselian rings, Cauchy sequences that converges to the root of a polynomial required to converge, but not every Cauchy sequences need to converge.

Lemma 2.2.5 (Hensel's Lemma III)**.** *Let A be a complete DVR with maximal ideal $\mathbf{p}$ and residue field k, let $f \in A[x]$ have image $\bar{f} \in k[x]$, and suppose $\bar{f} = \bar{g}\bar{h}$ for some coprime $\bar{g}, \bar{h} \in k[x]$. Then there exists a polynomials $g, h \in A[x]$ for which $f = gh, g \equiv \bar{g} \mod \mathbf{p}, h \equiv \bar{h} \mod \mathbf{p}$ s.t. $\deg g = \deg \bar{g}$.*

Lemma 2.2.6 (Hensel-Kurschak Lemma)**.** *Let A be a complete DVR with fraction field K, and let $f \in K[x]$ be an irreducible polynomial whose leading and constant coefficients lie in A. Then $f \in A[x]$.*

pf. Let $\mathbf{p} = (\pi)$ be the maximal ideal of A, let $k \equiv A/\mathbf{p}$ and write $f = \sum f_i x^i$, with $f_n \neq 0$. Then $n > 0, f_0 \neq 0$ since f is irreducible. Let $m \equiv \min(v_{\mathbf{p}}(f_i))$. Suppose the contrary. Then $m < 0$. Let $g \equiv \pi^{-m} f = \sum g_i x^i \in A[x]$, which is an irreducible polynomial in $A[x]$ with $g_0, g_n \in \mathbf{p}$ since $m < 0$, and $f_0, f_n \in A$ and g_i is a unit for some $0 < i < n$ by the minimality of m. The reduction $\bar{g}$ of g to $k[x]$ has positive degree and constant term 0 and is therefore divisible by x. If we let $\bar{u} \equiv x^d$ be the largest power of x dividing $\bar{g}$, $0 < d \leq \deg \bar{g} < n$ and $\bar{v} \equiv \bar{g}/x^d \in k[x]$ is coprime ot $\bar{u}$. By the Hensel's lemma III, $g = uv, u, v \in A[x]$ with $0 < \deg u = \deg \bar{u} < n$. This means g is reducible, a contradiction.

Corollary 2.2.5. *Let A be a complete DVR with fraction field K, and let L/K be a finite extension of degree n. Then $\alpha \in L$ is integral over A iff $N_{L/K}(\alpha) \in A$.*

pf. Let $f = \sum_{i=0}^{d} f_i x^i \in K[x]$ be the minimal polynomial of α. If α is integral over A, $f \in A[x]$, $N_{L/K}(\alpha) = (-1)^n f(0)^e$ where $e = [L : K(\alpha)]$.
Conversely, if $N_{L/K}(\alpha) = (-1)^n f(0)^e \in A$, $f(0) \in A$, since $f(0) \in K$ is a root of $x^e - (-1)^n N_{L/K}(\alpha) \in A[x]$ and A is integrally closed. The constant coefficient of f thus lies in A as does its leading coefficient 1, so $f \in A[x]$ by the previous lemma and α is integral over A.

Theorem 2.2.3. *Assume AKLB and that A is a complete DVR with maximal ideal $\mathbf{p}$. Then B is a DVR whose maximal ideal $\mathbf{q}$ is necessarily the unique prime ideal above $\mathbf{o}$.*

pf. At least one prime $\mathbf{q}$ of B lies above $\mathbf{p}$, since the factorisation of $\mathbf{p}B \subsetneq B$ is nontrivial. Suppose the contrary. Then $\exists \mathbf{q_1}, \mathbf{q_2} \in \{\mathbf{q}|\mathbf{p}\}$ with $\mathbf{q_1} \neq \mathbf{q_2}$. Choose $b \in \mathbf{q_1} \setminus \mathbf{q_2}$ and consider the ring $A[b] \subseteq B$. The ideals $\mathbf{q}_1 \cap A[b], \mathbf{q}_2 \cap A[b]$ are distinct prime ideals of $A[b]$ containing $\mathbf{p}A[b]$ and both are maximal, since they are nonzero and $\dim A[b] = \dim A = 1$. The quotient ring $A[b]/\mathbf{p}A[b]$ thus has at least two maximal ideals. Let $f \in A[x]$ be the minimal polynomial of b over K and let $\bar{f} \in (A/\mathbf{p})[x]$ be its reduction to the residue field $A/\mathbf{p} \to \frac{(A/\mathbf{p})[x]}{(\bar{f})} \cong \frac{A[x]}{(\mathbf{p}, f)} \cong \frac{A[b]}{\mathbf{p}A[b]}$ thus the ring $(A/\mathbf{p})[x]/(\bar{f})$ has at least two maximal ideals, i.e. $\bar{f}$ is divisible by two distinct irreducible polynomials. Thus we can factor $\bar{f} = \bar{g}\bar{h}$, where two factors are coprime. Then by the Hensel's Lemma III, we can lift this to a nontrivial

factorisation of $f = gh \in A[x]$, a contradiction. Hence there is only one prime $\mathbf{q} \in \{\mathbf{q}|\mathbf{p}\}$.
Every maximal ideal of B lies above a maximal ideal of A but A has only one maximal $\mathbf{p}$ and $\#\{\mathbf{q}|\mathbf{p}\} = 1$, and therefore B has a unique maximal ideal $\mathbf{q}$. So B is a local Dedekind domain, hence a local PID that is not a field, hence a DVR.

Definition 2.2.10. *Let K be a field with absolute value $|\cdot|$ and let V be a K-vector space. A norm on V is a function $||\cdot||"V \to \mathbb{R}_+$ s.t.*

$$||v|| = 0 \text{ iff } v = 0$$
$$||\lambda v|| = |\lambda|||v||, \ \forall \lambda \in K, v \in V$$
$$||v + w|| \leq ||v|| + ||w||, \ \forall w, v \in V$$

Each norm $||\cdot||$ induces sa topology on V via the distance metric $d(v,w) \equiv ||v - w||$.

e.g. A sup-norm $||v||_\infty \equiv \sup\{|v_i|\}$ is a norm on V where $v = \sum v_i e_i$ where K-basis for V is (e_i).

Proposition 2.2.5. *Let V be a vector space of finite dimension over a complete field K. Every norm on V induces the same topology in which V is a complete metric space.*

pf. **CLAIM : Equivalent norms induce the same topology.**
pf. The basis for the topology is a set of open balls $B_{<r}(x)$. For any open ball in a metric 1, $B^1_{<r}(x) = \{\forall y \in V : ||y - x||_1 < r\} = B^2_{<r'}(x) = \{\forall y \in V : ||y - x||_2 < r'\}$ since $\exists c \in \mathbb{R}$ s.t. $||\cdot||_1 < c||\cdot||_2, ||\cdot||_2 < c||\cdot||_1$. So they have the same basis and therefore their induced topologies are equivalent.
CLAIM : The sup-norm does not depend on the choice of basis.
pf. Let $(e_i),(d_i)$ are two different K-basis for V. Then $\exists M \in \mathbb{R}^{n\times n}$s.t. $eM = d$, with $\det M = 1$. Then for any d_i, $d_i = \sum_j a_{ij}e_j$, $\forall v \in V, v = \sum_i v_i e_i = \sum_i a'_i d_i = \sum a'_i \sum_j a_{ij}e_j = \sum_j e_j(\sum_i a'_i a_{ij})$. Then $||v||_\infty = \max_j \sum_i a'_i a_{ij} = \max_i v_i$.
CLAIM : For any norm on V is equivalent to the sup-norm.
pf. For any norm $||\cdot||$ on V, $\forall v = \sum v_i e_i \in V, ||v|| \leq \sum ||v_i e_i|| = \sum |v_i|||e_i|| = \sum |v_i| \leq n \max_i |v_i| = n||v||_\infty$. So $\exists n \in \mathbb{R} < \infty$ s.t. $||\cdot|| \leq n||\cdot||_\infty$.
Conversely, when $n = 1, ||v|| = ||v_1 e_1|| = |v_1||e_1|| = |v_1| \geq 1/c||v||_\infty$ for all $c \geq 1$.
When $n > 1$, let c' be the constant we get from the case $n - 1$.

- if $||v||_\infty = |v_n|$
  $||v||_\infty = |v_n| = r \max_{i\in[1,n-1]} |v_i| \equiv rv'$ with $r \geq 1$. Then $||v|| = ||v_1 e_1 + \cdots + v_n e_n|| \leq ||v_1 e_1 + \cdots + v_{n-1}e_{n-1}|| + |v_n| \leftrightarrow \frac{||v||}{||v||_\infty} \leq \frac{||v'||}{||v||_\infty} + 1 = \frac{||v'||}{rv'} + 1 = \frac{c'}{r} + 1. \leftrightarrow ||v||_\infty \leq (1 + \frac{c'}{r})||v||$ so we get a new constant.

- if $||v||_\infty \neq |v_n|$
  $c'||v|| = c'||v_1 e_1 + \cdots + v_n e_n|| \geq c'||v_1 e_1 + \cdots + v_{n-1}e_{n-1}|| - c'||v_n e_n|| \geq ||v_1 e_1 + \cdots + v_{n-1}e_{n-1}||_\infty - c'|v_n|$ by the inductive hypothesis. Then the LHS $= ||v||_\infty - r||v||_\infty = (1 - r)||v||_\infty$ where $r \leq 1$. Thus we get a new constant $c = \frac{c'}{1-r} \in \mathbb{R}$

Among the set of constants, we choose a minimum value so that $c||v|| \geq ||v||_\infty$. Thus they are equivalent and therefore the proposition holds.

Theorem 2.2.4. *Let A be a complete DVR with fraction field K, maximal ideal $\mathbf{p}$, discrete valuation $v_\mathbf{p}$, and absolute value $|x|_\mathbf{p} \equiv c^{v_\mathbf{p}(x)}$, with $0 < c < 1$. Let L/K be a finite extension of degree n. The following hold.*

1. *There is a unique absolute value $|x| \equiv |N_{L/K}(x)|_\mathbf{p}^{1/n}$ on L that extends $|\cdot|_\mathbf{p}$*
2. *The field L is complete w.r.t. $|\cdot|$, and its valuation ring $\{x \in L : |x| \leq 1\}$ is equal to the integral closure of B of A in L*
3. *If L/K is separable, then B is a complete DVR whose maximal ideal $\mathbf{q}$ induces $|x| = |x|_\mathbf{q} \equiv c^{\frac{1}{e_q} v_q(x)}$ where $e_\mathbf{q}$ is the ramification index of $\mathbf{q}$ that is $\mathbf{p}B = \mathbf{q}^{e_\mathbf{q}}$.*

pf. **CLAIM : $|\cdot|$ is an absolute value.**
pf. it is clearly multiplicative and $|x| = 0$ iff $x = 0$. So only the triangular inequality remains. Let $|x| \leq 1 \leftrightarrow |N_{L/K}(x)|_\mathbf{p} \leq 1 \leftrightarrow N_{L/K}(x) \in A \leftrightarrow x \in B \leftrightarrow x+1 \in B \leftrightarrow |x+1| \leq 1$ and then for $y, z \in L$, assume wlog $|y| \geq |z| \to |y+z| = |y||1+\frac{z}{y}| \leq |y|+|z| = |y|(\frac{|z|}{|y|}+1)$ so the triangular inequality holds and it is an absolute value.
$\forall x \in K, |x| = |N_{L/K}(x)|_\mathbf{p}^{1/n} = |x^n|_\mathbf{p}^{1/n} = |x|_\mathbf{p}$ and therefore it extends $|\cdot|_\mathbf{p}$ and is therefore a norm on L. Since it is nontrivial, $\exists x \in K^\times$ s.t. $|x|_\mathbf{p} \neq 1$ and $|x|^a = |x|_\mathbf{p} = |x|$ only for $a = 1$, which implies $|\cdot|$ is the unique absolute value extending $|\cdot|_\mathbf{p}$.
Assume L/K is separable. Then B is a DVR and it is complete because it is a valuation ring of L. Let $\mathbf{q}$ be the unique maximal ideal of B. The valuation $v_\mathbf{q}$ extends $v_\mathbf{p}$ with index $e_\mathbf{q}$ so $v_\mathbf{q}(x) = v_\mathbf{p}(x)e_\mathbf{q}$ for $x \in K^\times$. Since $0 < c^{1/e_q} < 1$, $|x|_\mathbf{q} \equiv (c^{1/e_q})^{v_q(x)} = c^{\frac{1}{e_q} e_q v_p(x)} = c^{v_p(x)} = |x|_\mathbf{p}$.

$\to$ The transitivity of $N_{L/K}$ implies we can uniquely extend the absolute value on the gractoin field K of a complete DVR to an algebraic closure $\bar{K}$.

Corollary 2.2.6. *Assume AKLB and A is a complete DVR with maximal ideal $\mathbf{p}$ and let $\mathbf{q}|\mathbf{p}$. Then $v_\mathbf{q}(x) = \frac{1}{f_\mathbf{q}} v_\mathbf{p}(N_{L/K}(x))$ for all $x \in L$.*

pf. $v_\mathbf{p}(N_{L/K}(x)) = v_\mathbf{p}(N_{L/K}((x))) = v_\mathbf{p}(N_{L/K}(\mathbf{q}^{v_q(x)})) = v_\mathbf{p}(\mathbf{p}^{f_q v_q(x)}) = f_\mathbf{q} v_\mathbf{q}(x)$.

$\to$ A valuation is a surjective homomorphism $v : K^\times \to \Gamma$ in which Γ is a totally ordered abelian group and $v(x+y) \geq \min(v(x), v(y))$; we extend v to K by defining $v(0) = \infty$ to be strictly greater than any element of Γ.
e.g. In AKLB with A a complete DVR, a valuation is defined $v(x) = \frac{1}{e_\mathbf{q}}(v_\mathbf{q}(x))$ with image $\frac{1}{e_\mathbf{q}}\mathbb{Z}$ that restricts to the discrete valuation $v_\mathbf{p}$ on K.

Lemma 2.2.7 (Nakayama's Lemma)**.** *Let A be a local ring with maximal ideal $\mathbf{p}$ and let M be a finitely generated A-module. If the images of $x_1 \dots x_n \in M$ generate $M/\mathbf{p}M$ as an $(A/\mathbf{p})$-vector space, then $x_1, \dots, x_n$ generate M as an A-module.*

Corollary 2.2.7. *Let A be a local noetherian ring with maximal ideal $\mathbf{p}$, let $g \in A[x]$ and let $B \equiv A[x]/(g)$. Every maximal ideal $\mathbf{m}$ of B contains the ideal $\mathbf{p}B$.*

pf. Assume the contrary. Then $\mathbf{m} + \mathbf{p}B = B$ for some maximal $\mathbf{m}$ of B. The ring B is finitely generated over a noetherian ring A, hence a noetherian A-module, so its A-submodules are all finitely generated. Let $z_1, \dots, z_n$ be A-module generator for $\mathbf{m}$. Every coset of $\mathbf{p}B$ in B can be wrttien as $z + \mathbf{p}B$ for some A-linear combination z of $z_1, \dots, z_n$ so the images of them generate $B/\mathbf{p}B$ as an $(A/\mathbf{p})$-vector space. Then by the Nakayama's lemma, $z_1, \dots, z_n$ generate B i.e. $\mathbf{m} = B$, a contradiction.

Corollary 2.2.8. *Let A be a local noetherian ring with maximal ideal $\mathbf{p}$, let $g \in A[x]$ be a polynomial with reduction $\bar{g} \in (A/\mathbf{p})[x]$ and let α be the image of x in the ring $B \equiv A[x]/(g) = A[\alpha]$. The maximal ideals of B are$(\mathbf{p}, g_i(\alpha))$, where $g_1, \dots, g_m \in A[x]$ are lifts of the distinct irreducible polynomials $\bar{g}_i \in (A/\mathbf{p})[x]$ that divides $\bar{g}$.*

pf. By the previous corollary, $B \to B/\mathbf{p}B$ gives a one-to-one correspondence between maximal ideals of B and maximal ideals of $B/\mathbf{p}B$ and $\frac{B}{\mathbf{p}B} \cong \frac{A[x]}{\mathbf{p},g)} \cong \frac{(A/\mathbf{p})[x]}{(\bar{g})}$. Each maximal of $(A/\mathbf{p})[x]/(\bar{g})$ is the reduction of an irreducible divisor of $\bar{g}$ hence one of the $\bar{g}_i$. Thus the corollary follows.

Theorem 2.2.5. *Assume AKLB with A, B DVRs with residue fields $k \equiv A/\mathbf{p}, l \equiv B/\mathbf{q}$. If l/k is separable, then $B = A[\alpha]$ for some $\alpha \in B$; if L/K is unramified this holds for every lift α of any generator $\bar{\alpha}$ for $l = k(\bar{\alpha})$.*

pf. Let $\mathbf{p}B = \mathbf{q}^e$ be the factorisation of $\mathbf{p}B$ and let $f = [l : k]$ be the residue field degree, so that $ef = n \equiv [L : K]$. The extension l/k is separable, so by the primitive element theorem, $l = k(\bar{\alpha}_0)$ for some $\bar{\alpha}_0 \in l$ whose minimal polynomial $\bar{g}$ is separable of degree f. Let $g \in A[x]$ be a monic lift of $\bar{g}$ and let α_0 be any lift of $\bar{\alpha}_0$ to B. If $v_{\mathbf{q}}(g(\alpha_0)) = 1$, then $\alpha = \alpha_0$. Otherwise let π_0 be any uniformiser for B and let $\alpha \equiv \alpha_0 + \pi_0 \in B$ so that $v_{\mathbf{q}}(g(\alpha)) = v_{\mathbf{q}}(g(\alpha_0 + \pi_0)) = v_{\mathbf{q}}(g(\alpha_0) + \pi_0 g'(\alpha_0) + \pi_0^2 h(\alpha_0)) = 1$ where $h \in A[x]$, i.e. $\pi \equiv g(\alpha)$ is a uniformiser for B.
CLAIM : $B = A[\alpha]$.
pf. $\mathbf{p} = \mathbf{q}^e$ so $\mathbf{p}B = (\pi^e)$. Represent each element of $B/\mathbf{p}B$ as a coset $b + \mathbf{p}B = b_0 + b_1\pi + \dots + b_{e-1}\pi^{e-1} + \mathbf{p}B$ where b_is are determined up to equivalence module πB. Let $1, \bar{\alpha}, \dots, \bar{\alpha}^{f-1}$ are a basis for $B/\mathbf{p}B = B/\mathbf{q}$ as a k-vector space, and $\pi = g(\alpha)$, so $b + \mathbf{p}B = (a_0 + a_1\alpha + \dots + a_{f-1}\alpha^{f-1}) + (a_f + a_{f+1}\alpha + \dots + a_{2f-1}\alpha^{f-1})g(\alpha) + \dots + (a_{ef-f+1} + \dots + a_{ef-1}\alpha^{f-1})g(\alpha)^{e-1} + \mathbf{p}B$. Since $\deg g = f$, $n = ef$, this expresses $b + \mathbf{p}B$ in the form $b' + \mathbf{p}B$ with b' in the A-span of $1, \dots, \alpha^{n-1}$ thus the claim follows.
Then by the Nakayama's lemma, $1, \dots, \alpha^{n-1}$ also span B.

If L/K is unraimified, l/k is separable and $e = 1, f = n$, in which case there is no need to require $g(\alpha)$ to be a uniformiser and we can just take $\alpha = \alpha_0$ to be any lift of any $\bar{\alpha}_0$ that generate l over k.

2.2.2 Extensions of Complete DVRs-Unramified

Define a category $\mathscr{C}_K^{unr}$ to be the finite unramified extensions L of K with morphisms K-algebra homomorphisms and a category $\mathscr{C}_k^{sep}$ to be the finite separable extensions l of k with morphisms k-algebra homomorphisms.

Theorem 2.2.6. *Let A be a complete DVR with fraction field K and residue field $k \equiv A/\mathbf{p}$. The categories $\mathscr{C}_K^{unr}, \mathscr{C}_k^{sep}$ are equivalent via functor $\mathscr{F} : \mathscr{C}_K^{unr} \to \mathscr{C}_k^{sep}$ That sends each unramified extension L of K to its residue field l, and each K-algebra homomorphism $\phi : L_1 \to L_2$ to the k-algebra homomorphism $\bar{\phi} : l_1 \to l_2$ defined by $\bar{\phi}(\bar{\alpha}) \equiv \overline{\phi(\alpha)}$, where α is any lift of $\bar{\alpha} \in l_1 \equiv B_1/\mathbf{q}_1$ to B_1 and $\overline{\phi(\alpha)}$ is the reduction of $\phi(\alpha) \in B_2$ to $l_2 \equiv B_2/\mathbf{q}_2$; here $\mathbf{q_1}, \mathbf{q_2}$ are the maximal ideals of the valuation rings B_1, B_2 of L_1, L_2 respectively. In particular, $\mathscr{F}$ gives a bijection between the isomorphism classes in $\mathscr{C}_K^{unr}$ and $\mathscr{C}_k^{sep}$ and if L_1, L_2 and have residue fields l_1, l_2, then $\mathscr{F}$ induces a bijection of finite sets $Hom_K(L_1, L_2) \to Hom_k(l_1, l_2)$.*

pf. **CLAIM : $\mathscr{F}$ is well-defined.**
pf. $\mathscr{F}$ maps finite unramified extensions L/K to a finite separable extensions l/k, but we need to check it does not depend on the choice of the lift α of $\bar{\alpha}$ we pick. So let $\phi : L_1 \to L_2$ be a K-algebra homomorphism and for $\bar{\alpha} \in l_!$, let α, α' be two lifts of $\bar{\alpha}$ to B_1. Then $\alpha - \alpha' \in \mathbf{q}_1$ and this implies $\phi(\alpha - \alpha') \in \phi(\mathbf{q}_1) = \phi(B_1) \cap \mathbf{q}_2 \subseteq \mathbf{q}_2$ and thereefore $\overline{\phi(\alpha)} = \overline{\phi(\alpha')}$. The identity $\phi(\mathbf{q}_1) = \phi(B_1) \cap \mathbf{q}_2 \subseteq \mathbf{q}_2$ follows from the fact that ϕ restricts to an injective ring homomorphism $B_1 \to B_2$ and $B_2/\phi(B_!)$ is a finite extension of DVRS in which $\mathbf{q}_2$ lies over the prime $\phi(\mathbf{q}_1)$ of $\phi(B_1)$. Then $\mathscr{F}$ sends identity morphisms to the identity morphisms and it is compatible with composition. Thus the claim follows.
CLAIM : $\mathscr{F}$ is essentially surjective, i.e. each separable l/k is isomorphic to the residue field of some unramified L/K
pf. Given a finite separable extension l/k, by the primitive element theorem $l \cong k(\bar{\alpha}) = \frac{k[x]}{(\bar{g})}$ for some $\bar{\alpha} \in l$, whose minimal polynomial $\bar{g} \in k[x]$ is monic irreducible separable and of degree $n \equiv [l : k]$. Let $g \in A[x]$ be any monic lift of $\bar{g}$, then g is also irreducible separable and of degree n. Let $L \equiv \frac{K[x]}{(g)} = K(\alpha)$ where α is the image of x in $K[x]/(g)$. Then L/K is a finite separable extension and by the previous corollary 2.3.6, $(\mathbf{p}, g(\alpha))$ is the unique maximal ideal of $A[\alpha]$ and $B/\mathbf{q} \cong \frac{A[\alpha]}{(\mathbf{p}, g(\alpha))} \cong \frac{A[x]}{(\mathbf{p}, g)} \cong \frac{(A/\mathbf{p})[x]}{(\bar{g})} \cong l$. Thus $[L : K] = \deg g = [l : k] = n$ so L/K is an unramified extension of degree n and the ramification index of $\mathbf{q}$ is necessarily $e = n/f = 1$ and the extension l/k is separable by assumption.
CLAIM : $\mathscr{F}$ is full and faithful; i.e. tie induced map $Hom_K(L_1, L_2) \to Hom_k(l_1, l_2)$

is a bijection.
pf. Given finite unramified extensions L_1,L_2 with valuation rings B_1,B_2, and residue fields l_1,l_2 we have induced maps $Hom_K(L_1,L_2) \to Hom_A(B_1,B_2) \to Hom_k(l_1,l_2)$. The first map is given by the restriction from L_1 to B_1 and since tensoring with K gives an inverse map in the other direction, it is a bijection. The second map sends $\phi : B_1 \to B_2$ to the k-homomorphism $\bar{\phi}$ that sends $\bar{\alpha} \in l_1 = B_1/\mathbf{q}_1$ to the reduction of $\phi(\alpha)$ modulo $\mathbf{q}_2$, where α is any lift of $\bar{\alpha}$. By the primitive element theorem $l_1 = k(\bar{\alpha}) = k[x]/(\bar{g})$ for some $\bar{\alpha} \in l_1$. If we lift $\bar{\alpha}$ to $\alpha \in B_1$, $L_1 = K(\alpha)$ since $[L_1 : K] = [l_1 : k]$ is equal to the degree of the minimal polynomial $\bar{g}$ of $\bar{\alpha}$ which cannot be less than the degree of the minimal polynomial g of α, where both are monic. Moreover $B_1 = A[\alpha]$ since it is true of the valuation ring of every finite unramified extensions in the category. Each A-module homomorphism in $Hom_A(B_1,B_2) = Hom_A(\frac{A[x]}{(g)},B_2)$ is uniquely determined by the image of x in B_2. This gives a bijection between $Hom_A(B_1,B_2)$ and the roots of g in B_2. Similarly each k-algebra homomorphism in $Hom_k(l_1,l_2) = Hom_k(\frac{k[x]}{(\bar{g})},l_2)$ is uniquely determined by the image of x in l_2 and there is a bijection between $Hom_k(l_1,l_2)$ and the roots of $\bar{g}$ in l_2. Now $\bar{g}$ is separable, so every root of $\bar{g}$ in $l_2 = B_2/\mathbf{q}_2$ lifts to a unique root of g in B_2 by the Hensel's Lemma. Thus the second map above is also a bijection and the claim follows.

$\to$ even if L_2/K is not unramifield, the map $Hom_K(L_1.L_2) \to Hom_l(l_1,l_2)$ is a bijection.

Corollary 2.2.9. *Assume AKLB with A a complete DVR with residue field k. Then L/K is unramified iff $B = A[\alpha]$ for some $\alpha \in L$ whose minimal polynomial $g \in A[x]$ has separable image $\bar{g}$ in $k[x]$.*

pf. $\bar{g}$ must be irreducible, since otherwise we could use Hensel's lemma to lift a nontrivial factorisation of $\bar{g}$ to a nontrivial factorisation of g so the residue field extension is separable and has the same degree as L/K. So L/K is unramified. The converse follows by the previous theorem.

Corollary 2.2.10. *Let A be a complete DVR with fraction field K and residue field k and let ζ_n be a primitive nth root of unity in some algebraic closure of K with n prime to the characteristic of k. The extension $K(\zeta_n)/K$ is unramified.*

pf. The field $K(\zeta_n)$ is the splitting field of $f(x) = x^n - 1$ over K. The image $\bar{f}$ of f in $k[x]$ is separable when $p \not| n$ since $gcd(\bar{f},f) \neq 1$ only when $\bar{f}' = nx^{n-1}$ is zero, equivalently only when $p|n$. When $\bar{f}$ is separable, so are all of its divisors, including the reduction of the minimal polynomial of ζ_n, which must be irreducible since otherwise we could obtains a contradiction by lifting a nontrivial factorisation via Hensel's lemma. Thus the residue field of $K(\zeta_n)$ is a separable extension of k so $K(\zeta_n)/K$ is unramified.

Theorem 2.2.7. *Assume AKLB with A a complete DVR withe finite residue field. Then L/K is unramified iff $N_{L/K}(B^\times) = A^\times$.*

pf.**CLAIM :** $N_{L/K}(B^\times) \subseteq A^\times$, $N_{l/k}(l^\times) = k^\times, T_{l/k}(l) = k$.
pf. $a \in N_{L/K}(B^\times) \leftrightarrow \exists b \in B^\times$ s.t. $N_{L/K}((b)) = a, \exists b^{-1} \in B$ s.t. $N_{L/K}((b^{-1})) = \hat{a}$. Since the norm map is a group homomorphism $N_{L/K}((1)) = N_{L/K}((bb^{-1})) = N_{L/K}((b))N_{L/K}((b^{-1})) = a\hat{a}$ implies $a \in A^\times$.
CLAIM : $\forall i \geq 0, U_i \equiv 1 + \mathbf{p}^i \equiv \{1 + a : a \in \mathbf{p}^i\}$. **Then U_i are distinct closed subgroup of $A^\times$ that form a basis of nhoods $1 \in A^\times$.**
pf. $\forall 1 + a, 1 + a' \in U_i, (1 + a)(1 + a') = 1 + (a + a' + aa') \in U_i$ subgroup of $A^\times$.
If $U_i \cap U_j \neq, \exists a \in U_i \cap U_j$ i.e. $a = 1 + rp = 1 + sq$ where $r, s \in A^\times, p \in {}^i, q \in {}^j, i \neq j$. Wlog assume $i > j$. Then $a = 1 + rp'q = 1 + sq \leftrightarrow rp' = s \in A^\times$ a contradiction. Hence U_i are distinct.
Pick an arbitrary open nhood of 1 in the topological group $A^\times$. Then it contains an open ball $B_{<r}(1) = \{x \in A^\times : |x - 1| < r\}$. For any $r \in \mathbb{R}, \exists i \in \mathbb{Z}_{++}$ s.t. $1/r < i$ and $U_i \subseteq B_{<r}(1)$ implies U_i form a base of nhoods $1 \in A^\times$.
CLAIM : If L/K is totally ramified, the norm of every $b \in B^\times$ lies in a coset of U_1 of the form $u^n U_1$ where $n = [L : K]$.
pf. Since L/K is totally ramified, for an uniformiser of K, π_A, $L = K(\pi_A^n)$, $B = A[\pi_A^{1/n}] = B[\pi]$ where n is defined above. Since $b \in B^\times = A[\pi]^\times, a = N_{L/K}(b) \in A^\times$ i.e. $a \notin \mathbf{p}$. d

By the previous claim, $\forall n > 1$, the norms of the cosets above do not cover $A^\times$. Thus if $N_{L/K}(B^\times) = A^\times, L/K$ must be unramified.
CLAIM : Assume L/K is unramified. Then for every $u \in A^\times, \exists \alpha_0 \in B^\times$ with $N_{L/K}(\alpha_0) \equiv u \mod \mathbf{p}$.
pf. Since L/K is unramified, there is a corresponding finite separable extension l/k with $N_{l/k}(l^\times) = k^\times$. So $\bar{u} \in k^\times, \exists \bar{v} \in l^\times$ s.t. $N_{l/k}(\bar{v}) = \bar{u}$. Then we can get a lift for $\bar{v}$, say v s.t. $N_{L/K}(v) \equiv u \mod \mathbf{p}$. $\exists \alpha_1 \in B^\times$ with $N_{L/K}(\alpha_0\alpha_1) \equiv u \mod \mathbf{p}^2$. Continuing by this fashion, we get $\alpha \in B^\times$ s.t. $N_{L/K}(\alpha) = u$.

Definition 2.2.11. *Let L/K be a separable extension. The maximal unramified extension of K in L is the subfield $\cup_{L \subseteq E \subseteq L, E/L\, fin.umr} E \subseteq L$ where the union is over finite unramified extensions E/K. When $L = K^{sep}$ is the separable closure of K, this is the maximal unramified extension of K, denoted K^{unr}.*

e.g. $\mathbb{Q}_p^{unr}$ is an infinite extension of $\mathbb{Q}_p$ with Galois group $Gal(\mathbb{Q}_p^{unr}/\mathbb{Q}_p) \cong Gal(\bar{\mathbb{F}}_p/\mathbb{F}_p) = \lim Gal(\mathbb{F}_{p^n}/\mathbb{F}_n) \cong \lim \mathbb{Z}/n\mathbb{Z} \equiv \hat{\mathbb{Z}}$, which is the profinite completion of .

2.2.3 Extensions of Complete DVRs-Totally Ramified

Definition 2.2.12. *Let A be a DVR with maximal ideal $\mathbf{p}$. A monic polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in A[x]$ is Eisenstein if $a_i \in \mathbf{p}, \forall 0 \leq i < n$ and $a_0 \notin \mathbf{p}^2$; equivalently, $v_{\mathbf{p}}(a_i) \geq 1, \forall 0 \leq i < n$ and $v_{\mathbf{p}}(a_0) = 1$.*

→ the definition implies that a_0 is a uniformiser.

Lemma 2.2.8 (Eisenstein Irreducibility)**.** *Let A be a DVR with fraction field K and let $f \in A[x]$ be Eisenstein. Then f is irreducible in both $A[x]$ and $K[x]$.*

pf. Suppose the contrary. Then $f = gh$ has degree $n \geq 2$ for some nonconstant monic $g, h \in A[x]$. Put $f(x) - \sum f_i x^i, g = \sum g_i x^i . h = \sum h_i x^i$. Then $v_{\mathbf{p}}(f_0) = v_{\mathbf{p}}(g_0 h_0) = v_{\mathbf{p}}(g_0) + v_{\mathbf{p}}(g_0) = 1$ where $\mathbf{p}$ is a maximalideal of A. Wlog assume $v_{\mathbf{p}}(h_0) = 1$. Let $i > 0$ be the least i for which $v_{\mathbf{p}}(h_i) = 0$; such an $i < n$ exists since h is monic and $\deg h < n$. Then we have $f_i = g_0 h_i + \cdots + g_i h_0$ with $v_{\mathbf{p}}(f_i) \geq 1$, given f Eisenstein and $i < n$. But the valuation of the RHS is zero since $v_{\mathbf{p}}(g_0 h_i) = 0, v_{\mathbf{p}}(g_i h) i - j) \geq 1, \forall 0 \leq j < i$ by the minimality of i, a contradiction. Thus f is irreducible in $A[x]$ and since A is a DVR, and therefore a PID and UFD. Thus by Gauss' lemma f is irreducible in $K[x]$.

→**EISENSTEIN's CRITERION :** Any polynomial $f(x)$ over a Dedekind domain A that is Eisenstein over a localisation $A_{\mathbf{p}}$, f is irreducible in $K[x]$, hence in $A[x]$, given $A_{\mathbf{p}}, A$ have the same fraction field K.

Lemma 2.2.9. *Let A be a DVR and let $f \in A[x]$ be an Eisenstein polynomial. Then $B = A[\pi] \equiv A[x]/(f)$ is a DVR with uniformiser π, where π is the image of x in $A[x]/(f)$.*

pf. Let $\mathbf{p}$ be the maximal of A. Then $f \equiv x^n \mod \mathbf{p}$ so by the Dedekind-Kummer theorem $\mathbf{q} = (\mathbf{p}, x) = (\mathbf{p}, \pi)$ is the only maximal ideal of B. Let $f = \sum f_i x^i$, then $\mathbf{p} = (f_0)$ and $\mathbf{q} = (f_0, \pi)$ and $f_0 = -\pi f_1 - \pi^2 f_2 - \cdots - \pi^n \in (\pi)$, so $\mathbf{q} = (\pi)$. The unique maximal (π) of B is nonzero and principal, so B is a DVR with uniformiser π.

Theorem 2.2.8. *Assume AKLB with A a complete DVR and π a uniformiser for B. The extension L/K is totally ramified iff $B = A[\pi]$ and the minimal polynomial of π is Eisentein.*

pf. Let $n = [L : K]$, $\mathbf{p}$ be the maximal ideal of A, $\mathbf{q}$ be the maximal ideal of B and let π be a uniformiser for B with minimal polynomial f. If $B = A[\pi]$ and f is Eisenstein, then by the previous lemma, we have $\mathbf{p} = \mathbf{q}^n$, so $v_{\mathbf{q}}$ extends $v_{\mathbf{p}}$ with index $e_{\mathbf{q}} = n$ and L/K is totally ramified.
Suppose L/K is totally ramified. Then $v_{\mathbf{q}}$ extends $v_{\mathbf{p}}$ with index n, which implies $v_{\mathbf{q}}(K) = n\mathbb{Z}$. The set $\{\pi^0, \ldots, \pi^{n-1}\}$ is linearly independent over K, since the valuation of $\pi^0, \ldots, \pi^{n-1}$ are distinct modulo $v_{\mathbf{q}}(K) = n\mathbb{Z}$. Thus $L = K(\pi)$. Let $f = \sum a_i x^i$ be the minimal polynomial of π. We have $v_{\mathbf{q}}(f(\pi)) = \infty$ and $v_{\mathbf{q}}(a_i \pi^i) \equiv i \mod n, \forall 0 \leq 9 \leq n$. This is possible only if $v_{\mathbf{q}}(a_0) = v_{\mathbf{q}}(a_0 \pi^0) = v_{\mathbf{q}}(a_n \pi^n) = v_{\mathbf{q}}(\pi^n) = n$, $v_{\mathbf{q}}(a_i) \geq n, \forall 0 \leq i < n$. This implies $v_{\mathbf{p}}(a_0) = 1$ since $v_{\mathbf{q}}$ extends $v_{\mathbf{p}}$ with index

n and $v_{\mathbf{p}}(a_i) \geq 1, \forall 1 \leq i < n$. Thus f is Eisenstein and by the previous lemma $A[\pi] \subseteq B$is a DVR and hence a maximal, so $B = A[\pi]$.

e.g.$K = \mathbb{Q}_3$ has distinct three quadratic extensions : $\begin{array}{ll} \mathbb{Q}_3(\sqrt{2}) & \text{unramified quadratic extension} \\ \mathbb{Q}_3(\sqrt{3}) & \text{ramified extension} \\ \mathbb{Q}_3(\sqrt{6}) & \text{ramified extension} \end{array}$

Definition 2.2.13. *Assume AKLB with A a complete DVR and separable residue field extension of characteristic $p \geq 0$. The extension L/K is tamely ramified if $p \nmid e_{L/K}$. Otherwise L/K is wildly ramified if $p | e_{L/K}$. A totally ramified extension L/Kis totally tamely ramified if $p \nmid e_{L/K}$ and it is totally wildly ramified if $e_{L/K}$ is a power of p.*

Proposition 2.2.6. *The properties of being unramified, tamely ramified, wildly ramified, totally ramified, totally tamely ramified and totally wildly ramified are all transitive in towers of extensions fraction fields of complete DVRs with separable residue field extensions.*

pf. follows by the transitivity of separability and the multiplicativity of ramification indices and degrees in towers.

A compositum of totally ramified extensions need not be totally ramified.(A compositum of $\mathbb{Q}_3(\sqrt{3}), \mathbb{Q}_3(\sqrt{6})$ of $\mathbb{Q}_3$ contains the unramified quadradic extension $\mathbb{Q}_3(\sqrt{2})$).

Theorem 2.2.9. *Assume AKLB with A a complete DVR and separable residue field extension of characteristic $p \geq 0$ not dividing $n \equiv [L : K]$. The extension L/K is totally ramified iff $L = K(\pi_A^{1/n})$ for some uniformiser π_A of A.*

pf. If $L = K(\pi_A^{1/n})$, then $\pi = \pi_A^{1/n}$ has minimal polynomial $x^n - \pi_A$, which is Eisenstein, so $A[\pi]$ is a DVR by the first lemma, and therefore $B = A[\pi]$, since DVRs are maximal and by the previous theorem L/K is totally ramified, given $p \nmid n$.
Conversely, assume L/K is totally tamely ramified, and let $\mathbf{p}, \mathbf{q}$ be the maximal ideals of A, B with uniformisers π_A, π_B respectively. Then $v_{\mathbf{q}}$ extends $v_{[\mathbf{p}}$ with index $e_{\mathbf{q}} = n$ and $v_{\mathbf{q}}(\pi_B^n) = n = v_{\mathbf{q}}(\pi_A) \to \pi_B^n = u\pi_A$ for some unit $u \in B\times$. We have $f_{\mathbf{q}} = 1$ so B, A have the same residue field and if we lift the image of u in $B/\mathbf{q} \cong A/\mathbf{p}$ to a unit u_A in A and replace π_A with $u_A^{-1}\pi_A$, we can assume $u \equiv 1 \mod \mathbf{q}$. Define $g(x) \equiv x^n - u \in B[x]$ with reduction $\bar{g} = x^n - 1$ in $(B/\mathbf{q})[x]$. Then $\bar{g}'(1) \neq 0$ so by the Hensel's Lemma I, we can lift the root 1 of $\bar{g}(x)$ in $B/\mathbf{q}$ to be a root r of $g(x)$ in B. Let $\pi \equiv \pi_B/r$. Then π is a uniformiser for B and $B = A[\pi]$ by the previous theorem, so $L = K(\pi)$ and $\pi^n = \pi_B^n/r^n = \pi_B^n/u = \pi_A$ so $L = K(\pi_A^{1/n})$ as desired.

Proposition 2.2.7. *Let L be a totally ramified extension of the fraction field K of a complete DVR. There is a unique intermediate field E s.t. E/K is totally tamely ramified and L/E is totally wildly ramified.*

pf. Let $e \equiv e_{L/K}$ be the ramification index and let $p \geq 0$ be the characteristic of the residue field. If $p \not| e$, the proposition holds with $E = L$, so we assume $p|e$ and put $e = mp^a$, where $p \not| m$. Let A be the valuation ring of K with maximal ideal $\mathbf{p}$ and let B be the valuation ring of L, which is also a complete DVR, with maximal ideal $\mathbf{q}$. Then by the previous theorem, we can choose the uniformiser π_A, π_B of A, B respectively, s.t. $\pi_B^n = u\pi_A$ with $u \in B^\times$, $u \equiv 1 \mod \mathbf{q}$. Let $g(x) = x^m - u \in B[x]$, then we can construct a root $r \in B$ of $g(x)$ by the Hensel's lemma, lifting the root 1 of $\bar{g} \in (B/\mathbf{q})[x]$. Consider the field extension $E \equiv K(\pi)$ where $\pi \equiv \pi_B^{o^a}/r$. Then we have $\pi^m = \pi_B^n/r^m = \pi_B^n/u = \pi_A \to E = K(\pi_A^{1/m}$ with $p \not| m$. The polynomial $x^m - \pi_A$ is Eisenstein, hence irreducible and has π as a root, so E/K has degree m. Then by the previous theorem E/K is totally tamely ramified, and L/E has degree p^a and is thus totally wildly ramified.
Suppose $E' \subseteq L$ is another totally tamely ramified extensions of K s.t. L/E' is totally wildly ramified. Then E' must be of the form $E' = K(\pi_A^{1/m})$ by the previous theorem. In other words, E, E' are both generated by the roots of $x^m - \pi_A$. The ratio of these roots is a mth root of unity $\zeta \in L$ that must lie in K because L/K is totally ramified and the extension $K(\zeta)/K$ is unramified. Thus $E = E'$.

Corollary 2.2.11. *Let L be a finite separable extension of the fraction field K of a complete DVR with separable field extension. There is a unique intermediate field E s.t. E/K is tamely ramified and L/E is totally wildly ramified.*

pf. Let F be the maximal uniramified extension of K in L. Then we can assume $F = K(\alpha)$ where α is integral, whose minimal polynomial g has separable image in $k[x]$ where k is the residue field of K. Then by the previous proposition to L/F, we get a tamely ramified extension E/F with L/E totally wildly ramified. Unramified extensions are tamely ramified, so $E/F/K$ is a tower of tamely ramified extensions, hence tamely ramified.
Any filed E' with L/E' totally wildly ramified must contain α, otherwise $E'(\alpha)$ would be a nontrivial unramified subextension L/E'. By the previous proposition, $E' = E$.

Let K be the fraction field of a complete DVR with absolute value $|\cdot|$. Then we can uniquely extend $|\cdot|$ to any finite extension L/K by defining $|x| \equiv |N_{L/K}(x)|^{1/n}$ where $n = [L:K]$. This induces a unique absolute value on $\bar{K}$ that restricts to the absolute value of K.

Lemma 2.2.10. *Let K be the fraction field of a complete DVR with algebraic closure $\bar{K}$ and absolute value $|\cdot|$ extended to $\bar{K}$. For all $\alpha \in \bar{K}$ and $\sigma \in Aut_K(\bar{K})$ we*

have $|\sigma(\alpha)| = |\alpha|$.

pf. The elements $\alpha, \sigma(\alpha)$ must have the same minimal polynomial $f \in K[x]$ since $f(\sigma(\alpha)) = \sigma(f(\alpha)) = \sigma(0) = 0$, so $N_{L/K}(\alpha) = f(0) = N_{K(\sigma(\alpha))/K}(\sigma(\alpha))$. It follows that $|\sigma(\alpha)| = |N_{K(\sigma(\alpha))/K}(\sigma(\alpha))|^{1/n} = |N_{K(\alpha)/K}(\alpha)|^{1/n} = |\alpha|$ where $n = \deg f$.

Definition 2.2.14. *Let K be the fraction field of a complete DVR with absolute value $|\cdot|$ extended to an algebraic closure $\bar{K}$. For $\alpha, \beta \in \bar{K}$, we say β belongs to α is $|\beta - \alpha| < |beta - \sigma(\alpha)|, \forall \sigma \in Aut_K(\bar{K})$ with $\sigma(\alpha) \neq \alpha$, that is β is strictly closer to α than to any of its conjugates. This is equivalent to requiring that $|\beta - \alpha| < |\alpha - \sigma(\alpha)|$, $\forall \sigma(\alpha) \neq \alpha$, since every nonarchimedean triangle is isosceles.*

Lemma 2.2.11 (Krasner's Lemma)**.** *Let K be the fraction field of a complete DVR and let $\alpha, \beta \in \bar{K}$ with α separable over K. If β belongs to α, then $K(\alpha) \subseteq K(\beta)$.*

pf. Suppose the contrary. Then β belongs to α, but $\alpha \notin K(\beta)$. The extension $K(\alpha, \beta)/K(\beta)$ is separable and nontrivial, so there is an automorphism $\sigma \in Aut_{K(\beta)}(\bar{K}/K(\beta))$ for which $\sigma(\alpha) \neq \alpha$. By the previous lemma to $\beta - \alpha \in \bar{K}, |\beta - \alpha| = |\sigma(\beta - \alpha)| = |\sigma(\beta) - \sigma(\alpha)| = |\beta - \sigma(\alpha)|$, a contradiction to the fact that β belongs to α.

Definition 2.2.15. *For a field K with absolute value $|\cdot|$, the L^1-norm of $f = \sum f_i x^i \in K[x]$ is $||f||_1 \equiv \sum_i |f_i|$.*

Lemma 2.2.12. *Let K be a field with absolute value $|\cdot|$ and let $f \equiv \Pi_i(x - \alpha_i) \in K[x]$ be a monic polynomial with roots $\alpha_1, \ldots, \alpha_n \in L$, where L/K is a field with an absolute value that extends $|\cdot|$. Then $|\alpha| \leq ||f||_1$ forevery root α of f.*

pf. When $n = 0, 1$, the lemma is clear, so assume $n \geq 2$. If $||f||_1 = 1$, $f = x^n$, $\alpha = 0$ so $|\alpha| = 0 < 1 = ||f||_1$ the lemma holds. Otherwise, $||f||_1 > 1$, and if $|\alpha| \leq 1$, also the lemma trivially holds, so assume $|\alpha| > 1$. Then we have $0 = |f(\alpha)| = |\alpha^n + \sum_{i=0}^{n-1} f_i \alpha^i| \geq |\alpha|^n - \sum_{i=0}^{n-1} |f_i||\alpha|^i \geq |\alpha|^n - |\alpha|^{n-1} \sum_{i=0}^{n-1} |f_i| \geq |\alpha| - (||f||_1 - 1)$. Thus $||f||_1 - 1 \geq |\alpha| \leftrightarrow ||f||_1 \geq |\alpha| + 1 > |\alpha|$.

Theorem 2.2.10 (Continuity of Roots)**.** *Let K be the fraction field of a complete DVR and $f \in K[x]$, a monic irreducible separable polynomial. There exists $\delta = \delta(f) \in \mathbb{R}_{++}$ s.t. for every monic polynomial $g \in K[x]$ with $||f - g||_1 < \delta$, the following holds:*

Every root β of g belongs to a root α of f for which $K(\beta) = K(\alpha)$.
In particular, every such g is separable, irreducible, and has the same splitting field as f.

pf. We can always pick $\delta < 1$, in which case any monic $g \in K[x]$ with $||f - g||_1 < \delta$ must have the same degree as f, so we can assume $\deg g = \deg f$. Fix an algebraic closure $\bar{K}$ of K with absolute value $|\cdot|$ extending the absolute value on K.

Let $\alpha_1,\dots,\alpha_n$ be the roots of f in $\bar{K}$ and write $f(x)=\Pi(x-\alpha_i)=\sum_{i=0}^n f_i x^i$. Let $\epsilon=\min(1,\min_{i\neq j}|\alpha_i-\alpha_j|)$. Define $\delta\equiv\delta(f)=(\frac{\epsilon}{2(||f||_1+1)})^n>0$ s.t. $\delta<1$, given $||f||_1\geq 1,\epsilon\leq 1$. Let $g(x)=\sum g_i x^i$ be a monic polynomial of degree n with $||f-g||_1<\delta$. Then $||g||_1\leq||f||_1+||g-f||_1=||f||_1+||f-g||_1<||f||_1+\delta$ and $\forall\beta\in\bar{K}$ any root of g, we have
$|f(\beta)|=|f(\beta)-g(\beta)|=|(f-g)(\beta)|=|\sum(f_i-g_i)\beta^i|\leq\sum|f_i-g_i||\beta|^i$. By the previous lemma
$|\beta|<||g||_1$, $||g||_1\geq 1$, so $|\beta|^i<||g||_1^i\leq||g||_1^n$ and therefore $|f(\beta)|<||f-g||_1\cdot||g||_1^n<\delta(||f||_1+\delta)^n<\delta(||f||_1+1)^n\leq(\epsilon/2)^n\to\Pi|\beta-\alpha_i|=|f(\beta)|<(\epsilon/2)^n$ and at least one $i,|\beta-\alpha_i|<\epsilon/2$. Also by the triangular inequality α_i is unique, given $|\alpha_i-\alpha_j|\geq\epsilon,\forall j\neq i$. This implies β belongs to $\alpha\equiv\alpha_i$.
By the Krasner's lemma, $K(\alpha)\subseteq K(\beta)$ and $n=[K(\alpha):K]\leq[K(\beta):K]\leq n$ so $K(\alpha)=K(\beta)$. Thus g is the minimal polynomial of β since $\deg g=[K(\beta):K]$. Thus g is irreducible and it is also separable, since $\beta\in K(\beta)=K(\alpha)$ lies in a separable extension of K. If a root β of g belongs to a root α of f, then for any $\tau\in Aut_K(\bar{K}),\forall\sigma\in Aut_K(\bar{K})$, s.t. $\sigma(\alpha)\neq\alpha$, $|\tau(\beta)-\tau(\alpha)|=|\tau(\beta-\alpha)|=|\beta-\alpha|<|\alpha-\sigma(\alpha)|=|\tau(\alpha-\sigma(\alpha))|=|\tau(\alpha)-\tau(\sigma(\alpha))|$ so $\tau(\beta)$ belongs to $\tau(\alpha)$. Since $Aut_K(\bar{K})$ acts transitively on the roots of f,g, so every root β of g belongs to a distinct root α of f for which $K(\beta)=K(\alpha)$. Therefore g has the same splitting field as f.

Theorem 2.2.11. *Let K be a global field with a nontrivial absolute value $|\cdot|$ and let $\hat{K}$ be the completion of K w.r.t. $|\cdot|$. Every finite separable extension $\hat{L}$ of $\hat{K}$ is the completion of a finite separable extension L of K w.r.t. an absolute value that restricts to $|\cdot|$. One can choose L so that $[L:K]=[\hat{L}:\hat{K}]$, in which case $\hat{L}=\hat{K}\cdot L$.*

pf. Let $\hat{L}/\hat{K}$ be a separable extension of degree n.

- $|\cdot|$ is archimedean, K is a number field and $\hat{K}$ is either $\mathbb{R}$ or $\mathbb{C}$; the only nontrivial case is $\hat{K}\cong\mathbb{R}$, $n=2$ and we may assume $\hat{L}=\hat{K}(\sqrt{d})$, $d\in\mathbb{Z}_{<0}$ is any nonsqure in K. Wlog assume $|\cdot|$ is the Euclidean absolute value on $\hat{K}\cong\mathbb{R}$ and uniquely extend $|\cdot|$ to $L\equiv K(\sqrt{d})$ by requiring $|\sqrt{d}|=\sqrt{-d}$. Then $\hat{L}$ is the completion of L w.r.t. $|\cdot|$ and $[L:K]=[\hat{L}:\hat{K}]=2$ and $\hat{L}=\hat{K}\cdot L$.

- $|\cdot|$ is nonarchimedean, the valuation ring of $\hat{K}$ is a complete DVR and $|\cdot|$ is induced by its discrete valuation. By the primitive element theorem, we may assume $\hat{L}=\hat{K}[x]/(f)$ where $f\in\hat{K}[x]$ is monic, irreducible and separable. The field K is dense in its completion $\hat{K}$, so we can find a monic $g\in K[x]\subseteq\hat{K}[x]$ s.t. $||g-f||_1\leq\delta,\forall\delta>0$. Then by the previous theorem $\hat{L}=\hat{K}[x]/(f)$ where g is separable. The field $\hat{L}$ is a finite separable extension of the fraction field of a complete DVR, so it is itself the fraction field of a complete DVR and has a unique absolute value that extends the absolute value $|\cdot|$ on $\hat{K}$. Let $L\equiv K[x]/(g)$. Then g is irreducible in $\hat{K}[x]$ hence in $K[x]$, so $[L:K]=$

$[\hat{L}:\hat{K}] = \deg g$. The field $\hat{L}$ contains both $\hat{K}, L$ and it is clearly the smallest field that does, so $\hat{L}$ is the compositum of $\hat{K}$ and L. The absolute value on $\hat{L}$ restricts to an absolute value on L, extending the absolute value on K, and $\hat{L}$ is complete, so $\hat{L}$ contains the completion of L w.r.t. $|\cdot|$. On the other hand, the completion of L w.r.t. $|\cdot|$ contains $L, \hat{K}$, so it must be $\hat{L}$.

Corollary 2.2.12. *For every finite Galois extension $\hat{L}/\hat{K}$ of local fields, there is a finite Galois extension of global fields L/K and an absolute value $|\cdot|$ on L s.t. $\hat{L}$ is the completion of L w.r.t. $|\cdot|$, $\hat{K}$ is the completion of K w.r.t. the restriction of $|\cdot|$ to K and $Gal(L/K) \cong Gal(\hat{L}/\hat{K})$.*

pf. The archimedean case follows by the previous theorem with $K = \mathbb{Q}$. So assume $\hat{L}$ is nonarchimedean and we take $|\cdot|$ to be the absolute value on $\hat{K}, \hat{L}$. The field $\hat{K}$ is an extension of either $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$, and by the previous theorem, we may assume $\hat{K}$ is the completion of a global field K w.r.t. the restriction of $|\cdot|$. Let $g \in K[x]$ be a monic separable polynomial irreducible in $\hat{K}[x]$ s.t. $\hat{L}[x] = \hat{K}[x]/(g)$ and define $L \equiv K[x]/(g)$ so that $\hat{L}$ is the compositum of $\hat{K}$ and L. Let M be the splitting field of g over K, the minimal extension of K that contains all the roots of g. The field $\hat{L}$ also contains these roots and $\hat{L}$ contains K, so $\hat{K}$ also contains a subextension of K isomorphic to M, which we now identify with M. Then $\hat{L}$ is also the completion of M w.r.t. the restriction of $|\cdot|$ to M. We have a group homomorphism $\phi : Gal(\hat{L}/\hat{K}) \to Gal(M/K)$ induced by restriction and ϕ is injective since $\forall \sigma \in Gal(\hat{L}/\hat{K})$ is determined by its action on any root of g in M. If we now replace K by the fixed field of the image of ϕ and replace L with M, the completion of K w.r.t. the restriction of $|\cdot|$ is still equal to $\hat{K}$ and similarly for L and $\hat{L}$. Now the corollary follows.

Under AKLB assumption, if $\mathbf{p}$ is a prime of K, each prime $\mathbf{q}|\mathbf{p}$ induces a valuation $v_{\mathbf{q}}$ of L that extends the valuation $v_{\mathbf{p}}$ of K with index $e_{\mathbf{q}}$, meaning $v_{\mathbf{q}}|_K = e_{\mathbf{q}} v_{\mathbf{p}}$. The embedding $K \to L$ induces an embedding $K_{\mathbf{p}} \to L_{\mathbf{q}}$ of local fields for each $\mathbf{q}|\mathbf{p}$.

Theorem 2.2.12. *Assume AKLB, let $\mathbf{p}$ be a prime of A, and let $\mathbf{p}B = \Pi_{\mathbf{q}|\mathbf{p}} \mathbf{q}^{e_{\mathbf{q}}}$ be the factorisation of $\mathbf{p}B$ in B. Let $K_{\mathbf{p}}$ be the completion of K w.r.t. $|\cdot|_{\mathbf{p}}$ and let $\hat{\mathbf{p}}$ be the maximal ideal of its valuation ring. For each $\mathbf{q}|\mathbf{p}$, let $L_{\mathbf{q}}$ denote the competion of L w.r.t. $|\cdot|_{\mathbf{q}}$, and $\hat{\mathbf{q}}$ the maximal ideal of its valuation ring. The following hold:*

1. *Each $L_{\mathbf{q}}$ is a finite separable extension of $K_{\mathbf{p}}$ with $[L_{\mathbf{q}} : K_{\mathbf{p}}] \leq [L : K]$*
2. *Each $\hat{\mathbf{q}}$ is the unique prime of $L_{\mathbf{q}}$ lying over $\hat{\mathbf{p}}$*
3. *Each $\hat{\mathbf{q}}$ has ramification index $e_{\hat{\mathbf{q}}} = e_{\mathbf{q}}$ and residue field degree $f_{\hat{\mathbf{q}}} = f_{\mathbf{q}}$*
4. $[L_{\mathbf{q}} : K_{\mathbf{p}}] = e_{\mathbf{q}} f_{\mathbf{q}}$

5. *the map* $\begin{array}{rcl} L \otimes_K K_{\mathbf{p}} & \to & \Pi_{\mathbf{q}|\mathbf{p}} L_{\mathbf{q}} \\ l \otimes x & \mapsto & (lx, lx, \ldots, lx) \end{array}$ *is an isomorphism of finite etale* $K_{\mathbf{p}}$*-algebras*

6. *If* L/K *is Galois, then each* $L_{\mathbf{q}}/K_{\mathbf{p}}$ *is Galois and we have isomorphisms of decomposition groups* $D_{\mathbf{q}} \cong D_{\hat{\mathbf{q}}} = Gal(L_{\mathbf{q}}/K_{\mathbf{p}})$ *and inertial groups* $I_{\mathbf{q}} \cong I_{\hat{\mathbf{q}}}$.

pf.

1. For each $\mathbf{q}|\mathbf{p}$, the embedding $K \to L$ induces an embedding $K_{\mathbf{p}} \to L_{\mathbf{q}}$ defined by $[(x_n)] \mapsto [(x_n)]$ on equivalence classes of Cauchy sequences; a sequence (x_n) that is Cauchy in K w.r.t. $|\cdot|_{\mathbf{p}}$ is also Cauchy in L w.r.t. $|\cdot|_{\mathbf{q}}$ because $v_{\mathbf{q}}$ extends $v_{\mathbf{p}}$. Thus we may view $K_{\mathbf{p}}$ as a topological subfield of $L_{\mathbf{q}}$ and it is clear that $[L_{\mathbf{q}} : K_{\mathbf{p}}] \leq [L : K]$ since any K-basis $b_1, \ldots, b_m$ for $L \subseteq L_{\mathbf{q}}$ spans $L_{\mathbf{q}}$ as a $K_{\mathbf{p}}$-vector space; given a Cauchy sequence $y \equiv (y_n)$ of elements of L, if we write each y_n as $\sum_i x_{in} b_i, x_{in} \in K$, we obtain Cauchy sequence $x_1 \equiv (x_{1n}), \ldots, x_m \equiv (x_{mn})$ of elements in K and we can write $[y] = [x_1]b_1 + \cdots + [x_m]b_m$ as a $K_{\mathbf{p}}$-linear combination of b_i. The field L is a finite etale K-algebra, since L/K is a separable extension so its base change $L \otimes_K K_{\mathbf{p}}$ to $K_{\mathbf{p}}$ is a finite etale $K_{\mathbf{p}}$-algebra. Consider the $K_{\mathbf{p}}$-algebra homomorphism $\begin{array}{rcl} \phi_{\mathbf{q}} : L \otimes_K K_{\mathbf{p}} & \to & L_{\mathbf{q}} \\ l \otimes x & \mapsto & lx \end{array}$. Then $\phi_{\mathbf{q}}(b_i \otimes 1) = b_i$ for each of the K-basis elements $b_i \in L$, and b_is span $L_{\mathbf{q}}$ as $K_{\mathbf{p}}$-vector space, thus $\phi_{\mathbf{q}}$ is surjective. As a finite etale $K_{\mathbf{p}}$-algebra, $L \otimes_K K_{\mathbf{p}}$ is isomorphic to a finite product of finite separable extensions of $K_{\mathbf{p}}$; $L_{\mathbf{q}}$ is isomorphic to a subproduct and thus also a finite etale $K_{\mathbf{p}}$-algebra. In particular $L_{\mathbf{q}}/K_{\mathbf{p}}$ is separable.

2. The valuation rings of $K_{\mathbf{p}}, L_{\mathbf{q}}$ are complete DVRs, so this follows by the previous theorem.

3. The valuation ring $v_{\hat{\mathbf{q}}}$ extends $v_{\mathbf{q}}$ with index 1, which in turn extends $v_{\mathbf{p}}$ with index $e_{\mathbf{q}}$. The valuation $v_{\hat{\mathbf{p}}}$ extends $v_{\mathbf{p}}$ with index 1, and therefore $v_{\hat{\mathbf{q}}}$ extends $v_{\hat{\mathbf{p}}}$ with index $e_{\mathbf{q}}$ and therefore $e_{\hat{\mathbf{q}}} = e_{\mathbf{q}}$. The residue field of $\hat{\mathbf{p}}$ is the same as that of $\mathbf{p}$: for any Cauchy sequence (a_n) over K, a_n eventually all have the same image in the residue field at $\mathbf{p}$ since $v_{\mathbf{p}}(a_n - a_m) > 0$ for all sufficiently large m, n. Similar statements apply to all $\hat{\mathbf{q}}$ and $\mathbf{q}$ and thus $f_{\hat{\mathbf{q}}} = f_{\mathbf{q}}$

4. By 2, $[L_{\mathbf{q}} : K_{\mathbf{p}}] = e_{\hat{\mathbf{q}}} f_{\hat{\mathbf{q}}}$ since $\hat{\mathbf{q}}$ is the only prime above $\hat{\mathbf{p}}$ and by 3, this implies $[L_{\mathbf{q}} : K_{\mathbf{p}}] = e_{\mathbf{q}} f_{\mathbf{q}}$.

5. Let $\phi \equiv \Pi_{\mathbf{q}|\mathbf{p}} \phi_{\mathbf{q}}$ where $\phi_{\mathbf{q}} : L \otimes_K K_{\mathbf{p}} \to \Pi_{\mathbf{q}|\mathbf{p}} L \otimes_K K_{\mathbf{p}} \to L_{\mathbf{q}}$ is surjective $K_{\mathbf{p}}$-algebra homomorphism, defined in the proof of 1. Then $\phi : L \otimes_K K_{\mathbf{p}} \to$

$\Pi_{\mathbf{q}|\mathbf{p}} L_{\mathbf{q}}$ is a $K_{\mathbf{p}}$-algebra homomorphism. By 4, and by the fact that taking the base change of a finite etale algebra does not change its dimension, we have $\dim_{K_{\mathbf{p}}}(L \otimes_K K_{\mathbf{p}}) = \dim_K L = [L : K] = \sum_{\mathbf{q}|\mathbf{p}} e_{\mathbf{q}} f_{\mathbf{q}} = \sum_{\mathbf{q}|\mathbf{p}} [L_{\mathbf{q}} : K_{\mathbf{p}}] = \dim_{K_{\mathbf{p}}} \Pi_{\mathbf{q}|\mathbf{p}} L_{\mathbf{q}}$. Pick a $K_{\mathbf{p}}$-basis $\{\beta_i\}$ for $\Pi_{\mathbf{q}|\mathbf{p}} L_{\mathbf{q}}$, fix $\epsilon > 0$, and for each basis element $\beta_i = (\beta_{i\mathbf{q}})_{\mathbf{q}|\mathbf{p}}$ use the weak approximation theorem to construct $\alpha_i \in L$ s.t. $|\alpha_i - \beta_{i\mathbf{q}}|_{\mathbf{q}} < \epsilon$, $\forall \mathbf{q}|\mathbf{p}$. In the metric space $\Pi_{\mathbf{q}|\mathbf{p}} L_{\mathbf{q}}$ with the sup-norm, each $\phi(\alpha_i \otimes 1)$ is close to β_i. The $K_{\mathbf{p}}$-matrix whose jth column expresses $\phi(\alpha_j \otimes 1)$ in terms of the basis $\{\beta_i\}$ is then close to the identity matrix w.r.t. $|\cdot|_{\mathbf{p}}$ and the determinant D of this matrix is close to 1. For sufficiently small ϵ, we must have $D \neq 0$ and then $\{\phi(\alpha_i \otimes 1)\}$ is a basis for $\Pi_{\mathbf{q}|\mathbf{p}} L_{\mathbf{q}}$. Thus ϕ is surjective and therefore an isomorphism, since its domain and codomain have the same dimension.

6. Assume L/K is Galois. Then $\forall \sigma \in D_{\mathbf{q}}$ acts on L and respects the valuation $v_{\mathbf{q}}$, since it fixes $\mathbf{q}(x \in \mathbf{q}^n \to \sigma(x) \in \sigma(\mathbf{q}^n) = \sigma(\mathbf{q})^n = \mathbf{q}^n)$ and therefore if (x_n) is a Cauchy in L, so is $(\sigma(x_n))$, thus σ is an automorphism of $L_{\mathbf{q}}$ and it fixes $K_{\mathbf{p}}$. Thus $\exists \phi : D_{\mathbf{q}} \to Aut_{K_{\mathbf{p}}}(L_{\mathbf{q}})$. If $\sigma \in D_{\mathbf{q}}$ acts trivially on $L_{\mathbf{q}}$, then it acts trivially on $L \subseteq L_{\mathbf{q}}$, so $\ker \phi$ is trivial. Also $e_{\mathbf{q}} f_{\mathbf{q}} = |D_{\mathbf{q}}| \leq \# Aut_{K_{\mathbf{p}}}(L_{\mathbf{q}}) \leq [L_{\mathbf{q}} : K_{\mathbf{p}}] = e_{\mathbf{q}} f_{\mathbf{q}}$ so $\# Aut_{K_{\mathbf{p}}}(L_{\mathbf{q}}) = [L_{\mathbf{q}} : K_{\mathbf{p}}]$ and $L_{\mathbf{q}}/K_{\mathbf{p}}$ is Galois, and this also shows ϕ is surjective and therefore an isomorphism. There is only one prime $\hat{\mathbf{q}}$ of $L_{\mathbf{q}}$, and it necessarily fixed by $\forall \sigma \in Gal(L_{\mathbf{q}}/K_{\mathbf{p}})$, $Gal(L_{\mathbf{q}}/K_{\mathbf{p}}) = D_{\hat{\mathbf{q}}}$. The inertial groups $I_{\mathbf{q}}$ and $I_{\hat{\mathbf{q}}}$ have order $e_{\mathbf{q}} = e_{\hat{\mathbf{q}}}$ and ϕ restricts to a homomorphism $I_{\mathbf{q}} \to I_{\hat{\mathbf{q}}}$, so the inertial groups are also isomorphic.

Corollary 2.2.13. *Assume AKLB and let* $\mathbf{p}$ *be a prime of* A. *For every* $\alpha \in L$, $N_{L/K}(\alpha) = \Pi_{\mathbf{q}|\mathbf{p}} N_{L_{\mathbf{q}}/K_{\mathbf{p}}}(\alpha)$, $T_{L/K}(\alpha) = \sum_{\mathbf{q}|\mathbf{p}} T_{L_{\mathbf{q}}/K_{\mathbf{p}}}(\alpha)$ *where we view* α *as an element of* $K_{\mathbf{q}}$ *and* $N_{L/K}(\alpha)$ *as an element of* $K_{\mathbf{p}}$ *via the canonical embeddings* $L \to L_{\mathbf{q}}, K \to K_{\mathbf{p}}$.

pf. The norm and trace are defined as the determinant and trace of K-linear maps $L \to_{\times \alpha} L$ that are unchanged upon tensoring with $K_{\mathbf{p}}$, and therefore by the previous theorem 5's isomorphism the corollary follows.

Corollary 2.2.14. *Assume AKLB and let* $\mathbf{p}$ *be a prime of* A. *Let* $\mathbf{p}B = \Pi_{\mathbf{q}|\mathbf{p}} \mathbf{q}^{e_q}$ *be the factorisation of* $\mathbf{p}B$ *in* B. *Let* $\hat{A}_{\mathbf{p}}$ *be the completion of* A *w.r.t.* $|\cdot|_{\mathbf{p}}$ *and for each* $\mathbf{q}|\mathbf{p}$, *let* $\hat{B}_{\mathbf{q}}$ *denote the completion of* B *w.r.t.* $|\cdot|_{\mathbf{q}}$. *Then* $B \otimes_A \hat{A}_{\mathbf{p}} \cong \Pi_{\mathbf{q}|\mathbf{p}} \hat{B}_{\mathbf{q}}$ *as* $\hat{A}_{\mathbf{p}}$*-algebras.*

pf. After localising, we can replace A with $A_{\mathbf{p}}$, B with $B_{\mathbf{p}}$, and we may assume A is a DVR, B/A os a free A-module of rank $n \equiv [L : K] = \sum_{\mathbf{q}|\mathbf{p}} e_{\mathbf{q}} f_{\mathbf{q}}$. Then $B \otimes_A \hat{A}_{\mathbf{p}}$ is a free $\hat{A}_{\mathbf{p}}$-module of rank n. Viewing $\hat{A}_{\mathbf{p}}$ and the $\hat{B}_{\mathbf{q}}$ as valuation rings of $K_{\mathbf{p}}, L_{\mathbf{q}}$, by the previous theorem 4, $\Pi \hat{B}_{\mathbf{q}}$ is a free $\hat{A}_{\mathbf{p}}$-module of rank

$\sum_{\mathbf{q}|\mathbf{p}}[L_{\mathbf{q}}:K_{\mathbf{p}}] = \sum e_{\mathbf{q}} f_{\mathbf{q}} = n$. These isomorphic $\hat{A}_{\mathbf{p}}$-modules lie in isomorphic finite etale $K_{\mathbf{p}}$-algebras $L \otimes_K K_{\mathbf{p}} \cong \Pi L_{\mathbf{q}}$ by the previous theorem 6, and this $K_{\mathbf{p}}$-algebra isomorphism restricts to an $\hat{A}_{\mathbf{p}}$-algebra isomorphisms.

2.2.4 Ramification of the Extensions of the Complete DVRs

The trace mapping $\begin{array}{ccc} L \times L & \to & K \\ (x,y) & \mapsto & T_{L/K}(xy) \end{array}$ is a perfect pairing under the AKLB set-up. An A-lattice M in L is a finitely generated A-module that spans L as a K-vector space. Every A-lattice M in L has a dual lattice $M^* \equiv \{x \in L : T_{L/K}(xm) \in A,\ \forall m \in M\}$ which is an A-lattice in L isomorphic to the dual A-module $M^\vee \equiv Hom_A(M,A)$. Every fractional ideal I of B is finitely generated as a B-module and therefore finitely generated as an A-module. If I is nonzero, it necessarily spans L, since B does. Thus every element of $\mathscr{I}_B$ of nonzero fractional ideals of B is an A-lattice in L.

Lemma 2.2.13. *Assume AKLB. If $I \in \mathscr{I}_B$, $I^* \in \mathscr{I}_B$.*

pf. The dual lattice I^* is a finitely generated A-module. Consider $b \in B, x \in I^*, m \in I$. Then we have $T_{L/K}((bx)m) = T_{L/K}(x(bm)) \in A$ since $x \in I^*, bm \in I$, so $bx \in I^*$ and it is closed by the multiplication by the element $b \in B$, hence a finitely generated B-module. Thus it is a fractional ideal of B.

Definition 2.2.16. *Assume AKLB. The different $\mathscr{D}_{L/K}$ of L/K is the inverse of B^* in $\mathscr{I}_B$. Explicitly $B^* \equiv \{x \in L : T_{L/K}(xb) \in A,\ \forall b \in B\}$ and we define*

$$\mathscr{D}_{L/K} \equiv \mathscr{D}_{B/A} = (B^*)^{-1} = (B : B^*) = \{x \in L : xB^* \subseteq B\}$$

$B \subseteq B^*$ since $T_{L/K}(ab) \in A,\ \forall a,b \in B \to \mathscr{D}_{B/A} = (B^*)^{-1}) \subseteq B^{-1} = B$ so the different is an ideal.

Proposition 2.2.8. *Assume AKLB and let S be a multiplicative subset of A. Then $S^{-1}\mathscr{D}_{B/A} = \mathscr{D}_{S^{-1}B/S^{-1}A}$.*

pf. by the fact that inverses and duals are both compatible with localisation

Proposition 2.2.9. *Assume AKLB and let $\mathbf{q}|\mathbf{p}$ be a prime of B. Then $\mathscr{D}_{\hat{B}_{\mathbf{q}}/\hat{A}_{\mathbf{p}}} = \mathscr{D}_{B/A}\hat{B}_{\mathbf{q}}$, where $\hat{A}_{\mathbf{p}}, \hat{B}_{\mathbf{q}}$ are the completions of A, B at $\mathbf{p}, \mathbf{q}$ respectively.*

pf. Let $\hat{L} \equiv L \otimes K_{\mathbf{p}}$ be the basechange of the finite etale K-algebra L from K to $K_{\mathbf{p}}$. Then $\hat{L} \cong \Pi_{\mathbf{q}|\mathbf{p}} L_{\mathbf{q}}$ by the theorem 2.3.12. (5). Even though $\hat{L}$ needs not be a field, it is a free $K_{\mathbf{p}}$-module of finite rank and thus is equipped with a trace map that necessarily satisfies $T_{\hat{L}/K_{\mathbf{p}}}(x) = \sum_{\mathbf{q}|\mathbf{p}} T_{L_{\mathbf{q}}/K_{\mathbf{p}}}(x)$ that defines a trance

pairing on $\hat{L}$. Let $\hat{B} \equiv B \otimes \hat{A}_{\mathbf{p}}$, which is an A-lattice in the $K_{\mathbf{p}}$-vector space $\hat{L}$> Then $\hat{B} \cong \Pi_{\mathbf{q}|\mathbf{p}} \hat{B}_{\mathbf{q}} \cong \oplus_{\mathbf{q}|\mathbf{p}} \hat{B}_{\mathbf{q}}$ and therefore $\hat{B}^* \oplus_{\mathbf{q}|\mathbf{p}} \hat{B}^*_{\mathbf{q}}$. Then $\hat{B}^* \cong B^* \otimes_A \hat{A}_{\mathbf{p}}$. In particular, B^* generates each fractional ideal $\hat{B}^*_{\mathbf{q}} \in \mathscr{I}_{\hat{B}_{\mathbf{q}}}$. Taking inverses, $\mathscr{D}_{B/A} = (B^*)^{-1}$ generates the $\hat{B}_{\mathbf{q}}$-ideal $(\hat{B}^*_{\mathbf{q}})^{-1} = \mathscr{D}_{\hat{B}_{\mathbf{q}}/\hat{A}_{\mathbf{p}}}$.

Definition 2.2.17. *Let S/R be a ring extension in which S is a free R-module of rank n. For any $x_1,\ldots,x_n \in S$, we define the discriminant $disc(x_1,\ldots,x_n) \equiv disc_{S/R}(x_1,\ldots,x_n) \equiv \det[T_{S/R}(x_i x_j)]_{i,j} \in R$.*

→ if x_is are not R-linearly independent, the discriminant will be zero by the linearity of the trace.

Under AKLB assumption where $(e_i) \in B$ is a basis for L as a K-vector space, $disc(e_1,\ldots,e_n) = \det[T_{L/K}(e_i e_j)]_{ij} \in A$.

Proposition 2.2.10. *Let L/K be a finite separable extension of degree n and let Ω/K be a field extension for which there are distinct $\sigma_1,\ldots,\sigma_n \in Hom_K(L,\Omega)$. For any $e_1,\ldots,e_n \in L$, $disc(e_1,\ldots,e_n) = \det[\sigma_i(e_j)]^2_{ij}$ and for any $x \in L$ $disc(1,x,\ldots,x^{n-1}) = \Pi_{i<j}(\sigma_i(x) - \sigma_j(x))^2$. Such a field extension Ω/K always exists, since L/K is separable.*

pf. For $1 \le i,j \le n, T_{L/K}(e_i e_j) = \sum_{k=1}^n \sigma_k(e_i e_j)$ and therefore $disc(e_1,\ldots,e_n) = \det[T_{L/K}(e_i e_j)]_{ij} = \det([\sigma_k(e_i)]_{ik}[\sigma_k(e_j)]_{kj}) = \det([\sigma_k(e_i)]_{il}[\sigma_k(e_j)]^y_{jk}) = \det[\sigma_i(e_j)]^2_{ij}$, since the determinant is multiplicative and $\det M = \det M^t$ for any matrix M. Let $x \in L$, and put $e_i \equiv x^{i-1}, \forall 1 \le i \le n$. Then $disc(1,x,\ldots,x^{n-1}) = \det[\sigma_i(x^{j-1})]^2_{ij} = \det[\sigma_i(x)^{j-1}]^2_{ij} = \Pi_{i<j}(\sigma_i(x) - \sigma_j(x))^2$ since $[\sigma_i(x)^{j-1}]_{ij}$ is a Vandermonde matrix.

Definition 2.2.18. *For a polynomial $f(x) = \Pi_i(x - \alpha_i)$, the discriminant of f is $disc(f) \equiv \Pi_{i<j}(\alpha_i - \alpha_j)^2$, equivalently if A is a Dedekind domain, $f \in A[x]$ is a monic polynomial, and α is the image of x in $A[x]/(f)$, then $disc(f) = disc(1,\alpha,\ldots,\alpha^{n-1}) \in A$.*

e.g. $disc(x^2 + bx + c) = b^2 - 4c$, $disc(x^3 + ax + b) = -4a^4 - 27b^2$.

Assume AKLB and let M be an A-lattice in L. Then M is a finitely generated A-module that contains a K-basis for L.
Consider the case where M is a free A-lattice. If $e_1,\ldots,e_n \in M \subseteq L$ and $e'_1,\ldots,e'_n \in M \subseteq L$ are two A-basis for M, then $disc(e'_1,\ldots,e'_n) = u^2 disc(e_1,\ldots,e_n)$ where $u \in A^\times$.
pf. $\exists P \in A^{n\times n}$ invertible and $\det P = u$ s.t. it changes the base matrix from e to e', i.e. $e' = eP$, where $e = (e_1,\ldots,e_n), e' = (e'_1,\ldots,e'_n)$. Then $disc(e') = \det[T_{L/K}(e'_i e'_j)]_{ij} =$

$\det[T_{L/K}((eP)_i(eP)_j)]_{ij} = \det[P^t(T_{L/K}(e_ie_j)]_{ij})P] = (\det P^t)disc(e)(\det P) = (\det P)^2 disc(e)$.
$\to A = \mathbb{Z}$, B is always a free $\mathbb{Z}$-lattice and the only units in $\mathbb{Z}$ is $u = \pm 1$, so $u^2 = 1$.

Definition 2.2.19. *Assume AKLB, let M be an A-lattice in L and let $n \equiv [L:K]$. Then discriminant $D(M)$ of M is the A-module generated by $\{disc(x_1,\ldots,x_n), x_i \in M\}$.*

Lemma 2.2.14. *Assume AKLB and let $M' \subseteq M$ be free A-lattices in L. The discriminants $D(M') \subseteq D(M)$ are nonzero principal fractional ideals. If $D(M') = D(M)$, $M' = M$.*

pf. Let $e \equiv (e_1,\ldots,e_n)$ be an A-basis for M. Then $disc(e) \in D(M)$ and for any row vector $x \equiv (x_1,\ldots,x_n)$ with entries in M there is a matrix $P \in A^{n\times n}$ for which $x = eP$ and we then have $disc(x) = (\det P)^2 disc(e)$ as above. Then $D(M) = (disc(e))$ is principal, and it is nonzero because e is a basis for L and the trace pairing is nondegenerate. If we let $e' \equiv (e'_1,\ldots,e'_n)$ be an A-basis for M', $D(M') = (disc(e'))$ is also nonzero and principal. $M' \subseteq M$ implies $e' = eP$ for some matrix $P \in A^{n\times n}$ and $disc(e') = (\det P)^2 disc(e)$. If $D(M') = D(M), \det P$ must be a unit, i.e. P is invertible and $e = e'P^{-1}$, which implies $M \subseteq M' \to M = M'$.

Proposition 2.2.11. *Assume AKLB and let M be an A-lattice in L. Then $D(M) \in \mathscr{I}_A$.*

pf. The A-module $D(M) \subseteq K$ is nonzero because M contains a K-basis $e = (e_1,\ldots,e_n)$ for L and $disc(e) \neq 0$ because the trace pairing is nondegenerate. Let N be the free A-lattice in L generated by e and then pick a nonzero $a \in A$ s.t. $M \subseteq a^{-1}N$. Then $D(M) \subseteq D(a^{-1}N)$, which is a principal fractional ideal of A, hence a noetherian A-module. Thus its submodule $D(M)$ is finitely generated, and therefore the proposition follows.

Definition 2.2.20. *Assume AKLB. The discriminant $D_{L/K}$ of L/K is the discriminant of B as an A-module:*

$$D_{L/K} \equiv D_{B/A} \equiv D(b) \in \mathscr{I}_A$$

which is an A-ideal, since $disc(x_1,\ldots,x_n) = \det[T_{B/A}(x_ix_j)]_{ij} \in A, \forall x_i \in B$.

e.g. $A = \mathbb{Z}, K = \mathbb{Q}, L = \mathbb{Q}(i), B = \mathbb{Z} = [i]$ where B is a free A-lattice with basis $(1,i)$. Then

$$D_{L/K} = \begin{array}{c} disc(1,i) = \det\begin{bmatrix} T_{L/K}(1\cdot 1) & T_{L/K}(1\cdot i) \\ T_{L/K}(i\cdot 1) & T_{L/K}(i\cdot i)\end{bmatrix} = -4 \\ disc(1,i) = (\det\begin{bmatrix} 1 & 1 \\ i & -i\end{bmatrix})^2 = -4 \\ B = \mathbb{Z}[i] = \mathbb{Z}[x]/(x^2+1) \to disc(x^2+1) = -4 \end{array}.$$

Proposition 2.2.12. *Assume AKLB and let S be a multiplicative subset of A. Then $S^{-1}D_{B/A} = D_{S^{-1}B/S^{-1}A}$.*

pf. Let $x = s^{-1}disc(e_1,\dots,e_n) \in S^{-1}D_{B/A}$ for some $s \in S, e_i \in B$. Then $x = s^{2n-1}disc(s^{-1}e_1,\dots,s^{-1}e_n)$ lies in $D_{S^{-1}B/S^{-1}A}$.
For any $e_1,\dots,e_n \in S^{-1}B$, $\exists s \in S \subseteq A$ s.t. $se_i \in B\ \forall i$. Then $disc(e_1,\dots,e_n) = s^{-2n}disc(se_1,\dots,se_n) \in S^{-1}D_{B/A}$.

Proposition 2.2.13. *Assuem AKLB and let $\mathbf{p}$ be a prime of A. Then $D_{B/A}\hat{A}_{\mathbf{p}} = \Pi_{\mathbf{q}|\mathbf{p}}D_{\hat{B}_{\mathbf{q}}/\hat{A}_{\mathbf{p}}}$ where $\hat{A}_{\mathbf{p}}, \hat{B}_{\mathbf{q}}$ are the completions of A, B at $\mathbf{p}, \mathbf{q}$ respectively.*

pf. After localising at $\mathbf{p}$, assume A is a DVR, B is a free A-module of rank n. Then we have a trace pairing on the finite etale $K_{\mathbf{p}}$-algebra $\hat{L} \equiv L \otimes K_{\mathbf{p}}$, $\hat{B} \equiv B \otimes \hat{A}_{\mathbf{p}} \cong \oplus_{\mathbf{q}|\mathbf{p}}\hat{B}_{\mathbf{q}}$ is an $\hat{A}_{\mathbf{p}}$-lattice in the $K_{\mathbf{p}}$-vector space $\hat{L}$ that is a direct sum of free $\hat{A}_{\mathbf{p}}$-module, and thus a free $\hat{A}_{\mathbf{p}}$-module of rank $n = \sum e_{\mathbf{q}}f_{\mathbf{q}}$. We can choose $\hat{A}_{\mathbf{p}}$ basis for each $\hat{B}_{\mathbf{q}}$ using elements in B by the weak approximation and the fact that B is dense in $\hat{B}_{\mathbf{q}}$. Then we can construct an $\hat{A}_{\mathbf{p}}$-basis $\hat{e}$ for the direct sum $\oplus\hat{B}_{\mathbf{q}} \cong \hat{B}$ whose elements have nonzero projections to exactly one of the $\hat{B}_{\mathbf{q}}$, along with a corresponding A-basis e for B obtained from $\hat{e}$ as the union of these projections. The matrix $[T_{\hat{L}/K_{\mathbf{p}}}(\hat{e}_i\hat{e}_j)]$ is a block diagonal; each block corresponds to a matrix whose determinant is the discriminant of the $\hat{A}_{\mathbf{p}}$-basis we chose for each $\hat{B}_{\mathbf{q}}$. Thus $D_{\hat{B}/\hat{A}_{\mathbf{p}}} = \Pi_{\mathbf{q}|\mathbf{p}}D_{\hat{B}_{\mathbf{q}}/\hat{A}_{\mathbf{p}}}$. Then $disc_{B/A}(e_1,\dots,e_n) = disc_{(B\otimes A_{\mathbf{p}})/\hat{A}_{\mathbf{p}}}(e_1 \otimes 1,\dots,e_n \otimes 1)$ generated $D_{B/A}$ as an A-ideal and also generate $D_{\hat{B}/\hat{A}_{\mathbf{p}}}$ as an $\hat{A}_{\mathbf{p}}$-ideal. Thus $D_{B/A}\hat{A}_{\mathbf{p}} = D_{\hat{B}/\hat{A}_{\mathbf{p}}} = \Pi_{\mathbf{q}|\mathbf{p}}D_{\hat{B}_{\mathbf{q}}/\hat{A}_{\mathbf{p}}}$.

Theorem 2.2.13. *Assume AKLB. Then $D_{B/A} = N_{B/A}(\mathscr{D}_{B/A})$.*

pf. The different and discriminant are both compatible with localisation by the previous propositions, and the A-module $D_{B/A}, N_{B/A}(\mathscr{D}_{B/A})$ of A are both determined by the intersection of their localisations at the maximal ideals, so we can reduce to the case where A is its localisation at a prime of A. Then A is a DVR, B is a free A-lattice in L; fix an A-basis $(e_1,\dots,e_n)$ for B. The dual A-lattice $B^* = \{x \in L : T_{L/K}(xb) \in A,\ \forall b \in B\} \in \mathscr{I}_B$ is also a free A-lattice in L with basis $(e_1^*,\dots,e_n^*)$ uniquely determined by $T_{L/K}(e_i^*e_j) = \delta_{ij}$. If we write $e_i = \sum_j a_{ij}e_j^*$ in terms of the K-basis $(e_1^*,\dots,e_n^*)$ for L, then $T_{L/K}(e_ie_j) = T_{L/K}(\sum_k a_{il}e_k^*e_j) = \sum_k a_{ik}T_{L/K}(e_k^*e_j) = \sum a_{ik}\delta_{kj} = a_{ij}$. Thus $P \equiv [T_{L/K}(e_ie_j)]_{ij}$ is the change of basis matrix from $e^* \equiv (e_1^*,\dots,e_n^*)$ to $e \equiv (e_1,\dots,e_n)$. If we let ϕ denote the K-linear transformation with matrix P. it is an isomorphism with matrix P, then ϕ is an isomorphism of free A-modules and $D_{B/A}(\det[T_{L/K}(e_ie_j)]_{ij}) = (\det\phi) = [B^* : B]_A$ which is a module index. Then $D_{B/A} = [B^* : B]_A = N_{B/A}((B : B^*)) = N_{B/A}((B^*)^{-1}) = N_{B/A}(\mathscr{D}_{B/A})$.

Under AKLB assumption, if $\mathbf{p}$ is a prime of A, $\mathbf{p}B = \mathbf{q}_1^{e_1} \dots \mathbf{q}_r^{e_r}$ is the factorisation of $\mathbf{p}B$. By the CRT, $B/\mathbf{p}B \cong B/\mathbf{q}_1^{e_1} \times \dots \times B/\mathbf{q}_r^{e_r}$, which is a commutative $A/\mathbf{p}$-algebra of dimension $\sum e_i f_i$ where $f_i = [B/\mathbf{q}_i : A/\mathbf{p}]$ is the residue degree. It is a product of fields iff $e_i = 1, \forall i$ and it is a finite etale algebra iff it is a product of fields that are separable extensions of $A/\mathbf{p}$.

Lemma 2.2.15. *Let k be a field and let R be a commutative k-algebra with k-basis $r_1, \dots, r_n$. Then R is a finite etale k-algebra iff $disc(r_1, \dots, r_n) \neq 0$.*

pf. R is a finite etale k-algebra iff the trace pairing on R is a prefect pairing, which is equivalent to being nondegenerate, given k a field. IF the trace pairing is degenerate, then for some nonzero $x \in R, T_{R/k}(xy) = 0, \forall y \in R$. If we write $x = \sum x_i r_i$, $\forall x_i \in k$, $T_{R/k}(xr_j) = \sum_i x_i T_{R/k}(r_i r_j) = 0$,
$\forall r_j$ and this implies that the column of the matrix $[T_{R/k}(r_i r_j)]_{ij}$ are linearly dependent and $disc(r_1, \dots, d_n) = \det[T_{R/k}(r_i r_j)]_{ij} = 0$.
Conversely, if $disc(r_1, \dots, r_n) = 0$, the columns of $[T_{R/k}(r_i r_j)]_{ij}$ are linearly dependent and for some $x_i \in k$, not identically zero, $\sum_i x_i T_{R/k}(r_i r_j) = 0 \ \forall j$. For $x \equiv \sum x_i r_i$, and for any $y = \sum y_j r_j \imath R$, $T_{R/k}(xy) = \sum_j y_j \sum_i x_i T_{R/k}(r_i r_j) = 0$ and the trace pairing is nondegenerate.

Theorem 2.2.14. *Assume AKLB, let $\mathbf{q}$ be a prime of B lying above a prime $\mathbf{p}$ of A s.t. $B/\mathbf{q}$ is a separable extension of $A/\mathbf{p}$. The extension L/K is unramified at $\mathbf{q}$ iff $\mathbf{q}$ does not divide $\mathscr{D}_{B/A}$, and it is unramified at $\mathbf{p}$ iff $\mathbf{p}$ does not divide $D_{B/A}$.*

pf. By the previous proposition, the different is compatible with completion, so it reduces to the case where A, B are complete DVRs. Then $[L : K] = e_{\mathbf{q}} f_{\mathbf{q}}$, where $\mathbf{p}B = \mathbf{q}^{e_{\mathbf{q}}}$. Since B is a DVR with maximal $\mathbf{q}$, we must have $\mathscr{D}_{B/A} = \mathbf{q}^m, m \geq 0$. By the previous theorem, $D_{B/A} = N_{B/A}(\mathscr{D}_{B/A}) = N_{B/A}(\mathbf{q}^m) = \mathbf{p}^{f_q m}$. Thus $\mathbf{q} | \mathscr{D}_{B/A}$ iff $\mathbf{p} | D_{B/A}$. Since A is a PID, B is a free A-module and we may choose an A-module bases $e_1, \dots, e_n$ for B that is also a K-basis for L. Let $k \equiv A/\mathbf{p}$, and $\bar{e}_i$ be the reduction of e_i to the k-algebra $R \equiv B/\mathbf{p}B$. Then $(\bar{e}_1, \dots, \bar{e}_n)$ is a k-basis for R, and $[R : k] = [B/\mathbf{q}^{e_{\mathbf{q}}} : A/\mathbf{p}] = e_{\mathbf{q}} f_{\mathbf{q}} = [L : K] = n$. Since B has an A-module basis, we may compute its discriminant as $D_{B/A}) = disc(e_1, \dots, e_n))$. Thus $\mathbf{p} | D_{B/A}$ iff $disc(e_1, \dots, e_n) \in \mathbf{p}$, i.e. $disc(\bar{e}_1, \dots, \bar{e}_n) = 0$. By the previous lemma, $disc(\bar{e}_1, \dots, \bar{e}_n) = 0$ iff the k-algebra $B/\mathbf{p}B$ is not finite etale, i.e. iff $\mathbf{p}$ is ramified. Thus $\mathbf{p} | D_{B/A}$ iff $\mathbf{p}$ is ramified. There is only one prime over $\mathbf{q}$ above $\mathbf{p}$, $\mathbf{q} | \mathscr{D}_{B/A}$ iff $\mathbf{q}$ is ramified.

Corollary 2.2.15. *Assume AKLB. Only finitely many primes of A ramify.*

pf. A, B are both Dedekind domains, so the ideals $D_{B/A}, \mathscr{D}_{B/A}$ have unique factorisation into prime ideals, in which only finitely many primes appear.

If $\mathscr{O}$ is an order of the form $A[\alpha]$, where $\alpha \in B$ generates $L = K(\alpha)$ with minimal polynomial $f \in A[x]$, $\mathscr{O}$ is a free A-lattice with basis $1, \alpha, \ldots, \alpha^{n-1}$, where $n = \deg f$ and $D_{\mathscr{O}/A} = (disc(1, \alpha, \ldots, \alpha^{n-1})) = (disc(f))$, which is a principal A-ideal contained in $D_{B/A}$. If B is also a free A-lattice, $D_{\mathscr{O}/A} = (\det P)^2 D_{B/A} = [B : \mathscr{O}]_A^2 D_{B/A}$ where P is the matrix of the A-linear map $\phi : B \to \mathscr{O}$ that sends an A-basis for B to an A-basis for $\mathscr{O}$ and $[B : \mathscr{O}]_A$ is the module index.

Theorem 2.2.15. *Assume AKLB and let $\mathscr{O}$ be an order with integral closure B and conductor $\rfloor$. Then $D_{\mathscr{O}/A} = N_{B/A}(\mathbf{c}) D_{B/A}$.*

pf. **CLAIM : $\mathbf{c} = (B^* : C^*) \equiv \{x \in L : xC^* \subseteq B^*\}$ where $C = A[\alpha]$ is an order.**
pf. Let A-basis for B, C be $(b_1, \ldots, b_n), (c_1, \ldots, c_n)$ s.t. $\exists (b_1^*, \ldots, b_n^*), (c_1^*, \ldots, c_n^*)$ with $b_i^*(b_j) = \delta_{ij} = c_i^*(c_j)$ respectively and since $C \subseteq B$, $B^* \subseteq C^*, c_i = \sum_j l_{ij} b_j$. If $c \in \mathbf{c}$ is a conductor, $cb_i \in C$. Then $(c \sum_i x_i c_i^*, \sum_j y_j b_j) = \sum_{i,j} c x_i y_j c_i^*(b_j)$. Since $c \in \mathbf{c}, \sum_{ij} c c_i^*(b_j)$ is well-defined and therefore it is in $(B^* : C^*)$.
If $\exists c$ s.t. $(c \sum_i x_i c_i^*, \sum_j y_j b_j) = \sum_{i,j} c x_i y_j c_i^*(b_j), \sum_{ij} c c_i^*(b_j)$ is a well-defined pairing. This implies $c \in \mathbf{c}$.
Then if we define $\mathscr{D}_{C/A} \equiv (B : C^*), D_{C/A} \equiv D(C)$,

$$\begin{cases} \mathscr{D}_{C/A} = (B : C^*) = (B^* : C^*)(B : B^*) = \mathbf{c}\mathscr{D}_{B/A}, \\ D_{C/A} = D(C) = N_{B/A}(\mathbf{c}\mathscr{D}_{B/A}) = N_{B/A}(\mathbf{c}) D_{B/A} \end{cases} \to D_{C/A} = N_{B/A}(\mathscr{D}_{C/A}).$$

Theorem 2.2.16 (Dedekind)**.** *Let $K = \mathbb{Q}(\alpha)$ be a number field, let f be the minimal polynomial of α, let p be a prime, and let $\bar{f} = \Pi_{i=1}^r \bar{f}_i^{e_i}$ be the factorisation of $\bar{f} \equiv f$ mod p in $\mathbb{F}_p[x]$. Let f_i be any monic lift of $\bar{f}_i$ to $\mathbb{Z}[x]$, let $u \equiv \Pi_i f_i$, let v be any monic lift of $\bar{f}/\bar{f}$ to $[x]$ and let $w = (uv - f)/p \in \mathbb{Z}[x]$. Then p divides $[\mathscr{O}_K : \mathbb{Z}[\alpha]]$ iff $\bar{u}, \bar{v}, \bar{w}$ have no common factor in $\mathbb{F}_p[x]$.*

Proposition 2.2.14. *Assume AKLB. If $B = A[\alpha]$ for some $\alpha \in L, f \in A[x]$ is the minimal polynomial of α, then $\mathscr{D}_{B/A} = (f'(\alpha))$ is the B-ideal generated by $f'(\alpha)$.*

pf. **CLAIM : $B^* = \{x \in L : T_{L/K}(xb) \in A \ \forall b \in B\}$ is the principal B-ideal $(1/f'(\alpha))$.**

pf. $T_{L/K}(\frac{\alpha^i}{f'(\alpha)}) = \begin{matrix} 0 & 0 \le i \le n-2 \\ 1 & i = n-1 \\ \in A & i \ge n \end{matrix}$ and therefore $(\frac{1}{f'(\alpha)}) \subseteq B^*$.
Assume there exists $a \in B^*$. Then $T_{L/K}(ab) \in A, \forall b \in B$. Since $f'(\alpha) \in B, T_{L/K}(af'(\alpha)) \in A \to a \in (1/f'(\alpha))$.
Thus $\mathscr{D}_{B/A} = (B^*)^{-1} = (f'(\alpha))$

Definition 2.2.21. *Assume AKLB and let $\alpha \in B$ have minimal polynomial $f \in A[x]$. The different of α is defined by $\delta_{B/A}(\alpha) \equiv \begin{matrix} f'(\alpha) & L = K(\alpha) \\ 0 & otherwise \end{matrix}$*

Proposition 2.2.15. *Assume AKLB. Then $\mathscr{D}_{B/A} = (\delta_{B/A}(\alpha) : \alpha \in B)$.*

$\rightarrow D_{B/A} = \{N_{L/K}(\delta_{B/A}(\beta)) : \beta \in B\}$

pf.

Theorem 2.2.17. *Assume AKLB and let* $\mathbf{q}$ *be a prime of* L *lying above* $\mathbf{p} = \mathbf{q} \cap A$ *for which the residue field extension* $(B/\mathbf{q})/(A/\mathbf{p})$ *is separable. Then*

$$e_{\mathbf{q}} - 1 \leq v_{\mathbf{q}}(\mathscr{D}_{B/A}) \leq e_{\mathbf{q}} - 1 + v_{\mathbf{q}}(e_{\mathbf{q}})$$

and the lower bound is an equality iff $\mathbf{q}$ *is tamely ramified.*

pf. $e_{\mathbf{q}} = v_{\mathbf{q}}(\mathscr{D}_{C/A}) = v_{\mathbf{q}}(\mathbf{c}\mathscr{C}_{B/A}) = v_{\mathbf{q}}(\mathbf{c}) + v_{\mathbf{q}}(\mathscr{D}_{B/A}) \leftrightarrow e_{\mathbf{q}} - 1 \leq e_{\mathbf{q}} - v_{\mathbf{q}}(\mathbf{c}) = v_{\mathbf{q}}(\mathscr{D}_{B/A})$, given $0 \leq v_{\mathbf{q}}(\mathbf{c}) \leq 1$. So the first inequality follows. Also $\min v_{\mathbf{q}}(\mathbf{c}) = 1 - v_{\mathbf{q}}(e_{\mathbf{q}})$ so the second inequality and the last statement follows.

Proposition 2.2.16. *Assume AKLB and let* M/L *be a finite separable extension and let* C *be the integral closure of* A *in* M*. Then* $\mathscr{D}_{C/A} = \mathscr{D}_{C/B} \cdot \mathscr{D}_{B/A}$ *and* $D_{C/A} = (D_{B/A})^{[M:L]} N_{B/A}(D_{C/B})$.

Chapter 3

Global Fields

3.1 Global Fields and Primes

3.1.1 Global Fields

Definition 3.1.1. *Let K be a field. A place of K is an equivalence class of nontrivial absolute values on K. There is an one-to-one correspondence between places of K and completions of K, given the completion of K at an absolute value depends only on its equivalence class. M_k denotes the set of places of K, and for each place v, we use $|\cdot|_v$ to denote any representative absolute value and K_v to denote the completion of K w.r.t. $|\cdot|_v$.*

Let K be a global field. Then for any place v of K, the completion K_v is a local field.

- If $K_v \cong \mathbb{R}, v$ is a real place
- If $K_v \cong \mathbb{C}, v$ is a complex place
- If $|\cdot|_v$ is induced by a DVR v_p corresponding to a prime $\mathbf{p}$ of K, then v is a finite place; otherwise v is an infinite place

e.g.

- $M_{\mathbb{Q}}$ consists of finite places p corresponding to $p$adic absolute values $|\cdot|_p$ and a single archimedean infinite place ∞ corresponding to the Euclidean absolute value $|\cdot|_\infty$.
- $M_{\mathbb{F}_q(t)}$ consists of finite places corresponding to irreducible polynomials in $\mathbb{F}_q[t]$ and a single nonarchimedean infinite place ∞ corresponding to the absolute value $|\cdot|_\infty \equiv q^{det(\cdot)}$

Definition 3.1.2. *If L/K is an extension of global fields, for every place w of L, any absolute value $|\cdot|_w$ that represents the equivalence class w restricts to an absolute value on K that represents a place v of K that is independent of the choice of $|\cdot|_w$. This is said that w extends v or w lies above v, denoted $w|v$.*

Theorem 3.1.1. *Let L/K be a finite separable extension of global fields and let v be a place of K. We have an isomorphism of finite etale K_v-algebras*

$$\begin{array}{rcl} L \otimes_K K_v & \to_{\sim} & \prod_{w|v} L_w \\ l \otimes x & \mapsto & (lx, \ldots, lx) \end{array}$$

pf. The separable extension L/K is a finite etale K-algebra, so the base change $L \otimes_K K_v$ is a finite etale K_v-algebra, and is therefore isomorphic to a finite product $\prod_{i \in I} L_i$ of finite separable extensions L_i of K_v, each of which is a local field, given the fact that any finite extension of a local field is a local field. Fix an absolute value $|\cdot|_v$ on K_v representing the place v. Each L_i is a local field extending K_v, and therefore has a unique absolute value $|\cdot|_w$ that restricts to $|\cdot|_v$, by the theorem for v nonarchimedean, and by the fact that $K_v \cong L_w$ or $K_v \cong \mathbb{R} \subseteq \mathbb{C} \cong L_w$ and the Euclidean absolute value on $\mathbb{R}$ is the restrction of the Euclidean absolute value on $\mathbb{C}$, for v archimedean. The map $L \to L \otimes_K K_v \cong \prod_i L_i \to L_i$ allows to veiw L as a subfield of each L_i, so the absolute value $|\cdot|_w$ on L_i restricts to an absolute value on L that uniquely determines a place $w|v$. This defines a map $\phi : \{L_i : i \in I\} \to \{L_w : w|v\}$.

CLAIM :ϕ above is a bijection, satisfying $\phi(L_i) \cong L_i$.

pf. $L \otimes_K K_v \cong \prod_i L_i$ can be viewed as an isomorphism of topological rings, since both sides are finite dimensional vector spaces over the complete field K_v. Then there exists a unique topology induced by the sup norm and this topology agrees with the product topology on $\prod_i L_i$. The image of the canonical embedding $\begin{array}{rcl} L & \to & L \otimes_K K_v \\ l & \mapsto & l \otimes 1 \end{array}$ is dense because $K \subseteq L$ is dense in K_v. The image of L is therefore dense in $\prod_i L_i$ and has dense image under the projections $\prod_i L_i \to L_i$ and $\prod_i L_i \to L_i \times L_j (i \neq j)$. If $\phi(L_i) = L_w, L_i \cong L_w$, since L is dense in the completion field L_i and L_w is the completion of L w.r.t. the restriction of the absolute value on L_i to L, by the universal property of the completions.

If $\phi(L_i) = \phi(L_j) = L_w, i \neq j$, because the image of the diagonal embedding $L \to L_w \times L_w$ is not dense in $L_w \times L_w$ but the image of L is dense in $L_i \times L_j$ we get the contradiction. So injectivity follows.

For each $w|v$, define a continuous homomorphism of finite etale K_v-algebras and topological rings $\begin{array}{rcl} \phi_w : L \otimes_K K_v & \to & L_w \\ l \otimes x & \mapsto & lx \end{array}$. The map ϕ_w is surjective because its image contains L and is complete, and L_w is the completion of L. Thus ϕ_w factors through the projection of $L \otimes_K K_v \cong \prod_i L_i$ onto one of its factors L_i, and induces a homeomorphism from L_i to L_w. So $\phi(L_i) = L_w$ and the map is surjective.

Corollary 3.1.1. *Let L/K be a finite separable extension of global fields, v be a place of K and $f \in K[x]$ be a monic irreducible polynomial s.t. $L \cong K[x]/(f)$. There is an one-to-one correspondence between the irreducible factors of f in $K_v[x]$ and the places of L lying above v. If $f = f_1 \cdots f_r$ is the factorisation of f in $K_v[x]$, then we can order the set $\{w|v\} = \{w_1, \ldots, w_r\}$ so that $L_{w_i} \cong K_v[x]/(f_i)$ for $i \in [1, r]$.*

pf. f_i are distinct since f is separable over K and therefore separable over every extension of K. Then corollary follows.

Definition 3.1.3. *Let X be a locally compact Hausdorff space. The σ-algebra $\sum$ of X is the collection of subsets of X generated by the open and closed sets under countable unions and countable intersections. Its elements are Borel sets or measurable sets. A Borel measure on X is a countably additive function $\mu : \sum \to \mathbb{R}_{\geq 0} \cup \{\infty\}$. A Radon measure on X is a Borel measure on X that additionally satisfies*

1. *$\mu(X) < \infty$ if S is compact*
2. *$\mu(X) = \inf\{\mu(U) : S \subseteq U, U$ is open $\}$*
3. *$\mu(S) = \sup\{\mu(C) : C \subseteq S, C$ is compact $\}$*

for all Borel sets S.

Definition 3.1.4. *A locally compact group is a topological group that is Hausdorff and locally compact. A Haar measure μ on a locally compact group G is a nonzero Radon measure that is translation invariant, meaning $\mu(S) = \mu(x + S), \forall x \in G$ and measurable $S \subseteq G$. A compact group is a locally compact group that is compact.*

Theorem 3.1.2 (Weil)**.** *Every locally compact group G has a Haar measure. If μ, μ' are two Haar measure on G, then there is a positive real number λ s.t. $\mu'(S) = \lambda\mu(S)$ for all measurable sets S.*

e.g. The standard Lebesgue measure on $\mathbb{R}^n$ with $\mu(\prod_i |a_i, b_i|) = \prod_i |b_i - a_i|$ is the unique Haar measure on $\mathbb{R}^n$ for which the unit cube has measure 1.

The additive group of a local field K is a locally compact group since it is a metric space and hence Hausdorff. For compact group G, normalise the Haar measure so that $\mu(G) = 1$, but local fields are never compact and therefore $\mu(K) = \infty$. For a nonarchimedean local fields, the valuation ring $A = B_{\leq 1}(0)$ is a compact group so it is natural to normalise the Haar measure on K so that $\mu(A) = 1$.

Proposition 3.1.1. *Let K is a local fiedl with discrete valuation v, residue field k and absolute value $|\cdot|_v \equiv (\#k)^{-v(\cdot)}$, and let μ be a Haar measure on K. $\forall x \in K$ and measurable $S \subseteq K$, we have $\mu(xS) = |x|_v \mu(S)$. Moreover the absolute value $|\cdot|_v$ is the unique absolute value compatible with the topology on K for which this is true.*

pf. Let A be a valuation ring of K with maximal $\mathbf{p}$. When $x = 0$, the proposition holds trivially. Assume $x \neq 0$, and the map $\phi_x : y \mapsto xy$ is an automorphism of the additive group of K and it follows that the composition $\mu_x = \mu \circ \phi_x$ is a Haar measure on K and hence a multiple of μ i.e. $\mu_x = \lambda_x \mu$ for some $\lambda_x \in \mathbb{R}_{>0}$. Define a function $\begin{array}{rcl} \chi : K^\times & \to & \mathbb{R}_{>0} \\ & \mapsto & \mu_x(A)/\mu(A) = \lambda_x \end{array}$. Then $\mu_x = \chi(x)\mu$ and for all $x, y \in K^\times, \chi(xy) = \frac{\mu_{xy}(A)}{\mu(A)} = \frac{\mu_x(yA)}{\mu(A)} = \frac{\chi(x)\chi(y)\mu(A)}{\mu(A)} = \chi(x)\chi(y)$ which implies χ is multiplicative.
CLAIM : $\chi(x) = |x|_v$ **for all** $x \in K^\times$
pf. Since both are multiplicative, it suffices to consider $x \in A \setminus \{0\}$. For any such x, the ideal xA is equal to $\mathbf{p}^{v(x)}$, since A is a DVR. The residue field $k \equiv A/\mathbf{p}$ is finite, hence A/xA is also finite. Writing A as a finite disjoint union of cosets of xA, we have $\mu(A) = [A : xA]\mu(xA) = (\#k)^{v(x)}\chi(x)\mu(A)$, which implies $\chi(x) = (\#k)^{-v(x)} = |x|_v$. Thus $\mu(xS) = \mu_x(S) = \chi(x)\mu(S) = |x|_v\mu(S), \forall x \in K, S \subseteq K$, measurable.
If $|\cdot|$ is an absolute value on K that induces the same topology as $|\cdot|_v$, then for some $0 < c < 1, |x| = |x|_v^c$ for all $x \in K^\times$. Fix $x \in K^\times$ with $|x|_v \neq 1$. If $|\cdot|$ satisfies $\mu(xS) = |x|\mu(S), \frac{\mu(xA)}{\mu(A)} = |x| = |x|_v^c = (\frac{\mu(xA)}{\mu(A)})^c$ which implies $c = 1, |\cdot| = |\cdot|_v$. Hence it is unique.

Definition 3.1.5. *Let K be a global field. For each place v of K, the normalised absolute value $||\cdot||_v : K_v \to \mathbb{R}_{\geq 0}$ on the completion of K at v is defined by $||x||_v \equiv \frac{\mu(xS)}{\mu(S)}$, where μ is a Haar measure on K_v and $S \subseteq K_v$ is a measurable set with finite nonzero measure.*

Then by the previous theorem, it does not depend on the choice of μ and S.

- If v is nonarchimedean, $||\cdot||_v$ is the absolute value $|\cdot|_v$ defined by the previous proposition
- If v is a real place, $||\cdot||_v$ is the Euclidean absolute value $|\cdot|_\mathbb{R}$
- If v is a complex place, $||\cdot||_v$ is the square of the Euclidean absolute value $|\cdot|_\mathbb{C}$

Lemma 3.1.1. *Let L/K be a finite separable extension of global fields, let v be a place of K and let $w|v$ be a place of L. Then $||x||_w = ||N_{L_w/K_v}(x)||_v$.*

pf. If $[L_w : K_v] = 1$, the lemma trivially holds.
Otherwise, if v is archimedean, $L_w \cong \mathbb{C}$, $K_v \cong \mathbb{R}$, in which case $\forall x \in L_w, ||x||_w = \mu(xS)/\mu(S) = |x|_\mathbb{C}^2 = |x\bar{x}|_\mathbb{R} = |N_{\mathbb{C}/\mathbb{R}(x)}|_\mathbb{R} = ||N_{L_w/K_v}(x)||_v$. If v is nonarchimedean, let π_v, π_w be the uniformisers for the local fields K_v, L_w respectively, and let f be the degree of the corresponding residue field extension k_w/k_v. Wlog, assume $x = \pi_w^{w(x)}$ since $||x||_v = |x|_v$ depends only on $w(x)$. Then $||N_{L_w/K_v}(\pi_w)||_v = ||\pi_v^f||_v = (\#k)^{-f}$, so $||N_{L_w/K_v}(x)||_v = (\#k)^{-fw(x)}$. By the previous proposition $||x||_w = (\#k)^{-w(x)} = (\#k_v)^{-fw(x)} = ||N_{L_w/K_v}(x)||_v$.

Theorem 3.1.3 (Product Formula)**.** *Let L be a global field. $\forall x \in L^\times$, we have $\prod_{v \in M_L} ||x||_v = 1$ where $||\cdot||_v$ denotes the normalised absolute value for each place $v \in M_L$.*

pf. The global field L is a finite separable extension of $K = \mathbb{Q}$ or $K = \mathbb{F}_q(t)$. Let p be a place of K. Then by the theorem, any basis for L as a K-vector space is also a basis for $L \otimes_K K_p \cong \prod_{v|p} K_v$ as a K_v-vector space. Thus $N_{L/K}(x) = N_{(L\otimes_K K_p)/K_p}(x) = \prod_{v|p} N_{L_v/K_p}(x)$. Taking normalised absolute values on both sides yields $||N_{L/K}(x)||_p = \prod_{v|p} ||N_{L_v/K_p}(x)||_p = \prod_{v|p} ||x||_v$. By taking the product of both sides over all places $p \in M_K$, $1 = \prod_{p \in M_K} ||N_{L/K}(x)||_p = \prod_{p \in M_K} \prod_{v|p} ||x||_v = \prod_{v \in M_L} ||x||_v$.

Definition 3.1.6. *A global field is a field K with at least one place whose completion at each of its place $v \in M_K$ is a local field K_v and which has a product formula of the form*

$$\prod_{v \in M_k} ||x||_v = 1$$

where each normalisedabsolute value $||\cdot||_v : K_v \to \mathbb{R}_{\geq 0}$ satisfies $||\cdot||_v = |\cdot|_v^{m_v}$ for some absolute value $|\cdot|_v$ representing v and some fixed $m_v \in \mathbb{R}_{>0}$.

By the above result, we can define a global field K if it has a product formula and the completion K_v of K at each place $v \in M_K$ is a local field.

Definition 3.1.7. *Define an M_K-divisor to be a sequence of positive real numbers $c = (c_v)$ indexed by $v \in M_K$ with all but finitely many $c_v = 1$ s.t. $\forall v \in M_K, \exists x \in K_v^\times$ for which $c_v = ||x||_v$.*

Theorem 3.1.4 (Artin-Whaples)**.** *Every global field is a finite extension of $\mathbb{Q}$ or $\mathbb{F}_q(t)$.*

pf. **CLAIM : Let E/F be a finite extension. Then E is a global field iff F is.**
pf. Given E/F is Galois, if F is a global field, its finite Galois extension has absolute value extending the absolute value induced by the place $v \in M_F \to \Pi_{w \in M_E} ||x||_w = (\Pi_{v \in M_F} ||x||_v)^{m_v} = 1$ so the product formula holds. Also E_w is a finite etale F_v-algebra and therefore it is also a local field. Thus E is a global field. Conversely, if E is a global field, for any place w has $v \in M_F$ and therefore $1 = \Pi_w ||x||_w = (\Pi_v ||x||_v)^{m_v}$ so the product formula holds. Also for any local field L_w it is reduced to K_v which is also a local field since its closed ball is compact. Thus F is also a global field.
If E/F is not Galois, pick an intermediate Galois field extension $E/\bar{F}/F$. Then by the previous statement, we can reduce to the case $E/\bar{F}$, which is totally ramified. By the similar argument, we get the equality.
CLAIM : M_K is infinite but contains only finitely many archimedean places.

pf. Assume the contrary. Then there are finite discrete value induced places and finite archimedean places. Since the product formula holds, $1 = \Pi_v ||x||_v = \Pi_{v|\not\infty}||x||_v \Pi_{v|\infty}||x||_v \leftrightarrow (\Pi_{v|\not\infty}||x||_v)^{-1} = (\Pi \# k_v)^{-v(x)})^{-1} = \Pi_{v|\infty}||x||_v$. Then for $x \in K$ s.t. every discrete valuation induced absolute value $v(x) = 1$. Then $\Pi_{v|\not\infty}\# k_v = \Pi_{v|\infty}||x||_v$. Here LHS is nonarchiedean but RHS is archimedean, so contradiction. Hence M_K is infinite.
Assume the contrary. Set c s.t. $\forall v|\infty, c_v = 1 \to 1 = \Pi_{v|\not\infty}||c||_v \Pi_{v|\infty}||c||_v = \Pi_{v|\not\infty}||c||_v = I_c$ a contradiction since 1 is a unit. Hence the archimedean places are finitely many.
CLAIM : $L(c) = \{x \in K : ||x||_v \leq c_v, \forall v \in M_K\}$ is finite for every M_K-divisor c.
pf. for v, archimedean, $L(c) = ||x||_v \leq c_v$ for $c_v = 1$ for all nonarchimedean places. $\{x \in K_v : ||x||_v \leq c_v\}$ is a compact set and given K dense in K_v, $K \cap \{x \in K_v : ||x||_v \leq c_v\}$ is discrete and compact and therefore finite.
It there is no archimedean place, pick an arbitrary discrete valuation induced place v and for other places, set $c_{v'} = 1$. Then $L(c) = \{x \in K : ||x||_v \leq c_v\}$ which is dense in K_v. $L(\bar{c})$ is a compact ball since K_v is a local field and $L(c)$ is again an intersection of discrete set and compact set and therefore finite.
CLAIM : If M_K contains an Archimedean place, then K is a finite extension of $\mathbb{R}$.
pf. For any $a/b \in \mathbb{Q}$, $a, b \in \mathbb{Z}$ s.t. $(a, b) = 1$. Then $\Pi_v ||a/b||_v = \Pi ||a||_v \Pi_v ||b^{-1}||_v = 1$ hence $a/b \in K$. Pick c s.t. other than place induced by $\mathbb{Q}$, $c_v = 1$. Then $L(c) = \{x \in K : ||x||_v \leq c_v\} \leq \infty$ for all $c_v \in \mathbb{R}$. Thus K is a finite extension of $\mathbb{Q}$.
CLAIM : If M_K does not have an archimedean place, then K is a finite extension of $\mathbb{F}_q(t)$ for some finite field $\mathbb{F}_q$.
pf. Pick c s.t. for all places v, $c_v = 1$. Then $L(c) < \infty$. For $x \in L(c), ||x^2||_v = \# k_v^{-2v(x)} = (\# k_v^{-v(x)})^2 \leq \# k_v^{-v(x)} \leq c_v$ hence $x^2 \in L(c)$. But it is finite and therefore $\exists n \in \mathbb{Z}_{++}$ s.t. $x^n = 1$ and therefore $L(c)$ is a field. Let $k = L(c)$. $\forall t \in K \setminus k$, assume it is not transcendental. Then $\exists f(x) \in k[x]$ s.t. $f(t) = 0 = t^n + a_{n-1}t^{n-1} + \cdots + a_0 \leftrightarrow a_0 = t(t^{n-1} + \cdots + a_1) \in k$ a contradiction. Hence $\forall t \in K \setminus k$, it is transcendental. Also $k(t) \subseteq K$, and K is an extension of $k(t)$. Pick c s.t. for v induced by $(k(t))$, $c_v \neq 1$ and 1 everywhere else. Then $L(c) = \{x \in K : ||x||_v \leq c_v\} < \infty$. For any c_v it is finite and therefore K is a finite extension.

3.1.2 Number Fields

Definition 3.1.8. *A subgroup H of a topological group G is discrete if the subspace topology on H is the discrete topology and cocompact if H is a normal subgroup of G and the quotient G/H is compact.*

Lemma 3.1.2. *A subgroup $G \subseteq V \cong \mathbb{R}^n$ is finitely generated iff it is discrete, in which case $G \cong \mathbb{Z}^m$ for some $m \leq n$, and G is then cocompact iff $m = n$.*

pf, If G is finitely generated, it is a free $\mathbb{Z}$-module, since it lies in the torsion free abelian group V and the rank of G must be equal to the dimension $m \leq n$ of the subspace it spans. A subgroup $G \cong \mathbb{Z}^m$ of $V \cong \mathbb{R}^n$ is discrete.
Let $v_1, \ldots, v_n$ be a basis for the subspace W of V spanned by G and let $\Lambda \cong \mathbb{Z}^m$ be the subgroup of G generated by this basis. The quotient $W/\Lambda \cong \mathbb{R}^m/\mathbb{Z}^m$ is compact, so the image of the discrete group G under the quotient map $\pi : W \to W/\Lambda$ must be finite. Thus we can choose a finite $S \subseteq G$ s.t. $\pi(S) = \pi(G)$ so G is generated by $v_1, \ldots, v_m$ and S.

Definition 3.1.9. *A lattice in $V \cong \mathbb{R}^n$ is a $\mathbb{Z}$-submodule generated by an $\mathbb{R}$-basis, equivalently a discrete cocompact subgroup.*

A discrete subgroup of a Hausforff topological group is always closed and therefore the quotient of a Hausdorff topological group by a normal discrete subgroup is Hausdorff. $\to$The quotient of a Hausdorff topological group by a discrete cocompact subgroup is a compact group.

Any basis $v_1, \ldots, v_n$ for V determines a parallelepipied $F(v_1, \ldots, v_n) \equiv \{\sum_i t_i v_i : \forall t_i \in [0,1)\}$ which can be viewed as the unit cube by fixing an isomorphism $\phi : V \to_\sim \mathbb{R}^n$ that maps $(v_1, \ldots, v_n)$ to the standard basis of unit vectors of $\mathbb{R}^n$. Normalise the Haar measure μ so that $\mu(F(v_1, \ldots, v_n)) = 1$ so that we have $\mu(S) = \mu_{\mathbb{R}^n}(\phi(S))$ for every measurable set $S \subseteq V$, where $\mu_{\mathbb{R}^n}$ is the standard Lebesgue measure on $\mathbb{R}^n$. For any other basis $e_1, \ldots, e_n$ of V, if we let $E = [e_{ij}]$ be the matrix whose jth column expresses $e_j = \sum_i e_{ij} v_i$ in therms of the normalised basis $v_1, \ldots, v_n$, then $\mu(F(e_1, \ldots, e_n)) = |detE| = \sqrt{detE'detE} = \sqrt{det(E'E)} = \sqrt{det(< e_i, e_j >)}$ where $< e_i, e_j >$ is the canonical inner product on $\mathbb{R}^n$.

Proposition 3.1.2. *Let $T : V \to V$ be a linear transformation of $V \cong \mathbb{R}^n$. For any Haar measure μ on V, and every measurable set $S \subseteq V, \mu(T(S)) = |detT|\mu(S)$.*

Definition 3.1.10. *Let Λ be a lattice in $V \cong \mathbb{R}^n$. A fundamental domain for Λ is a measurable set $F \subseteq V$ s.t. $V = \coprod_{\lambda \in \Lambda}(F + \lambda)$, i.e. F is a measurable set of coset representatives for V/Λ.*

Proposition 3.1.3. *Let Λbe a lattice in $V \cong \mathbb{R}^n$ and let μ be a Haar measure on V. Every fundamental domain for Λ has the same measure, and this measure is finite and nonzero.*

pf. Let F, G be two fundamental domains for Λ. By the translation invariance and countable additivity of $\mu, \mu(F) = \mu(F \cap V) = \mu(F \cap \coprod_{\lambda \in \Lambda}(G + \lambda)) = \mu(\coprod_{\lambda \in \Lambda}(F \cap (G + \lambda)) = \sum_{\lambda \in \Lambda} \mu(F \cap (G + \lambda)) = \sum_{\lambda \in \Lambda} \mu((F - \lambda) \cap G) = \sum \mu(G \cap (F + \lambda)) = \mu(G)$. If we fix a $\mathbb{Z}$-basis $e_1, \ldots, e_n$ for Λ, the parallelepiped $F(e_1, \ldots, e_n)$ is a fundamental domain for Λ and its closure is compact, so $\mu(F(e_1, \ldots, e_n))$ is finite and it is nonzero since

there is an isomorphism $V \cong \mathbb{R}^n$ that maps the closure of $F(e_1, \ldots, e_n)$ to the unit cube in $\mathbb{R}^n$ whose Lebesgue measure is nonzero.

Definition 3.1.11. *Let Λ be a lattice in $V \cong \mathbb{R}^n$ and fix a Haar measure μ on V. Then covolume $covol(\Lambda) \in \mathbb{R}_{>0}$ of Λ is the measure $\mu(F)$ of any fundamental domain F for Λ.*

Proposition 3.1.4. *If $\Lambda' \subseteq \Lambda$ are lattices in $V \cong \mathbb{R}^n$, then $covol(\Lambda') = [\Lambda : \Lambda']covol(\Lambda)$.*

pf. Fix a fundamental domain F for Λ and a set of coset representatives S for Λ/Λ'. Then $F' \equiv \coprod_{\lambda \in S}(F + \lambda)$ is a fundamental domain for Λ' and $\#S = [\Lambda : \Lambda'] = \mu(F')/\mu(F)$ is finite. Then $covol(\Lambda') = \mu(F') = \sum_{\lambda \in S} \mu(F + \lambda) = (\#S)\mu(F) = [\Lambda : \Lambda']covol(\Lambda)$ since every translation $F + \lambda$ is a fundamental domain for Λ.

Definition 3.1.12. *Let S be a subset of a real vector space. The set S is symmetric if it is closed under negation, and convex if $\forall x, y \in S, \{tx + (1-t)y : t \in [0,1]\} \subseteq S$*

Theorem 3.1.5 (Minkowski's Lattice Point Theorem)**.** *Let Λ be a lattice in $V \cong \mathbb{R}^n$ and μ a Haar measure on V. If $S \subseteq V$ is a symmetric convex measurable set that satisfies $\mu(S) > 2^n covol(\Lambda)$, then S contains a nonzero element of Λ.*

pf. **CLAIM : For any measurable $S \subseteq \mathbb{R}^n$ with measure $\mu(S) > 1$, $\exists$ distinct $s, t \in S$ s.t. $s - t \in \mathbb{Z}^n$.**
pf. $\mu(S) = \pi_{i=1}^n |a_i - b_i| > 1 \leftrightarrow \exists i \in [1, n]$ s.t. $|a_i - b_i| > 1$. Then we draw a line from a_i to b_i with the coordinate $j \neq i$ fixed. Then on the line $\exists b_i'$ s.t. $|a_i - b_i'| = 1$. Then we set those two points a_i, b_i' as s, t and the claim follows.
Since S is symmetric, we set $\mu(S) > 2$ and apply the above claim $\forall i \in [1, n]$ to. get a_i, b_i', $\forall i$ and the theorem follows.

$\rightarrow$ the theorem is tight in the sense that convexity(consider two circles around zero with $r_1 = 1.9, r_2 = 1$. Then between them area is symmetric and $\mu(S) > 2^2$, but not convex and it does not contain the point in $\mathbb{Z}^2$ and symmetricity and $\mu(S) > 2^n$(consider two circle with the same radius r; $(x-r)^2 + y^2 = r^2, (x+r)^2 + y^2 = r^2$ s.t. $\exists \theta r^2\theta/2\pi - 1/2r(r-1)cos\theta = 1$. Set S' =two circles$\cap((-1,1) \times (-\infty, \infty))$ and set $S = \bar{S}' \setminus \partial(S')$. Then a symmetric and convex with $\mu(S) = 4$, S does not contain a point in $\mathbb{Z}^n$) are all necessary.

Let $K/\mathbb{Q}$ be a number field of degree n with r real places and s complex places. Then $n = r + 2s$ is the total degree. By the base changes of K, $\begin{array}{r} K_{\mathbb{R}} \equiv K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^r \times \mathbb{C}^s \\ K_{\mathbb{C}} \equiv K \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C}^n \end{array}$. where the second isomorphism follows from the fact that $\mathbb{C}$ is separably closed. Note $K_{\mathbb{R}}$ is an $\mathbb{R}$-vector space of dimension n, thus $K_{\mathbb{R}} \cong \mathbb{R}^n$. Then we have a sequence of injective homomorphisms of topological rings

$$\mathcal{O}_K \rightarrow K \rightarrow K_{\mathbb{R}} \rightarrow K_{\mathbb{C}} \tag{3.1}$$

where $\begin{array}{lll} \mathcal{O}_K \to K & & \text{inclusion} \\ K \to K_{\mathbb{R}} = K \otimes_{\mathbb{Q}} \mathbb{R} & & \alpha \mapsto \alpha \otimes 1 \\ K_{\mathbb{R}} \cong \mathbb{R}^r \times \mathbb{C}^s \to \mathbb{C}^r \times \mathbb{C}^{2s} \cong K_{\mathbb{C}} & & (x,z) \mapsto (x,z,\bar{z}) \end{array}$. Each $\mathbb{C}$ in $\mathbb{C}^s$ arises as $\mathbb{R}[\alpha] = \mathbb{R}[x].(f) \cong \mathbb{C}$ for some monic irreducible polynomial $f \in \mathbb{R}[x]$ with degree 2, but when we base-change to $\mathbb{C}$, the field $\mathbb{R}[\alpha]$ splits into the etale algebra $\mathbb{C}[x]/(x-\alpha) \times \mathbb{C}[x]/(x-\bar{\alpha}) \cong \mathbb{C} \times \mathbb{C}$. So the composition $K \to K_{\mathbb{R}} \to K_{\mathbb{C}}$ is given by the map $x \mapsto (\sigma_1(x), \ldots, \sigma_n(x))$ where $Hom_{\mathbb{Q}}(K\mathbb{C}) = \{\sigma_1, \ldots, \sigma_n\}$. If we put $K = \mathbb{Q}(\alpha) \equiv K[x]/(f)$ and let $\alpha_1, \ldots, \alpha_n \in \mathbb{C}$ be the roots of f in $\mathbb{C}$, each σ_i is the $\mathbb{Q}$-algebra homomorphism $\begin{array}{ccc} K & \to & \mathbb{C} \\ \alpha & \mapsto & \alpha_i \end{array}$. The ring of integers $\mathcal{O}_K$ is a lattice in the real vector space $K_{\mathbb{R}} \cong \mathbb{R}^n$, which inherits an inner product from the canonical Hermitian inner product on $K_{\mathbb{C}} \cong \mathbb{C}^n$ defined by $< z, z' > \equiv \sum z_i \bar{z}'_i \in \mathbb{C}$. For elements $x, y \in K \to K_{\mathbb{R}} \to K_{\mathbb{C}}$, the Hermitian inner product can be computed as $< x, y > \equiv \sum_{\sigma \in Hom_{\mathbb{Q}}(K,\mathbb{C})} \sigma(x)\bar{\sigma}(y) \in \mathbb{R}$, which is a real number because the non-real embeddings in $Hom_{\mathbb{Q}}(K, \mathbb{C})$ come in complex conjugate pairs. Also the inner product defined above agrees with the restriction of the Hermitian inner product on $K_{\mathbb{R}} \to K_{\mathbb{C}}$.

If $z \in K_{\mathbb{C}} \cong \mathbb{C}^n$ is written as vectors (z_σ) indexed by the set $\sigma \in Hom_{\mathbb{Q}}(K, \mathbb{C}), K_{\mathbb{R}}$ as the image in $K_{\mathbb{C}}$ is the set $K_{\mathbb{R}} = \{z \in K_{\mathbb{C}} : \bar{z}_\sigma = z_{\bar{\sigma}}, \forall \sigma \in Hom_{\mathbb{Q}}(K, \mathbb{C})\}$. Then each $x \in K_{\mathbb{R}}$ is uniquely written in the form $(w_1, \ldots, w_r, x_1 + iy_i, x_1 - iy_1, \ldots, x_s + iy_s, x_s - iy_s), w_i, x_j, y_j \in \mathbb{R}$, where each w_i corresponds to a z_σ with $\sigma = \bar{\sigma}$ and each $(x_j - iy_j, x_j + iy_j)$ corresponds to a complex conjugate pair $(z_\sigma, z_{\bar{\sigma}})$ with $\sigma \neq \bar{\sigma}$. Then the canonical inner product on $K_{\mathbb{R}}$ is $< z, z' > = \sum_i w_i w'_i + 2\sum_j (x_j x'_j + y_j y'_j)$. Thus if we take w_i, x_j, y_j as coordinates for $K_{\mathbb{R}} \cong \mathbb{R}^n$, in order to nomralise the Haar measure μ on $K_{\mathbb{R}}$, so that it is consistent with the Lebesgue measure $\mu_{\mathbb{R}^n}$, define $\mu(S) \equiv 2^s \mu_{\mathbb{R}^n}(S)$ for any measurable set $S \subseteq K_{\mathbb{R}}$. Given this normalisation, $\mu(F(e_q, \ldots, e_n)) = \sqrt{|det < \sigma_i, \sigma_j >|}$ with e_i for any $\mathbb{R}$-basis of $K_{\mathbb{R}}$. Using the Hermitian inner product on $K_{\mathbb{R}} \subseteq K_{\mathbb{C}}$, the above equality still holds.

Recall the discriminant of a number field K is $D_K \equiv disc\mathcal{O}_K \equiv disc(e_!, \ldots, e_n) \in \mathbb{Z}$

Proposition 3.1.5. *Let K be a number field. Using the normalised Haar measure on $K_{\mathbb{R}}$ defined previously,*

$$covol(\mathcal{O}_K) = \sqrt{|D_K|}$$

pf. Let $e_1, \ldots, e_n \in \mathcal{O}_K$ be a $\mathbb{Z}$-basis for $\mathcal{O}_K$, let $Hom_{\mathbb{Q}}(K, \mathbb{C}) = \{\sigma_1, \ldots, \sigma_n\}$ and define $A \equiv [\sigma_i(e_j)] \in \mathbb{C}^{n \times n}$. Then $D_K = disc(e_1, \ldots, e_n) = (detA)^2$. Viewing $\mathcal{O}_K \to K_{\mathbb{R}}$ as a lattice in $K_{\mathbb{R}}$ with basis $e_i, covol(\mathcal{O}_K) = \mu(F(e_1, \ldots, e_n)) = \sqrt{|det[< e_i, e_j >]|} = \sqrt{|det[\sum_j \sigma_k(e_i)\bar{\sigma}_k(e_j)|} = \sqrt{|det(A'\bar{A})|} = \sqrt{|det(A)||det(\bar{A})|}$. Since $detA$ is the square root of an integer, $covol(\mathcal{O}_K)^2 = |(detA)^2| = |D_K|$.

For a number field K, we view the absolute norm $\begin{array}{rcl} N:\mathscr{I}_{\mathscr{O}} & \to & \mathscr{I}_{\mathbb{Z}} \\ I & \mapsto & [\mathscr{O}_K:I]_{\mathbb{Z}} \end{array}$ as having image in $\mathbb{Q}_{++}$ by identifying $N(I)\in\mathscr{I}_{\mathbb{Z}}$ with a positive generator for $N(I)$, given $\mathbb{Z}$ is a PID. Also $[\mathscr{O}_K:I]_{\mathbb{Z}}$ is a module index of $\mathbb{Z}$-lattices in the $\mathbb{Q}$-vector space K, and for ideal $I\subseteq\mathscr{O}_K$, it is the positive integer $[\mathscr{O}_K:I]_{\mathbb{Z}}=[\mathscr{O}_K:I]$. When $I=(a), N(x)\equiv N((a))=|N_{K/\mathbb{Q}}(a)|$.

Corollary 3.1.2. *Let K be a number field and let I be a nonzero fractional ideal of $\mathscr{O}_K$. Then $covol(I)=N(I)\sqrt{|D_K|}$.*

pf. Let $n=[K:\mathbb{Q}]$. Since $covol(bI)=b^n covol(I)$ and $N(bI)=b^nN(I)$ for any $b\in\mathbb{Z}_{++}$, wlog we may assume $I\subseteq\mathscr{O}_K$. Then $covol(I)=[\mathscr{O}_K:I]covol(\mathscr{O}_K)=N(I)covol(\mathscr{O}_K)=N(I)\sqrt{|D_K|}$.

Lemma 3.1.3. *Let K be a number field of degree n with r real and s complex places. For each $t\in\mathbb{R}_{++}$, the measure of the convex symmetric set $S_t\equiv\{(z_\sigma)\in K_{\mathbb{R}}:\sum|z_\sigma|\le t\}\subseteq K_{\mathbb{R}}$ w.r.t. the normalised Haar measure μ on $K_{\mathbb{R}}$ is $\mu(S_t)=2^r\pi^s\frac{t^n}{n!}$.*

pf. Write each $z=(z_\sigma)\in K_{\mathbb{R}}$ in the form $(w_1,\dots,w_r,x_1+iy_1,\dots,x_s-iy_s)$ with $w_i,x_j,y_j\in\mathbb{R}$. Then $\sum_\sigma|z_\sigma|\le t$ iff $\sum_i|w_i|+\sum_j 2\sqrt{|x_j|^2+|y_j|^2}\le t$. Compute the volume of this region in $\mathbb{R}^n$ by relating it to the volume of the simplex $U_t\equiv\{(u_1,\dots,u_n)\in\mathbb{R}^n_+:\sum u_i\le t\}\subseteq\mathbb{R}^n$, which is $\mu_{\mathbb{R}^n}t^n/n!$. If we view all w_i,x_j,y_j as fixed except the last pair (x_s,y_s), the (x_s,y_s) ranges over a disk of some radius $d\in[0,t/2]$ determined by the previous inequality. If we replace (x_s,y_s) with (u_{n-1},u_n) ranging over the triangular region bounded by $u_{n-1}+u_n\le 2d, u_{n-1},u_n\in\mathbb{R}_+$, we need to incorporate a factor of $\pi/2$ to account for the difference between $(2d)^2/2=2d^2$ and πd^2. Repeat this s times. Similarly, if we hold everything but w_r fixed, w_r ranges over $[-d,d]$ and we can replace it with $\exists d\in[0,t]$ wirh u_r ranging over $[0,d]$ by incorporating a factor of 2 to account for the change of variables. Repeat this r times. Then $\mu(S_t)=2^s\mu_{\mathbb{R}^n}(S_t)=2^s(\frac{\pi}{2})^s2^r\mu_{\mathbb{R}^n}(U)=2^r\pi^s\frac{t^n}{n!}$.

Theorem 3.1.6 (Minkowski Bound)**.** *Let K be a number field of degree n with s complex places. Define the Minkowski constant m_K for K as the positive real number $m_K\equiv\frac{n!}{n^n}(\frac{4}{\pi})^s\sqrt{|D_K|}$. For every nonzero fractional ideal I of $\mathscr{O}_K$, there is a nonzero $a\in I$ for which $N(a)\le m_KN(I)$.*

pf. Let I be a nonzero fractional ideal of $\mathscr{O}_K$. By the Minkowski lattice point theorem, if we choose t so that $\mu(S_t)>2^n covol(I)$, then S_t will contain a nonzero $a\in I$. By the previous lemma, and previous corollary, it suffices to choose t so that $(\frac{t}{n})^n=\frac{n!\mu(S_t)}{n^n2^r\pi^s}>\frac{n!2^n}{n^n2^r\pi^s}covol(I)=\frac{n!}{n^n}(\frac{4}{\pi})^s\sqrt{|D_K|}N(I)=m_KN(I)$. Pick t so that $(\frac{t}{n})^n>m_KN(I)$. Then S_t contains $a\in I$ with $\sum_\sigma|\sigma(a)|\le t$. By the geometric-arithmetic inequality, $N(a)=(N(a)^{1/n})^n=(\pi_\sigma|\sigma(a)|^{1/n})^n\le(\frac{q}{n}\sum_\sigma|\sigma(a)|)^n\le(\frac{t}{n})^n$. By taking $(\frac{t}{n})^n\to M_KN(I)$, $N(a)\le m_KN(I)$.

$cl\mathcal{O}_K$ is the quotient of the ideal group $\mathcal{I}_K$ of $\mathcal{O}_K$ by its subgroup of principal fractional ideals.

Theorem 3.1.7. *Let K be a number field. Every ideal class in $cl\mathcal{O}_K$ contains an ideal $I \subseteq \mathcal{O}_K$ of absolute norm $N(I) \leq m_K$, where m_K is the Minkowski constant for K.*

pf. Let $[J]$ be an ideal class of $\mathcal{O}_K$ represented by the nonzero fractional idea J. By the Minkowski bound, the fractional ideal J^{-1} contains a nonzero element a for which $N(a) \leq m_K N(J^{-1}) = m_K N(J)^{-1}$. Thus $N(aJ) = N(a)N(J) \leq m_K$. We have $a \in J^{-1} \to aJ \subseteq J^{-1}J = \mathcal{O}_K$ so $I = aJ$ is an $\mathcal{O}_K$-ideal in the ideal class $[J]$ with $N(I) \leq m_K$.

Lemma 3.1.4. *Let K be a number field of degree n and let $M \in \mathbb{R}_{++}$. The number of $\mathcal{O}_K$-ideals of norm $N(I) \leq M$ is bounded by $(n <)^{\log_2 M}$.*

pf. Let I be an ideal of absolute norm $N(I) \leq M$, and let $I = \mathbf{p_1} \dots \mathbf{p_k}$ be its factorisation into prime ideals, Then $N(\mathbf{p}_i) \geq 2$ for each $\mathbf{p}_i$, so $k \leq \log_2 M$. There are less than M primes $p \leq M$ and at most n primes $\mathbf{p}$ of $\mathcal{O}_K$ above each $p \leq M$. Thus there are less than $(nM)^{\log_2 M}$ $\mathcal{O}_K$-ideals with norm $N(I) \leq M$.

Corollary 3.1.3. *Let K be a number field. The ideal class group of $\mathcal{O}_K$ is finite.*

pf. By the previous theorem, each ideal class is represented by an ideal of norm at most m_K, and by the previous lemma, the number of such ideals is finite.

Theorem 3.1.8. *Let A be the ring $\mathbb{Z}$ or $\mathbb{F}_q[t]$ and let B the integral closure of A in a finite separable extension of its fraction field. The ideal class group of B is finite.*

pf.
$\to K = \mathbb{Z}(\sqrt{d})$ has class number $h_K \equiv \#cl\mathcal{O}_K \to \infty$ as $d \to \infty$. This implies there are only a finite number of imaginary quadratic fields with any particular class number.

Corollary 3.1.4. *Let K be a number field of degree n with s complex places. Then*

$$|D_K| \geq (\frac{n^n}{n!})^2 (\frac{\pi}{4})^{2s} > \frac{1}{e^2 n} (\frac{\pi e^2}{4})^n$$

pf. If I is an deal and $a \in I, a \neq 0$, $N(a) \geq N(I)$ so by the previous theorem $m_K = \frac{n!}{n^n}(\frac{4}{\pi})^2 \sqrt{|D_K|} \geq 1$ and the first inequality follows.
By the Stirling's approximation $n! \leq e\sqrt{n}(\frac{n}{e})^n$. Also $2s \leq n$, the second inequality follows.

Corollary 3.1.5. *If K is a number field other than $\mathbb{Q}$, then $|D_K| > 1$, equivalently there are no nontrivial unramified extensions of $\mathbb{Q}$.*

A commutative ring R with finite quotients by all nonzero ideals is a finite quotient domain. Assume R is not a field. For such G, define the absolute ideal norm $N_R \equiv \#R/I$ for each nonzero R-ideal I, let $N_R((0)) =\equiv 0$, and put $N_R(r) \equiv N_R((r)), \forall r \in R$. Assume $AKLB$. IF A is a finite quotient domain, then so is B and we have the identity $N_B(\alpha) = N_A(N_{L/K}(\alpha))$ for all $\alpha \in B$.
pf.

Definition 3.1.13. *A PID is basic if it is a finite quotient domain and $\exists c_1, c_2 \in \mathbb{Z}_{++}$ s.t.*

$$\begin{array}{c} \#\{x \in A : N_A(x) \leq c_1 m\} \geq m, \forall m \in \mathbb{Z} \\ N_A(x+y) \leq c_2(N_A(x) + N_A(y)), \forall x, y \in A \end{array}.$$

CLAIM : $\mathbb{Z}$ is a basic PID but $\mathbb{Z}[\sqrt{5}]$ is not. Also $\mathbb{F}_q[t]$ is a basic PID.
pf. $\#\mathbb{Z}/(x) = x \to \forall c > 1, \#\{x \in \mathbb{Z} :, N_{\mathbb{Z}}(x) \leq cm\} \geq m$. Also $\#\mathbb{Z}/(x+y) = x+y, \mathbb{Z}/(x) = x, \mathbb{Z}/(y) = y \to \forall c \geq 1, N(x+y) \leq c(N(x) + N(y))$. Hence $\mathbb{Z}$ is a basic PID.
Let f be the monic irreducible polynomial in $\mathbb{F}_q[t]$. $\#\mathbb{F}_q[t]/(f) = \begin{array}{cc} \alpha \leq q & \deg f = 1 \\ n \leq q^2 & \deg f = 2 \end{array}$ and so on. This implies $\#\{N_{\mathbb{F}_q[t]}(f) \leq cm\} \geq m$ where $c = q^n < \infty$. Also $N(f+g) = \#\mathbb{F}_q[t]/(f+g) \leq \#\mathbb{F}_q[t]/(f) + \#\mathbb{F}_q[t]/(g) \to \forall c \geq 1$, inequality holds and therefore it is a basic PID.

Definition 3.1.14. *A global Dedekind domain over A is a Dedekind domain that is free of finite rank as a module over a subring A that is a basic PID.*

CLAIM : Given AKLB, if A is a basic PID, then B is a global Dedekind domain.
pf. Given AKLB, B is a Dedekind domain with finite rank. $\forall a \in A, \forall \phi \in Aut(B).\phi(ab) = a\phi(b)$ implies it is an A-module. Hence the claim follows.

CLAIM : Let B be a global Dedekind domain over a basic PID A and let e_i be an A-basis for B. Then there exists a homogeneous $f \in A[x_1, \ldots, x_n]$ of degree n s.t. $\forall \alpha \in B, N_{L/K}(\alpha) = f(a_1, \ldots, a_n)$, where $\alpha = \sum a_i e_i$ with $a_i \in A$. Also $\exists c \in \mathbb{Z}_{++}$ s.t.
$N_B(\alpha) \leq c \max(N_A(a_1), \ldots, N_A(a_n))^n$ **for all** $\alpha = \sum a_i e_i \in B$.
pf. The determinant for the matrix of base of change gives such f, which is homogeneous of degree n. Then $N_{L/K}(\alpha) = \det(L \to_{\times \alpha} L) = f(a_1, \ldots, a_n) \to N_B(\alpha) = \frac{B}{(\alpha)} = \Pi_{i=1}^n A/(a_i) \leq c \max(N_A(a_1), \ldots, N_A(a_n))^n$ with $c \geq 1$.

CLAIM : Let B be a global Dedekind domain. Then there is a real number m s.t. $\forall B$-ideal I, nonzero, there is a nonzero $\alpha \in I$ for which $N_B(\alpha) \leq m N_B(I)$.
pf. $\mathcal{O}_K \subseteq B$ and by the Minkowski bound, for every nonzero fractional ideal I of

$\mathcal{O}_K$, $\exists a \in I$ for which $N(a) \leq m_L N(I)$. Also $(disc(f')) = D_{\mathcal{O}_K/A} = N_{B/A}(\mathbf{c})D_{B/A} = (\mathbf{c})(disc(f))$. For any B-ideal I containing $\mathbf{c}$,

Theorem 3.1.9. *Let A be the ring $\mathbb{Z}$ or $\mathbb{F}_q[t]$ and let B be the integral closure of A in a finite separable extension of its fraction field. The ideal class group of B is finite.*

pf. Since $\mathbb{Z}, \mathbb{F}_q[t]$ are basic PID, B is a global Dedekind domain.
$\rightarrow K = \mathbb{Z}(\sqrt{d})$ has class number $h_K \equiv \#cl\mathcal{O}_K \rightarrow \infty$ as $d \rightarrow \infty$. This implies there are only a finite number of imaginary quadratic fields with any particular class number.

Corollary 3.1.6. *Let K be a number field of degree n with s complex places. Then $|D_K| \geq (\frac{n^n}{n!})^2(\frac{\pi}{4})^{2s} > \frac{1}{e^2 n}(\frac{\pi e^2}{4})^n$.*

pf. If I is an ideal and $a \in I$ is nonzero, $N(a) \geq N(1)$ so $m_K = \frac{n!}{n^n}(\frac{4}{\pi})^s\sqrt{|D_K|} \geq 1$. By Stirling's approximation, $n! \leq e\sqrt{n}(\frac{n}{e})^n$ and $2s \leq n$ gives the second inequality.

Corollary 3.1.7. *If K is a number field other than $\mathbb{Q}$, then $|D_K| > 1$, equivalently there are no nontrivial unramified extensions of $\mathbb{Q}$.*

Let K be a global field, M_K be the set of places of K. For each place $v \in M_K$, use K_v to denote the completion of K w.r.t. v and we have a normalised absolute value $\begin{array}{rcl} ||\cdot|| : K_v & \rightarrow & \mathbb{R}_+ \\ x & \mapsto & \frac{\mu(xS)}{\mu(S)} \end{array}$, where μ is a Haar measure and S is any measurable set of positive finite measure. Denote k_v to denote the residue fieldm which is a finite field. $\begin{array}{ll} ||x|| : |x|_v = (\#k_v)^{-v(x)} & v \text{ nonarchimedean} \\ |x|_{\mathbb{R}} & K_v \cong \mathbb{R} \\ |x|_{\mathbb{C}}^2 & K_v \cong \mathbb{C} \end{array}$.

Definition 3.1.15. *Let K be a global field. A multiplicative Arakelov divisor is a sequence of positive real numbers $c = (c_v)$ indexed by $v \in M_K$ with all but finitely many $c_v = 1$ and $c_v \in ||K_v^\times|| \equiv \{||x||_v : x \in K_v^\times\}$. The set of Arakelov divisors $DivK$ forms an abelian group under pointwise multiplication $(c_v)(d_v) \equiv (c_v d_v)$. The multiplicative group $K^\times$ is canonically embedded in $DivK$ via the map $x \mapsto (||x||_v)$, where it forms the subgroup $PrincK$ of principal Arakelov divisors.*

Definition 3.1.16. *Let K be a global field. The size of an Arakelov divisor c is $||c|| \equiv \Pi_{v \in M_K} c_v \in \mathbb{R}_{++}$. The map $\begin{array}{rcl} DivK & \rightarrow & \mathbb{R}_{++}^\times \\ c & \mapsto & ||c|| \end{array}$ is a group homomorphism whose kernel contains $PrincK$. Corresponding to each Arakelov divisor c is a subset $L(c)$ of K defined by $L(c) \equiv \{x \in K : ||x||_v \leq c_v \ \forall v \in M_K\}$ and a nonzero fractional ideal of $\mathcal{O}_K$ defined by $I_c \equiv \Pi_{v \nmid \infty} \mathbf{q}_v^{v(c)}$ where $\mathbf{q}_v \equiv \{a \in \mathcal{O}_K : v(a) > 0\}$ is the prime ideal corresponding to the discrete valuation v that induces $||\cdot||_v$ and $v(c) \equiv -\log_{\#k_v}(c_v) \in \mathbb{Z}$.*

$\to L(c) \subseteq I_c \subseteq K$ and the map $c \mapsto I_c$ defines a group homomorphism $DivK \to \mathscr{I}_K$. The quotient of $DivK$ by the subgroup $PrincK$ is denoted $PicK$. Then we have a commutative diagram $\begin{array}{ccc} DivK & \longrightarrow & \mathscr{I}_K \\ \downarrow & & \downarrow \\ PicK & \longrightarrow & cl\mathscr{O}_K \end{array}$. The Arakelov divisors of size 1 form a subgroup of $DivK$, denoted Div^0K that contains $PrincK$ and surjects onto $\mathscr{I}_K$ via the map $DivK \to \mathscr{I}_K$. The quotient of Div^0K by $PrincK$ is the Arakelov class group Pic^0K, which also admits the ideal class group $cl\mathscr{O}_K$ as a quotient.

Let K be a number field. Then the absolute norm $N(I)$ of a fractional ideal $\mathscr{O}_K$ is the unique $t \in \mathbb{Q}_{++}$ for which $N_{\mathscr{O}_K/\mathbb{Z}}(I) = (t)$. Then $N(I_c) = \Pi_{v|\not\infty} N(\mathbf{q}_v)^{v(c)} = \Pi_{v|\not\infty}(\#k_v)^{v(c)} = \Pi_{v|\not\infty} c_v^{-1} \to ||c|| = N(I_c)^{-1}\Pi_{v|\infty} c_v$. Define $R_c \equiv \{x \in K_{\mathbb{R}} : |x|_v \leq c_v, \forall v|\infty\}$, which is a compact symmetric subset of the real vector space $K_{\mathbb{R}} \equiv K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^r \times \mathbb{C}^s$, where r is the number of real places of K and s is the number of complex places. If we view I_c,$L(c)$ as subsets of $K_{\mathbb{R}}$ via the canonical embedding $K \to K_{\mathbb{R}}$, $L(c) = I_c \cap R_c$.

Lemma 3.1.5. *Let c be an Arakelov divisor of a global field K. The set $L(c)$ is finite.*

pf. Assume K is a number field. The fractional ideal I_c is a lattice in $K_{\mathbb{R}}$ and is thus a closed discrete subset of $K_{\mathbb{R}}$. In $K_{\mathbb{R}}$ we may view $L(c) = I_c \cap R_c$ as the intersection of a discrete closed set with a compact set, which is a compact discrete set and therefore finite.

Corollary 3.1.8. *Let K be a global field, and let μ_K denote the torsion subgroup of $K^\times$(equivalently, the roots of unity in K). The group μ_K is finite and equal to the kernel of the map $\begin{array}{ccc} K^\times & \to & DivK \\ x & \mapsto & (||x||_v) \end{array}$. It is also the torsion subgroup of $\mathscr{O}_K^\times$.*

pf. Each $\zeta \in \mu_K$ satisfies $\zeta^n = 1$ for some positive integer n. For every place $v \in M_K$, $||\zeta^n||_v = ||\zeta||_v^n = 1$ and therefore $||\zeta||_v = 1$. Thus $\mu_k \subseteq \ker(K^\times \to DivK)$.
Let c be the Arakelov divisor with $c_v = 1$ for all $v \in M_K$. Then $\ker(K^\times \to DivK) \subseteq L(c)$ is a finite subgroup of $K^\times$ and is therefore contained in the torsion subgroup μ_K. Every element of μ_K is an algebraic integer(since they are roots of $x^n - 1$), so $\mu_K \subseteq \mathscr{O}_K^\times$.

$\to 0 \longrightarrow \mu_K \longrightarrow K^\times \longrightarrow DivK \longrightarrow PicK \longrightarrow 0$ an exact sequence of abelian groups.

Proposition 3.1.6. *Let K be a number field with s complex places, define $B_K \equiv (\frac{2}{\pi})^s\sqrt{|D_K|}$. If c is an Arakelov divisor of size $||c|| > B_K$, then $L(c)$ contains an element of $K^\times$.*

pf. Normalise the Haar measure μ on the locally compact group $K_{\mathbb{R}} \cong \mathbb{R}^r \times \mathbb{C}^s \cong \mathbb{R}^n$ so that $\mu(S) = 2^s \mu_{\mathbb{R}^n}(S)$ for measurable $S \subseteq K_{\mathbb{R}}$. For each real place v, the constraint $||x||_v = |x|_{\mathbb{R}} \leq c_v$ contributes a factor of $2c_v$ to $\mu(R_c)$ and for each complex place v the constraint $||x||_v = |x|_{\mathbb{C}}^2 \leq c_v$ contributes a factor of πc_v. Then $\frac{\mu(R_c)}{covol(I_c)} = \frac{2^n \mu_{\mathbb{R}^n}(R_c)}{covol(I_c)} = \frac{2^s(\Pi_{v\ real} 2c_v)(\Pi_{v\ complex} \pi c_v)}{covol(I_c)} = \frac{2^r(2\pi)^s \Pi_{v|\infty} c_v}{\sqrt{|D_K|} N(I_c)} = \frac{2^{r+s}\pi^s}{\sqrt{|D_K|}} ||c|| = \frac{||c||}{B_K} 2^n > 2^n$. Then by the Minkowski lattice point theorem, $L(c) = R_c \cap I_c$, a convex discrete set, contains an element $K^\times$.
$\rightarrow$ the above bound can be an asymptotic one, $\forall c \in DivK, ||c|| \rightarrow \infty$, $\#L(x) = (\frac{2^r(2\pi)^s}{\sqrt{|D_K|}} + o(1))||c||$.

Let K be a number field with ring of integers $\mathcal{O}_K$. The multiplicative group $\mathcal{O}_K^\times$ is the unit group of $\mathcal{O}_K$. As a ring, the finite etale $\mathbb{R}$-algebra $K_{\mathbb{R}} = K \otimes_{\mathbb{Q}} \mathbb{R}$ also has a unit group and we have an isomorphism of topological groups $K_{\mathbb{R}}^\times \cong \Pi_{v|\infty} K_v^\times \cong \Pi_{rea\ v|\infty} \mathbb{R}^\times \Pi_{complex\ v|\infty} \mathbb{C}^\times = (\mathbb{R}^\times)^r \times (\mathbb{C}^\times)^s$. Writing elements of $K_{\mathbb{R}}^\times$ as vectors $x = (x_v)$ indexed by the infinite places v of K, define a surjective homomorphism of locally compact groups $\begin{array}{rcl} Log : K_{\mathbb{R}}^\times & \rightarrow & \mathbb{R}^{r+s} \\ (x_v) & \mapsto & (\log ||x_v||_v) \end{array}$. It is surjective and continuous because each of the maps $x_v \mapsto \log ||x_v||_v$ is, and it is a group homomorphism, because $Log(xy) = (\log ||x_v y_v||_v) = (\log ||x_v||_v + \log ||y_v||_v) = (\log ||x_v||_v) + (\log ||y_v||_v) = Logx + Logy$. There is an one-to-one correspondence between the infinite places of K and the $Gal(\mathbb{C}/\mathbb{R})$-orbits of $Hom_{\mathbb{Q}}(K, \mathbb{C})$. For each $v|\infty$, pick a representative σ_v of its corresponding $Gal(\mathbb{C}/\mathbb{R})$-orbit in $Hom_{\mathbb{Q}}(K, \mathbb{C})$;
$\begin{array}{ll} \sigma_v & v\ real \\ \sigma_v, \bar{\sigma}_v & v\ complex \end{array}$. Then we have $\begin{array}{ll} ||x||_v = |\sigma_v(x)|_{\mathbb{R}} & v|\infty\ real \\ |\sigma_v(x)\bar{\sigma}_v(x)|_{\mathbb{R}} & v|\infty\ complex \end{array}$.

The absolute norm $N : K^\times \rightarrow \mathbb{Q}_{++}^\times$ extends naturally to a continuous homomorphism of locally compact groups $\begin{array}{rcl} N : K_{\mathbb{R}}^\times & \rightarrow & \mathbb{R}_{++}^\times \\ (x_v) & \mapsto & \Pi_{v|\infty} ||x_v||_v \end{array}$, which is compatible with the canonical embedding $K^\times \rightarrow K_{\mathbb{R}}^\times$. Then we have $N(x) = |N_{K/\mathbb{Q}}(x)| = |\Pi_\sigma \sigma(x)|_{\mathbb{R}} = \Pi_{v|\infty} ||x||_v$ and we have a commutative diagram

$$\begin{array}{ccccc} K^\times & \longrightarrow & K_{\mathbb{R}}^\times & \xrightarrow{Log} & \mathbb{R}^{r+s} \\ \downarrow N & & \downarrow N & & \downarrow T \\ \mathbb{Q}_{++}^\times & \longrightarrow & \mathbb{R}_{++}^\times & \xrightarrow{log} & \mathbb{R} \end{array}$$

where $\begin{array}{rcl} T : \mathbb{R}^{r+s} & \rightarrow & \mathbb{R} \\ x & \mapsto & \sum_i x_i \end{array}$. The commutativity can be expressed as $T(Logx) = \log(N(x))$, which holds for all $x \in K^\times, \forall x \in K_{\mathbb{R}}^\times$. The norm of a unit in $\mathcal{O}_K$ must be a unit in $\mathbb{Z}$, hence have absolute value 1. Thus $\mathcal{O}_K^\times$ lies in the kernel of the map $x \mapsto \log N(x)$ and therefore also lies in the kernel of the map $x \mapsto T(Logx)$. Thus $Log(\mathcal{O}_K^\times)$ is a subgroup of the trace zero hyperplane $\mathbb{R}_0^{r+s} \equiv \{x \in \mathbb{R}^{r+s} : T(x) = 0\} \subseteq \mathbb{R}^{r+s}$ a subgroup and an $\mathbb{R}$-vector subspace of dimension $r+s-1$.

Proposition 3.1.7. *Let K be a number field with r real and s complex places and let Λ_K be the image of the unit group $\mathcal{O}_K^\times$ in $\mathbb{R}_0^{r+s}$ under the Log map. The following holds:*

1. *We have a split exact sequence of finitely generated abelian groups*

$$0 \to \mu_K \to \mathcal{O}_K^\times \to \Lambda_L \to 0$$

2. *Λ_K is a lattice in the trace zero hyperplane $\mathbb{R}_0^{r+s}$.*

Here μ_K is not a Haar measure, but the group of roots of unity in K, all of which are clearly torsion elements of $\mathcal{O}_K^\times$.

pf.

1. Let Z be the kernel of $\mathcal{O}_K^\times \to \Lambda_K$. Then $\mu_K \subseteq Z$ since $\Lambda_K \subseteq \mathbb{R}_0^{r+s}$ is torsion free. Let x be the Arakelov divisor with $I_c = \mathcal{O}_K$ and $c_v = 2$ for $v|\infty\}$ so that $L(c) = \{x \in \mathcal{O}_K : ||x||_v \leq 2\ \forall v|\infty\}$. For $x \in \mathcal{O}_K^\times$, $x \in L(c) \leftrightarrow Log(x) \in LogR_c = \{z \in \mathbb{R}^{fr+s} : z_i \leq \log 2\}$. The set on the RHS includes the zero vector, thus $Z \subseteq L(c)$, which is a finite set by the previous lemma. As a finite subgroup of $\mathcal{O}_K^\times$, $Z \subseteq \mu_K$ thus the sequence is exact at $\mathcal{O}_K^{times}$. By the definition Log map is surjective and the exactness follows.
$\Lambda_K \cap Log(R_c) = Log(\mathcal{O}_K^\times \cap L(c)) < \infty$ since $L(c) < \infty$. Thus 0 is an isolated point of Λ_K in $\mathbb{R}^{r+s}$ and in $\mathbb{R}_0^{r+s}$, so λ_K is a discrete subgroup of $\mathbb{R}_0^{r+s} \cong \mathbb{R}^{r+s-1}$ and must be finitely generated. It follows that $\mathcal{O}_K^\times$ is finitely generated, since it lies in a s.e.s. whose left and right terms are finitely generated. By the structure theorem for finitely generated abelian groups, the sequence must split, since μ_K is the torsion subgroup of $\mathcal{O}_K^\times$.

2. By the previous statement, we know Λ_K is a finitely generated $\mathbb{R}_0^{r+s}$-module. So it is a lattice if it spans $\mathbb{R}_0^{r+s}$. Let V be the subspace of $\mathbb{R}_0^{r+s}$ spannced by Λ_K and suppose the contrary; $\dim V < \dim \mathbb{R}_0^{r+s}$. The orthogonal subspace $V^\perp$ then contains a unit vector u and $\forall \lambda \in \mathbb{R}_{++}$, the open ball $B_{<\lambda}(\lambda_u)$ does not intersect Λ_K. Thus $\mathbb{R}_0^{s+r}$ contains points arbitrarily far away from every point in Λ_K. Fix a real number $B > B_K$, where B_K is as in the previous proposition, so that for every $c \in DivK$ with $||c|| \geq B$, $L(c)$ contains a nonzero element and fix a vector $b \in \mathbb{R}^{r+s}$ with nonnegative components b_i s.t. $T(b) = \sum b_i = \log B$. Let $(\alpha_1), \ldots, (\alpha_m)$ be the list of all nonzero principal ideals with $N(\alpha_J) \leq B$, which is finite. LEt M be the twice the maximum of $(r+s)B$ and $\max_j ||Log(\alpha_j)||$. Let $h \in \mathbb{R}_0^{r+s}$ and define $c \in DivK$ by $I_c \equiv \mathcal{O}_K, c_v \equiv \exp(h_i + b_i)$ for $v|\infty$ where i is the coordinate in $\mathbb{R}^{r+s}$ corresponding to v under the Log map. Then

$$||c|| = \Pi||c_v|| = \exp(\sum(h_i + b_i)) = \exp T(h+b) = \exp(T(h) + T(b)) = \exp(T(b)) = B > B_K$$

thus $L(c)$ contains a nonzero $\gamma \in I_c \cap K = \mathcal{O}_K$ and $g = Log(\gamma)$ satisfies $g_i \leq \log c_v = h_i + b_i$. Also $T(g) = T(Log\gamma) = \log N(\gamma) \geq 0$ since $N(\gamma) \geq 1$, $\forall \gamma \in \mathcal{O}_K$ nonzero. The vector $w \equiv g - h \in \mathbb{R}^{r+s}$ satisfies $\sum w_i = T(v) = T(g) - T(h) = T(g) \geq 0$, $w_i \leq b_i \leq B$ with together implies $|w_i| \leq (r+s)B$, $||g-h|| = ||w|| \leq M/2$. Additionally, $\log N(\gamma) = T(\log\gamma) \leq T(h+b) = T(b) = \log B \to N(\gamma) \leq B, (\gamma) = (\alpha_j)$ for one of the α_j fixed above. Thus $\gamma/\alpha_j \in \mathcal{O}_K^\times$ is a unit and $l \equiv Log(\gamma/\alpha_j) = Log(\gamma) - Log(\alpha_j) \in \Lambda_K$ satisfies $||g-l|| = ||Log(\alpha_j)|| \leq M/2$. Thus $||h-l|| \leq ||h-g|| + ||g-l|| \leq M$, which contradicts that $\mathbb{R}_0^{r+s}$ contains a points arbitrarily far away from every point in Λ_K so $\dim V = \dim \mathbb{R}_0^{r+s}$.

Theorem 3.1.10 (Dirichlet's Unit Theorem)**.** *Let K be a number field with r real and s complex places. Then $\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}^{r+s-1}$ is a finitely generated abelian group.*

pf. The image of the torsion-free part of the unit group $\mathcal{O}_K^\times$ under the Log map is the lattice Λ_K in the trace-zero hyperplane $\mathbb{R}_0^{r+s}$, which has dimension $r+s-1$.

Theorem 3.1.11 (Unit Theorem for Global Fields)**.** *Let K/F be a finite separable extension, with $F = \mathbb{Q}$ or $F = \mathbb{F}_q(t)$, let $S \subseteq M_K$ be the set of places of K lying above the unique infinite place of F and define $\mathcal{O}_K^\times \equiv \{x \in K^\times : v(x) = 0,\ \forall v \in M_K \setminus S\}$. Then $\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}^{\#S-1}$ is a finitely generated abelian group.*

pf. When $F = \mathbb{Q}$, $\#S = r+s$ and this is Dirichlet's unit theorem.

Chapter 4

Local Class Field Theory

4.1 Artin Map and Local Class Field Theory

4.1.1 The Prime Number Theorem

Definition 4.1.1. *The Riemann zeta function is the complex function defined by the series*

$$\zeta(s) \equiv \sum_{n\geq 1} n^{-s}$$

for $Re(s) > 1$, where n varies over positive integers.

$\rightarrow$ it converges absolutely and locally uniformly on $\mathrm{Re}(s) > 1$.

Theorem 4.1.1 (Euler Product)**.** *For $Re(s) > 1$, $\zeta(x) = \sum_{n\geq 1} n^{-s} = \Pi_p(1-p^{-s})^{-1}$ where the product converges absolutely. In particular, $\zeta(s) \neq 0$, $Re(s) > 1$.*

pf. $\sum n^{-s} = \sum \Pi p^{-v_p(n)s}$. Define the partial zeta function $\zeta_m(s) \equiv \sum_{n\in S_m} n^{-s} = \sum_{e_1,\ldots,e_k\geq 0}(p_1^{e_1}\ldots p_k^{e_k})^{-s} = \Pi_{1\leq i\leq k}\sum_{e_i\geq 0}(p_i^s)^{e_i} = \Pi_{p\leq m}(1-p^{-s})^{-1}$ where S_m is the set of m-smooth integers (those with no prime factors $p > m$) and p_i are the primes upto m. For any $\delta > 0$, $zeta_m(s)$ converges uniformly on $\mathrm{Re}(s) > 1+\delta$ to $\zeta(s) \rightarrow$ $\forall\epsilon > 0$, $|\zeta_m(s)-\zeta(s)| \leq |\sum_{n\geq m} n^{-s}| \leq \sum |n^{-s}| = \sum n^{-\mathrm{Re}(s)} \leq \int_m^\infty x^{-1-\delta}dx \leq \frac{1}{\delta}m^{-\delta} < \epsilon$ for all sufficiently large m. Thus the sequence $\zeta_m(s)$ converges locally uniformly to $\zeta(s)$ on $\mathrm{Re}(s) > 1$. The sequence of functions $P_m \equiv \Pi_{p\leq m}(1-p^{-s})^{-1}$ also converges locally uniformly to $\Pi(1-p^{-s})^{-1}$ on any region in which the latter function is absolutely convergent. For any s in $\mathrm{Re}(s) > 1$,

$$\begin{aligned}\sum_p \log(1-p^{-s})^{-1}| &= \sum_p |\sum_{e\geq 1}\tfrac{1}{e}p^{-es}| \quad \text{by } \log(1-z) = -\sum_{n\geq 1}\tfrac{1}{n}z^n,\ \forall |z| < 1 \\ &\leq \sum_p \sum_{e\geq 1}|p^{-s}|^e = \sum_p(|p|^s-1)^{-1} < \infty\end{aligned}$$

. Thus $\Pi_p(1-p^{-s})^{-1}$ is absolutely convergent on $\mathrm{Re}(s) > 1$.

Theorem 4.1.2 (Analytic Continuation I)**.** *For $Re(s) > 1$,*

$$\zeta(s) = \frac{1}{s-1} + \phi(s)$$

where $\phi(s)$ is a holomorphic function on $Re(s) > 0$. Thus $\zeta(s)$ extends to a meromorphic function on $Re(s) > 0$ that has a simple pole at $s = 1$ with residue 1 and no other poles.

pf. For $\mathrm{Re}(s) > 1$, $\zeta(s) - \frac{1}{s-1} = \sum n^{-s} - \int_1^\infty x^{-s} ds = \sum(n^{-s} - \int_n^{n+1} x^{-s} dx) = \sum \int_n^{n+1}(n^{-s} - x^{-s})dx$. For each $n \geq 1$, $\phi_n(s) \equiv \int_n^{n+1}(n^{-s} - x^{-s})dx$ is holomorphic on $\mathrm{Re}(s) > 0$. For each fixed s in $\mathrm{Re}(s) > 0$, $\forall x \in [n, n+1]$, $|n^{-s} - x^{-s}| = |\int_n^x st^{-s-1}dt| \leq \int_n^x \frac{|s|}{|t^{s+1}|}dt = \int_n^x \frac{|s|}{t^{1+\mathrm{Re}(s)}}dt \leq \frac{|s|}{n^{1+\mathrm{Re}(s)}} \to |\phi_n(s)| \leq \int_n^{n+1} |n^{-s} - x^{-s}|dx \leq \frac{|s|}{n^{1+\mathrm{Re}(s)}}$. For any s_0 in $\mathrm{Re}(s) > 0$, $\begin{cases} \epsilon \equiv \mathrm{Re}(s_0)/2 \\ U \equiv B_{<\epsilon}(s_0) \end{cases}$, $\forall n \geq 1$, $\sup_{s \in U} |\phi_n(s)| \leq \frac{|s_0|+\epsilon}{n^{1+\epsilon}} \equiv M_n$ and $\sum M_n = (|s_0| + \epsilon)\zeta(1+\epsilon)$ converges. The series $\sum \phi_n$ thus converges locally uniformly on $\mathrm{Re}(s) > 0$. By the Weierstrass test $\sum_n \phi_n$ converges to a function $\phi(s) = \zeta(s) - \frac{1}{s-1}$ which is holomorphic on $\mathrm{Re}(s) > 0$.

Lemma 4.1.1 (Mertens). *$\forall x, y \in \mathbb{R}$ with $z > 1$,*

$$|\zeta(x)^2\zeta(x+iy)^4\zeta(x+2iy)| \geq 1$$

pf. $\begin{aligned} \log|\zeta(s)| &= -\textstyle\sum_p \log|1-p^{-s}| && \text{by the Euler product} \\ &= -\textstyle\sum \mathrm{Re}\log(1-p^{-s}) && \text{by } \log|z| = \mathrm{Re}\log z \\ &= \textstyle\sum_p \sum_{n\geq 1} \frac{\mathrm{Re}(p^{-ns})}{n} \end{aligned}$. Let $s = x + iy$. Then $\log|\zeta(x+iy)| = \sum\sum \frac{\cos(ny\log p)}{np^{nx}} = \sum\sum \frac{2(1+\cos ny\log p)^2}{np^{nx}} \geq 0$ by the trigonometric identity. So the lemma follows.

Corollary 4.1.1. *$\zeta(s)$ has no zeros on $Re(s) \geq 1$.*

pf. By the analytic continuity I, $\zeta(s)$ has no zeros on $\mathrm{Re}(s) > 1$. Suppose $\zeta(1+iy) = 0$ for some $y \in \mathbb{R}$. Then $y \neq 0$ since when $s = 1$, $\zeta(s)$ has a simple pole. Thus $\lim_{x\to 1} |\zeta(x)^3\zeta(x+iy)^4\zeta(x+i2y)| = 0$, a contradiction to the previous lemma, and the corollary follows.

The prime counting function $\begin{array}{rcl} \pi : \mathbb{R} & \to & \mathbb{Z}_+ \\ x & \mapsto & \sum_{p\leq x} 1 \end{array}$ counts the number of primes upto x. Define the log-weighted prime counting function $\vartheta(x) \equiv \sum_{p\leq x} \log p$.

Theorem 4.1.3 (Chebyshev). *$\pi(x) \sim \frac{x}{\log x}$ iff $\vartheta(x) \sim x$.*

pf. $0 \leq \vartheta(x) \leq \pi(x)\log x \to \frac{\vartheta(x)}{x} \leq \frac{\pi(x)\log x}{x}$. Then we have $\forall \epsilon \in (0,1)$, $\vartheta(x) \geq \sum_{x^{1-\epsilon}<p\leq x} \log p \geq (1-\epsilon)(\log x)(\pi(x) - \pi(x^{1-\epsilon})) \geq (1-\epsilon)(\log x)(\pi(x) - x^{1-\epsilon}) \Leftrightarrow \pi(x) \leq \frac{1}{1-\epsilon}\frac{\vartheta(x)}{\log x} + x^{1-\epsilon}$. Thus $\forall \epsilon \in (0,1)$, $\frac{\vartheta(x)}{x} \leq \frac{\pi(x)\log x}{x} \leq (\frac{1}{1-\epsilon})\frac{\vartheta(x)}{x} + \frac{\log x}{x^\epsilon}$. Then $\lim_{x\to\infty} \frac{\log x}{x^\epsilon} = 0$. By choosing ϵ sufficiently small, we can make the ratios of $\vartheta(x)$ to x and $\pi(x)$ to $x/\log x$ arbitrarily close together as $x \to \infty$. Thus the lemma follows.

Lemma 4.1.2 (Chebyshev). *For $x \geq 1$*

$$\vartheta(x) \leq (4\log 2)x$$

pf. For any positive integer n, $2^{2n} = (1+1)^{2n} = \sum_{m=0}^{2n}\binom{2n}{m} \geq \binom{2n}{n} = \frac{(2n)!}{n!n!} \geq \Pi_{n<p\leq 2n}p - \exp(\vartheta(2n)-\vartheta(n))$ since $(2n)!$ is divisible by every prime $p \in (n, 2n]$, but $n!$ is not. Taking logarithms on both sides yields $\vartheta(2n)-\vartheta(n) \leq 2n\log 2$, $\forall n$. For any integer $m \geq 1$, $\vartheta(2^m) = \sum_{n=1}^{m}(\vartheta(2^n)-\vartheta(2^{n-1})) \leq \sum 2^n \log s \leq 2^{m+1}\log 2$. For any real $x \geq 1$, $\exists m \in \mathbb{Z}_{++}$ s.t. $2^{m-1} \leq x < 2^m$, which implies $\vartheta(x) \leq \vartheta(2^m) \leq 2^{m+1}\log 2 = (4\log 2)2^{m-1} = (4\log 2)x$ and the lemma follows.
$\rightarrow \vartheta(x) = O(x)$.

Lemma 4.1.3. *Lef $f : \mathbb{R}_{\geq 1} \rightarrow \mathbb{R}$ be a nondecreasing function. If the integral $\int_1^\infty \frac{f(t)-t}{t^2}dt$ converges, then $f(x) \sim x$.*

pf. Let $F(x) \equiv \int_1^x \frac{f(t)-t}{t^2}dt$. By the hypothesis, $\lim_{x\rightarrow\infty}F(x)$ exists. Then $\forall\lambda > 1$, $\forall\epsilon > 0$, $|F(\lambda x)-F(x)| < \epsilon$ for sufficiently large x. Fix $\lambda > 1$ and suppose there is. an unbounded sequence (x_n) s.t. $f(x_n) \geq \lambda x_n$ for all n. For each x_n,

$$\begin{aligned} F(\lambda x_n)-F(x_n) &= \int_{x_n}^{\lambda x_n}\frac{f(t)-t}{t^2}dt \geq \int\frac{\lambda x_n - t}{t^2}dt \quad \text{by } f \text{ being nondecreasing} \\ &= \int_1^\lambda \frac{\lambda - t}{t^2}dt = c > 0 \text{ constant} \end{aligned}$$

. Taking $\epsilon < c$. Then $|F(\lambda x_n)-F(x_n)| = c > \epsilon$ for arbitrarily large x_n, a contradiction. Thus $f(x) < \lambda x$ for all sufficiently large x. Similarly we can get $f(x) > \frac{1}{\lambda}x$ for all sufficiently large x. These inequalities hold for all $\lambda >!$, by letting λ sufficiently close to 1, we get $\lim f(x)/x = 1$, i.e. $f(x) \sim x$.
$\rightarrow$ for $f = \vartheta$, $H(t) = \vartheta(e^t)e^{-t}-1$ gives $\int_1^\infty \frac{\vartheta(t)-t}{t^1}dt$ converges$\Leftrightarrow \int_0^\infty H(u)du$ converges where $t = e^u$.

Definition 4.1.2. *Let $h : \mathbb{R}_{++} \rightarrow \mathbb{R}$ be a piecewise continuous function. The Laplace transform $\mathscr{L}h$ is the complex function defined by*

$$\mathscr{L}h(s) \equiv \int_0^\infty e^{-st}h(t)dt$$

which is holomorphic on $Re(s) > c$ for any $c \in \mathbb{R}$ for which $h(t) = O(e^{ct})$.

Define the auxiliary function $\Phi(x) \equiv \sum_p p^{-s}\log p$.

Lemma 4.1.4. *$\mathscr{L}\vartheta(e^t)(x) = \frac{\Phi(s)}{x}$ is holomorphic on $Re(s) > 1$.*

pf. By the Chebyshev lemma, $\vartheta(e^t) = O(e^t)$, so $\mathscr{L}(\vartheta(e^t))$ is holomorphic on $\mathrm{Re}(s) > 1$. Let p_n be the nth prime and put $p_0 = 0$. The function $\vartheta(e^t)$ is constant on $t \in (\log p_n, \log p_{n+1}) \rightarrow \int_{\log p_n}^{\log p_{n+1}} e^{-st}\vartheta(e^t)dt = \vartheta(p_n)\int e^{-st}dt = \frac{1}{s}\vartheta(p_n)(p_n^{-s} - p_{n+1}^{-s})$. Thus we get $(\mathscr{L}\vartheta(e^t))(s) = \int_0^\infty e^{-st}\vartheta(e^t)dt = \frac{1}{s}\sum_{n=1}^\infty \vartheta(p_n)(p_n^{-s}-p_{n+1}^{-s}) = \frac{1}{s}\sum\vartheta(p_n)p_n^{-s}-$

$\frac{1}{s}\sum\vartheta(p_{n-1})p_n^{-s} = \frac{1}{s}\sum(\vartheta(p_n)-\vartheta(p_{n-1}))p_n^{-s} = \frac{1}{s}\sum p_n^{-s}\log p_n = \frac{\Phi(s)}{s}$.
$\to$ for $H(t) \equiv \vartheta(e^t)e^{-t} - 1$,

$$\mathscr{L}H(s) = \mathscr{L}(\vartheta(e^t)e^{-1})(s) - \mathscr{L}1(s) = \mathscr{L}(\vartheta(e^t)(s+1)) - \frac{1}{s} = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$$

Lemma 4.1.5. *The function* $\Phi(s) - \frac{1}{s-1}$ *extends to a meromorphic function on* $Re(s) > 1/2$ *that is holomorphic on* $Re(s) \geq 1$.

pf. $\zeta(s)$ extends to a meromorphic function on $\mathrm{Re}(s) > 0$, which we also denote $\zeta(x)$ with one simple pole at $s = 1$ and no zeros on $\mathrm{Re}(s) \geq 1$. The logarithmic derivative $\zeta'(s)/\zeta(s)$ is meromorphic on $\mathrm{Re}(s) > 0$ with no zeros on $\mathrm{Re}(s) \geq 1$ and only one simple pole at $s = 1$ with residue -1. Then by the Euler product, for $\mathrm{Re}(s) > 1$, $-\frac{\zeta'(s)}{\zeta(s)} = (-\log\zeta(x))' = (-\log\Pi(1-p^{-s})^{-1})' = (\sum\log(1-p^{-s}))' = \sum\frac{p^{-s}\log p}{1-p^{-s}} = \sum\frac{\log p}{p^s-1} = \sum(\frac{1}{p^s} + \frac{1}{p^s(p^s-1)})\log p = \Phi(s) + \sum\frac{\log p}{p^s(p^s-1)}$. Then $\sum\frac{\log p}{p^s(p^s-1)}$ converges absolutely and locally uniformly to a holomorphic function on $\mathrm{Re}(s) > 1/2$. $\frac{\zeta'(s)}{\zeta(s)}$ is meromorphic on $\mathrm{Re}(s) > 0$ and on $\mathrm{Re}(s) \geq 1$, it has only a simple pole at $s = 1$ with residue 1. Thus $\Phi(s) - \frac{1}{s-1}$ extends to a meromorphic function on $\mathrm{Re}(s) > 1/2$ that is holomorphic on $\mathrm{Re}(s) \geq 1$.

Corollary 4.1.2. *The function* $\Phi(s+1) - \frac{1}{s}$ *and* $(\mathscr{L}H)(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$ *both extends to meromorphic functions on* $Re(s) > -1/2$ *that are holomorphic on* $Re(s) \geq 0$.

pf. By the previous lemma, the first stmt follows immediately. $\frac{\Phi(s+1)}{s+1} - \frac{1}{s} = \frac{1}{s+1}(\Phi(s+1) - \frac{1}{s}) - \frac{1}{s+1}$ is meromorphic on $\mathrm{Re}(s) > -1/2$ and holomorphic on $\mathrm{Re}(s) \geq 0$ since it is a sum of products of such functions.

Theorem 4.1.4. *Let* $f : \mathbb{R}_+ \to \mathbb{R}$ *be a bounded piecewise continuous function and suppose its Laplace transform extends to a holomorphic function* $g(s)$ *on* $Re(s) \geq 0$. *Then the integral* $\int_0^\infty f(t)dt$ *converges and is equal to* $g(0)$.

pf. Wlog assume $f(t) \leq 1$, $\forall t \geq 0$. For $\tau \in \mathbb{R}_{++}$, define $g_\tau(s) \equiv \int_0^\tau f(t)e^{-st}dt$. Then since $\int_0^\infty f(t)dt - \lim_{\tau\to\infty} g_\tau(0)$, if $\lim g_\tau(0) = g(0)$, the theorem follows. For $r > 0$, let γ_r be the boundary of the region $\{s : |s| \leq r,\ \mathrm{Re}(s) \geq -\delta\}$ with $\delta_r > 0$, chosen so that g is holomorphic on γ_r. Such δ_r exists because g is holomorphic on $\mathrm{Re}(s) \geq 0$, hence on some open ball $B_{\leq 2\delta(y)}(iy)$, $\forall y \in [-r, r]$ we may take $\delta_r \equiv \inf\{\delta(y) : y \in -[r, r]\}$ which is positive, given $[-r, r]$ being compact. Each γ_r is a simple closed curve and $\forall\tau > 0$, $h(s) \equiv (g(s) - g_\tau(s))e^{s\tau}(1 + \frac{s^2}{r^2})$ is holomorphic on a region containing γ_r. By the Cauchy's integral formula, $g(0) - g_\tau(0) = h(0) = \frac{1}{2\pi i}\int_{\gamma_r}(g(s) - g_\tau(s))e^{s\tau}(\frac{1}{s} + \frac{s}{r^2})ds$. Set $r = 3/\epsilon > 0$.

- let γ_r^+ denote the part of γ_r in $\mathrm{Re}(s) > 0$, a semicircle of radius r.

The integrand is absolutely bounded by $1/r$ on γ_r^+, since $|s| = r$ and $\mathrm{Re}(s) > 0$, $|g(s) - g_\tau(s)| \cdot |e^{s\tau}(\frac{1}{s} + \frac{s}{r^2})| = |\int_\tau^\infty f(t)e^{-st}dt| \cdot \frac{e^{\mathrm{Re}(s)\tau}}{r} \cdot |\frac{r}{s} + \frac{s}{r}| \le \int_\tau^\infty e^{-\mathrm{Re}(s)t}dt \cdot \frac{e^{\mathrm{Re}(s)\tau}}{r} \cdot \frac{2\mathrm{Re}(s)}{r} = \frac{e^{-\mathrm{Re}(s)\tau}}{\mathrm{Re}(s)} \cdot \frac{e^{\mathrm{Re}(s)\tau}}{r} \cdot \frac{2\mathrm{Re}(s)}{r} = 2/r^2 \to |\frac{1}{2\pi i}\int_{\gamma_r^+}(g(s) - g_\tau(s))e^{s\tau}(\frac{1}{s} + \frac{s}{r^2})ds| \le \frac{1}{2\pi} \cdot \pi r \cdot \frac{2}{r^2} = \frac{1}{r}$.

- let γ_r^- be the part of γ_r in $\mathrm{Re}(s) < 0$, a truncated semicircle.

  For any fixed r, the first term $g(s)e^{s\tau}(s^{-1} + sr^{-2})$ in the integrand tends to 0 as $\tau \to \infty$ for $\mathrm{Re}(s) < 0$ and $|s| \le r$. For the second term, since $g_\tau(s)$ is holomorphic on $\mathbb{C}$, it makes no difference if we instead integrate over the semicircle of radius r in $\mathrm{Re}(s) < 0$. For $\begin{cases} |s| = r \\ \mathrm{Re}(s) < 0 \end{cases}$, $|g_\tau(s)e^{s\tau}(\frac{1}{s} + \frac{s}{r^2})| = |\int_0^\tau f(t)e^{-st}dt| \cdot \frac{e^{\mathrm{Re}(s)\tau}}{r}\frac{-2\mathrm{Re}(s)}{r} = (1 - \frac{e^{-\mathrm{Re}(s)\tau}}{\mathrm{Re}(s)})\frac{e^{\mathrm{Re}(s)\tau}}{r}\frac{-2\mathrm{Re}(s)}{r} = \frac{2}{r^2} \cdot (1 - e^{\mathrm{Re}(s)\tau}\mathrm{Re}(s))$ where $\lim_{\tau\to\infty}(1 - e^{\mathrm{Re}(s)\tau}\mathrm{Re}(s)) = 1$ since $\mathrm{Re}(s_< 0$. Thus we obtain the bound $\frac{1}{r} + o(1)$ when we replace γ_r^+ with γ_r^{-1} under γ_r^+ case above.

Then $g(0) - g_\tau(0) = h(0) = \frac{1}{2\pi i}\int_{\gamma_r}(g(s) - g_\tau(s))e^{s\tau}(\frac{1}{s} + \frac{s}{r^2})ds < \frac{2}{r} + o(1)$ as $\tau \to \infty$. Thus $\forall \epsilon > 0$, $\forall r = 3/\epsilon > 0$, $|g(0) - g_\tau(0)| < 3/r = \epsilon$ for all sufficiently large τ. So the theorem follows.
$\to$ for a piecewise continuous function $f : \mathbb{R}_+ \to \mathbb{R}$, its Laplace transform $\mathscr{L}f(s) \equiv \int_0^\infty e^{-st}f(t)dt$ is typically not defined on $\mathrm{Re}(s) \le c$ where c is the least c for which $f(t) = O(e^{ct})$. But $\mathscr{L}f$ may have an analytic continuation to a larger domain. The theorem says when $\mathscr{L}f$ extends to a holomorphic function on the entire half plane $\mathrm{Re}(s) \ge 0$, its value at $s = 0$ is exactly what we would get by simply plugging 0 into the integral defining $\mathscr{L}f$.

Theorem 4.1.5 (Prime Number Theorem)**.**

$$\pi(x) \sim \frac{x}{\log x}$$

pf. $H(t) = \vartheta(e^t)e^t - 1$ is piecewise continuous and bounded by the Chebyshev lemma. Its Laplace transform extends to a holomorphic function on $\mathrm{Re}(s) \ge 0$ by the previous corollary. Then by the previous theorem, $\int_0^\infty H(t)dt - \int_0^\infty(\vartheta(e^t)e^{-t}01)dt$ converges. Replacing t with $\log x$ gives $\int_1^\infty(\vartheta(x)\frac{1}{x} - 1)\frac{dx}{x} = \int_1^\infty \frac{\vartheta(x) - x}{x^2}dx$ converges.Then by the lemma 4.1.3, $\vartheta(x) \sim x \Leftrightarrow \pi(x) \sim \frac{x}{\log x}$ by the Chebyshev theorem.

4.1.2 Dirichlet Theorem

Theorem 4.1.6 (Mertens)**.** *As* $x \to \infty$, $\begin{cases} i) & \sum_{p\le x}\frac{\log p}{p} = \log x + R(x),\ |R(x)| \le 2 \\ ii) & \sum_{p\le x}\frac{1}{p} = \log\log x + B + O(\frac{1}{\log x}) \\ iii) & \sum_{p\le x}\log(1 - \frac{1}{p}) = -\log\log x - \gamma + O(\frac{1}{\log x}) \end{cases}$.

pf.

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Definition 4.1.3. *A function $f : \mathbb{Z} \to \mathbb{C}$ is called an arithmetic function. The function f is multiplicative if $f(1) = 1$ and $f(mn) = f(m)f(n)$ for all $m \perp n \in \mathbb{Z}$. It is totally multiplicative if $f(1) = 1$ and $f(mn) = f(m)f(n)$, $\forall m, n \in \mathbb{Z}$. For $m \in \mathbb{Z}_{++}$, we say f is m-periodic if $f(n+m) = f(n)$, $\forall n \in \mathbb{Z}$ and we call m the period of f if it is the least $m > 0$ for which this holds.*

Definition 4.1.4. *A Dirichlet character is a periodic totally multiplicative arithmetic function $\chi : \mathbb{Z} \to \mathbb{C}$.*

$\to$ the image of a Dirichlet character is a finite multiplicative closed subset of $\mathbb{C}$, hence the union of a finite subgroup of $U(1)$ and a subset of $\{0\}$.

e.g. the trivial Dirichlet character is $\mathbb{1}(n) \equiv 1$, which is the unique Dirichlet character of period 1.

Each m-periodic Dirichlet character χ restricts to a group character $\chi \in (\mathbb{Z}/m\mathbb{Z})^\times$ and each group character χ of $(\mathbb{Z}/m\mathbb{Z})^\times$ can be extended to a Dirichlet character by defining $\chi(n) = 0$, $\forall n \not\perp m$, i.e. by the extension by zero.

Definition 4.1.5. *A Dirichlet character of modulus m is an m-periodic Dirichlet character χ that is the extension by zero of a group character on $(\mathbb{Z}/m\mathbb{Z})^\times$, i.e. an m-periodic Dirichlet character for which $\chi(n) \neq 0 \Leftrightarrow m \perp n$.*

Lemma 4.1.6. *Let χ be a Dirichlet character of period m. Then χ is a Dirichlet character of modulus m' iff $m|m'|m^k$ for some k.*

pf.

- Let $m|m'|m^k$ for some k. Suppose $\chi(n) \neq 0$ with $m \not\perp n$. Let $p = gcd(m,n)$. Then $\chi(p)\chi(n/p) = \chi(n) \neq 0 \to \chi(p) \neq 0$. For any $r \in \mathbb{Z}$, $\chi(r)\chi(p) = \chi(rp) = \chi(rp+m) = \chi(r+m/p)\chi(p)$ so χ is (m/p)-periodic, a contradiction to the m-period hypothesis. Thus $\chi(n) \neq 0 \to m \perp n$. For any $n \perp m$, $\exists a = n^e \equiv 1 \mod m$ so that $\chi(1) = \chi(a) = \chi(n^e) = \chi(n)^e \neq 0$ in which case $\chi(n) \neq 0$. Thus $n \perp m \to \chi(n) \neq 0$. Thus χ is a Dirichlet character of modulus m. If $m|m'|m^k$, the prime divisors of m' coincides with those of m. Thus $n \perp m' \Leftrightarrow n \perp m \Leftrightarrow \chi(n) \neq 0$ and therefore χ is also m'-periodic, so χ is a Dirichlet character of modulus m'.

- If χ is a Dirichlet character of modulus m', then χ is m'-periodic and therefore $m|m'$ by the definition of the period. Since χ is a Dirichlet character of modulus m and of modulus m', $\forall p$, prime, $p|m \Leftrightarrow \chi(p) = 0 \Leftrightarrow p|m'$ thus the prime divisors of m, m' coincide and m' must divide some power m^k of m.

Definition 4.1.6. *Let χ_1, χ_2 be Dirichlet characters of modulus m_1, m_2 respectively with $m_1|m_2$. If $\chi_2(n) = \chi_1(n)$, $\forall n \in (\mathbb{Z}/m_2\mathbb{Z})^\times$, then χ_2 is induced by χ_1. A Dirichlet character that is not induced by any character other than itself if called primitive.*

Lemma 4.1.7. *A Dirichlet character χ_2 of modulus m_2 is induced by a Dirichlet character of modulus $m_1|m_2$ iff χ_2 is constant on residue classes in $(\mathbb{Z}/m_2\mathbb{Z})^\times$ that are congruent module m_1. When this holds, the Dirichlet character χ_1 of modulus m_1 that induces χ_2 is uniquely determined.*

pf.

- if χ_2 is induced by χ_1, it must be constant on the residue classes in $(\mathbb{Z}/m_2\mathbb{Z})^\times$ that are congruent module m_1 since χ_1 is.

- there is a surjective ring homomorphism $\mathbb{Z}/m_2\mathbb{Z} \to \mathbb{Z}/m_1\mathbb{Z}$ given by reduction modulo m_1. Suppose $u_1 \in \mathbb{Z} \mod m_1$. Let a be the product of all primes dividing m_2/m_1 but not u_1. Then $u_2 = u_1 + m_1 a$ is not divisible by any prime $\begin{cases} p|m_1 & \text{since } u_1 \text{ is not} \\ p|(m_2/m_1) & \text{since such } p \text{ divides exactly one of } u_1 \text{ and } m_1 \end{cases}$. Thus u_2 is a unit modulo m_2 that reduces to u_1 module m_1 and therefore $\pi : (\mathbb{Z}/m_2\mathbb{Z})^\times \to (\mathbb{Z}/m_1\mathbb{Z})^\times$ induced by the above surjective ring homomorphism is also surjective. If χ_2 is a Dirichlet character of modulus m_2 constant on fibers of π, we can define a Dirichlet character χ_1 of modulus m_1 via $\chi_1(n_1) \equiv \chi_2(n_2)$, $\forall n_1 \in (\mathbb{Z}/m_1\mathbb{Z})^\times$ with $n_2 \in \pi^{-1}(n_1)$. Thus χ_1 induces χ_2.

If $\exists \chi_1'$ also induces χ_2, it must satisfy the same condition $\chi_1(n_1 = \chi_2(n_2)$, which gives the uniqueness.

Definition 4.1.7. *A Dirichlet character χ induced by $\mathbb{1}$ is called principal. For $m \in \mathbb{Z}_{++}$, $\mathbb{1}_m$ denotes the principal Dirichlet character of modulus m; it corresponds to the trivial character of $(\mathbb{Z}/m\mathbb{Z})^\times$.*

Lemma 4.1.8. *Let χ be a Dirichlet character of modulus m. Then*

$$\sum_{n \in \mathbb{Z}/m\mathbb{Z}} \chi(n) \neq 0 \Leftrightarrow \chi = \mathbb{1}_m$$

pf. $\chi(n) = 0$, $\forall n \notin (\mathbb{Z}/m\mathbb{Z})^\times$ and the sum over $(\mathbb{Z}/m\mathbb{Z})^\times$ is nonzero iff χ restricts to the trivial character on $(\mathbb{Z}/m\mathbb{Z})^\times$ by the orthogonality of characters.
$\to$ the principal Dirichlet characters $\mathbb{1}_m$ and $\mathbb{1}_{m'}$ necessarily coincide when $m|m'|m^k$.

Theorem 4.1.7. *Every Dirichlet character χ is induced by a primitive Dirichlet character $\tilde{\chi}$ that is uniquely determined by χ.*

pf. Define a partial ordering $\precsim$ on the set of all Dirichlet characters by defining $\chi_1 \precsim \chi_2$ iff χ_1 induces χ_2. Then it is reflexive and transitive by the lemma 4.1.7. Let χ be a Dirichlet character of period m and consider the set $X = \{\chi' : \chi' \precsim \chi\}$. Each $\chi' \in X$ necessarily has period $m'|m$ and there is at most one χ' of period m' for each divisor m' of m by the lemma 4.1.7. Thus X is finite and nonempty ($\chi \in X$). Suppose $\chi_1, \chi_2 \in X$ with period m_1, m_2 respectively. Then $m_1, m_2|m$ as does $m_3 = gcd(m_1, m_2)$. Then we have a commutative diagram of surjective unit groups $\begin{array}{ccc} (\mathbb{Z}/m\mathbb{Z})^\times & \longrightarrow & (\mathbb{Z}/m_1\mathbb{Z})^\times \\ \downarrow & & \downarrow \\ (\mathbb{Z}/m_2\mathbb{Z})^\times & \longrightarrow & (\mathbb{Z}/m_2\mathbb{Z})^\times \end{array}$ by pushout. By the lemma 4.1.7., χ is constant on residue classes on $(\mathbb{Z}/m\mathbb{Z})^\times$ that are congruent modulo m_1, m_2 and therefore it is constant on residue classes in $(\mathbb{Z}/m\mathbb{Z})^\times$ congruent modulo m_3 as are χ_1, χ_2. Thus there is a unique Dirichlet character χ_3 of modulus m_3 that induces χ, χ_1, χ_2. This implies for every pair $\chi_1, \chi_2 \in X$ has a lower bound χ_3 under the partial ordering $\precsim$ that is compatible with the total ordering of X by period. Thus X contains a unique element $\tilde{\chi}$ that is minimal both w.r.t. $\precsim$ and w.r.t. the total ordering by period. By the transitivity of $\precsim$, it must be primitive.

Definition 4.1.8. *The conductor of a Dirichlet character χ is the period of the unique primitive Dirichlet character $\tilde{\chi}$ that induces χ.*

Corollary 4.1.3. *For a Dirichlet character χ of modulus m we have $\sum_{n \in \mathbb{Z}/m\mathbb{Z}} \chi(n) \neq 0$ iff χ has conductor 1.*

Corollary 4.1.4. *Let $M(m)$ be the set of Dirichlet characters of moudulus m, $X(m)$ be the set of primitive Dirichlet characters of conductor dividing m and $\hat{G}(m)$ be the character group of $(\mathbb{Z}/m\mathbb{Z})^\times$. Then we have canonical bijections*

$$\begin{array}{ccccc} M(m) & \to & X(m) & \to & \hat{G}(m) \\ \chi & \mapsto & \tilde{\chi} & \mapsto & (n \mapsto (\tilde{\chi}(n)) \end{array}$$

pf. By the previous theorem, $\chi \mapsto \tilde{\chi}$ is injective. Also $\forall \tilde{\chi} \in X(m)$, it induces the character $\chi \in M(m)$ by setting $\chi(n) \equiv \tilde{\chi}(n)$, $\forall n (\mathbb{Z}/m\mathbb{Z})^\times$ and extending by zero, i.e. it is surjective. The map $\begin{array}{ccc} M & \to & \hat{G}(m) \\ \chi & \mapsto & (m \mapsto \chi(m)) \end{array}$ defines a bijection and it factors through the map $\chi \mapsto \tilde{\chi}$ since $\tilde{\chi} = \chi(n)$, $\forall n \in (\mathbb{Z}/m\mathbb{Z})^\times$.

Definition 4.1.9. *The Dirichlet L-function associated to a Dirichlet character χ is*

$$L(s, \chi) \equiv \Pi_p (1 - \chi(p) p^{-s})^{-1} = \sum_{n \geq 1} \chi(n) n^{-s}$$

→ summand converges absolutely on $\mathrm{Re}(s) > 1$ since $|\chi(n)| \leq 1$ and therefore $L(s,\chi)$ is holomorphic on $\mathrm{Re}(s) > 1$.

e.g. for the trivial Dirichlet character $\mathbb{1}$, $L(s,\mathbb{1}) = \zeta(s)$ and for the principal character $\mathbb{1}_m$ of modulus m, induced by $\mathbb{1}$, $\zeta(x) = L(s,\mathbb{1}_m)\Pi_{p|m}(1-p^{-s})^{-1}$.
→ $L(s,\mathbb{1}_m)$ has a simple pole at $s=1$ with residue $\mathrm{res}_{s=1}L(s,\mathbb{1}_m) = \lim_{s\to 1^+}(s-1)\zeta(s)\Pi_{p|m}(1-p^{-s}) = \Pi(1-p^{-1}) = \frac{\phi(m)}{m}$.

Proposition 4.1.1. *Let χ be a nonprincipal Dirichlet character of modulus m. Then $L(s,\chi)$ extends to a holomorphic function on $Re(s) > 0$.*

pf. Define the function $\begin{array}{rcl} T:\mathbb{R}_+ & \to & \mathbb{C} \\ x & \mapsto & \sum_{0<n\leq x}\chi(n) \end{array}$. For any $x \in \mathbb{R}_+$, $T(x+m) - T(x) = \sum_{x<n\leq x+m}\chi(n) = \sum_{x\in\mathbb{Z}/m\mathbb{Z}}\chi(n) = 0$, given χ being nonprincipal. Thus $T(x)$ is periodic modulo m and therefore bounded. Then

$$\begin{aligned} L(s,\chi) &= \textstyle\sum_{n\geq 1}\chi(n)n^{-s} = \int_0^\infty x^{-s}dT(x) && \text{by writing as the Stieljtes integral} \\ &= x^{-s}T(x)|_0^\infty - \int_0^\infty T(x)d(x^{-s}) = 0 - \int_0^\infty T(x)(-sx^{-s-1})dx = s\int_0^\infty T(x)x^{-s-1}dx \end{aligned}$$

where $s\int_0^\infty T(x)x^{-s-1}dx$ is holomorphic on $\mathrm{Re}(s) > 0$ since it is the limit of the uniformly converging sequence of functions $\phi_n(s) \equiv s\int_0^n T(x)x^{-s-1}dx$ and is thus the analytic continuation of $L(s,\chi)$ on $\mathrm{Re}(s) > 0$.

Definition 4.1.10. *The Dedekind zeta function of a number field K is defined as*

$$\zeta_K(s) \equiv \sum_\alpha N(\alpha)^{-s} = \Pi_{\mathbf{p}}(1-N(\mathbf{p})^{-s})^{-1}$$

where α ranges over nonzero $\mathcal{O}_K$-ideals and $\mathbf{p}$ ranges over nonzero prime ideals.

→ it converges absolutely on $\mathrm{Re}(s) > 1$ since $\{\mathbf{p}|p\} \leq n \equiv [K:\mathbb{Q}]$ and $N(\mathbf{p}) \equiv [\mathcal{O}_K : \mathbf{p}] \geq p$ imply $\sum_{\mathbf{p}}|\log(1-N(\mathbf{p})^{-s})| \leq n\sum_p|\log(1-p^{-s})|$ and the RHS converges on $\mathrm{Re}(s) > 1$.

Definition 4.1.11. *Let X,Y be metric spaces. A function $f: X \to Y$ is Lipschitz continuous if $\exists c \in \mathbb{R}_{++}$ s.t. $\forall x_1,x_2 \in X$, $d(f(x_1),f(x_2)) \leq c\cdot d(x_1,x_2)$.*

→ every Lipschitz continuous function is uniformly continuous, but not vice versa.

Definition 4.1.12. *A set B in a metric space X is d-Lipschitz parametrizable if it is the union of the images of a finite number of Lipschitz continuous functions $f_i : [0,1]^d \to X$.*

Lemma 4.1.9. *Let $S \subseteq \mathbb{R}^n$ be a measurable set whose boundary $\partial S \equiv \bar{S}\backslash S^0$ is $(n-1)$-Lipschitz parametrizable. Then $\#(tS\cap\mathbb{Z}^n) = \mu(S)t^n + O(t^{n-1})$ as $t\to\infty$, where μ is the standard Lebesgue measure on $\mathbb{R}^n$.*

pf. Partition $\mathbb{R}^n$ as the disjoint union of half-open cubes of the form $C(a_1,\ldots,a_n) = \{(x_1,\ldots,x_n) \in \mathbb{R}^n : x_i \in [a_i, a_i+1)\}$, with $(a_1,\ldots,a_n) \in \mathbb{Z}^n$. Let $\mathscr{C}$ be the set of all such half-open cubes C. For each $t > 0$, define $\begin{cases} B_0(t) = \#\{C \in \mathscr{C} : C \subseteq tS\} \\ B_1(t) \equiv \#\{C \in \mathscr{C} : C \cap tS \neq \emptyset\} \end{cases}$. Then $\forall t > 0$, $B_0(t) \leq \#(tS \cap \mathbb{Z}^n) \leq B_1(t)$. We can bound $B_1(t) - B_0(t)$ by noting that each $C(a_1,\ldots,a_n)$ counted by this dfference contains a point $(a_1,\ldots,a_n) \in \mathbb{Z}^n$ within a distance $\sqrt{n} = O(1)$ of a point in $\partial tS = t\partial S$. Let $f_1,\ldots,f_m$ be Lipschitz functions $[0,1]^{n-1} \to \partial S$ whose images cover ∂S and let $c_1,\ldots,c_m$ be constants s.t. $d(f_i(x_1), f_i(x_2)) \leq c_i d(x_1,x_2)$, $\forall x_1,x_2 \in [0,1]^{n-1}$. Then $\forall y \in \partial S$, $\exists i$ s.t. $y = f_i(x_1,\ldots,x_{n-1}$. If we put $r_j = \lfloor tx_j \rfloor \in \mathbb{Z}$ so that $0 \leq x_j - r_j/t \leq 1/t$, $d(y, f_i(\frac{r_1}{t},\ldots,\frac{r_{n-1}}{t})) \leq c_i d((x_1,\ldots,x_{n-1}),(\frac{r_1}{t},\ldots,\frac{r_{n-1}}{t})) < c_i\sqrt{n}/t \leq c/t$ where $c \equiv \max_i \sqrt{n} c_i$. Thus $\forall y \in \partial S$ lies within a distance c/t of a point in the set $\mathscr{P} = \{f_i(\frac{r_1}{t},\ldots,\frac{r_{n-1}}{t})) : 0 \leq r_i \leq t,\ 1 \leq i \leq m\}$ which has cardinality $m(t+1)^{n-1} = O(t^{n-1})$. Thus every point of $t\partial S$ is within a distance c of one of the $O(t^{n-1})$ points in $t\mathscr{P}$. The number of integer lattice points within a distance $\sqrt{n}$ of a point in $t\partial S$ is also $O(t^{n-1})$ and therefore $B_1(t) - B_0(t) = O(t^{n-1})$. Also $B_0(t) \leq \mu(tS) = t^n\mu(S) \leq B_1(t)$ and therefore the lemma follows.

Corollary 4.1.5. *Let Λ be a lattice in an $\mathbb{R}$-vector space $V \cong \mathbb{R}^n$ and let $S \subseteq V$ be a measurable set whose boundary is $(n-1)$-Lipschitz parametrizable. Then*

$$\#(tS \cap \Lambda) = \frac{\mu(S)}{covol(\Lambda)} t^n + O(t^{n-1})$$

pf.

- the case $\Lambda \subseteq \mathbb{Z}^n$ is followed by the previous lemma.
- for general Λ, $\exists x > 0$ s.t. $s\Lambda$ is arbitrarily close to an integer lattice, for example take s to be the lcm of all denominators appearing in rational approximations of the coordinates of the basis for Λ. This necessarily have a finite index. Then by the previous case, the corollary hold for $s\Lambda$. But $tS \cap s\Lambda = (t/s)S \cap \Lambda$ and therefore the corollary holds for any Λ.

$\to$ covol$(\Lambda) = \mu(F)$ for any fundamental region F for Λ, so the ratio $\mu(S)/$covol$(\Lambda) = \mu(S)/\mu(F)$ and therefore it odes not depend on the normalisation of the Haar measure μ.

By the chapter 3, the unit group $K_{\mathbb{R}}^\times \equiv K \otimes_{\mathbb{Q}} \mathbb{R}$ is the locally compact group $K_{\mathbb{R}}^\times \cong \Pi_{v|\infty} K_v^\times \cong \Pi_{v|\infty,\text{real}}\mathbb{R}^\times \times \Pi_{v|\infty,\text{complex}}\mathbb{C}^\times$. There is a continuous homomorphism $\begin{array}{rcl} \text{Log}: K_{\mathbb{R}}^\times & \to & \mathbb{R}^{r+s} \\ (x_v) & \mapsto & (\log||x_v||_v) \end{array}$ and we have an exact sequence of abelian groups

$1 \longrightarrow \mu_K \longrightarrow \mathcal{O}_K^\times \xrightarrow{Log} \Lambda_K \longrightarrow 0$ in which Λ_K is a lattice in the trace-zero hyperplane $\mathbb{R}_0^{r+s} \equiv \{x \in \mathbb{R}^{r+s} : T(x) = 0\}$. The regulator $R_K = \text{covol}(\Lambda_K)$ in $\mathbb{R}_0^{r+s}$ where it is equipped with the Euclidean measure induced by any coordinate projection $\mathbb{R}^{r+1} \to \mathbb{R}^{r+s-1}$. Then by the Dirichlet unit theorem, $\mathcal{O}_K^\times = U \otimes \mu_K$ where $U \subseteq \mathcal{O}_K^\times$ is free of rank $r+s-1$.

Consider the principal ideals of $\mathcal{O}_K$. Then $\forall \alpha, \alpha' \in K$. $(\alpha) = (\alpha')$ iff $\alpha/\alpha' \in \mathcal{O}_K^\times$ and therefore $\{(\alpha) : N(\alpha) \le t\} = \{\alpha \in K^\times \cap \mathcal{O}_K : N(\alpha) \le t\}/\mathcal{O}_K^\times$. If we define $K_{\mathbb{R},\le t}^\times \equiv \{x \in K_{\mathbb{R}}^\times : N(x) \le t\} \subseteq K_{\mathbb{R}}^\times \subseteq K_{\mathbb{R}}$, the cardinality of $\{\alpha \in K^\times \cap \mathcal{O}_K : N(\alpha) \le t\}/\mathcal{O}_K^\times$ is equivalent to the cardinality of the finite set $(K_{\mathbb{R},\le t}^\times \cap \mathcal{O}_K)/\mathcal{O}_K^\times$ where the intersection takes place in $K_{\mathbb{R}}$ and produces a subset of $K_{\mathbb{R}}^\times$ that we partition into equivalence classes modulo $\mathcal{O}_K^\times$. We have a w_K to 1 map $(K_{\mathbb{R},\le t}^\times \cap \mathcal{O}_K)/U \to (K_{\mathbb{R},\le t}^\times \cap \mathcal{O}_K)/\mathcal{O}_K^\times$ where U is defined above. Then we can get the cardinality by getting the one for $(K_{\mathbb{R},\le t}^\times \cap \mathcal{O}_K)/U$ and divide it by w_K. For $x = (x_v) \in K_{\mathbb{R}}^\times$, the norm map $\begin{array}{rcl} N : K_{\mathbb{R}}^\times & \to & \mathbb{R}_{++}^\times \\ x & \mapsto & \Pi_{v\text{real}}|x_v|_{\mathbb{R}} \Pi_{v\text{complex}}|x_v|_{\mathbb{C}}^2 \end{array}$ satisfies $\mathrm{T}(\mathrm{Log}(x)) = \log N(x)$, $\forall x \in K_{\mathbb{R}}^\times$. Define a surjective homomorphism $\begin{array}{rcl} \nu : K_{\mathbb{R}}^\times & \to & K_{\mathbb{R},1}^\times \\ x & \mapsto & xN(x)^{-1/n} \end{array}$. Then the image of $K_{\mathbb{R},1}^\times$ under the Log map is precisely the trace zero hyperplane $\mathbb{R}_0^{r+s}$ in $\mathbb{R}^{r+s}$ in which $\mathrm{Log}(U) = \mathrm{Log}(\mathcal{O}_K^\times) = \Lambda_K$ is a lattice. Fix a fundamental domain F for the lattice Λ_K in $\mathbb{R}_0^{r+2}$ so that $S \equiv \nu^{-1}(\mathrm{Log}^{-1}(F))$ is a set of unique coset representatives for the quotient $K_{\mathbb{R}}^\times/U$. If we define $S_{\le t} \equiv \{x \in S : N(x) \le t\} \subset K_{\mathbb{R}}$, we want to estimate the cardinality of the finite set $S_{\le t} \cap \mathcal{O}_K$. Here $tS_{\le 1} = S_{\le t^n}$ so we can estimate the cardinality of $S_{\le t} = t^{1/n} S_{\le 1}$ via the previous corollary with $S = S_{\le 1}$ and $\Lambda = \mathcal{O}_K$ by replacing t with $t^{1/n}$, provided that the boundary of $S_{\le 1}$ is $(n-1)$-Lipschitz parametrisable.

CLAIM : $\partial S_{\le 1}$ is $(n-1)$-Lipshcitz parametrisable.
pf. The Kernel of the Log map is $\{\pm 1\}^r \times U(1)^s$, where $U(1) \equiv \{z \in \mathbb{C} : z\bar{z} = 1\}$ is the unit circle in $\mathbb{C}$. Thus $\begin{array}{rcl} K_{\mathbb{R}}^\times = (\mathbb{R}^\times)^r \times (\mathbb{C}^\times)^s & \to & \mathbb{R}^{r+s} \times \{\pm 1\}^r \times [0, 2\pi)^s \\ x & \mapsto & (Log x) \times (sgn x_1, \ldots, sgn x_n) \times (\arg z_1, \ldots, \arg z_s) \end{array}$ is a continuous isomorphism of locally compact groups where $\left\{ \begin{array}{ll} \to \mathbb{R}^{r+s} & \text{Log map} \\ \to \{\pm 1\}^r & \text{vector of signs of the real components} \\ \to [0.2\pi)^s & \text{the vector of angles s.t. } z/|z| = e^{i \arg z} \end{array} \right.$. The set $S_{\le 1}$ consists of 2^r connected components, one for each element of $\{\pm 1\}^r$. We can parametrise each of these using n real parameters; $r+s-1$ parameters in $[0,1)$ that encode a point in F as an $\mathbb{R}$-linear combination of $Log\epsilon_1 \ldots, Log\epsilon_{s+r-1}$ where $\{\epsilon_i\}$ is a basis for U, and a parameter in $(0,1]$ that encodes the nth root of the norm. Additionally $\exists s$ parameters in $[0,1)$ that encode an element of $U(1)^s$. These define a continu-

ously differentiable bijection from the set $C = [0,1)^{n-1} \times (0,1] \subseteq [0,1]^n$ s.t. ∂C is the boundary of the unit n-cube and therefore it is $(n-1)$-Lipschitz parametrisable. Thus each component of $S_{\leq 1}$ and itself is clearly $(n-1)$-Lipschitz parametrisable. Thus by the previous corollary, $\#S_{\leq t} \cap \mathcal{O}_K) = \frac{\mu(S_{\leq 1})}{\mathrm{covol}\mathcal{O}_K}(t^{1/n})^n + O((t^{1/n})^{n-1}) = \frac{\mu(S_{\leq 1}}{|D_K|^{1/2}}t + O(t^{1-1/n})$.

- $\mu(S_{\leq 1}) = 2^r(2\pi)^s R_K$

  We want to get μ on $K_{\mathbb{R}}$ the Haar measure with the parametrisation. In terms of the standard Lebesgue measure $\begin{cases} dx & \text{on } \mathbb{R} \\ dA & \text{on } \mathbb{C} \end{cases}$, $\mu = (dx)^r(2dA)^s$.

  For each factor of $K_{\mathbb{R}}^\times \cong (\mathbb{R}^\times)^r \times (\mathbb{C}^\times)^s$, define the maps $\begin{array}{rcl} \mathbb{R}^\times & \to & \mathbb{R} \times \{\pm 1\} \\ x & \mapsto & (\log|x|, sgnx) \\ \pm e^l & \mapsto & (l, \pm 1) \\ dx & \mapsto & e^l dl \mu_{\pm 1} \end{array}$,

  $\begin{array}{rcl} \mathbb{C}^\times & \to & \mathbb{R} \times [0, 2\pi) \\ z & \mapsto & (2\log|z|, argz) \\ e^{l/2+i\theta} & \mapsto & (l, \theta) \\ adA & \mapsto & 2e^{l/2}(de^{l/2})d\theta = d^l d; d\theta \end{array}$ where

  $\begin{cases} dl \text{ is the Lebesgue measure on } \mathbb{R} \\ \mu_{\pm 1} \text{ is the counting measure on } \{\pm 1\} \\ d\theta \text{ is the Lebesgue measure on } [0, 2\pi) \end{cases}$. Thus there exists a new parametrisation $\begin{array}{rcl} \mathbb{R}^{r+s} & \to & \mathbb{R}^{r+s-1} \times \mathbb{R} \\ x & \mapsto & (x_1, \ldots, x_{r+s-1}, y \equiv T(x)) \\ e^{T(x)}\mu_{\mathbb{R}^{r+s}} & \mapsto & e^y \mu_{\mathbb{R}^{r+s-1}} dy \end{array}$. If we let $\pi_0 : \mathbb{R}^{r+s} \to \mathbb{R}^{r+s-1}$ denote the coordinate projection, then the Lebesgue measure of $\pi_0(F)$ in $\mathbb{R}^{r+s-1}$ is the regulator R_K. The Log map gives a bijection $\begin{array}{rcl} R_{\leq 1} & \to & F + (-\infty, 0](\frac{1}{,}1, \ldots, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{2}{n}) \\ x = N(x)^{1/n}v(x) & \mapsto & \mathrm{Lod}v(x) + \log N(x)(\frac{1}{n}, \ldots, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{2}{n}) \end{array}$. Since $y = T(\mathrm{Log}x) = \log N(x)$, we can view $S_{\leq 1}$ as an infinite union of cosets of $\mathrm{Log}^{-1}(F)$ parametrised by $e^y = N(x) \in (0,1]$, Then under the change of coordinate, $\begin{array}{rcl} K_{\mathbb{R}}^\times & \to & \mathbb{R}^{r+s-1} \times \{\pm 1\}^r \times [0, 2\pi)^s \\ S_{\leq 1} & \to & \pi_0(F) \times (-\infty, 0] \times \{\pm 1\}^r \times [0, 2\pi)^s \end{array}$. $\mu(S_{\leq 1}) = \int_{-\infty}^0 e^y R_K 2^r(2\pi)^s dy = 2^r(2\pi)^s R_K$.

$\to \#(S_{\leq t} \cap \mathcal{O}_K) = \frac{\mu(S_{\leq 1})}{\mathrm{covol}(\mathcal{O}_K)}(t^{1/n})^n + O((t^{1/n})^{1-\frac{1}{n}}) = (\frac{\mu(S_{\leq 1})}{|D_K|^{1/2}})t + O(t^{1-1/n})$.

Theorem 4.1.8. *Let K be a number field of degree n. As $t \to \infty$, the number of nonzero $\mathcal{O}_K$-ideals α of absolute norm $N(\alpha) \leq t$ is*

$$(\frac{2^r(2\pi)^s h_K R_K}{w_K |D_K|^{1/2}})t + O(t^{1-1/n})$$

where r,s are the number of real and complex places of K, respectively, $h_K \equiv \#cl\mathcal{O}_K$ is the class number, R_K is the regulator, $w_K = \#\mu_K$ is the number of roots of unity and $D_K \equiv disc\mathcal{O}_K$.

pf. Group the nonzero $\mathcal{O}_K$-ideals by ideal class.

- For the trivial class, we just need to count nonzero principal ideals (α), equivalently, the number of nonzero $\alpha \in \mathcal{O}_K$ with $N(\alpha) \le t$, $\mod \mathcal{O}_K^\times$. Dividing $\#(S_{\le t} \cap \mathcal{O}_K) = (\frac{2^r(2\pi)^s R_K}{|D_K|^{1/2}})t + O(t^{1-1/n})$ by w_K to account for the w_k to 1 map $S_{\le t} \cap \mathcal{O}_K \to (K^\times_{\mathbb{R},\le t} \cap \mathcal{O}_K)/\mathcal{O}_K^\times$, we get $\#\{(\alpha) \subseteq \mathcal{O}_K : N(\alpha) \le t\} = (\frac{2^r(2\pi)^s R_K}{w_K|D_K|^{1/2}})t + O(t^{1-1/n})$.

- For a general ideal class $[\alpha]$, fix one with $\alpha \subseteq \mathcal{O}_K$ nonzero. Multiplication by α gives a bijection $\{$ ideals $\beta \in [\alpha^{-1}] : N(\beta) \le t\} \to \{$ nonzero principal ideals $(\alpha) \subseteq \alpha : N(\alpha) \le tN(\alpha)\}$ where the RHS is equivalent to the set $\{$ nonzero $\alpha \in \alpha : N(\alpha) \le tN(\alpha)\}/\mathcal{O}_K^\times$. Let $S_{[\alpha],\le t}$ denote the set of the RHS. The above estimate and the previous corollary gives $\#S_{[\alpha],\le t} = (\frac{2^r(2\pi)^s R_K}{w_K \mathrm{covol}(\alpha)})tN(\alpha) + O(t^{1-1/n}) = (\frac{2^r(2\pi)^s R_K}{w_K \mathrm{covol}(\mathcal{O}_K)N(\alpha)})tN(\alpha) + O(t^{1-1/n}) = (\frac{2^r(2\pi)^s R_K}{w_K|D_K|^{1/2}})t + O(t^{1-1/n})$ since $\mathrm{covol}(\alpha) = N(\alpha)\mathrm{covol}(\mathcal{O}_K)$. Then the RHS does not depend on the ideal class $[\alpha]$. Summing over the ideal classes gives $\#\{$ nonzero ideals $\beta \subseteq \mathcal{O}_K : N(\beta) \le t\} = \sum_{[\alpha]\in \mathrm{cl}(\mathcal{O}_K)} \#S_{[\alpha],\le t} = (\frac{2^r(2\pi)^s h_K R_K}{w_K|D_K|^{1/2}})t + O(t^{1-1/n})$.

Lemma 4.1.10. *Let $a_1, a_2, \ldots$ be a sequence of complex numbers and let σ be a real number. Suppose $\sum_{i=1}^t a_i = O(t^\sigma)$ as $t \to \infty$. Then the Dirichlet series $\sum a_n n^{-s}$ defines a holomorphic function on $Re(s) > \sigma$.*

pf. Let $A(x) \equiv \sum_{0<n\le x} a_n$. Writing the Dirichlet sum as a Stieljtes integral gives $\sum a_n n^{-s} = \int_{1^-}^\infty x^{-s} dA(x) = \frac{A(x)}{x^s}|_{1^{-1}}^\infty - \int_{1^{-1}}^\infty A(x) dx^{-s} = (0-0) - \int_{1^-}^\infty A(x)(-sx^{-s-1})dx = s\int_{1^{-1}}^\infty \frac{A(x)}{x^{s+1}}dx$ converges locally uniformly on $\mathrm{Re}(s) > \sigma$ and the lemma follows.

Lemma 4.1.11. *Let $a_1, a_2, \ldots,$ be a sequence of complex numbers that satisfies $\sum_{i=1}^t a_i = \rho t + O(t^\sigma)$ as $t \to \infty$ for some $\sigma \in [0,1)$ and $\rho \in \mathbb{C}^\times$. The Dirichlet series $\sum a_n n^{-s}$ converges on $Re(s) > 1$ and has a meromorphic continuation to $Re(s) > \sigma$ that is holomorphic except for a simple pole at $s = 1$ with residue ρ.*

pf. Define $b_n \equiv a_n - \rho$. Then $b_1 + \cdots + b_t = O(t^\sigma)$ and $\sum a_n n^{-s} = \rho \sum n^{-s} + \sum b_n n^{-s} = \rho\zeta(s) + \sum b_n n^{-s}$. Here $\zeta(s)$ is holomorphic on $\mathrm{Re}(s) > 1$ and has a meromorphic continuation ot $\mathrm{Re}(s) >$ that is holomorphic except for a simple pole at 1 with residue 1. By the previous lemma, $\sum b_n n^{-s}$ is holomorphic on $\mathrm{Re}(s) > \sigma$ and since $\sigma < 1$, it is holomorphic at $s = 1$. SO the RHS has a meromorphic continuation to $\mathrm{Re}(s) > \sigma$ that is holomorphic except for the simple pole at $s = 1$ coming from $\zeta(s)$ and the residue at $s = 1$ is $\rho \cdot 1 + 0 = \rho$.

Theorem 4.1.9 (Analytic Class Number Formula)**.** *Let K be a number field of degree n. The Dedekind zeta function $\zeta_K(z)$ extends to a meromorphic function on $Re(z) > 1 - \frac{1}{n}$ that is holomorphic except for a simple pole at $z = 1$ with residue*

$$\lim_{z \to 1^+} (z-1)\zeta_K(z) = \frac{2^r (2\pi)^s h_K R_K}{w_K |D_K|^{1/2}}$$

where $\begin{cases} r & \text{the number of real places of } K \\ s & \text{the number of complex places of } K \end{cases}$, *$R_k$ is the regulator, $h_K \equiv \#cl\mathcal{O}_K$, $w_K \equiv \#\mu_K$, $D_K \equiv disc\mathcal{O}_K$.*

pf. $\zeta_K(z) = \sum_\alpha N(\alpha)^{-z} = \sum_{t \geq 1} a_t t^{-z}$ where α ranges over nonzero ideals of $\mathcal{O}_K$ and $a_t \equiv \#\{\alpha : N(\alpha) = t\}$ with $t \in \mathbb{Z}_{++}$. If we define $\rho_K \equiv \frac{2^r (2\pi)^s h_K R_K}{w_K |D_K|^{1/2}}$, by the previous theorem, $\sum a_i = \#\{\alpha : N(\alpha) \leq t\} = \rho_K t + O(t^{1-1/n})$ as $t \to \infty$. By the previous lemma with $\sigma = 1 - 1/n$, $\zeta_K(z) = \sum a_t t^{-z}$ extends to a meromorphic function on $\mathrm{Re}(s) > 1 - 1/n$ that is holomorphic except for a simple pole at $z = 1$ with residue ρ_K.

Given any finite set of primitive Dirichlet characters, if we let m be the lcm of their conductors and consider the subgroup H of $X(m)$ they generate, we may associate to H the subfield $K \equiv \mathbb{Q}(\zeta_m)^{\phi(H)}$ where $\phi(H) \equiv \{\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) : \chi(\sigma) = 1, \forall \chi \in H\}$. Then we may regard H as the character group of $\mathrm{Gal}(K/\mathbb{Q})$. The same applies if we replace m with any multiple m', since $H \subseteq X(m) \subseteq X(m')$ for all $m|m'$ and we will get the same field $K \subseteq \mathbb{Q}(\zeta_m) \subseteq \mathbb{Q}(\zeta_{m'})$. Conversely, for each subfield K of a cyclotomic field $\mathbb{Q}(\zeta_m)$, there is a corresponding subgroup $H \equiv \{\chi \in X(m) : \chi(\sigma) = 1, \ \forall \sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_m)/K)\}$ for which $K = \mathbb{Q}(\zeta_m)^{\phi(H)}$. Then Since $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ is abelian, $K/\mathbb{Q}$ is Galois and we may view H as the character group of $\mathrm{Gal}(K/\mathbb{Q})$.
→There is an one-to-one correspondence between subgroups $H \subseteq X(m)$ and subfields of $K \subseteq \mathbb{Q}(\zeta_m)$ in which H corresponds to the character group of $\mathrm{Gal}(K/\mathbb{Q})$ and $K = \mathbb{Q}(\zeta_m)^{\phi(H)}$.

Proposition 4.1.2. *Let p be a prime, m be a positive integer and $m' = m/p^{v_p(m)}$. Then $\mathbb{Q}(\zeta_{m'})$ is the maximal extension of $\mathbb{Q}$ in $\mathbb{Q}(\zeta_m)$ unramified at p. In particular, if p odes not divide m, then $\mathbb{Q}(\zeta_m)$ is unramified at p.*

pf. The extension $\mathbb{Q}_p(\zeta_{m'})/\mathbb{Q}_p$ is unramified. Thus $\mathbb{Q}(\zeta_{m'})/\mathbb{Q}$ is unramified at p. This applies to any prime $q \neq p$ dividng m. Thus the extension $\mathbb{Q}(\zeta_{p^{v_p(m)}})$ is ramified only at p. Then there are no nontrivial unramified extensions of $\mathbb{Q}$, so every subfield of $\mathbb{Q}(\zeta_{p^{v_p(m)}})$ that properly contains $\mathbb{Q}$ is ramified at p. Now $\mathbb{Q}(\zeta_m)$ is the compositum of $\mathbb{Q}(\zeta_{p^{v_p(m)}})$ and $\mathbb{Q}(\zeta_{m'})$, which intersect in $\mathbb{Q}$, so any nontrivial

extension $\mathbb{Q}(\zeta_{m'})$ in $\mathbb{Q}(\zeta_m)$ contains a subfield of $\mathbb{Q}(\zeta_{p^{v_p(m)}})$ properly containing $\mathbb{Q}$ which must be ramified at p. Thus the proposition follows.

Theorem 4.1.10. *Let $H \subseteq X(m)$ be a group of primitive Dirichlet characters and let $K = \mathbb{Q}(\zeta_m)^{\phi(H)}$ be the corresponding subfield of $\mathbb{Q}(\zeta_m)$ with $\phi(H)$ defined as above. Then*

$$\zeta_K(x) = \Pi_{\chi\in H} L(s,\chi)$$

pf. $\begin{cases} \zeta_K(x) = \Pi_{\mathbf{p}}(1-N(\mathbf{p})^{-s})^{-1} = \Pi_p \Pi_{\mathbf{p}|p}(1-N(\mathbf{p})^{-s})^{-1} \\ \Pi_{\chi\in H} L(s,\chi) = \Pi_{\chi\in H}\pi_p(1-\chi(p)p^{-s})^{-1} = \Pi_p \Pi_{\chi\in H}(1-\chi(p)p^{-s})^{-1} \end{cases}$. Since $K/\mathbb{Q}$ is Galois, $[K:\mathbb{Q}] = e_p f_p g_p$. $\Pi_{\mathbf{p}|p}(1-N(\mathbf{p})^{-s}) = (1-(p^{f_p})^{-s})^{g_p} = (1-(p^{-s})^{f_p})^{g_p}$ which does not change if we replace K with the maximal subfield K' of K in which p is unramified. Also on the $\Pi_{\chi\in H}(1-\chi(p)p^{-s})$, any $\chi \in H$ with conductor divisible by p, $\chi(p) = 0$ and therefore we can replace H with the subgroup H' of Dirichlet characters with conductors prime to p. Then by the previous proposition, $K' = \mathbb{Q}(\zeta_m)^{\phi(H')}$. Thus wlog assume $p \nmid m$ so K is unramified at p and $\#H = [K:\mathbb{Q}] = f_p g_p$. Since $K/\mathbb{Q}$ is abelian, and unramified at p, the Artin map gives a Frobenius element σ_p corresponding to the Frobenius automorphism $x \mapsto x^p$ of the residue field, which by definition has order f_p, so σ_p has order f_p in $\mathrm{Gal}(K/\mathbb{Q})$. Viewing H as a character group of $\mathrm{Gal}(K/\mathbb{Q})$, the map $\chi \mapsto \chi(\sigma_p)$ defines a surjective homomorphism from H to the group of f_pth roots of unity $\alpha \in U(1)$ and the kernel of this map has cardinality $\#H/f_p = g_p$. Thus $\Pi_{\chi\in H}(1-\chi(p)p^{-s}) = \Pi_{\alpha^{f_p}=1}(1-\alpha p^{-s})^{g_p} = (1-(p^{-s})^{f_p})^{g_p}$ by the identity $\Pi_{\alpha^{f_p}=1}(1-\alpha T) = 1-T^{f_p} \in \mathbb{C}[T]$. Thus they are equivalent.

Theorem 4.1.11. *Let ψ be any nonprincipal Dirichlet character. Then $L(1,\psi) \neq 0$.*

pf. Let ψ be a nonprincipal Dirichlet character, say of modulus m. Then ψ is induced by a nontrivial primitive Dirichlet character $\tilde{\psi}$ of conductor $\tilde{m}$ dividing m. The L-functions of $\psi, \tilde{\psi}$ differ at only finitely many Euler factors $(1-\psi(p)p^{-s})^{-1}$, which corresponds to primes p dividing $m/\tilde{m}$, and these factors are clearly nonzero at $s = 1$,given $p > 1$. Thus assume wlog $\psi = \tilde{\psi}$ is primitive. Let K be the mth cyclotomic field $\mathbb{Q}(\zeta_m)$. Then by the previous theorem, $\zeta_K(x) = \Pi_\chi L(s,\chi)$ where χ ranges over the primitive Dirichlet characters of conductor dividing m, including ψ. By the analytic class number formula, the LHS has a simple pole at $s = 1$, and the same must be true for the RHS. Then $\mathrm{ord}_{s=1}\zeta_K(s) = \mathrm{ord}_{s-1}\Pi_\chi L(s,\chi) \Leftrightarrow -1 = \mathrm{ord}_{s=1}L(s,\mathbb{1})\Pi_{\chi\neq\mathbb{1}}L(s,\chi) = \mathrm{ord}_{s=1}\zeta(s)\Pi_{\chi\neq\mathbb{1}}L(s,\chi) = -1 + \sum_{\chi\neq\mathbb{1}}\mathrm{ord}_{s=1}L(s,\chi)$. For nonprincipal χ, $L(s,\chi)$ is holomorphic on $\mathrm{Re}(s) > -$ thus $\mathrm{ord}_{s=1}L(s,\chi) \geq 0$, $\forall\chi$ appearing in the sum, which can therefore be zero iff every term is zero. So $L(1,\chi) \neq 0$ for every nontrivial primitive Dirichlet character χ of conductor dividing m, including ψ.

Theorem 4.1.12 (Dirichlet)**.** *For all coprime integers a, m, there are infinitely many primes $p \equiv a \mod m$.*

pf. Construct the indicator function $\frac{1}{\phi(m)} \sum_{\chi \in X(m)} \chi(p/a) = \begin{cases} 1 & p \equiv a \mod m \\ 0 & otherwise \end{cases}$ where p/a is computed modulo m and χ ranges over primitive Dirichlet characters of conductor dividing m, which we identify with the character group of $(\mathbb{Z}/m\mathbb{Z})^\times$ via the corollary 4.1.4. As $s \to 1^+$,

$$\sum_{p \equiv a \mod m} p^{-s} = \sum_p p^{-s} \frac{1}{\phi(m)} \sum_{\chi \in X(m)} \chi(p/a) = \sum_{\chi \in X(m)} \frac{\chi(1/a)}{\phi(m)} \sum_p \chi(p) p^{-s}$$
$$= \sum_{\chi \in X(m)} \frac{\chi(1/a)}{\phi(m)} (\log L(s,\chi) + O(1)) = \frac{\log \zeta(s)}{\phi(m)} + \sum_{\chi \in X(m), \chi \neq \mathbb{1}} \frac{\chi(1/a)}{\phi(m)} \log L(s,\chi) + O(1)$$

So long as χ is not principal, $L(1,\chi) \neq 0$ by the previous theorem. Then $\log L(s,\chi) = O(1)$ as $s \to 1^+$ ad therefore $\sum_{p \equiv a \mod m} p^{-s} = \frac{\log \zeta(s)}{\phi(m)} + O(1)$ is unbounded as $s \to 1^+$, since $\zeta(s)$ is. Moreover, by the Merten's second theorem implies $\sum_{p \leq x, p \equiv a \mod m} \frac{1}{p} \sim \frac{\log \log x}{\phi(m)}$ which proves there are infinitely many primes $p \equiv a \mod m$.

Definition 4.1.13. *The Dirichlet density of a set of primes S is given by*

$$d(S) \equiv \lim_{s \to 1^+} \frac{\sum_{p \in S} p^{-s}}{\sum_p p^{-s}}$$

defined whenever this limit exists.

$\Leftrightarrow \delta(S) \equiv \lim_{x \to \infty} \frac{\#\{p \leq x : p \in S\}}{\#\{p \leq x\}}$ the natural density.

The Dirichlet density of the set $S = \{p \equiv a \mod m\}$ is $d(S) = \lim \frac{\sum_{p \in S} p^{-s}}{\sum_p p^{-s}} = \lim \frac{\log \zeta(s)/\phi(m)}{\log \zeta(s)} = \frac{1}{\phi(m)}$, thus the primes are equidistributed modulo $m$in the sense that $\forall a \perp m$, $\sum_{p \leq x, p \equiv a \mod m} \frac{1}{p} \sim \frac{1}{\phi(m)} \sum_{p \leq x} \frac{1}{p} \sim \frac{1}{\phi(m)} \log \log x$.

4.1.3 Kronecker-Weber Theorem

Suppose $K/\mathbb{Q}$ is any arbitrary finite abelian extension. Then the character group of $\mathrm{Gal}(K/\mathbb{Q})$ correspond to a group of Dirichlet characters and we can factor the Dedekind zeta function $\zeta_K(x)$ as a product of Dirichlet L-functions since every finite abelian extension of $\mathbb{Q}$ lies in a cyclotomic filed by the Kronecker-Weber theorem.

Theorem 4.1.13 (Kronecker-Weber)**.** *Every abelian extension of $\mathbb{Q}$ is contained in $\mathbb{Q}(\zeta_n)$ for some n, where ζ_n is a primitive nth root of unity.*

pf.

CLAIM : Every fnite abelian extension of $\mathbb{Q}_p$ lies in a cyclotomic field $\mathbb{Q}_p(\zeta_m)$.

pf.

- When $K/\mathbb{Q}_p$ of prime power degree l, $l \neq p$
  Let F be the maximal unramified extension of $\mathbb{Q}_p$ in K. Then $F = \mathbb{Q}_p(\zeta_n)$ for some $n \in \mathbb{Z}_{\geq 1}$. The extension K/F is totally ramified, and it must be tamely ramified, given the hypothesis $l \neq p$. Then $K = F(\pi^{1/e})$ for some uniformiser $\pi, e = [K : F]$. We may assume $\pi = -pu$, $u \in \mathcal{O}_K^\times$ since $F/\mathbb{Q}_p$ is unramified. If $q|p$ is the maximal of $\mathcal{O}_F$, the valuation v_q extends v_p with index 1, so $v_q(-pu) = v_p(-p) = 1$. Then the field $K = F(\pi^{1/e})$ lies in the compositum of $F((-p)^{1/e})$ and $F(u^{1/e})$. The extension $F(u^{1/e})/F$ is unramified since $v_q(disc(x^e - u)) = 0$, $p \not| e$, so $F(u^{1/e})/\mathbb{Q}_p$ is unramified and $F(u^{1/e}) = \mathbb{Q}_p(\zeta_k) \to K(u^{1/e}) = K \cdot \mathbb{Q}_p(\zeta_k)$ is a compositum of abelian extension so $K(u^{1/e})/\mathbb{Q}_p$ is abelian and it contains the subextension $\mathbb{Q}_p((-p)^{1/e})/\mathbb{Q}_p$ which must be Galois, since it lies in an abelian extension and totally ramified, since it is an Eisenstein extension. The field $\mathbb{Q}_p((-p)^{1/e})$ contains ζ_e and is totally ramified, but $\mathbb{Q}_p(\zeta_e)/\mathbb{Q}_p$ is unramified, given $p \not| e$, so $\mathbb{Q}_p(\zeta_e) = \mathbb{Q}_p$. Thus $e|(p-1)$ and $\mathbb{Q}_p((-p)^{1/e}) \subseteq \mathbb{Q}_p((-p)^{1/(p-1)}) = \mathbb{Q}_p(\zeta_p)$. Thus $F((-p)^{1/e}) = F \cdot \mathbb{Q}_p((-p)^{1/e}) \subseteq \mathbb{Q}_p(\zeta_n) \cdot \mathbb{Q}_p(\zeta_p) \subseteq \mathbb{Q}_p(\zeta_{np})$. Thus $K \subseteq \mathbb{Q}_p(\zeta_{knp})$.

  For any prime p, $\mathbb{Q}_p((-p)^{1/(p-1)}) = \mathbb{Q}_p(\zeta_p)$.
  pf. Let $\alpha = (-p)^{1/(p-1)}$. Then it is a root of the Eisenstein polynomial $x^{p-1} + p$, so extension $\mathbb{Q}_p((-p)^{1/(p-1)}) = \mathbb{Q}_p(\alpha)$ is totally ramified of degree $p-1$ and α is a uniformizer. Let $\pi = \zeta_p - 1$. The minimal polynomial of π is $f(x) \equiv \frac{(x+1)^p - 1}{x} = x^{p-1} + px^{p-2} + \cdots + p$, which is Eisenstein, so $\mathbb{Q}_p(\pi) = \mathbb{Q}_p(\zeta_p)$ is also totally ramified of degree $p-1$ and π is a uniformizer. We have $u \equiv -\pi^{p-1}/p \equiv 1 \mod \pi$ so u is a unit in $\mathbb{Q}_p(\zeta_p)$. Let $g(x) = x^{p-1} - u$, $g(1) \equiv 0 \mod \pi, g'(1) \equiv p - 1 \not\equiv 0 \mod \pi$, so by Hensel's Lemma I, we can lift 1 to a root β of $g(x)$ in $\mathbb{Q}_p(\zeta_p)$. Then $p\beta^{p-1} = pu = -\pi^{p-1} \to (\pi/\beta)^{p-1} + p = 0$ and therefore $\pi/\beta \in \mathbb{Q}_p(\zeta_p)$ is a root of the minimal polynomial of α. Since $\mathbb{Q}_p(\zeta_p)$ is Galois, this implies $\alpha \in \mathbb{Q}_p(\zeta_p)$, given $\mathbb{Q}_p(\alpha), \mathbb{Q}_p(\zeta_p)$ having degree $p-1$ s.t. two fields coincide.

- $l = p > 2$
  There are two candidates for K, the cyclotomic field $\mathbb{Q}_p(\zeta_{p^{p^r}-1})$ which is an unramified extension of degree p^r and the index $p-1$ subfield of the cyclotomic field $\mathbb{Q}_p(\zeta_{p^{r+1}})$ which is a totally ramified extension of degree p^r. If K is contained in the compositum of these two fields, then $K \subseteq \mathbb{Q}_p(\zeta_m)$, $m = (p^{p^r} - 1)p^{r+1}$ and the claim follows. Otherwise, the field $K(\zeta_m)$ is a Galois

extension of $\mathbb{Q}_p$ with $Gal(K(\zeta_m)/\mathbb{Q}_p) \cong \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/p^s\mathbb{Z}$, where the first factor comes from the Galois group of $\mathbb{Q}_p(\zeta_{p^{p^r}-1})$, the second two factor come from the Galois group of $\mathbb{Q}_p(\zeta_{p^{r+1}})$ and the last factor comes from the hypothesis $K \not\subseteq \mathbb{Q}_p(\zeta_m)$, so $Gal(K(\zeta_m)/\mathbb{Q}_p(\zeta_m))$ is nontrivial and must have order $p^s, s \geq 1$. The abelian group $Gal(K(\zeta_m)/\mathbb{Q}_p)$ has a quotient isomorphic to $(\mathbb{Z}/p\mathbb{Z})^3$ and the subfield of $K(\zeta_m)$ corresponding to this quotient is an abelian extension of $\mathbb{Q}_p$ with Galois group $(\mathbb{Z}/p\mathbb{Z})^3$, which is impossible, given it is not cyclic.

For odd p, every totally wildly ramified Galois extension of $\mathbb{Q}_p$ is cyclic.
pf.
If $Gal(K/\mathbb{Q}_p) \cong (\mathbb{Z}/p\mathbb{Z})^3$, $G \equiv Gal(K/\mathbb{Q}_p)$ as the internal direct sum of the inertial subgroup $I \leq G$ and a cyclic subgroup $H \leq G$, since L^I is an unramified, and hence cyclic extension of $\mathbb{Q}_p$ with Galois group isomorphic to $G/I \cong H$. But then L^H is totally wildly ramified abelian extension of $\mathbb{Q}_p$ whose Galois group G/H is not cyclic, a contradiction.

- $l = p = 2$
The unramified cyclotomic field $\mathbb{Q}_2(\xi_{2^{2^r}-1})$ has Galois group $\mathbb{Z}/2^r\mathbb{Z}$, and the totally ramified cyclotomic field $\mathbb{Q}_2(\xi_{2^{r+2}})$ has Galois group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^r\mathbb{Z}$. Let $m = (2^{2^r}-1)2^{r+2}$. If K is not contained in $\mathbb{Q}_2(\xi_m)$, $Gal(K(\xi_m)/\mathbb{Q}_2) \cong$ $\begin{array}{ll} \mathbb{Z}/2\mathbb{Z} \times (\mathbb{Z}/2^r\mathbb{Z})^2 \times \mathbb{Z}/2^s\mathbb{Z} & 1 \leq s \leq r \\ (\mathbb{Z}/2^r\mathbb{Z})^2 \times \mathbb{Z}/2^s\mathbb{Z} & 2 \leq s \leq r \end{array}$ Thus it admits a quotient isomorphic to $(\mathbb{Z}/2\mathbb{Z})^4$ or $(\mathbb{Z}/4\mathbb{Z})^3$. Then since no extension of $\mathbb{Q}_2$ has either of these Galois groups, K must lie in $\mathbb{Q}_2(\xi_m)$.
* **No extension of $\mathbb{Q}_2$ has Galois group isomorphic to $(\mathbb{Z}/2\mathbb{Z})^4$ or $(\mathbb{Z}/4\mathbb{Z})^3$**
pf. There are 7 quadratic extensions of $\mathbb{Q}_2$, which has no Galois group $(\mathbb{Z}/2\mathbb{Z})^4$ since this group has 15 subgroups of index 2, whose fixed fields would yield 15 distinct quadratic extension of $\mathbb{Q}_2$.
Also there are only finitely many extensions of $\mathbb{Q}_2$ of any fixed degree d and these can be enumerated by considering Eisenstein polynomials in $\mathbb{Q}_2[x]$ of degrees dividing d up to an equivalence relation by Krasner's lemma. Then no extension of $\mathbb{Q}_2$ has Galois group $(\mathbb{Z}/4\mathbb{Z})^3$, since it has 28 subgroups whose fixed fields would yield 28 distinct cyclic quartic extensions of $\mathbb{Q}_2$.

4.2 Cohomologies and Local Class Field Theory

4.2.1 Cohomologies

Lemma 4.2.1. *For each $x \in L$,* $\begin{array}{rcl} H_x : L & \to & L \\ y & \mapsto & x \cdot \sigma(y) \end{array}$ *which is a linear map of K-vector spaces. Then the characteristic polynomial of H_x is $t^n - N(x) \in K[t]$, where we have normalised the definition of the characteristic polynomial to be monic.*

pf. H_x^n corresponds to multiplication by $N(x)$ since $H_x^n(y) = x \cdot \sigma(x \cdot \sigma(x \dots \sigma(y) \dots) = x \cdot \sigma(x) dots \sigma^{n-1}(x) \sigma^n(y) = N(x)y$ for any $y \in L$. Thus the minimal polynomial of H_x divides $t^n - N(x)$. The minimal polynomial of the linear operator T always divides its characteristic polynomial, which has degree n. Since $\lambda \in \bar{K}$, $L \to L \otimes_K \bar{K} \cong \Pi_{g \in G} \bar{K}$ so that we can work on this. We can compute the characteristic polynomial after extension of scalars $\begin{array}{rcl} L \otimes_K \bar{K} & \to & \Pi_{i=0}^{n-1} \bar{K} \\ a \otimes b & \mapsto & ((\sigma^i a) \cdot b)_i \end{array}$. This extends noncanonically to an automorphism of $\bar{K}$. The map $\sigma \otimes id : L \otimes_K \bar{K} : L \otimes_K \bar{K} \to L \otimes_K \bar{K}$ corresponds to the permutation of coordinates and there exists a map $\begin{array}{rcl} \mu_x \otimes id : L \otimes_K \bar{K} & \to & L \otimes_K \bar{K} \\ (y_0, \dots, y_{n-1}) & \mapsto & (xy_0, \dots, \sigma^{n-1} x y_{n-1}) \end{array}$ where μ_x denote the multiplication by x. Then for any eigenvalue $\lambda \in \bar{K}$ of H_x with corresponding eigenvector $(y_0, \dots, y_{n-1})$, $H_x(y) = (xy_1, (\sigma x) y_2, \dots, (\sigma^{n-1} x) y_0) = (\lambda y_0, \dots, \lambda y_{n-1}) \Leftrightarrow y_i = \frac{\lambda^i}{x \dots \sigma^{i-1} x} y_0$ i.e. all coordinates are uniquely determined by y_0, so $(y_0, \dots, y_{n-1}) = y_0 (1, \frac{\lambda}{x}, \dots, \frac{\lambda^{n-1}}{\Pi \sigma^i x})$. So the eigenspace has dimension one as desired and this is valid if $\frac{\lambda^n}{x \dots \sigma^{n-1} x} = \frac{\lambda^n}{N(x)} = 1$ which is indeed the case.

Definition 4.2.1. *A cochain complex X of abelian groups is a sequence* $\dots \longrightarrow X^{-1} \xrightarrow{d^{-1}} X^0 \xrightarrow{d^0} X^1 \xrightarrow{d^1} \dots$ *s.t. the differential satisfies $d^{i+1} d^i = 0$, $\forall i$.*

Definition 4.2.2. *A map f s.t. the diagram*

$$\begin{tikzcd} \dots \arrow[r] & X^{-1} \arrow[r, "d^{-1}"] & X^0 \arrow[r, "d^0"] \arrow[d, "f^0"] & X^1 \arrow[r] \arrow[d, "f^1"] & \dots \\ \dots \arrow[r] & Y^{-1} \arrow[r, "d^{-1}"] & Y^0 \arrow[r, "d^0"] & Y^1 \arrow[r] & \dots \end{tikzcd}$$

commutes is a map of complexes. Then f induces a map of cohomologies because both the kernel and image of the differentials in $X^\bullet$ are preserved in $Y^\bullet$ by commutativity. A map h as in the following diagram

$$\begin{tikzcd} \dots \arrow[r] & X^{-1} \arrow[r, "d^{-1}"] \arrow[d, "f^{-1}"] \arrow[dl] & X^0 \arrow[r, "d^0"] \arrow[d, "f^0"] \arrow[dl, "h^0"] & X^1 \arrow[r] \arrow[d, "f^1"] \arrow[dl, "h^1"] & \dots \\ \dots \arrow[r] & Y^{-1} \arrow[r, "d^{-1}"] & Y^0 \arrow[r, "d^0"] & Y^1 \arrow[r] & \dots \end{tikzcd}$$

s.t. $dh + hd = f$ is a nullhomotopy of f.

Definition 4.2.3. *The homotopy cokernel or cone hCoker(f) = Cone(f) has the universal property that maps of chain complexes hCoker$(f) \to Z$ are equivalent to*

$$\left\{ \begin{array}{r} Y \to Z \text{ a map} \\ X \to Z \text{ a nullhomotopy} \end{array} \right. \text{ s.t. } \begin{tikzcd} X \arrow[r] \arrow[d, "f"] & Z \\ Y \arrow[ur] & \end{tikzcd} \text{ commutes.}$$

CLAIM : If $f : X \to Y$ is a map of complexes, then hCoker(f), characterised by the universal property that maps hCoker$(f) \to Z$ of chain complexes are equivalent to $\left\{ \begin{array}{rr} g : Y \to Z & \textbf{a map} \\ h = g \circ f : Z \to Z & \textbf{a nullhomotopy} \end{array} \right.$ exist.

pf. hCoker(f) is $\ldots X^0 \oplus Y^{-1} \longrightarrow X^1 \oplus Y^0 \longrightarrow \ldots$ s.t. $\begin{array}{rcl} d : X^{i+1} \oplus Y^i & \to & X^{i+2} \oplus Y^i \\ \begin{bmatrix} x \\ y \end{bmatrix} & \mapsto & \begin{bmatrix} -dx \\ f(x) + dy \end{bmatrix} \end{array}$

$\to \begin{bmatrix} -d & 0 \\ f & d \end{bmatrix} \begin{bmatrix} -d & 0 \\ f & d \end{bmatrix} = \begin{bmatrix} d^2 & 0 \\ -fd + df & d^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ implies $d^2 = 0$.

Suppose we have a map hCoker$(f) \to Z$ s.t.

$$\begin{tikzcd} \ldots \arrow[r] & X^{i+1} \oplus Y^i \arrow[r] \arrow[d] & X^{i+2} \oplus Y^{i+1} \arrow[r] \arrow[d] & \ldots \\ \ldots \arrow[r] & Z^i \arrow[r] & Z^{i+1} \arrow[r] & \ldots \end{tikzcd}$$

commutes. If we write such a map as $(x, y) \mapsto h(x) + g(y)$, then $dh(x) + dg(y) = d(h(x) + g(y)) = h(-dx) + g(f(x) + dy) = -hdx + gfx + gdy$. Taking $x = 0$ gives $dg = gd$ so $hd + dh = g \circ f$ a nullhomotopy of $g \circ f$ as desired and the universal property satisfied.
$\to X \to Y \to$ hCoker(f) **is canonically null-homotpic.**

e.g. Let $\left\{ \begin{array}{l} X \equiv (\cdots \to 0 \to A \to 0 \to \ldots) \\ Y \equiv (\cdots \to 0 \to B \to 0 \to \ldots) \end{array} \right.$ for finite abelian groups A, B in degree 0. Let $f : A \to B$. Then hCoker$(f) = (\cdots \to A \to_f B \to 0 \to \ldots)$ with B in degree 0. Then $\left\{ \begin{array}{l} H^0 \text{hCoker}(f) = \text{Coker}(f) \\ H^{-1} \text{hCoker}(f) = \ker(f) \end{array} \right.$.

Lemma 4.2.2. *For all maps $f : X \to Y$, the sequence $H^i X \to H^i Y \to H^i hCoker(f)$ is exact for all i.*

pf. The composition is zero since $X \to Y \to$ hCoker(f) is nullhomotopic. Let $y \in Y^i$ s.t. $dy = 0$ and suppose its image in H^ihCoker(f) is zero so that $\begin{bmatrix} 0 \\ y \end{bmatrix} = \begin{bmatrix} -d & 0 \\ f & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -d\alpha \\ f(\alpha) + d\beta \end{bmatrix}$, $\exists \left\{ \begin{array}{r} \alpha \in X^i \text{ with } d\alpha = 0 \\ \beta \in Y^{i-1} \end{array} \right.$. Then $f(\alpha) + d\beta = y$ implies $f(\alpha) = y$ in $H^i Y$ so the exactness follows.

There is a notion of the homotopy kernel hKer(f) defined by the universal property that maps $Z \to$ hKer(f) are equivalent to $\begin{cases} Z \to X & \text{a map} \\ Z \to X \to Y & \text{nullhomotopy of the composition} \end{cases}$**. In particular, hKer($f$) = hCoker($f$)[−1].** $\to X \to_f Y \to \mathrm{hCoker}(f)$ is nullhomotopic so there exists a canonical map $X \to \mathrm{hKer}(Y \to \mathrm{hCoker}(f))$ where we say the latter term as the mapping cylinder.

Definition 4.2.4. *The coinvariants of A are $A_G \equiv A/\sum_{g\in G}(g-1)A$, which satisfies the universal property that it is the maximal quotient of A with $gx = x$ holding for all $x \in A,\ g \in G$.*

Definition 4.2.5. *Let X, Y be complexes of A-modules. Then the mapping complex $\underline{hom}_A(X,Y)$ is the complex of abelian groups defined by $\underline{\mathrm{hom}}^i_A(X,Y) \equiv \Pi_{j\in\mathbb{Z}} \mathrm{hom}_A(X^j, Y^{j+i})$, with differential $d^i f \equiv df - (-1)^i f d$. This can be visualised as*

$$\begin{tikzcd}
\dots \arrow[r] & \Pi_{i\in\mathbb{Z}} \mathrm{hom}(X^i, Y^{i-1}) \arrow[r, "h\mapsto dh+hd"] \arrow[d, "\sim"] & \Pi_{i\in\mathbb{Z}} \mathrm{hom}(X^i, Y^i) \arrow[r, "f\mapsto df-fd"] \arrow[d, "\sim"] & \Pi_{i\in\mathbb{Z}} \mathrm{hom}(X^i, Y^{i+1}) \arrow[r] \arrow[d, "\sim"] & \dots \\
& \underline{\mathrm{hom}}^{-1}(X,Y) & \underline{\mathrm{hom}}^{0}(X,Y) & \underline{\mathrm{hom}}^{1}(X,Y) &
\end{tikzcd}.$$

$\to \forall X, Y$ complexes of A-modules, $\underline{\mathrm{hom}}_A(X,Y)$ is a complex.

Definition 4.2.6. *A complex P of A-modules is projective if for every acyclic complex Y of A-modules, $\underline{\mathrm{hom}}_A(P,Y)$ is also acyclic.*

Definition 4.2.7. *An A-module P^i is projective as an A-module if any of the following holds*

1. $\mathrm{hom}_A(P^i, -)$ *preserves surjections*
2. $\mathrm{hom}_A(P^i, -)$ *is an exact functor*
3. *P^i is a direct summand of a free module*
4. *given any surjection $N \to M$, every map $P^i \to M$ of A-modules lifts to a map*

$$P^i \to N \text{ s.t. } \begin{tikzcd}
& N \arrow[d] \\
P^i \arrow[r] \arrow[ur, dotted] & M \arrow[d] \\
& 0
\end{tikzcd} \text{ commutes}$$

An A-module P is projective as an A-module iff it is projective as a complex in degree 0.

pf.

- if P is projective as a complex in degree 0, let $f : N \to M$ be a surjection and form the acyclic complex $X \equiv (\cdots \to 0 \to \ker f \to N \to M \to 0 \to 0)$. Then $\underline{\hom}_A(P,X)$ is $\cdots \to 0 \to \hom_A(P,\ker f) \to \hom_A(P,N) \to \hom_A(P,M) \to 0 \to \ldots$ so $\hom_A(P^i,-)$ preserves surjections and therefore P is projective as an A-module.

- if $X \equiv (\cdots \to X^1 \to X^0 \to X^1 \to \ldots)$ si an acyclic complex and P is projective as an A-module, then $\underline{\hom}_A(P,A) = (\cdots \to \hom_A(P,X^{-1}) \to \hom_A(P,A^0) \to \hom_A(P,X^1) \to \ldots)$ which is acyclic (if $X^{i-1} \to \ker(X^i \to X^{i+1})$ is a surjection, then $\hom(P,X^{i-1}) \to \hom(P,\ker(X^i \to X^{i+1})) = \ker(\hom(P,X^i) \to \hom(P,X^{i+1}))$ is a surjection since $\hom(P,-)$ is exact so it preserves kernels by assumption. Thus $\underline{\hom}_A(P,X)$ is also acyclic and P is projective as a complex in degree 0.

Lemma 4.2.3. *If P is a complex of A-modules with $P^i = 0$ for all $I \gg 0$ and P^i is projective as an A-module for all i, then P is projective as a complex.*

pf. Let Y be an acyclic complex of A-modules.

- **CLAIM : Every map $P \to Y$ is nullhomotopic.**
  pf. Let $f : P \to Y$. Construct a nullhomotopy h of f by descending induction. For $i \gg 0$, $P^i = 0$ so $f^i = 0 \to h^i \equiv 0$.
  Assume we have maps $h : P^j \to Y^{j-1}$, $\forall j > i$ so that
  $$\begin{tikzcd} \ldots \arrow[r, "d"] & P^{i-1} \arrow[r, "d"] \arrow[d, "f"] & P^i \arrow[r, "d"] \arrow[d, "f"] \arrow[dl, "?"] & P^{i+1} \arrow[r, "d"] \arrow[d, "f"] \arrow[dl, "h"] & \ldots \\ \ldots \arrow[r, "d"] & Y^{i-1} \arrow[r, "d"] & Y^i \arrow[r, "d"] & Y^{i+1} \arrow[r, "d"] & \ldots \end{tikzcd}$$
  . $\forall x \in P^i$, $(d(f-hd))(x) = ((df-(f-hd)d)(x) = (df-fd)(x) = 0$ by the inductive hypothesis. Since $(f-hd) : P^i \to \ker(d : Y^i \to Y^{i+1})$ by the above statement. There is a surjection $d : Y^{i-1} \to \ker(d : Y^i \to Y^{i+1})$. Therefore the map $f - hd$ lifts to a map $h : P^i \to Y^{i-1}$ s.t. $dh = f - hd$ as P^i is projective by hypothesis. Thus $hd + df = f$ and h defines a nullhomotopy of f for i.
  By induction, the claim follows.

Then $H^0(\underline{\hom}_A(P,A)) = \{P \to Y\} = 0$ by the previous claim and $H^i = (\underline{\hom}_A(P,Y)) = H^0(\underline{\hom}_A(P,Y[i])) = 0$ as $Y[i]$ is also acyclic for each i. Thus the cohomologies vanish for each i and therefore $\underline{\hom}_A(P,Y)$ is acyclic, i.e. P is projective.

Definition 4.2.8. *A projective resolution of X, an A-module complex, is $P \xrightarrow{qis} X$ where P is a projective complex.*

$\to \underline{\hom}_A^{der}(X,Y) \equiv \underline{\hom}_A(P,Y)$ for $P \xrightarrow{qis} X$ a projective resolution.

e.g. When $A \equiv \mathbb{Z}[G]$, choose any projective resolution $P \xrightarrow{qis} \mathbb{Z}$, and define $X^{hG} \equiv \underline{\hom}_G(P,X)$. Then if X is acyclic, X^{hG} is acyclic and X^{hG} commutes with cones. Also if $X = (\cdots \to 0 \to M \to 0 \to \ldots)$ is in degree 0 only, then $H^0(X^{hG}) = M^G$.

Proposition 4.2.1. *Let X be a complex of A-modules and suppose X is bounded above. Then there exits a projective resolution $P \xrightarrow{qis} X$.*

pf. Wlog, assume X is bounded above at index 0. Let P^0 be a free module surjecting onto X^0 via a map α^0. Then take P^{-1} to be a free module surjecting onto $P^0 \times_{X^0} X^{-1}$ so that

$$\begin{tikzcd} P^{-1} \arrow[r] \arrow[d, dotted] & P^0 \arrow[r] \arrow[d, "\alpha_0"] & 0 \arrow[r] \arrow[d] & \ldots \\ X^{-1} \arrow[r] & X^0 \arrow[r] & 0 \arrow[r] & \ldots \end{tikzcd}$$

commutes. This preserves cohomologies as $H^0X = X^0/im(X^{-1}) = P^0/P^{-1} = H^0P$ since P^{-1} surjects onto $\ker\alpha^0$ and has image in $P^0/\ker\alpha^0 \cong X^0$ equal to X^{-1} as $P^0 \to X^0$ is a surjection. Since $P^{-1} \to X^{-1}$ is a surjection, we may iterate the process to construct a projective resolution P of X as desired.

Lemma 4.2.4. *Suppose P_1, P_2 are projective resolutions of a complex X of A-modules. Then there exists a homotopy equivalence γ s.t.*

$$\begin{tikzcd} P_1 \arrow[rr, "\gamma"] \arrow[dr, "\alpha"'] & & P_2 \arrow[dl, "\beta"] \\ & X & \end{tikzcd}$$

commutes upto homotopy, i.e. $\beta\gamma \cong \alpha$.

pf. Consider the diagram

$$\begin{tikzcd} P_! \arrow[r, dotted, "\gamma"] \arrow[dr, "\alpha"] \arrow[ddr, "\delta"'] & P_2 \arrow[d, "\beta"] \\ & X \arrow[d] \\ & \mathrm{hCoker}(\beta) \end{tikzcd}$$

. Since β is a quasi-isomorphism, $\mathrm{hCoker}(\beta)$ is acyclic. Then the composition $\delta = P \to X \to \mathrm{hCoker}(\beta)$ is nullhomotopic, hence there is a canonical map $P_1 \xrightarrow{\gamma} \mathrm{hKer}(X \to \mathrm{hCoker}(\beta)) \cong P_2$ via homotopy equivalence. By symmetry, such a map exists in the opposite direction, hence γ is a homotopy equivalence and the diagram commutes upto homotopy.
$\to \gamma$ is all homotopic.

Definition 4.2.9. *The ith Ext group of two chain complexes of A-modules M,N is defined as $Ext^i_A(M,N) \equiv H^i\underline{\hom}(P,N)$, where P is some projective resolution of M.*

Define the derived functor $\underline{\hom}^{der}(X,-)$ via

$$\underline{\hom}^{der}(X,Y) \equiv \lim_{P \to_{qis} X, \text{projective}} \underline{\hom}(P,Y)$$

where P are ordered by maps of chain complexes $\begin{tikzcd} P' \arrow[r] \arrow[rr, bend right, "qis"'] & P \arrow[r, "qis"] & X \end{tikzcd}$ which forcibly removes the choice P.

CLAIM : Suppose we have a map of chain complexes $f : X_1 \to X_2$, which have projective resolutions P_1, P_2 respectively. Then we have a map $\phi : P_1 \to P_2$ s.t. the diagram below commutes. Moreover such a map is unique upto homotopy.

$$\begin{tikzcd} X_1 \arrow[r, "f"] & X_2 \\ P_1 \arrow[u, "qis"] \arrow[r, dotted, "\phi"] & P_2 \arrow[u, "qis"'] \end{tikzcd}$$

pf. Since the derived functor preserves quasi-isomorphisms, the induced map of complexes of maps $\underline{\hom}(P_1, P_2) \to \underline{\hom}(P_1, X_2)$ is a quasi-isomorphism. So there exists a map $P_1 \to X_1 \to X_2$, which is killed by the differential since it is a map of chain complexes. Thus it defines a cohomology class in $H^0\underline{\hom}(P_1, X_2)$. This implies there is some cohomology class in $H^0\underline{\hom}(P_1, P_2)$ which is a lift of the map through P_2.
$\to$ for every chain complex Y, $\underline{\hom}_A^{der}(X_2, Y) \to \underline{\hom}_A^{der}(X_1, Y)$ by pullbacking along ϕ.

CLAIM : If $H \subseteq G$ is a subgroup and X is a complex of G-modules, then we get a restriction map $X^{hG} \to X^{hH}$.
pf. Consider $\mathbb{Z}[G/H] = \{f : G/H \to \mathbb{Z} | f \text{ nonzero at finitely many points}\}$, which has a G-action via translations and is equivalent to the induced module from H to G by the trivial module $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} \mathbb{Z}$. Let $P_H \to \mathbb{Z}$ be a projective resolution for $\mathbb{Z}$ as a H-modules. Then we have an induced G-module $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} P_H$. Then $\mathbb{Z}[G]$ is free as a $\mathbb{Z}[H]$-module, as choosing coset representatives for G/H yields a basis. Therefore $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} -$ preserves quasi-isomorphisms. This implies $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} P_H \to \mathbb{Z}[G/H]$ is a projective resolution for $\mathbb{Z}[G/H]$ as a $\mathbb{Z}[G]$-modules. This is because both $\mathbb{Z}[G]$ and P_H are bounded so it will be bounded and inducing upto $\mathbb{Z}[G]$ preserves projective modules as we will obtain a direct summand of a free module. Then $\underline{\hom}_G^{der}(\mathbb{Z}[G/H], X) \equiv \underline{\hom}_G(\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} P_H, X) = \underline{\hom}_H(P_H, X) \equiv X^{hH}$ by definition. This implies $\exists \begin{array}{ccc} \epsilon : \mathbb{Z}[G/H] & \to & \mathbb{Z} \\ \sum_{g_i \in G/H} n_i g_i & \mapsto & \sum n_i \end{array}$ gives a map $X^{hG} \to X^{hH}$, which is clearly G-equivariant when we equip $\mathbb{Z}$ with the trivial action. Then $X^{hG} = \underline{\hom}_G^{der}(\mathbb{Z}, X) \to \underline{\hom}_G^{der}(\mathbb{Z}[G/H], X) = X^{hH}$ is well-defined upto homotopy.

If $H \leq G$ is finite index, we have a G-equivariant map $\mathbb{Z} \xrightarrow{\kappa} \mathbb{Z}[G/H] \xrightarrow{\epsilon} \mathbb{Z}$ defined as $1 \mapsto \sum_{g \in G/H} g$. Then it corresponds to multiplication by the index

$[G:H]$, which gives an **inflation map** $X^{hH} \to X^{hG}$ s.t. the composition $X^{hG} \to X^{hH} \to X^{hG}$ is homotopic to multiplication by $[G:H]$.

Definition 4.2.10. *$H^i(G,X) \equiv H^i(X^{hG})$ is the hypercohomology of G with coefficients in X.*

CLAIM : If X is only in nonnegative degrees, then $H^0(G,X) = H^0(X)^G$ and $H^i(G,X) = 0$, $\forall i < 0$.

pf. Choose any projective resolution P of $\mathbb{Z}$ as a G-module where it has only nonpositive degrees only. By definition, $H^0(G,X)$ is equivalent to the homotopy classes of maps s.t. $f : P \to X$

$$\begin{tikzcd}
\dots \arrow[r] & P^{-2} \arrow[r] \arrow[d] & P^{-1} \arrow[r, "d"] \arrow[d] & P^0 \arrow[r] \arrow[d, "f"] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \dots \\
\dots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & X^0 \arrow[r, "d"] & X^1 \arrow[r] & X^2 \arrow[r] & \dots
\end{tikzcd}$$

commutes. Then any homotopy of f is 0 and $H^i(G,X) = 0$, $\forall i < 0$. By commutativity, $df = fd = 0$ and therefore such maps f are equivalent to G-equivariant maps $\mathbb{Z} = P^0/dP^{-1} = \mathrm{Coker}(P^{-1} \to P^0) \to \ker(X^0 \to X^1) = H^0(X)$ by quasi-isomorphism. That is equivalent to a G-invariant vector in the cohomology $H^0(X)$.

- **A Projective Resolution of $\mathbb{Z}$ as a G-module**

  Let $G \equiv \mathbb{Z}/n\mathbb{Z}$ with generator σ. If M is a G-module, then M^{hG} is quasi-isomorphic to the complex

  $$\dots \longrightarrow 0 \longrightarrow 0 \longrightarrow M \xrightarrow{1-\sigma} M \xrightarrow{\Sigma\sigma^i} M \xrightarrow{1-\sigma} \dots$$

  where the first M is in degree 0.

  Definition 4.2.11. *Let A be a commutative ring, and B an A-algebra. Then the bar complex $Bar_A(B)$ is*

  $$\cdots \to B \otimes_A B \otimes_A B \xrightarrow{b_1 \otimes b_2 \otimes b_3 \mapsto b_1 b_2 \otimes b_3 - b_1 \otimes b_2 b_3} B \otimes_A B \xrightarrow{b_1 \otimes b_2} B \to 0 \to \dots$$

  with B in degree 0. In general, $Bar_A^{-n}(B) \equiv B^{\otimes_A n+1}$ with differential $b_1 \otimes \cdots \otimes b_{n+1} \mapsto b_1 b_2 \otimes b_3 \otimes \cdots \otimes b_{n+1} - b_1 \otimes b_2 b_3 \otimes \cdots \otimes b_{n+1} + \dots$.

  $\to$ homotopy equivalent to zero.

  pf.
  $$\begin{tikzcd}
  \dots \arrow[r] & B \otimes_A B \otimes_A B \arrow[r] \arrow[d, "id"] \arrow[dl, "h^2"] & B \otimes_A B \arrow[r] \arrow[d, "id"] \arrow[dl, "h^1"] & B \arrow[r] \arrow[d, "id"] \arrow[dl, "h^0"] & 0 \arrow[r] \arrow[d] \arrow[dl] & \dots \\
  \dots \arrow[r] & B \otimes_A B \otimes_A B \arrow[r] & B \otimes_A B \arrow[r] & B \arrow[r] & 0 \arrow[r] & \dots
  \end{tikzcd}$$

  Define

$\begin{cases} h^0(b) \equiv 1 \otimes b \\ h^1(b_1 \otimes b_2) \equiv 1 \otimes b_1 \otimes b_2 \end{cases}$. Then $(dh^1 + h^0 d)(b_1 \otimes b_2) = d(1 \otimes b_1 \otimes b_2) + 1 \otimes b_1 b_2 = b_1 \otimes b_1 - 1 \otimes b_1 b_2 + 1 \otimes .b_1 b_2 = b_1 \otimes b_2$. We can construct h^n in the similar fashion, and such h is a nullhomotopy we wanted.

We can rewrite the diagram as
$$\begin{tikzcd} \dots \arrow[r] & B \otimes_A B \otimes_A B \arrow[r] \arrow[d] & B \otimes_A B \arrow[r] \arrow[d, "d"] & 0 \arrow[r] \arrow[d] & \dots \\ \dots \arrow[r] & 0 \arrow[r] & b \arrow[r] & 0 \arrow[r] & \dots \end{tikzcd}$$
where d is the multiplication map in the differential. Then its cone is $\mathrm{Bar}_A(B)$ and therefore it is a homotopy equivalent. Given a B-module M, we can tensor with M over B, which yields a diagram
$$\begin{tikzcd} \dots \arrow[r] & B \otimes_A B \otimes_A M \arrow[r] \arrow[d] & B \otimes_A M \arrow[r] \arrow[d, "b \otimes m \mapsto bm"] & 0 \arrow[r] \arrow[d] & \dots \\ \dots \arrow[r] & 0 \arrow[r] & M \arrow[r] & 0 \arrow[r] & \dots \end{tikzcd}$$
that is also a homotopy equivalence. The differentials are the same except the last term is replaced with an element of m. In words, M is canonically equivalent to a complex where every term is of the form $B \otimes_A N$, where N stands for $B \otimes_A \cdots \otimes_A \otimes_A M$.

Let $A = \mathbb{Z}$, $B \equiv \mathbb{Z}[G]$, $M \equiv \mathbb{Z}$. Then $\begin{matrix} B \otimes_A M & \to & M & \to & B \otimes_A M \\ b \otimes m & \mapsto & bm & \mapsto & 1 \otimes bm \end{matrix}$ s.t. $1 \otimes bm \neq b \otimes m = b(1 \otimes m)$ since the action is on B not on M and it is not an action of B-modules. Thus we obtain a canonical projective resolution of $\mathbb{Z}$ by free G-modules
$$\begin{tikzcd} \dots \arrow[r] & \mathbb{Z}[G^3] \arrow[r] \arrow[d] & \mathbb{Z}[G \times G] \arrow[r] \arrow[d] & \mathbb{Z}[G] \arrow[r] \arrow[d, "\epsilon"] & 0 \arrow[r] \arrow[d] & \dots \\ \dots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \mathbb{Z} \arrow[r] & 0 \arrow[r] & \dots \end{tikzcd}$$
with differentials $(g_1, g_2) \mapsto g_1 g_2 - g_1$ and so forth, since G acts trivially on $\mathbb{Z}$. Here $\mathbb{Z}[G \times G] \cong \mathbb{Z}[G] \otimes_{\mathbb{Z}} \mathbb{Z}[G]$ since both have a basis by the elements of the product group. Then we have a complex of the form

$$\cdots \to 0 \to M \to \mathbb{Z}[G] \otimes M \to \mathbb{Z}[G \times G] \otimes M \to \dots$$

with M in degree 0 and G finite.

Definition 4.2.12. *Elements in $\mathbb{Z}[G] \otimes M$ in the kernel of the differential are called group n-cocycles with coefficients in M; elements in the image of differential are called n-coboundaries.*

Denote P_G^{can} for the resolution we defined. Then $X^{hG} \cong \underline{\hom}_G(P_G^{can}, X)$.

For every G-module M, $\cdots \to 0 \to M \xrightarrow{m \mapsto (gm-m)} \Pi_{g\in G} M \to \Pi_{g,h\in G} M \to \dots$ where $\{\phi : G \to M \in \Pi_{g\in G} M$. Then ϕ is killed by this differential if it i s a 1-cocycle, that is $\phi(gh) = \phi(g) + g\cdot\phi(h)$, $\forall g,h \in G$ via the group acton. Similarly ϕ is a 1-coboundary if $\exists m \in M$ s.t. $\phi(g) = g\cdot m - m$, $\forall g \in G$. Then

$$H^1(G,M) \equiv H^1(M^{hG}) = \{\text{1-cocycles}\}/\{\text{1-coboundaries}\}$$

If M is a G-module, $M_G = M \otimes_{\mathbb{Z}[G]} \mathbb{Z}$. Define $I_G \equiv \ker\epsilon$ so that we have a s.e.s.

$$0 \longrightarrow I_G \longrightarrow \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \longrightarrow 0$$

where $\begin{array}{ccc} \epsilon : \mathbb{Z}[G] & \to & \mathbb{Z} \\ \sum n_i g_i & \mapsto & \sum n_i \end{array}$.

Definition 4.2.13. *If X,Y are chain complexes, then*

$$(X\otimes Y)^i \equiv \bigoplus_{j\in\mathbb{Z}} X^j \otimes Y^{i-j}, \quad d(x\otimes y) \equiv dx\otimes y + (-1)^j x \otimes dy$$

If X is a complex of right A-modules, and Y is a complex of left A-modules, then $X\otimes_A Y$ is defined similarly.

Definition 4.2.14. *A complex F of left A-modules is flat if for every acyclic complex Y of right A-modules, $Y\otimes_A F$ is also acyclic, i.e. $-\otimes_A F$ preserves injections.*

CLAIM : Any project complex is flat.

pf.

- F is in degree 0 only and F^0 is flat.

  For every complex $Y = Y^\bullet$, $\cdots \to Y^i \otimes_A F \xrightarrow{d^i \otimes id} Y^{i+1} \otimes_A F \to \dots$ for $Y\otimes_A F$. Since F is flat, $H^i(Y\otimes_A F) = H^i(Y)\otimes_A(F)$, $\forall i$, so if Y is acyclic, so is $Y\otimes_A F$.

- F is in degree 0 and -1 only and F^0, F^{-1} are flat.

  Then $F^\bullet = \mathrm{hCoker}(F^{-1} \to F^0)$. Then since tensor products commute with homotopy cokernels, $Y\otimes_A F = \mathrm{hCoker}(Y\otimes_A F^{-1} \to Y\otimes_A F^0)$ so by the previous case, if Y is acyclic, so are $Y\otimes_A F^0, Y\otimes_A F^{-1}$. Thus $Y\otimes_A F$ is as well by l.e.s. on cohomology. Similarly, for any bounded F, if F^I are flat for all i, so is F.

- General

$$\begin{tikzcd}
F_0 \arrow[d] & \cdots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & F^0 \arrow[r] \arrow[d, "id"] & 0 \arrow[r] \arrow[d] & \cdots \\
F_1 \arrow[d] & \cdots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & F^1 \arrow[r, "d"] \arrow[d, "id"] & F^0 \arrow[r] \arrow[d, "id"] & 0 \arrow[r] \arrow[d] & \cdots \\
F_2 \arrow[d] & \cdots \arrow[r] & 0 \arrow[r] \arrow[d] & F^2 \arrow[r, "d"] \arrow[d, "id"] & F^1 \arrow[r, "d"] \arrow[d, "id"] & F^0 \arrow[r] \arrow[d, "id"] & 0 \arrow[r] \arrow[d] & \cdots \\
\vdots & \cdot & \cdots & \cdots & \cdots & \cdots & \cdots & \cdot
\end{tikzcd}$$

Then all squares in the diagram commutes, hence there are all morphisms of complexes and $F = \lim F_i$. Since direct limits commute with tensor products, $Y \otimes_A F = \lim Y \otimes_A F_i$. Then by the previous case, $Y \otimes_A F_i$ is acyclic $\forall i$. Since cohomology commutes with direct limits, if Y is acyclic, so is $Y \otimes_A F$.

$\rightarrow$ for any A-modules Y, choose a projective resolution $F \rightarrow Y$. Then since F is projective, F is flat. Define $Y \otimes_A^{der} X \equiv F \otimes_A X$, which is well-defined upto quasi-isomorphism.

Definition 4.2.15. *The ith torsion group of Y against X is*

$$Tor_i^A(Y,X) \equiv H^{-i}(Y \otimes_A^{der} X)$$

Definition 4.2.16. *The homotopy coinvariants of a chain complex X is the complex $X_{hG} \equiv X \otimes_{\mathbb{Z}[G]}^{der} \mathbb{Z} \cong X \otimes_{\mathbb{Z}[G]} P_G$.*

$\rightarrow$ if X is acyclic, so is X_{hG} and it commute with cones. Also $H^0(X_{hG}) = M_G = \mathbb{Z} \otimes_{\mathbb{Z}[G]} M = M/\sum_{g \in G}(g-1)M$.

Definition 4.2.17. $H_i(G,X) \equiv H^{-i}(X_{hG})$.

CLAIM : $H_1(G,\mathbb{Z}) = G^{ab}$ where G^{ab} denotes the abelianisation of G.

pf. $0 \rightarrow I_G \rightarrow \mathbb{Z}[G] \rightarrow_\epsilon \mathbb{Z} \rightarrow 0$ is a s.e.s. which gives a l.e.s. on cohomologies $H_1(G,\mathbb{Z}[G]) \rightarrow H_1(G,\mathbb{Z}) \rightarrow H_0(G,I_G) \rightarrow H_0(G,\mathbb{Z}[G]) \rightarrow H_0(G,\mathbb{Z})$. Then $H_0(G,\mathbb{Z}[G]) = H^0(\mathbb{Z}[G])_G = \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} \mathbb{Z} = \mathbb{Z}$ and $H_1(G,\mathbb{Z}[G]) = 0$ since $\mathbb{Z}[G]_{hG} \equiv \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} P_G = P_G \cong \mathbb{Z}$ is a quasi-isomorphism. Thus the above exact sequence becomes $0 \rightarrow H_1(G,\mathbb{Z}) \rightarrow H_0(G,I_G) \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}$. We have an isomorphism $H_0(G,I_G) \cong H_1(G,\mathbb{Z}) = (I_G)_G = I_G/I_G^2$ since $M_G = M/I_G \cdot M$. Then $\begin{array}{ccc} \mathbb{Z}[G]/I_G^2 & \rightarrow & G^{ab} \otimes \mathbb{Z} \\ g & \mapsto & (\bar{g},1) \end{array}$: $\forall [g] \in \mathbb{Z}[G]$, $[g]+[h] \mapsto (\bar{g}\bar{h},2)$

$\begin{array}{l} g \mapsto (\bar{g},1) \\ h \mapsto (\bar{h},1) \end{array}$, $\forall g,h \in G$, which is a homomorphism. It has an inverse induced by the map $\begin{array}{ccc} G \otimes \mathbb{Z} & \rightarrow & \mathbb{Z}[G]/I_G^2 \\ (g,n) & \mapsto & [g]+n-1 \end{array}$, a homomorphism $([g]-1)([h]-1) = [gh]-[g]-[h]+1 \in I_G^2 \rightarrow ([g]-1)+([h]-1) \equiv [gh]-1 \mod I_G^2$. Thus the claim follows.

4.2.2 Tate Cohomologies

Define the norm map

$$\begin{array}{ccccccc} & N: X_{hG} & & & \to & & X^{hG} \\ X_{hG} = X \otimes_{\mathbb{Z}[G]} P_G & \to X \otimes_{\mathbb{Z}[G]} \mathbb{Z} \to & \underline{\hom}_{\mathbb{Z}[G]}(\mathbb{Z}, X) & \to & & \underline{\hom}_{\mathbb{Z}[G]}(P_G, X) = & X^{hG} \end{array}$$

. Then set

$$X^{tG} \equiv \mathrm{hCoker}(N : X_{hG} \to X^{hG})$$

$\to$ the **Tate cohomology** is defined as $\hat{H}^i(G,X) \equiv H^i(X^{tG})$ and X^{tG} is said to be the **Tate complex**.

For the Tate cohomologies for modules, suppose G acts on an A-module M. Then giving a map $f : N \to M^G$ for an abelian group N is the same as giving a map $f : N \to M$ s.t. $g \cdot f(x) = f(x)$, $\forall g \in G$. Dually, giving a map $f : M_G \to N$ is the same as giving a map $f : M \to N$ s.t. $f(g \cdot x) = f(x)$. Then $\begin{array}{ccc} M & \xrightarrow{N \equiv \sum g} & M \\ \downarrow & & \uparrow \\ M_G & \xrightarrow{N} & M^G \end{array}$ given $\begin{cases} N(g \cdot x) = N(x) \\ g \cdot N(x) = N(x) \end{cases}$. If $\#G$ is invertible in M, the norm map is an isomorphism. For G-modules, we have

$$X_{hG} = P \otimes_{\mathbb{Z}[G]} X \to \mathbb{Z} \otimes_{\mathbb{Z}[G]} X \to \underline{\hom}_G(\mathbb{Z}, X) \to \underline{\hom}_G(P, X) = X^{hG}$$

where the last map is via pullback and the middle map is the norm map applied termwise via the norm on modules. This map gives a map we will call $N : X_{hG} \to X^{hG}$.

Proposition 4.2.2. *Let M be a G-module, thought of as a complex in degree 0.*

Then
$$\begin{array}{lll} i) & \hat{H}^i(G,M) = H^i(G,M) & i \geq 1 \\ ii) & \hat{H}^i(G,M) = M^G/N(M) & i = 0 \\ iii) & \hat{H}^i(G,M) = \ker N/(g-1) & i = -1 \\ iv) & \hat{H}^i(G,M) = H_{-i-1}(G,M) & i \leq -2 \end{array}.$$

pf. $M_{hG} \to M^{hG} \to M^{tG} = \mathrm{hCoker}(N)$ yields a l.e.s. on cohomology

$$\cdots \to H_{-i}(M_{hG}) \to H^i(M^{hG}) \to \hat{H}^i(G,M) \to H_{-i-1}(M_{hG}) \to \dots$$

- $i \geq 1$

  $H_{-i}(M_{hG}), H_{-i-1}(M_{hG})$ vanish, so $H^i(G,M) \cong \hat{H}^i(G,M)$ by exactness.

- $i = 0, -1$

$H_{-1}(M_{hG}), H^{-1}(G,M)$ vanish, so $\exists$ a s.e.s. $0 \to \hat{H}^{-1}(G,M) \to M_G \to M^G \to \hat{H}^0(G,M) \to 0$. Thus ii and iii follow.

- $i \leq -2$

  $H^{-i-1}(G,M), H^{-i}(G,M)$ vanish for $i \geq 1$. Thus $\hat{H}^i(G,M) \cong H_{-i-1}(G,M)$ by exactness.

Given a s.e.s. of G-modules $0 \to M \to E \to N \to 0$, we have l.e.s.

$$\begin{cases} \hat{H}^0(G,M) \to \hat{H}^0(G,E) \to \hat{H}^0(G,N) \to_\delta \hat{H}^1(G,M) \to \dots \\ \hat{H}^1(G,M) \to \hat{H}^1(G,E) \to \hat{H}^1(G,N) \to_\partial \hat{H}^0(G,M) \to \dots \end{cases}, \text{ where}$$

$$\begin{cases} \delta & \text{lifts } x \in \hat{H}^0(N) = N^G/N(N) \text{ to } \tilde{x} \in E \text{ s.t. } (1-\sigma)\tilde{x} \in \ker N \\ \partial & \text{lifts } x \in \hat{H}^1(G,N) \text{ to } \tilde{x} \in E \text{ s.t. } N\tilde{x} = \sum \sigma^i \tilde{x} \in M^G \end{cases}.$$

Lemma 4.2.5. *Suppose* $0 \longrightarrow X^0 \xrightarrow{d^1} X^1 \xrightarrow{d^2} \dots \xrightarrow{d^{n-1}} X^{n-1} \xrightarrow{d^n} X^n \longrightarrow 0$ *is exact and* $\forall X^i$ *are finite. Then*

$$\#X^0 \cdot \#X^2 \cdots = \#X^1 \cdot \#X^3 \dots$$

pf.

- $n = 1$

  $0 \to X^0 \to X^1 \to 0$ is exact so $X^0 \cong X^1$ and therefore $\#X^0 = \#X^1$.

- $n > 1$, even

  We can separate the exact sequence into two $\begin{cases} 0 \to X^0 \to \dots \to X^{n-1} \to imd^{n-1} \to 0 \\ 0 \to imd^{n-1} \to X^{n-1} \to X^n \to 0 \end{cases}$.
  Then $\#X^0 \cdot \#X^2 \dots \#X^n = \#X^0 \dots \#X^{n-2} \cdot \frac{\#X^{n-1}}{\#imd^{n-1}} = \#X^1 \cdot \dots \#imd^{n-1} \cdot \frac{\#X^{n-1}}{\#imd^{n-1}} = \#X^1 \dots \#X^{n-1}$ by inductive hypothesis.

- similarly for $n > 1$, odd

Definition 4.2.18. *The Herbrand quotient or Euler characteristic of a G-module M is*

$$\chi(M) \equiv \frac{\#\hat{H}^0(G,M)}{\#\hat{H}^1(G,M)}$$

which is only defined when both are finite.

Lemma 4.2.6. *Let* $0 \to M \to E \to N \to 0$ *be a s.e.s. of G-modules. If* χ *is defined for two of the three G-modueles, then it is defined for all three, in which case* $\chi(M) \cdot \chi(N) = \chi(E)$.

pf. Construct a l.e.s. $0 \to \ker\alpha \to \hat{H}^0(M) \to_\alpha \hat{H}^0(E) \to \hat{H}^0(N) \to_\delta \hat{H}^1(M) \to \hat{H}^1(E) \to_\beta \hat{H}^1(N) \to \mathrm{Coker}(\beta) \to 0$. Since $\hat{H}^1(E) \to_\beta \hat{H}^1(N) \to_\partial \hat{H}^0(M) \to \hat{H}^0(E)$ is an exact sequence, $\ker\alpha = im\partial = \hat{H}^1(N)/\ker\partial = \hat{H}^1(N)/im\beta = \mathrm{Coker}(\beta)$. Then by the previous lemma we get the result.

Theorem 4.2.1 (Hilbert's Theorem 90)**.** *If L/K is a cyclic extension and $x \in L^\times$ with $N(x) = 1$, then $x = y/\sigma y$, $,\exists y \in L^\times$.*

pf. Let $x \in L$ s.t. $N(x) = 1$. Then the characteristic polynomial of $\hat{H}_x$ is $t^n - 1$, implying 1 is a root and hence an eigenvalue of H_x. Thus $\ker((H_x - 1) \otimes_K \bar{K}) \neq 0$. Since tensor products commute with taking kernels for fields, $\ker(H_x - 1) \neq 0$. Thus $\exists y \in L^\times$ s.t. $H_x(y) = x \cdot \sigma y = y \Leftrightarrow x = y/\sigma y$.
$\to \hat{H}^1(G, K^\times) = 0$.

e.g. $L \equiv \mathbb{Q}(i)$, $K = \mathbb{Q}$. Choose $x \in \mathbb{Q}(i)$ with $N(x) = 1$. then $x = a/c + (b/c)i$, $a, b, c \in \mathbb{Z}$ satisfying $a^2 + b^2 = c^2$. Then by the Hilbert theorem 90, $(r - s)^2 + (2rs)^2 = (r + s)^2$.

4.2.3 Local Class Field Theory

Previously, constructed $H_1(G, \mathbb{Z}) \cong G^{ab}$ by followings;
$\mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} P \cong \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} \mathbb{Z} = \mathbb{Z}$ is a quasi-isomorphism, given $P \to_{qis} \mathbb{Z}$ being a projective resolution. All of the lower homology groups vanish since $\mathbb{Z}$ is in degree 0. Thus $\exists$ a s.e.s. $0 \to I_G \to \mathbb{Z}[G] \to_\epsilon \mathbb{Z} \to 0$ where $\begin{array}{rcl} \epsilon : \mathbb{Z}[G] & \to & \mathbb{Z} \\ g & \mapsto & 1 \end{array}$ and $I_g = \ker\epsilon$.
This gives a l.e.s. on cohomologies, $H_1(G, \mathbb{Z}[G]) \to H_1(G, \mathbb{Z}) \to I_G/I_G^2 \to \mathbb{Z} \to \mathbb{Z} \to 0$ where $H_1(G, \mathbb{Z}[G]) = 0$ and $\mathbb{Z} \to \mathbb{Z}$ is an isomorphism. Then since I_G/I_G^2 is abelian, $G \to I_G/I_G^2$ factors through G^{ab},

$$\begin{tikzcd} G \arrow[rr, "g \mapsto g-1"] \arrow[dr] & & I_G/I_G^2 \\ & G^{ab} \arrow[ur, dotted] & \end{tikzcd}$$

is a homomorphism modulo I_G^2. For the inverse $I_G/I_G^2 \hookrightarrow \mathbb{Z}[G]/I_G^2 \to G^{ab}$ where $\sum_g n_g g \in \mathbb{Z}[G]$ maps to Πg^{n_g}.

Definition 4.2.19. *K^{unr} is the p-adic completion of the maximal unramified extension of $K \subseteq \bar{K}$.*

e.g.

- $K \equiv \mathbb{F}_q((t)) \to K_n = \mathbb{F}_{q^n}((t))$ is the nth unramified extension of K. Thus the maximal unramified extension of K is $\cup_{n \geq 1} \mathbb{F}_{q^n}((t)) \subseteq \bar{\mathbb{F}}_q((t)) = K^{unr}$
- $K \equiv \mathbb{Q}_p$. Then the maximal unramified extension $K^{unr} = W(\bar{\mathbb{F}}_p)[1/p]$ where $W(-)$ denotes the ring of Witt vectors.

Thus the basic structure of K^{unr} is a local field with residue field $\bar{\mathbb{F}}_q$. Let π be a uniformiser of K, and $\mathcal{O}_{K^{unr}}$ be the π-adic completion of $\cup_n \mathcal{O}_{K_n}$. Then $\exists$ a s.e.s. $0 \to \mathcal{O}_{K^{unr}}^\times \to K^{unr,\times} \to \mathbb{Z} \to 0$ where $v : K^{unr,\times} \to \mathbb{Z}$ is the normalised valuation on K^{unr}. Define $\mathrm{Gal}(K^{unr}/K) \equiv \mathrm{Aut}_{cts}(K^{unr}/K)$ where the latter is the set of continuous K-automorphisms of K^{unr}. Let k denote the residue field of K and σ denote the Frobenius element of $\mathrm{Gal}(\bar{k}/k)$. Then $\begin{array}{ccc} \mathrm{Gal}(K^{unr}/K) \cong \mathrm{Gal}(\bar{k}/k) & \to & \hat{\mathbb{Z}} \\ \sigma & \mapsto & 1 \end{array}$ is an isomorphism. Since $K = \{x \in K^{unr} : x = \sigma x\}$, there exists a resolution $0 \to K \to K^{unr} \to_{1-\sigma} K^{unr}$ and further $0 \to K^\times \to K^{unr,\times} \to_{x \mapsto x/\sigma x} K^{unr,\times} \to \mathbb{Z} \to 0$.

Theorem 4.2.2 (Main Theorem of Cohomological LCFT)**.** *Let L/K be an extension of nonarchimedean local fields with finite Galois group G. Then*

$$(L^\times)^{tG} \cong (\mathbb{Z}[-2])^{tG}$$

e.g. L/K unramified extension of local fields with $G \equiv \mathrm{Gal}(L/K) = \mathbb{Z}/n\mathbb{Z}$ cyclic. Then $\hat{H}^0(G, \mathcal{O}_L^\times) = 0$ i.e. $N : \mathcal{O}_L^\times \to \mathcal{O}_K^\times$ is surjective. Also the Herbrand quotient is $\chi(\mathcal{O}_L^\times) = 1 \to \hat{H}^1(G, \mathcal{O}_L^\times) = 0$, i.e. $(\mathcal{O}_L^\times)^{tG} \cong 0$ is a quasi-isomorphism. Then $\exists$ a s.e.s. $0 \to \mathcal{O}_L^\times \to L^\times \to \mathbb{Z} \to 0$ where $v : L^\times \to \mathbb{Z}$ is the normalised valuation. Taking Tate cohomology gives a quasi-isomorphism $\mathbb{Z}^{tG}[-2] \cong (L^\times)^{tG} \cong \mathrm{hCoker}(0 \to (L^\times)^{tG}) \cong \mathrm{hCoker}((\mathcal{O}_L^\times)^{tG} \to (L^\times)^{tG}) \cong \mathbb{Z}^{tG}$. Thus $(L^\times)^{tG}$ is 2-periodic.

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